6 Control Charts for Variables

Chapter Outline

6.1 INTRODUCTION

6.2 CONTROL CHARTS FOR $\bar{x}$ AND R

- 6.2.1 Statistical Basis of the Charts
- 6.2.2 Development and Use of $\bar{x}$ and R Charts
- 6.2.3 Charts Based on Standard Values
- 6.2.4 Interpretation of $\bar{x}$ and R Charts
- 6.2.5 The Effect of Non-normality on $\bar{x}$ and R Charts
- 6.2.6 The Operating-Characteristic Function
- 6.2.7 The Average Run Length for the $\bar{x}$ Chart

6.3 CONTROL CHARTS FOR $\bar{x}$ AND s

- 6.3.1 Construction and Operation of $\bar{x}$ and s Charts
- 6.3.2 The $\bar{x}$ and s Control Charts with Variable Sample Size
- 6.3.3 The s^2 Control Chart

6.4 THE SHEWHART CONTROL CHART FOR INDIVIDUAL MEASUREMENTS

6.5 SUMMARY OF PROCEDURES FOR $\bar{x}$, R, AND s CHARTS

6.6 APPLICATIONS OF VARIABLES CONTROL CHARTS

Supplemental Material for Chapter 6

S6.1 s^2 IS NOT ALWAYS AN UNBIASED ESTIMATOR OF σ^2

S6.2 SHOULD WE USE d_2 OR d_2^* IN ESTIMATING σ VIA THE RANGE METHOD?

S6.3 DETERMINING WHEN THE PROCESS HAS SHIFTED

S6.4 MORE ABOUT MONITORING VARIABILITY WITH INDIVIDUAL OBSERVATIONS

S6.5 DETECTING DRIFTS VERSUS SHIFTS IN THE PROCESS MEAN

S6.6 THE MEAN SQUARE SUCCESSIVE DIFFERENCE AS AN ESTIMATOR OF σ^2

The supplemental material is on the textbook Website www.wiley.com/college/montgomery.

Chapter Overview and Learning Objectives

A quality characteristic that is measured on a numerical scale is called a **variable.** Examples include dimensions such as length or width, temperature, and volume. This chapter presents Shewhart control charts for these types of quality characteristics. The $\bar{x}$ and R control charts are widely used to monitor the mean and variability of variables. Several variations of the $\bar{x}$ and R charts are also given, including a procedure to adapt them to individual measurements. The chapter concludes with typical applications of variables control charts.

After careful study of this chapter you should be able to do the following:

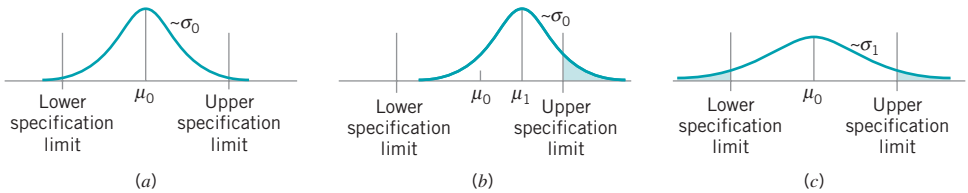
1. Understand the statistical basis of Shewhart control charts for variables
2. Know how to design variables control charts
3. Know how to set up and use $\bar{x}$ and R control charts
4. Know how to estimate process capability from the control chart information
5. Know how to interpret patterns on $\bar{x}$ and R control charts
6. Know how to set up and use $\bar{x}$ and s or s^2 control charts
7. Know how to set up and use control charts for individual measurements
8. Understand the importance of the normality assumption for individuals control charts and know how to check this assumption
9. Understand the rational subgroup concept for variables control charts
10. Determine the average run length for variables control charts

6.1 Introduction

Many quality characteristics can be expressed in terms of a **numerical measurement.** For example, the diameter of a bearing could be measured with a micrometer and expressed in millimeters. A single measurable quality characteristic, such as a dimension, weight, or volume, is called a **variable.** Control charts for variables are used extensively. Control charts are one of the primary tools used in the analize and control steps of DMAIC.

When dealing with a quality characteristic that is a variable, it is usually necessary to monitor both the mean value of the quality characteristic and its variability. Control of the process average or mean quality level is usually done with the control chart for means, or the ***x̄* control chart.** Process variability can be monitored with either a control chart for the standard deviation, called the ***s* control chart,** or a control chart for the range, called an ***R* control chart.** The R chart is more widely used. Usually, separate $\bar{x}$ and R charts are maintained for each quality characteristic of interest. (However, if the quality characteristics are closely related, this can sometimes cause misleading results; refer to Chapter 12 of Part IV.) The $\bar{x}$ and R (or s) charts are among the most important and useful on-line statistical process monitoring and control techniques.

It is important to maintain control over both the process mean and process variability. Figure 6.1 illustrates the output of a production process. In Fig. 6.1*a* both the mean μ and standard deviation σ are in control at their nominal values (say, μ_0 and σ_0); consequently, most of the process output falls within the specification limits. However, in Fig. 6.1*b* the mean has shifted to a value $\mu_1 > \mu_0$, resulting in a higher fraction of nonconforming product. In Fig. 6.1*c* the process standard deviation has shifted to a value $\sigma_1 > \sigma_0$. This also results in higher process fallout, even though the process mean is still at the nominal value.



■ **FIGURE 6.1** The need for controlling both process mean and process variability. (*a*) Mean and standard deviation at nominal levels. (*b*) Process mean $\mu_1 > \mu_0$. (*c*) Process standard deviation $\sigma_1 > \sigma_0$.

6.2 Control Charts for $\bar{x}$ and R

6.2.1 Statistical Basis of the Charts

Suppose that a quality characteristic is normally distributed with mean μ and standard deviation σ, where both μ and σ are known. If $x_1, x_2, \ldots, x_n$ is a sample of size n, then the average of this sample is

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

and we know that $\bar{x}$ is normally distributed with mean μ and standard deviation $\sigma_{\bar{x}} = \sigma/\sqrt{n}$. Furthermore, the probability is $1 - \alpha$ that any sample mean will fall between

$$\mu + Z_{\alpha/2}\sigma_{\bar{x}} = \mu + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \quad \text{and} \quad \mu - Z_{\alpha/2}\sigma_{\bar{x}} = \mu - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \tag{6.1}$$

Therefore, if μ and σ are known, equation (6.1) could be used as upper and lower control limits on a control chart for sample means. As noted in Chapter 5, it is customary to replace $Z_{\alpha/2}$ by 3, so that three-sigma limits are employed. If a sample mean falls outside of these limits, it is an indication that the process mean is no longer equal to μ.

We have assumed that the distribution of the quality characteristic is normal. However, the above results are still approximately correct even if the underlying distribution is nonnormal, because of the central limit theorem. We discuss the effect of the normality assumption on variables control charts in Section 6.2.5.

In practice, we usually will not know μ and σ. Therefore, they must be estimated from preliminary samples or subgroups taken when the process is thought to be in control. These estimates should usually be based on at least 20 to 25 samples. Suppose that m samples are available, each containing n observations on the quality characteristic. Typically, n will be small, often either 4, 5, or 6. These small sample sizes usually result from the construction of rational subgroups and from the fact that the sampling and inspection costs associated with variables measurements are usually relatively large. Let $\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_m$ be the average of each sample. Then the best estimator of μ, the process average, is the grand average—say,

$$\bar{\bar{x}} = \frac{\bar{x}_1 + \bar{x}_2 + \cdots + \bar{x}_m}{m} \tag{6.2}$$

Thus, $\bar{\bar{x}}$ would be used as the center line on the $\bar{x}$ chart.

To construct the control limits, we need an estimate of the standard deviation σ. Recall from Chapter 4 (Section 4.2) that we may estimate σ from either the standard deviations or the ranges of the m samples. For the present, we will use the range method. If $x_1, x_2, \ldots, x_n$ is a sample of size n, then the range of the sample is the difference between the largest and smallest observations; that is,

$$R = x_{\max} - x_{\min}$$

Let $R_1, R_2, \ldots, R_m$ be the ranges of the m samples. The average range is

$$\bar{R} = \frac{R_1 + R_2 + \cdots + R_m}{m} \tag{6.3}$$

We may now give the formulas for constructing the control limits on the $\bar{x}$ chart. They are as follows:

Control Limits for the $\bar{x}$ Chart

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + A_2\bar{R} \\ \text{Center line} &= \bar{\bar{x}} \\ \text{LCL} &= \bar{\bar{x}} - A_2\bar{R} \end{aligned} \tag{6.4}$$

The constant A_2 is tabulated for various sample sizes in Appendix Table VI.

Process variability may be monitored by plotting values of the sample range R on a control chart. The center line and control limits of the R chart are as follows:

Control Limits for the R Chart

$$\begin{aligned} \text{UCL} &= D_4\bar{R} \\ \text{Center line} &= \bar{R} \\ \text{LCL} &= D_3\bar{R} \end{aligned} \tag{6.5}$$

The constants D_3 and D_4 are tabulated for various values of n in Appendix Table VI.

Development of Equations (6.4) and (6.5). The development of the equations for computing the control limits on the $\bar{x}$ and R control charts is relatively easy. In Chapter 4 (Section 4.2) we observed that there is a well-known relationship between the range of a sample from a normal distribution and the standard deviation of that distribution. The random variable $W = R/\sigma$ is called the **relative range.** The parameters of the distribution of W are a function of the sample size n. The mean of W is d_2. Consequently, an estimator of σ is $\hat{\sigma} = R/d_2$. Values of d_2 for various sample sizes are given in Appendix Table VI. Therefore, if $\bar{R}$ is the average range of the m preliminary samples, we may use

$$\hat{\sigma} = \frac{\bar{R}}{d_2} \tag{6.6}$$

to estimate σ. This is an unbiased estimator of σ.

If we use $\bar{\bar{x}}$ as an estimator of μ and $\bar{R}/d_2$ as an estimator of σ, then the parameters of the $\bar{x}$ chart are

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + \frac{3}{d_2\sqrt{n}}\bar{R} \\ \text{Center line} &= \bar{\bar{x}} \\ \text{LCL} &= \bar{\bar{x}} - \frac{3}{d_2\sqrt{n}}\bar{R} \end{aligned} \tag{6.7}$$

If we define

$$A_2 = \frac{3}{d_2\sqrt{n}} \tag{6.8}$$

then equation (6.7) reduces to equation (6.4).

Now consider the R chart. The center line will be $\bar{R}$. To determine the control limits, we need an estimate of σ_R. Assuming that the quality characteristic is normally distributed, $\hat{\sigma}_R$ can be found from the distribution of the relative range $W = R/\sigma$. The standard deviation of W, say d_3, is a known function of n. Thus, since

$$R = W\sigma$$

the standard deviation of R is

$$\sigma_R = d_3\sigma$$

Since σ is unknown, we may estimate σ_R by

$$\hat{\sigma}_R = d_3\frac{\bar{R}}{d_2} \tag{6.9}$$

Consequently, the parameters of the R chart with the usual three-sigma control limits are

$$\begin{aligned} \text{UCL} &= \bar{R} + 3\hat{\sigma}_R = \bar{R} + 3d_3\frac{\bar{R}}{d_2} \\ \text{Center line} &= \bar{R} \\ \text{LCL} &= \bar{R} - 3\hat{\sigma}_R = \bar{R} - 3d_3\frac{\bar{R}}{d_2} \end{aligned} \tag{6.10}$$

If we let

$$D_3 = 1 - 3\frac{d_3}{d_2} \quad \text{and} \quad D_4 = 1 + 3\frac{d_3}{d_2}$$

equation (6.10) reduces to equation (6.5).

Phase I Application of $\bar{x}$ and R Charts. In **phase I control chart usage,** when preliminary samples are used to construct $\bar{x}$ and R control charts, it is customary to treat the control limits obtained from equations (6.4) and (6.5) as **trial control limits.** They allow us to determine whether the process was in control when the m initial samples were selected. To determine whether the process was in control when the preliminary samples were collected, plot the values of $\bar{x}$ and R from each sample on the charts and analyze the resulting display. If all points plot inside the control limits and no systematic behavior is evident, we conclude

that the process was in control in the past, and the trial control limits are suitable for controlling current or future production. It is highly desirable to have 20–25 samples or subgroups of size n (typically n is between 3 and 5) to compute the trial control limits. We can, of course, work with fewer data but the control limits are not as reliable.

Suppose that one or more of the values of either $\bar{x}$ or R plot out of control when compared to the trial control limits. Clearly, if control limits for current or future production are to be meaningful, they must be based on data from a process that is in control. Therefore, when the hypothesis of past control is rejected, it is necessary to **revise** the trial control limits. This is done by examining each of the out-of-control points, looking for an assignable cause. If an assignable cause is found, the point is discarded and the trial control limits are recalculated, using only the remaining points. Then these remaining points are reexamined for control. (Note that points that were in control initially may now be out of control, because the new trial control limits will generally be tighter than the old ones.) This process is continued until all points plot in control, at which point the trial control limits are adopted for current use.

In some cases, it may not be possible to find an assignable cause for a point that plots out of control. There are two courses of action open to us. The first of these is to eliminate the point, just as if an assignable cause had been found. There is no analytical justification for choosing this action, other than that points that are outside of the control limits are likely to have been drawn from a probability distribution characteristic of an out-of-control state. The alternative is to retain the point (or points) considering the trial control limits as appropriate for current control. Of course, if the point really does represent an out-of-control condition, the resulting control limits will be too wide. However, if there are only one or two such points, this will not distort the control chart significantly. If future samples still indicate control, then the unexplained points can probably be safely dropped.

Occasionally, when the initial sample values of $\bar{x}$ and R are plotted against the trial control limits, many points will plot out of control. Clearly, if we arbitrarily drop the out-of-control points, we will have an unsatisfactory situation, as few data will remain with which we can recompute reliable control limits. We also suspect that this approach would ignore much useful information in the data. On the other hand, searching for an assignable cause for *each* out-of-control point is unlikely to be successful. We have found that when many of the initial samples plot out of control against the trial limits, it is better to concentrate on **the patterns on control charts** formed by these points. Such a pattern will almost always exist. Usually, the assignable cause associated with the pattern of out-of-control points is fairly easy to identify. Removal of this process problem usually results in a major process improvement.

6.2.2 Development and Use of $\bar{x}$ and R Charts

In the previous section we presented the statistical background for $\bar{x}$ and R control charts. We now illustrate the construction and application of these charts. We also discuss some guidelines for using these charts in practice.

EXAMPLE 6.1 $\bar{x}$ and R Charts for a Manufacturing Process

A hard-bake process (see Section 5.3.1) is used in conjunction with photolithography in semiconductor manufacturing. We wish to establish statistical control of the flow width of the resist in this process using $\bar{x}$ and R charts. Twenty-five samples, each of size five wafers, have been taken when we think the process is in control. The interval of time between samples or subgroups is one hour. The flow width measurement data (in microns) from these samples are shown in Table 6.1.

(*continued*)

■ **TABLE 6.1**

Flow Width Measurements (microns) for the Hard-Bake Process

Sample Number	Wafers						
	1	2	3	4	5	$\bar{x}_i$	R_i
1	1.3235	1.4128	1.6744	1.4573	1.6914	1.5119	0.3679
2	1.4314	1.3592	1.6075	1.4666	1.6109	1.4951	0.2517
3	1.4284	1.4871	1.4932	1.4324	1.5674	1.4817	0.1390
4	1.5028	1.6352	1.3841	1.2831	1.5507	1.4712	0.3521
5	1.5604	1.2735	1.5265	1.4363	1.6441	1.4882	0.3706
6	1.5955	1.5451	1.3574	1.3281	1.4198	1.4492	0.2674
7	1.6274	1.5064	1.8366	1.4177	1.5144	1.5805	0.4189
8	1.4190	1.4303	1.6637	1.6067	1.5519	1.5343	0.2447
9	1.3884	1.7277	1.5355	1.5176	1.3688	1.5076	0.3589
10	1.4039	1.6697	1.5089	1.4627	1.5220	1.5134	0.2658
11	1.4158	1.7667	1.4278	1.5928	1.4181	1.5242	0.3509
12	1.5821	1.3355	1.5777	1.3908	1.7559	1.5284	0.4204
13	1.2856	1.4106	1.4447	1.6398	1.1928	1.3947	0.4470
14	1.4951	1.4036	1.5893	1.6458	1.4969	1.5261	0.2422
15	1.3589	1.2863	1.5996	1.2497	1.5471	1.4083	0.3499
16	1.5747	1.5301	1.5171	1.1839	1.8662	1.5344	0.6823
17	1.3680	1.7269	1.3957	1.5014	1.4449	1.4874	0.3589
18	1.4163	1.3864	1.3057	1.6210	1.5573	1.4573	0.3153
19	1.5796	1.4185	1.6541	1.5116	1.7247	1.5777	0.3062
20	1.7106	1.4412	1.2361	1.3820	1.7601	1.5060	0.5240
21	1.4371	1.5051	1.3485	1.5670	1.4880	1.4691	0.2185
22	1.4738	1.5936	1.6583	1.4973	1.4720	1.5390	0.1863
23	1.5917	1.4333	1.5551	1.5295	1.6866	1.5592	0.2533
24	1.6399	1.5243	1.5705	1.5563	1.5530	1.5688	0.1156
25	1.5797	1.3663	1.6240	1.3732	1.6887	1.5264	0.3224
						$\Sigma\bar{x}_i = 37.6400$	$\Sigma R_i = 8.1302$
						$\bar{\bar{x}} = 1.5056$	$\bar{R} = 0.32521$

SOLUTION

When setting up $\bar{x}$ and R control charts, it is best to begin with the R chart. Because the control limits on the $\bar{x}$ chart depend on the process variability, unless process variability is in control, these limits will not have much meaning. Using the data in Table 6.1, we find that the center line for the R chart is

$$\bar{R} = \frac{\sum_{i=1}^{25} R_i}{25} = \frac{8.1302}{25} = 0.32521$$

For samples of $n = 5$, we find from Appendix Table VI that $D_3 = 0$ and $D_4 = 2.114$. Therefore, the control limits for the R chart are, using equation (6.5),

$$\text{LCL} = \bar{R}D_3 = 0.32521(0) = 0$$
$$\text{UCL} = \bar{R}D_4 = 0.32521(2.114) = 0.68749$$

The R chart is shown in Fig. 6.2*b*. Both control charts in Fig. 6.2 were constructed by Minitab. Note that the UCL for the R chart is reported as 0.6876, because Minitab uses more decimal

places in the calculation than we did. When the 25 sample ranges are plotted on the R chart in Fig. 6.2*b* there is no indication of an out-of-control condition.

Since the R chart indicates that process variability is in control, we may now construct the $\bar{x}$ chart. The center line is

$$\bar{\bar{x}} = \frac{\sum_{i=1}^{25} \bar{x}_i}{25} = \frac{37.6400}{25} = 1.5056$$

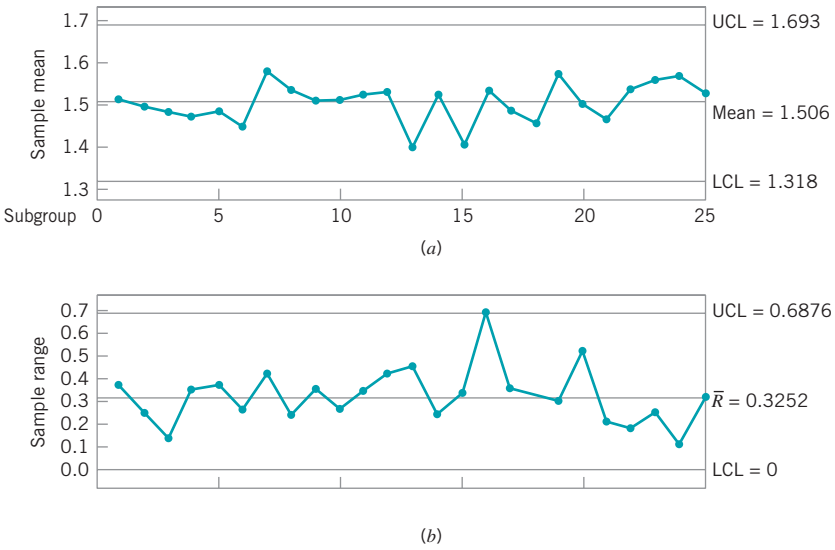
To find the control limits on the $\bar{x}$ chart, we use $A_2 = 0.577$ from Appendix Table VI for samples of size $n = 5$ and equation (6.4) to find

$$\text{UCL} = \bar{\bar{x}} + A_2\bar{R} = 1.5056 + (0.577)(0.32521) = 1.69325$$

and

$$\text{LCL} = \bar{\bar{x}} - A_2\bar{R} = 1.5056 - (0.577)(0.32521) = 1.31795$$

The $\bar{x}$ chart is shown in Fig. 6.2*a*. When the preliminary sample averages are plotted on this chart, no indication of an out-of-control condition is observed. Therefore, since both the $\bar{x}$ and R charts exhibit control, we would conclude that the process is in control at the stated levels and adopt the trial control limits for use in phase II, where monitoring of future production is of interest.



■ **FIGURE 6.2** $\bar{x}$ and R charts (from Minitab) for flow width in the hard-bake process.

Estimating Process Capability. The $\bar{x}$ and R charts provide information about the performance or **process capability** of the process. From the $\bar{x}$ chart, we may estimate the mean flow width of the resist in the hard-bake process as $\bar{\bar{x}} = 1.5056$ microns. The process standard deviation may be estimated using equation (6.6); that is,

$$\hat{\sigma} = \frac{\bar{R}}{d_2} = \frac{0.32521}{2.326} = 0.1398 \text{ microns}$$

where the value of d_2 for samples of size five is found in Appendix Table VI. The specification limits on flow width are 1.50 ± 0.50 microns. The control chart data may be used to describe the capability of the process to produce wafers relative to these specifications. Assuming that flow width is a normally distributed random variable, with mean 1.5056 and

standard deviation 0.1398, we may estimate the fraction of nonconforming wafers produced as

$$\begin{aligned} p &= P\{x < 1.00\} + P\{x > 2.00\} \\ &= \Phi\left(\frac{1.00 - 1.5056}{0.1398}\right) + 1 - \Phi\left(\frac{2.00 - 1.5056}{0.1398}\right) \\ &= \Phi(-3.61660) + 1 - \Phi(3.53648) \\ &\simeq 0.00015 + 1 - 0.99980 \\ &\simeq 0.00035 \end{aligned}$$

That is, about 0.035 percent [350 parts per million (ppm)] of the wafers produced will be outside of the specifications.

Another way to express process capability is in terms of the **process capability ratio (PCR)** C_p, which for a quality characteristic with both upper and lower specification limits (USL and LSL, respectively) is

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma} \tag{6.11}$$

Note that the 6σ spread of the process is the basic definition of process capability. Since σ is usually unknown, we must replace it with an estimate. We frequently use $\hat{\sigma} = \bar{R}/d_2$ as an estimate of σ, resulting in an estimate $\hat{C}_p$ of C_p. For the hard-bake process, since $\bar{R}/d_2 = \hat{\sigma} = 0.1398$, we find that

$$\hat{C}_p = \frac{2.00 - 1.00}{6(0.1398)} = \frac{1.00}{0.8388} = 1.192$$

This implies that the "natural" tolerance limits in the process (three-sigma above and below the mean) are inside the lower and upper specification limits. Consequently, a moderately small number of nonconforming wafers will be produced. The PCR C_p may be interpreted another way. The quantity

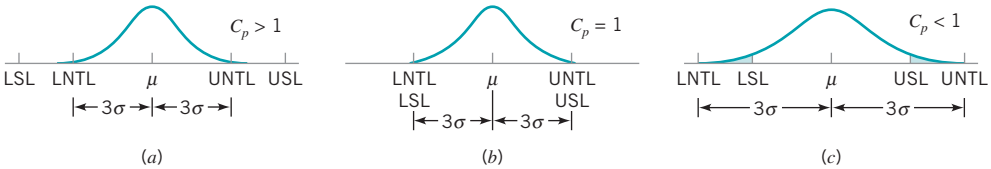
$$P = \left(\frac{1}{C_p}\right)100\%$$

is simply the percentage of the specification band that the process uses up. For the hard-bake process an estimate of P is

$$\hat{P} = \left(\frac{1}{\hat{C}_p}\right)100\% = \left(\frac{1}{1.192}\right)100\% = 83.89$$

That is, the process uses up about 84% of the specification band.

Figure 6.3 illustrates three cases of interest relative to the PCR C_p and process specifications. In Fig. 6.3*a* the PCR C_p is greater than unity. This means that the process uses up much less than 100% of the tolerance band. Consequently, relatively few non-conforming units



■ **FIGURE 6.3** Process fallout and the process capability ratio C_p.

will be produced by this process. Figure 6.3*b* shows a process for which the PCR $C_p = 1$; that is, the process uses up all the tolerance band. For a normal distribution this would imply about 0.27% (or 2700 ppm) nonconforming units. Finally, Fig. 6.3*c* presents a process for which the PCR $C_p < 1$; that is, the process uses up more than 100% of the tolerance band. In this case the process is very yield-sensitive, and a large number of nonconforming units will be produced.

Note that all the cases in Fig. 6.3 assume that the process is centered at the midpoint of the specification band. In many situations this will not be the case, and as we will see in Chapter 8 (which is devoted to a more extensive treatment of process capability analysis), some modification of the PCR C_p is necessary to describe this situation adequately.

Revision of Control Limits and Center Lines. The effective use of any control chart will require **periodic revision** of the control limits and center lines. Some practitioners establish regular periods for review and revision of control chart limits, such as every week, every month, or every 25, 50, or 100 samples. When revising control limits, remember that it is highly desirable to use at least 25 samples or subgroups (some authorities recommend 200–300 individual observations) in computing control limits.

Sometimes the user will replace the center line of the $\bar{x}$ chart with a **target value,** say $\bar{\bar{x}}_0$. If the R chart exhibits control, this can be helpful in shifting the process average to the desired value, particularly in processes where the mean may be changed by a fairly simple adjustment of a manipulatable variable in the process. If the mean is not easily influenced by a simple process adjustment, then it is likely to be a complex and unknown function of several process variables and a target value $\bar{\bar{x}}_0$ may not be helpful, as use of that value could result in many points outside the control limits. In such cases, we would not necessarily know whether the point was really associated with an assignable cause or whether it plotted outside the limits because of a poor choice for the center line.

When the R chart is out of control, we often eliminate the out-of-control points and recompute a revised value of $\bar{R}$. This value is then used to determine new limits and center line on the R chart and new limits on the $\bar{x}$ chart. This will tighten the limits on both charts, making them consistent with a process standard deviation σ consistent with use of the *revised* $\bar{R}$ in the relationship $\bar{R}/d_2$. This estimate of σ could be used as the basis of a preliminary analysis of process capability.

Phase II Operation of the $\bar{x}$ and R Charts. Once a set of reliable control limits is established, we use the control chart for monitoring future production. This is called **phase II control chart usage.**

Twenty additional samples of wafers from the hard-bake process were collected after the control charts were established and the sample values of $\bar{x}$ and R plotted on the control charts immediately after each sample was taken. The data from these new samples are shown in Table 6.2, and the continuations of the $\bar{x}$ and R charts are shown in Fig. 6.4. The control charts indicate that the process is in control, until the $\bar{x}$-value from the 43rd sample is plotted. Since this point (as well as the $\bar{x}$-value from sample 45) plots above the upper control limit, we would suspect that an assignable cause has occurred at or before that time. The general pattern of points on the $\bar{x}$ chart from about subgroup 38 onward is indicative of a shift in the process mean.

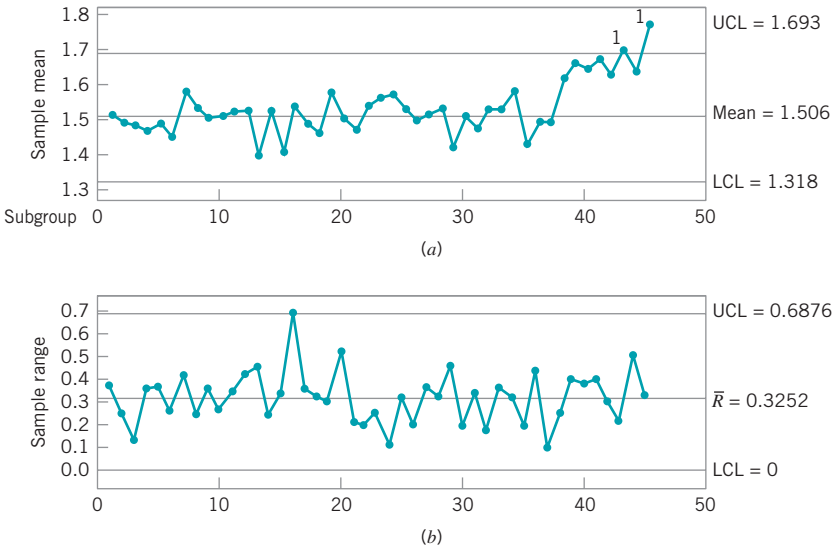
Once the control chart is established and is being used in on-line process monitoring, one is often tempted to use the sensitizing rules (or Western Electric rules) discussed in Chapter 5 (Section 5.3.6) to speed up shift detection. Here, for example, the use of such rules would likely result in the shift being detected around sample 40. However, recall the discussion from Section 5.3.3 in which we discouraged the routine use of these sensitizing rules

■ TABLE 6.2
Additional Samples for Example 6.1

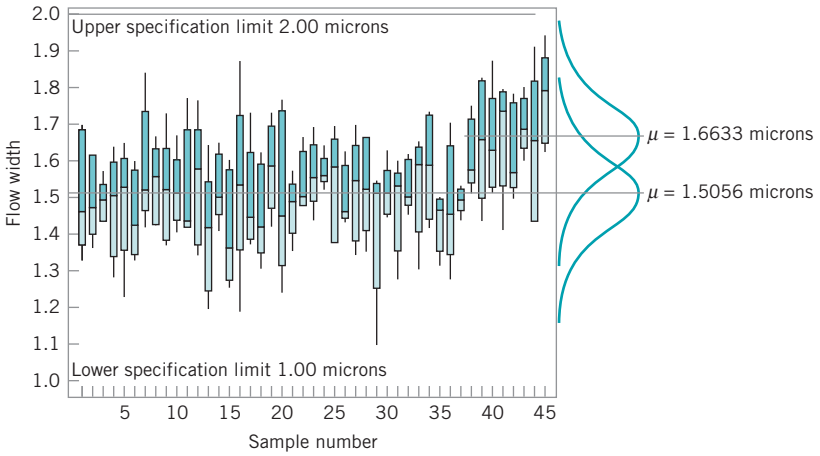
Sample Number	Wafers						
	1	2	3	4	5	$\bar{x}_i$	R_i
26	1.4483	1.5458	1.4538	1.4303	1.6206	1.4998	0.1903
27	1.5435	1.6899	1.5830	1.3358	1.4187	1.5142	0.3541
28	1.5175	1.3446	1.4723	1.6657	1.6661	1.5332	0.3215
29	1.5454	1.0931	1.4072	1.5039	1.5264	1.4152	0.4523
30	1.4418	1.5059	1.5124	1.4620	1.6263	1.5097	0.1845
31	1.4301	1.2725	1.5945	1.5397	1.5252	1.4724	0.3220
32	1.4981	1.4506	1.6174	1.5837	1.4962	1.5292	0.1668
33	1.3009	1.5060	1.6231	1.5831	1.6454	1.5317	0.3445
34	1.4132	1.4603	1.5808	1.7111	1.7313	1.5793	0.3181
35	1.3817	1.3135	1.4953	1.4894	1.4596	1.4279	0.1818
36	1.5765	1.7014	1.4026	1.2773	1.4541	1.4824	0.4241
37	1.4936	1.4373	1.5139	1.4808	1.5293	1.4910	0.0920
38	1.5729	1.6738	1.5048	1.5651	1.7473	1.6128	0.2425
39	1.8089	1.5513	1.8250	1.4389	1.6558	1.6560	0.3861
40	1.6236	1.5393	1.6738	1.8698	1.5036	1.6420	0.3662
41	1.4120	1.7931	1.7345	1.6391	1.7791	1.6716	0.3811
42	1.7372	1.5663	1.4910	1.7809	1.5504	1.6252	0.2899
43	1.5971	1.7394	1.6832	1.6677	1.7974	1.6970	0.2003
44	1.4295	1.6536	1.9134	1.7272	1.4370	1.6321	0.4839
45	1.6217	1.8220	1.7915	1.6744	1.9404	1.7700	0.3187

for on-line monitoring of a stable process because they greatly increase the occurrence of false alarms.

In examining control chart data, it is sometimes helpful to construct a run chart of the individual observations in each sample. This chart is sometimes called a **tier chart** or **tolerance diagram.** This may reveal some pattern in the data, or it may show that a particular



■ FIGURE 6.4 Continuation of the $\bar{x}$ and R charts in Example 6.1.

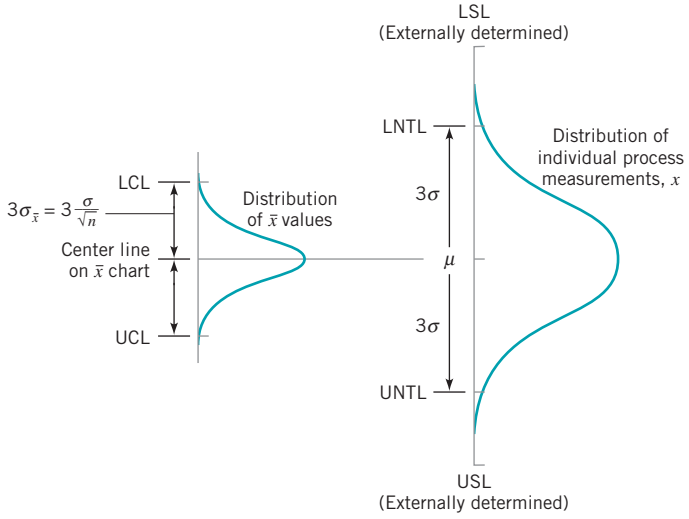


■ **FIGURE 6.5** Tier chart constructed using the Minitab box plot procedure for the flow width data.

value of $\bar{x}$ or R was produced by one or two unusual observations in the sample. A box plot is usually a very simple way to construct the tier diagram.

A tier chart of the flow width data observations is shown in Fig. 6.5. This chart does not indicate that the out-of-control signals were generated by unusual individual observations, but instead, they probably resulted from a shift in the mean around the time that sample 38 was taken. The average of the averages of samples 38 through 45 is 1.6633 microns. The specification limits of 1.50 ± 0.50 microns are plotted in Fig. 6.5, along with a sketch of the normal distribution that represents process output when the process mean equals the in-control value 1.5056 microns. A sketch of the normal distribution representing process output at the new apparent mean diameter of 1.6633 microns is also shown in Fig. 6.5. It is obvious that a much higher percentage of nonconforming wafers will be produced at this new mean flow rate. Since the process is out of control, a search for the cause of this shift in the mean must be conducted. The out-of-control-action plan **(OCAP)** for this control chart, shown in Fig. 5.6, would play a key role in these activities by directing operating personnel through a series of sequential activities to find the assignable cause. Often additional input and support from engineers, management, and the quality engineering staff is necessary to find and eliminate assignable causes.

Control Limits, Specification Limits, and Natural Tolerance Limits. A point that should be emphasized is that there is no connection or relationship between the **control limits** on the $\bar{x}$ and R charts and the **specification limits** on the process. The control limits are driven by the natural variability of the process (measured by the process standard deviation σ), that is, by the **natural tolerance limits of the process.** It is customary to define the upper and lower natural tolerance limits, say UNTL and LNTL, as 3σ above and below the process mean. The specification limits, on the other hand, are determined externally. They may be set by management, the manufacturing engineers, the customer, or by product developers/designers. One should have knowledge of inherent process variability when setting specifications, but **remember that there is no mathematical or statistical relationship between the control limits and specification limits.** The situation is summarized in Fig. 6.6. We have encountered practitioners who have plotted specification limits on the $\bar{x}$ control chart. This practice is completely incorrect and should not be done. When dealing with plots of *individual* observations (not averages), as in Fig. 6.5, it is helpful to plot the specification limits on that chart.



■ **FIGURE 6.6** Relationship of natural tolerance limits, control limits, and specification limits.

Rational Subgroups. **Rational subgroups** play an important role in the use of $\bar{x}$ and R control charts. Defining a rational subgroup in practice may be easier with a clear understanding of the functions of the two types of control charts. The $\bar{x}$ chart monitors the average quality level in the process. Therefore, samples should be selected in such a way that maximizes the chances for shifts in the process average to occur between samples, and thus to show up as out-of-control points on the $\bar{x}$ chart. The R chart, on the other hand, measures the variability *within* a sample. Therefore, samples should be selected so that variability within samples measures only chance or random causes. Another way of saying this is that the $\bar{x}$ chart monitors **between-sample variability** (variability in the process over time), and the R chart measures **within-sample variability** (the instantaneous process variability at a given time).

An important aspect of this is evident from carefully examining how the control limits for the $\bar{x}$ and R charts are determined from past data. The estimate of the process standard deviation σ used in constructing the control limits is calculated from the variability **within** each sample (i.e., from the individual sample ranges). Consequently, the estimate of σ reflects only **within-sample variability.** It is not correct to estimate σ based on the usual quadratic estimator—say,

$$s = \sqrt{\frac{\sum_{i=1}^{m}\sum_{j=1}^{n}\left(x_{ij} - \bar{\bar{x}}\right)^2}{mn-1}}$$

where x_{ij} is the jth observation in the ith sample, because if the sample *means* differ, then this will cause s to be too large. Consequently, σ will be overestimated. Pooling all of the preliminary data in this manner to estimate σ is not a good practice because it potentially combines *both between*-sample and *within*-sample variability. The control limits must be based on only **within-sample variability.** Refer to the supplemental text material for more details.

Guidelines for the Design of the Control Chart. To design the $\bar{x}$ and R charts, we must specify the sample size, control limit width, and frequency of sampling to be used. It is not possible to give an exact solution to the problem of control chart design, unless the analyst has detailed information about both the statistical characteristics of the control chart

tests and the economic factors that affect the problem. A complete solution of the problem knowledges of the cost of sampling, the costs of investigating and possibly correcting the process in response to out-of-control signals, and the costs associated with producing a product that does not meet specifications. Given this kind of information, an economic decision model could be constructed to allow economically optimum control chart design. In Chapter 10 (Section 10.6) we briefly discuss this approach to the problem. However, it is possible to give some general guidelines now that will aid in control chart design.

If the $\bar{x}$ chart is being used primarily to detect moderate to large process shifts—say, on the order of 2σ or larger—then relatively small samples of size $n = 4$, 5, or 6 are reasonably effective. On the other hand, if we are trying to detect small shifts, then larger sample sizes of possibly $n = 15$ to $n = 25$ are needed. When smaller samples are used, there is less risk of a process shift occurring while a sample is taken. If a shift does occur while a sample is taken, the sample average can obscure this effect. Consequently, this is an argument for using as small a sample size as is consistent with the magnitude of the process shift that one is trying to detect. An alternative to increasing the sample size is to use warning limits and other sensitizing procedures to enhance the ability of the control chart to detect small process shifts. However, as we discussed in Chapter 5, we do not favor the routine use of these sensitizing rules. If you are interested in small shifts, use the CUSUM or EWMA charts in Chapter 9.

The R chart is relatively insensitive to shifts in the process standard deviation for small samples. For example, samples of size $n = 5$ have only about a 40% chance of detecting on the first sample a shift in the process standard deviation from σ to 2σ. Larger samples would seem to be more effective, but we also know that the range method for estimating the standard deviation drops dramatically in efficiency as n increases. Consequently, for large n—say, $n > 10$ or 12—it is probably best to use a control chart for s or s^2 instead of the R chart. Details of the construction of these charts are shown in Sections 6.3.1 and 6.3.2.

From a statistical point of view, the operating-characteristic curves of the $\bar{x}$ and R charts can be helpful in choosing the sample size. They provide a feel for the magnitude of process shift that will be detected with a stated probability for any sample size n. These operating-characteristic curves are discussed in Section 6.2.6.

The problem of choosing the sample size and the frequency of sampling is one of **allocating sampling effort.** Generally, the decision maker will have only a limited number of resources to allocate to the inspection process. The available strategies will usually be either to take small, frequent samples or to take larger samples less frequently. For example, the choice may be between samples of size 5 every half hour or samples of size 20 every two hours. It is impossible to say which strategy is best in all cases, but current industry practice favors small, frequent samples. The general belief is that if the interval between samples is too great, too much defective product will be produced before another opportunity to detect the process shift occurs. From economic considerations, if the cost associated with producing defective items is high, smaller, more frequent samples are better than larger, less frequent ones. Variable sample interval and variable sample size schemes could, of course, be used. Refer to Chapter 10.

The rate of production also influences the choice of sample size and sampling frequency. If the rate of production is high—say, 50,000 units per hour—then more frequent sampling is called for than if the production rate is extremely slow. At high rates of production, many nonconforming units of product will be produced in a very short time when process shifts occur. Furthermore, at high production rates, it is sometimes possible to obtain fairly large samples economically. For example, if we produce 50,000 units per hour, it does not take an appreciable difference in time to collect a sample of size 20 compared to a sample of size 5. If per unit inspection and testing costs are not excessive, high-speed production processes are often monitored with moderately large sample sizes.

The use of three-sigma control limits on the $\bar{x}$ and R control charts is a widespread practice. There are situations, however, when departures from this customary choice of control limits

are helpful. For example, if false alarms or type I errors (when an out-of-control signal is generated when the process is really in control) are very expensive to investigate, then it may be best to use wider control limits than three-sigma—perhaps as wide as 3.5-sigma. However, if the process is such that out-of-control signals are quickly and easily investigated with a minimum of lost time and cost, then narrower control limits, perhaps at 2.5-sigma, are appropriate.

Changing Sample Size on the $\bar{x}$ and R Charts. We have presented the development of $\bar{x}$ and R charts assuming that the sample size n is constant from sample to sample. However, there are situations in which the sample size n is not constant. One situation is that of **variable sample size on control charts;** that is, each sample may consist of a different number of observations. The $\bar{x}$ and R charts are generally not used in this case because they lead to a changing center line on the R chart, which is difficult to interpret for many users. The $\bar{x}$ and s charts in Section 6.3.2 would be preferable in this case.

Another situation is that of making a **permanent** (or **semipermanent**) **change** in the sample size because of cost or because the process has exhibited good stability and fewer resources are being allocated for process monitoring. In this case, it is easy to recompute, the new control limits directly from the old ones without collecting additional samples based on the new sample size. Let

$$\begin{aligned}
\bar{R}_{\text{old}} &= \text{average range for the old sample size} \\
\bar{R}_{\text{new}} &= \text{average range for the new sample size} \\
n_{\text{old}} &= \text{old sample size} \\
n_{\text{new}} &= \text{new sample size} \\
d_2(\text{old}) &= \text{factor } d_2 \text{ for the old sample size} \\
d_2(\text{new}) &= \text{factor } d_2 \text{ for the new sample size}
\end{aligned}$$

For the $\bar{x}$ chart the new control limits are

$$\begin{aligned}
\text{UCL} &= \bar{\bar{x}} + A_2\left[\frac{d_2(\text{new})}{d_2(\text{old})}\right]\bar{R}_{\text{old}} \\
\text{LCL} &= \bar{\bar{x}} - A_2\left[\frac{d_2(\text{new})}{d_2(\text{old})}\right]\bar{R}_{\text{old}}
\end{aligned} \tag{6.12}$$

where the center line $\bar{\bar{x}}$ is unchanged and the factor A_2 is selected for the *new* sample size. For the R chart, the new parameters are

$$\begin{aligned}
\text{UCL} &= D_4\left[\frac{d_2(\text{new})}{d_2(\text{old})}\right]\bar{R}_{\text{old}} \\
\text{CL} &= \bar{R}_{\text{new}} = \left[\frac{d_2(\text{new})}{d_2(\text{old})}\right]\bar{R}_{\text{old}} \\
\text{LCL} &= \max\left\{0, D_3\left[\frac{d_2(\text{new})}{d_2(\text{old})}\right]\bar{R}_{\text{old}}\right\}
\end{aligned} \tag{6.13}$$

where D_3 and D_4 are selected for the *new* sample size.

EXAMPLE 6.2 Changing Sample Size

To illustrate the above procedure, consider the $\bar{x}$ and R charts developed for the hard-bake process in Example 6.1. These charts were based on a sample size of five wafers. Suppose that since the process exhibits good control, the process engineering personnel want to reduce the sample size to three wafers. Set up the new control charts.

SOLUTION

From Example 6.1, we know that

$$n_{\text{old}} = 5 \qquad \bar{R}_{\text{old}} = 0.32521$$

and from Appendix Table VI we have

$$d_2(\text{old}) = 2.326 \qquad d_2(\text{new}) = 1.693$$

Therefore, the new control limits on the $\bar{x}$ chart are found from equation (6.12) as

$$\begin{aligned}\text{UCL} &= \bar{\bar{x}} + A_2\left[\frac{d_2(\text{new})}{d_2(\text{old})}\right]\bar{R}_{\text{old}} \\ &= 1.5056 + (1.023)\left[\frac{1.693}{2.326}\right](0.32521) \\ &= 1.5056 + 0.2422 = 1.7478\end{aligned}$$

and

$$\begin{aligned}\text{LCL} &= \bar{\bar{x}} - A_2\left[\frac{d_2(\text{new})}{d_2(\text{old})}\right]\bar{R}_{\text{old}} \\ &= 1.5056 - (1.023)\left[\frac{1.693}{2.326}\right](0.32521) \\ &= 1.5056 - 0.2422 = 1.2634\end{aligned}$$

For the R chart, the new parameters are given by equation (6.13):

$$\begin{aligned}\text{UCL} &= D_4\left[\frac{d_2(\text{new})}{d_2(\text{old})}\right]\bar{R}_{\text{old}} \\ &= (2.574)\left[\frac{1.693}{2.326}\right](0.32521) \\ &= 0.6093\end{aligned}$$

$$\begin{aligned}\text{CL} = \bar{R}_{\text{new}} &= \left[\frac{d_2(\text{new})}{d_2(\text{old})}\right]\bar{R}_{\text{old}} \\ &= \left[\frac{1.693}{2.326}\right](0.32521) \\ &= 0.2367\end{aligned}$$

$$\begin{aligned}\text{LCL} &= \max\left\{0, D_3\left[\frac{d_2(\text{new})}{d_2(\text{old})}\right]\bar{R}_{\text{old}}\right\} \\ &= 0\end{aligned}$$

Figure 6.7 shows the new control limits. Note that the effect of reducing the sample size is to *increase* the width of the limits on the $\bar{x}$ chart (because $\sigma/\sqrt{n}$ is smaller when $n = 5$ than when $n = 3$) and to *lower* the center line and the upper control limit on the R chart (because the expected range from a sample of $n = 3$ is smaller than the expected range from a sample of $n = 5$).

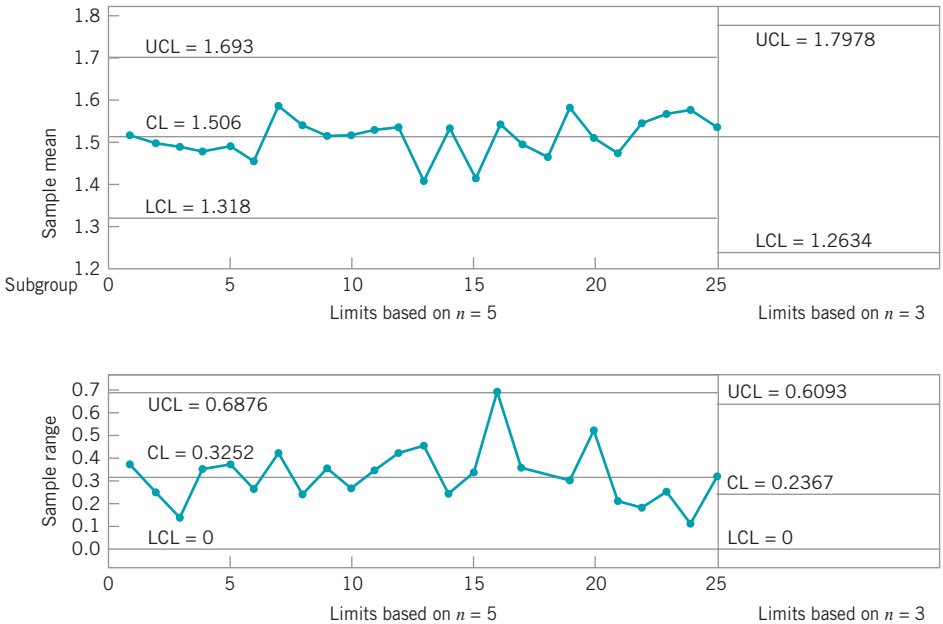


Figure 6.7 Recalculated control limits for the hard-bake process in Example 6.1 to reflect changing the sample size from $n = 5$ to $n = 3$.

Probability Limits on the $\bar{x}$ and R Charts. It is customary to express the control limits on the $\bar{x}$ and R charts as a multiple of the standard deviation of the statistic plotted on the charts. If the multiple chosen is k, then the limits are referred to as k-sigma limits, the usual choice being $k = 3$. As mentioned in Chapter 5, however, it is also possible to define the control limits by specifying the type I error level for the test. These are called **probability limits for control charts** and are used extensively in the United Kingdom and some parts of Western Europe.

It is easy to choose probability limits for the $\bar{x}$ chart. Since $\bar{x}$ is approximately normally distributed, we may obtain a desired type I error of α by choosing the multiple of sigma for the control limit as $k = Z_{\alpha/2}$, where $Z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution. Note that the usual **three-sigma control limits** imply that the type I error probability is $\alpha = 0.0027$. If we choose $\alpha = 0.002$, for example, as most writers who recommend probability limits do, then $Z_{\alpha/2} = Z_{0.001} = 3.09$. Consequently, there is very little difference between such control limits and three-sigma control limits.

We may also construct R charts using probability limits. If $\alpha = 0.002$, the 0.001 and 0.999 percentage points of the distribution of the relative range $W = R/\sigma$ are required. These points obviously depend on the subgroup size n. Denoting these points by $W_{0.001}(n)$ and $W_{0.999}(n)$, and estimating σ by $\bar{R}/d_2$, we would have the 0.001 and 0.999 limits for R as $W_{0.001}(n)(\bar{R}/d_2)$ and $W_{0.999}(n)(\bar{R}/d_2)$. If we let $D_{0.001} = W_{0.001}(n)/d_2$ and $D_{0.999} = W_{0.999}(n)/d_2$, then the probability limits for the R chart are

$$\text{UCL} = D_{0.999}\bar{R}$$
$$\text{LCL} = D_{0.001}\bar{R}$$

Tables of pairs of values $(D_{0.001}, D_{0.999})$, $(D_{0.005}, D_{0.995})$, and $(D_{0.025}, D_{0.975})$ for $2 \le n \le 10$ are in Grant and Leavenworth (1980). These control limits will not differ substantially from the customary three-sigma limits. However, for sample sizes $3 \le n \le 6$, they will produce a positive lower control limit for the R chart whereas the conventional three-sigma limits do not.

6.2.3 Charts Based on Standard Values

When it is possible to specify standard values for the process mean and standard deviation, we may use these standards to establish the control charts for $\bar{x}$ and R without analysis of past data. Suppose that the standards given are μ and σ. Then the parameters of the $\bar{x}$ chart are

$$\begin{aligned} \text{UCL} &= \mu + 3\frac{\sigma}{\sqrt{n}} \\ \text{Center line} &= \mu \\ \text{LCL} &= \mu - 3\frac{\sigma}{\sqrt{n}} \end{aligned} \tag{6.14}$$

The quantity $3/\sqrt{n} = A$, say, is a constant that depends on n, which has been tabulated in Appendix Table VI. Consequently, we could write the parameters of the $\bar{x}$ chart as

$$\begin{aligned} \text{UCL} &= \mu + A\sigma \\ \text{Center line} &= \mu \\ \text{LCL} &= \mu - A\sigma \end{aligned} \tag{6.15}$$

To construct the R chart with a standard value of σ, recall that $\sigma = R/d_2$, where d_2 is the mean of the distribution of the relative range. Furthermore, the standard deviation of R is $\sigma_R = d_3\sigma$, where d_3 is the standard deviation of the distribution of the relative range. Therefore, the parameters of the control chart are

$$\begin{aligned} \text{UCL} &= d_2\sigma + 3d_3\sigma \\ \text{Center line} &= d_2\sigma \\ \text{LCL} &= d_2\sigma - 3d_3\sigma \end{aligned} \tag{6.16}$$

It is customary to define the constants

$$\begin{aligned} D_1 &= d_2 - 3d_3 \\ D_2 &= d_2 + 3d_3 \end{aligned}$$

These constants are tabulated in Appendix Table VI. Thus, the parameters of the R chart with standard σ given are

$$\begin{aligned} \text{UCL} &= D_2\sigma \\ \text{Center line} &= d_2\sigma \\ \text{LCL} &= D_1\sigma \end{aligned} \tag{6.17}$$

One must exercise considerable care when standard values of μ and σ are given. It may be that these standards are not really applicable to the process, and as a result, the $\bar{x}$ and R charts will produce many out-of-control signals relative to the specified standards. If the process is really in control at some *other* mean and standard deviation, then the analyst may spend considerable effort looking for assignable causes that do not exist. Standard values of σ seem to give more trouble than standard values of μ. In processes where the mean of the quality characteristic is controlled by adjustments to the machine, standard or target values of μ are sometimes helpful in achieving management goals with respect to process performance.

6.2.4 Interpretation of $\bar{x}$ and R Charts

We have noted previously that a control chart can indicate an out-of-control condition even though no single point plots outside the control limits, if the pattern of the plotted points exhibits nonrandom or systematic behavior. In many cases, the pattern of the plotted points will provide useful diagnostic information on the process, and this information can be used to make process modifications that reduce variability (the goal of statistical process control). Furthermore, these patterns occur fairly often in phase I (retrospective study of past data), and their elimination is frequently crucial in bringing a process into control.

In this section, we briefly discuss **interpretation of control charts,** focusing on some of the more common patterns that appear on $\bar{x}$ and R charts and some of the process characteristics

that may produce the patterns. To effectively interpret $\bar{x}$ and R charts, the analyst must be familiar with both the statistical principles underlying the control chart and the process itself. Additional information on the interpretation of patterns on control charts is in the Western Electric *Statistical Quality Control Handbook* (1956, pp. 149–183).

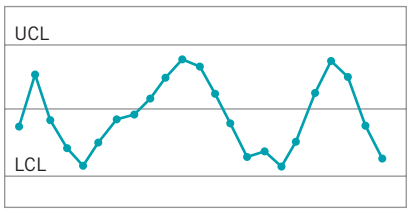
In interpreting patterns on the $\bar{x}$ chart, we must first determine whether or not the R chart is in control. Some assignable causes show up on *both* the $\bar{x}$ and R charts. If both the $\bar{x}$ and R charts exhibit a nonrandom pattern, the best strategy is to eliminate the R chart assignable causes first. In many cases, this will automatically eliminate the nonrandom pattern on the $\bar{x}$ chart. Never attempt to interpret the $\bar{x}$ chart when the R chart indicates an out-of-control condition.

Cyclic patterns occasionally appear on the control chart. A typical example is shown in Fig. 6.8. Such a pattern on the $\bar{x}$ chart may result from systematic environmental changes such as temperature, operator fatigue, regular rotation of operators and/or machines, or fluctuation in voltage or pressure or some other variable in the production equipment. R charts will sometimes reveal cycles because of maintenance schedules, operator fatigue, or tool wear resulting in excessive variability. In one study in which this author was involved, systematic variability in the fill volume of a metal container was caused by the on–off cycle of a compressor in the filling machine.

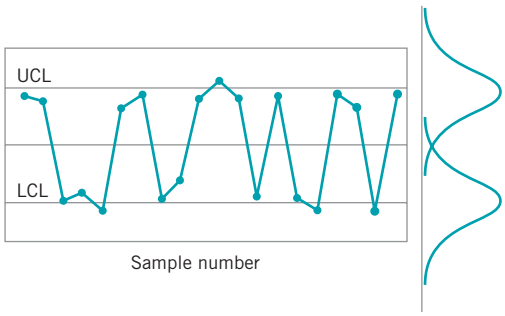
A **mixture** is indicated when the plotted points tend to fall near or slightly outside the control limits, with relatively few points near the center line, as shown in Fig. 6.9. A mixture pattern is generated by two (or more) overlapping distributions generating the process output. The probability distributions that could be associated with the mixture pattern in Fig. 6.9 are shown on the right-hand side of that figure. The severity of the mixture pattern depends on the extent to which the distributions overlap. Sometimes mixtures result from "overcontrol," where the operators make process adjustments too often, responding to random variation in the output rather than systematic causes. A mixture pattern can also occur when output product from several sources (such as parallel machines) is fed into a common stream which is then sampled for process monitoring purposes.

A **shift in process level** is illustrated in Fig. 6.10. These shifts may result from the introduction of new workers; changes in methods, raw materials, or machines; a change in the inspection method or standards; or a change in either the skill, attentiveness, or motivation of the operators. Sometimes an improvement in process performance is noted following introduction of a control chart program, simply because of motivational factors influencing the workers.

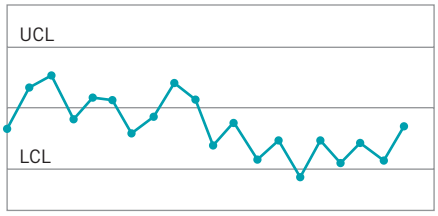
A **trend,** or continuous movement in one direction, is shown on the control chart in Fig. 6.11. Trends are usually due to a gradual wearing out or deterioration of a tool or some other critical process component. In chemical processes they often occur because of settling



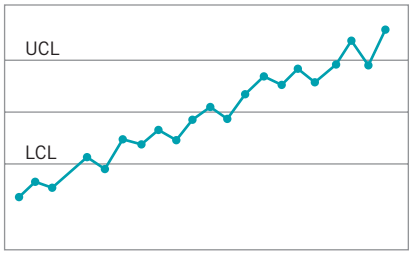
■ **FIGURE 6.8** Cycles on a control chart.



■ **FIGURE 6.9** A mixture pattern.



■ **FIGURE 6.10** A shift in process level.

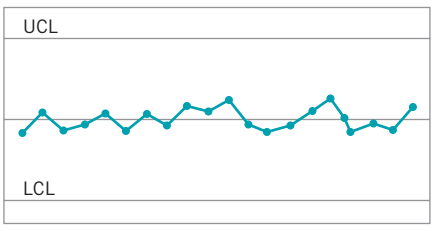


■ **FIGURE 6.11** A trend in process level.

or separation of the components of a mixture. They can also result from human causes, such as operator fatigue or the presence of supervision. Finally, trends can result from seasonal influences, such as temperature. When trends are due to tool wear or other systematic causes of deterioration, this may be directly incorporated into the control chart model. A device useful for monitoring and analyzing processes with trends is the **regression control chart** [see Mandel (1969)]. The **modified control chart**, discussed in Chapter 9, is also used when the process exhibits tool wear.

Stratification, or a tendency for the points to cluster artificially around the center line, is illustrated in Fig. 6.12. We note that there is a marked lack of natural variability in the observed pattern. One potential cause of stratification is incorrect calculation of control limits. This pattern may also result when the sampling process collects one or more units from several different underlying distributions within each subgroup. For example, suppose that a sample of size 5 is obtained by taking one observation from each of five parallel processes. If the largest and smallest units in each sample are relatively far apart because they come from two different distributions, then R will be incorrectly inflated, causing the limits on the $\bar{x}$ chart to be too wide. In this case R incorrectly measures the variability *between* the different underlying distributions, in addition to the chance cause variation that it is intended to measure.

In interpreting patterns on the $\bar{x}$ and R charts, one should consider the two charts jointly. If the underlying distribution is normal, then the random variables $\bar{x}$ and R computed from the same sample are statistically independent. Therefore, $\bar{x}$ and R should behave independently on the control chart. If there is correlation between the $\bar{x}$ and R values—that is, if the points on the two charts "follow" each other—then this indicates that the underlying distribution is skewed. If specifications have been determined assuming normality, then those analyses may be in error.



■ **FIGURE 6.12** A stratification pattern.

6.2.5 The Effect of Non-normality on $\bar{x}$ and R Charts

There often is an assumption that links **normality** and **control charts** in the development of the performance properties of $\bar{x}$ and R control charts; that is, that the underlying distribution of the quality characteristic is **normal.** In many situations, we may have reason to doubt the validity of this assumption. For example, we may know that the underlying distribution is not normal, because we have collected extensive data that indicate the normality assumption is inappropriate. Now if we know the form of the underlying distribution, it is possible to derive the sampling distributions of $\bar{x}$ and R (or some other measure of process variability) and to obtain exact probability limits for the control charts. This approach could be difficult in some cases, and most analysts would probably prefer to use the standard approach based on the normality assumption if they believed that the effect of departure from this assumption was not serious. However, we may know nothing about the form of the underlying distribution, and then our only choice may be to use the normal theory results. Obviously, in either case, we would be interested in knowing the effect of departures from normality on the usual control charts for $\bar{x}$ and R.

Several authors have investigated the effect of departures from normality on control charts. Burr (1967) notes that the usual normal theory control limit constants are very robust to the normality assumption and can be employed unless the population is extremely non-normal. Schilling and Nelson (1976), Chan, Hapuarachchi, and Macpherson (1988), and Yourstone and Zimmer (1992) have also studied the effect of non-normality on the control limits of the $\bar{x}$ chart. Schilling and Nelson investigated the uniform, right triangular, gamma (with $\lambda = 1$ and $r = \frac{1}{2}$, 1, 2, 3, and 4), and two bimodal distributions formed as mixtures of two normal distributions. Their study indicates that, in most cases, samples of size 4 or 5 are sufficient to ensure reasonable robustness to the normality assumption.

The worst cases observed were for small values of r in the gamma distribution [$r = \frac{1}{2}$ and $r = 1$ (the exponential distribution)]. For example, they report the actual α-risk to be 0.014 or less if $n \geq 4$ for the gamma distribution with $r = \frac{1}{2}$, as opposed to a theoretical value of 0.0027 for the normal distribution.

While the use of three-sigma control limits on the $\bar{x}$ chart will produce an α-risk of 0.0027 if the underlying distribution is normal, the same is not true for the R chart. The sampling distribution of R is not symmetric, even when sampling from the normal distribution, and the long tail of the distribution is on the high or positive side. Thus, symmetric three-sigma limits are only an approximation, and the α-risk on such an R chart is *not* 0.0027. (In fact, for $n = 4$, it is $\alpha = 0.00461$.) Furthermore, the R chart is more sensitive to departures from normality than the $\bar{x}$ chart.

Once again, it is important to remember the role of theory and assumptions such as normality and independence. These are fundamental to the study of the performance of the $\bar{x}$ control chart which is very useful for assessing its suitability for phase II, but plays a much less important role in phase I. In fact, these considerations are not a primary concern in phase I.

6.2.6 The Operating-Characteristic Function

The ability of the $\bar{x}$ and R charts to detect shifts in process quality is described by their operating-characteristic (OC) curves. In this section, we present these OC curves for charts used for phase II monitoring of a process.

Consider the **operating-characteristic (OC) curve for an $\bar{x}$ control chart.** The standard deviation σ is assumed known and constant. If the mean shifts from the in-control value—say, μ_0—to another value $\mu_1 = \mu_0 + k\sigma$, the probability of *not* detecting this shift on the first subsequent sample or the β-risk is

$$\beta = P\left\{\text{LCL} \leq \bar{x} \leq \text{UCL} \middle| \mu = \mu_1 = \mu_0 + k\sigma\right\} \tag{6.18}$$

Since $\bar{x} \sim N(\mu, \sigma^2/n)$, and the upper and lower control limits are UCL $= \mu_0 + L\sigma/\sqrt{n}$ and LCL $= \mu_0 - L\sigma/\sqrt{n}$, we may write equation (6.18) as

$$\beta = \Phi\left[\frac{\text{UCL} - (\mu_0 + k\sigma)}{\sigma/\sqrt{n}}\right] - \Phi\left[\frac{\text{LCL} - (\mu_0 + k\sigma)}{\sigma/\sqrt{n}}\right]$$
$$= \Phi\left[\frac{\mu_0 + L\sigma/\sqrt{n} - (\mu_0 + k\sigma)}{\sigma/\sqrt{n}}\right] - \Phi\left[\frac{\mu_0 - L\sigma/\sqrt{n} - (\mu_0 + k\sigma)}{\sigma/\sqrt{n}}\right]$$

where Φ denotes the standard normal cumulative distribution function. This reduces to

$$\beta = \Phi\left(L - k\sqrt{n}\right) - \Phi\left(-L - k\sqrt{n}\right) \tag{6.19}$$

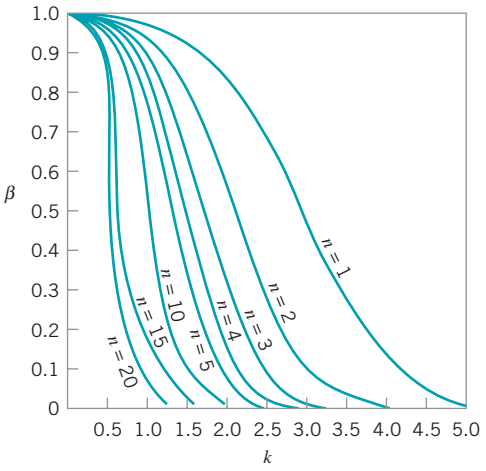
To illustrate the use of equation (6.19), suppose that we are using an $\bar{x}$ chart with $L = 3$ (the usual three-sigma limits), the sample size $n = 5$, and we wish to determine the probability of detecting a shift to $\mu_1 = \mu_0 + 2\sigma$ on the first sample following the shift. Then, since $L = 3$, $k = 2$, and $n = 5$, we have

$$\beta = \Phi\left[3 - 2\sqrt{5}\right] - \Phi\left[-3 - 2\sqrt{5}\right]$$
$$= \Phi(-1.47) - \Phi(-7.37)$$
$$\cong 0.0708$$

This is the β-risk, or the probability of not detecting such a shift. The probability that such a shift *will* be detected on the first subsequent sample is $1 - \beta = 1 - 0.0708 = 0.9292$.

To construct the OC curve for the $\bar{x}$ chart, plot the β-risk against the magnitude of the shift we wish to detect expressed in standard deviation units for various sample sizes n. These probabilities may be evaluated directly from equation (6.19). This OC curve is shown in Fig. 6.13 for the case of three-sigma limits ($L = 3$).

Figure 6.13 indicates that for the typical sample sizes 4, 5, and 6, the $\bar{x}$ chart is not particularly effective in detecting a small shift—say, those on the order of 1.5σ or less—on the first sample following the shift. For example, if the shift is 1.0σ and $n = 5$, then from Fig. 6.13, we have $\beta = 0.75$, approximately. Thus, the probability that the shift will be detected on the first sample is only $1 - \beta = 0.25$. However, the probability that the shift is detected on



■ **FIGURE 6.13** Operating-characteristic curves for the $\bar{x}$ chart with three-sigma limits. $\beta = P$ (not detecting a shift of $k\sigma$ in the mean on the first sample following the shift).

the second sample is $\beta(1-\beta) = 0.75(0.25) = 0.19$, whereas the probability that it is detected on the third sample is $\beta^2(1-\beta) = (0.75^2)0.25 = 0.14$. Thus, the probability that the shift will be detected on the rth subsequent sample is simply $1-\beta$ times the probability of not detecting the shift on each of the initial $r-1$ samples, or

$$\beta^{r-1}(1-\beta)$$

In general, the expected number of samples taken before the shift is detected is simply the **average run length,** or

$$\text{ARL} = \sum_{r=1}^{\infty} r\beta^{r-1}(1-\beta) = \frac{1}{1-\beta}$$

Therefore, in our example, we have

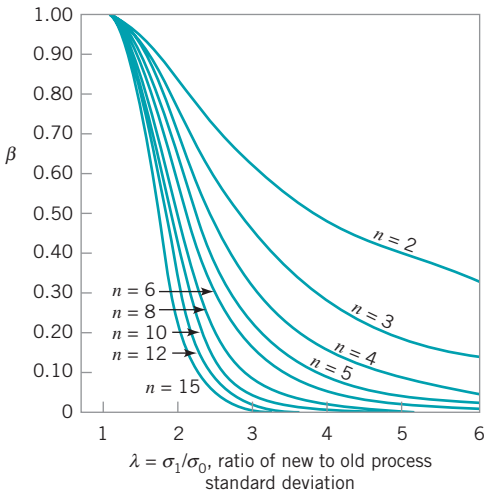
$$\text{ARL} = \frac{1}{1-\beta} = \frac{1}{0.25} = 4$$

In other words, the expected number of samples taken to detect a shift of 1.0σ with $n = 5$ is four.

The above discussion provides a supportive argument for the use of small sample sizes on the $\bar{x}$ chart. Even though small sample sizes often result in a relatively large β-risk, because samples are collected and tested periodically, there is a very good chance that the shift will be detected reasonably quickly, although perhaps not on the first sample following the shift.

To construct the OC curve for the R chart, the distribution of the relative range $W = R/\sigma$ is employed. Suppose that the in-control value of the standard deviation is σ_0. Then the OC curve plots the probability of not detecting a shift to a new value of σ—say, $\sigma_1 > \sigma_0$—on the first sample following the shift. Figure 6.14 presents the OC curve, in which β is plotted against $\lambda = \sigma_1/\sigma_0$ (the ratio of new to old process standard deviation) for various values of n.

From examining Fig. 6.14, we observe that the R chart is not very effective in detecting process shifts for small sample sizes. For example, if the process standard deviation doubles (i.e., $\lambda = \sigma_1/\sigma_0 = 2$), which is a fairly large shift, then samples of size 5 have only about a 40% chance of detecting this shift on each subsequent sample. Most quality engineers believe that the R chart is insensitive to small or moderate shifts for the usual subgroup sizes of $n = 4$, 5, or 6. If $n > 10$ or 12, the s chart discussed in Section 6.3.1 should generally be used instead of the R chart.



■ **FIGURE 6.14** Operating-characteristic curves for the R chart with three-sigma limits. (Adapted from A. J. Duncan, "Operating Characteristics of R Charts," *Industrial Quality Control*, vol. 7, no. 5, pp. 40–41, 1951, with permission of the American Society for Quality Control.)

The OC curves in Figs. 6.13 and 6.14 assume that the $\bar{x}$ and R charts are used for online process monitoring; that is, phase II process monitoring. It is occasionally useful to study the statistical performance of a chart used to analyze past data (phase I). This can give some indication of how the number of preliminary subgroups used to establish the control chart affects the chart's ability to detect out-of-control conditions that existed when the data were collected. It is from such analytical studies, as well as practical experience, that the recommendation to use at least 20 to 25 preliminary subgroups in establishing $\bar{x}$ and R charts has evolved.

6.2.7 The Average Run Length for the $\bar{x}$ Chart

For any **Shewhart control chart,** we have noted previously that the ARL can be expressed as

$$\text{ARL} = \frac{1}{P(\text{one point plots out of control})}$$

or

$$\text{ARL}_0 = \frac{1}{\alpha} \tag{6.20}$$

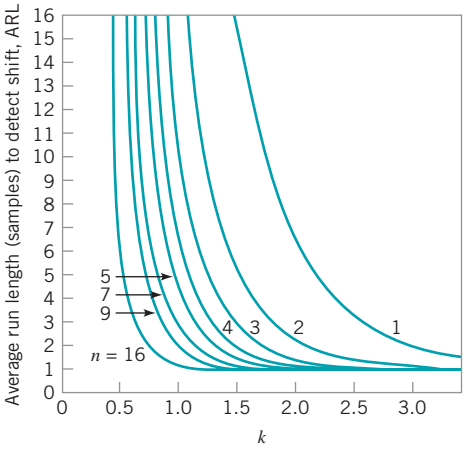
for the in-control ARL and

$$\text{ARL}_1 = \frac{1}{1-\beta} \tag{6.21}$$

for the out-of-control ARL. These results are actually rather intuitive. If the observations plotted on the control chart are independent, then the number of points that must be plotted until the first point exceeds a control limit is a geometric random variable with parameter p (see Chapter 3). The mean of this geometric distribution is simply $1/p$, the average run length.

Since it is relatively easy to develop a general expression for β for the $\bar{x}$ control chart to detect a shift in the mean of $k\sigma$ [see equation (6.19)], then it is not difficult to construct a set of ARL curves for the $\bar{x}$ chart. Figure 6.15 presents these ARL curves for sample sizes of n = 1, 2, 3, 4, 5, 7, 9, and 16 for the $\bar{x}$ control chart, where the ARL is in terms of the expected number of *samples* taken in order to detect the shift. To illustrate the use of Fig. 6.15, note that if we wish to detect a shift of 1.5σ using a sample size of $n = 3$, then the average number of samples required will be $\text{ARL}_1 = 3$. Note also that we could reduce the ARL_1 to approximately 1 if we increased the sample size to $n = 16$.

Recall our discussion in Chapter 5 (Section 5.3.3) indicating that ARLs are subject to some criticism as performance measures for control charts. We noted there (and above as well) that the distribution of run length for a Shewhart control chart is geometric and that this can be a very skewed distribution, so the average (that is, the ARL) may not be the best measure of a typical run length. Another issue concerning ARL relates to the fact that the computations for a specific control chart are usually based on **estimates** of the process



■ **FIGURE 6.15** Average run length (samples) for the $\bar{x}$ chart with three-sigma limits, where the process mean shifts by $k\sigma$. (Adapted from *Modern Methods for Quality Control and Improvement*, by H. M. Wadsworth, K. S. Stephens, and A. B. Godfrey, 2nd edition, John Wiley & Sons, 2002.)

parameters. This results in inflation of both ARL_0 and ARL_1. For example, suppose that the center line of the chart is estimated perfectly, but the process standard deviation is overestimated by 10%. This would result in $ARL_0 = 517$, considerably longer than the nominal or "theoretical" value of 370. With a normally distributed process, we are just as likely to underestimate the process standard deviation by 10%, and this would result in $ARL_0 = 268$, a value considerably shorter than 370. The average is $(268 + 517)/2 = 392.5$, suggesting that errors in estimating the process standard deviation result in ARLs that are overestimated. There is a very comprehensive literature review and discussion of this topic in the paper by Jensen et al. (2006).

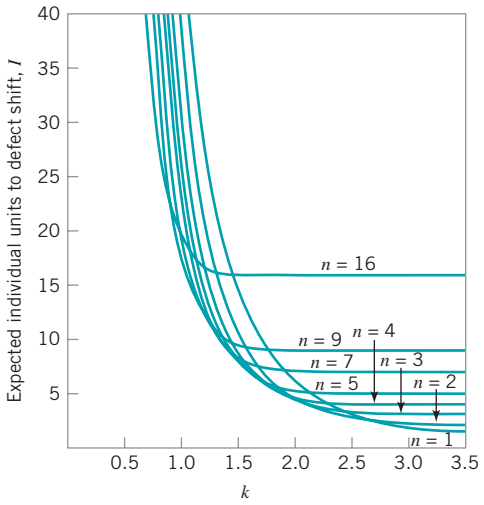
Two other performance measures based on ARL are sometimes of interest. The average time to signal is the number of time periods that occur until a signal is generated on the control chart. If samples are taken at equally spaced intervals of time h, then the **average time to signal** or the ATS is

$$\text{ATS} = \text{ARL}\, h \tag{6.22}$$

It may also be useful to express the ARL in terms of the expected number of individual *units* sampled—say, I—rather than the number of samples taken to detect a shift. If the sample size is n, the relationship between I and ARL is

$$I = n\,\text{ARL} \tag{6.23}$$

Figure 6.16 presents a set of curves that plot the expected number of individual units I that must be sampled for the $\bar{x}$ chart to detect a shift of $k\sigma$. To illustrate the use of the curve, note that to detect a shift of 1.5σ, an $\bar{x}$ chart with $n = 16$ will require that approximately 16 units be sampled, whereas if the sample size is $n = 3$, only about 9 units will be required, on the average.



■ **FIGURE 6.16** Average run length (individual units) for the $\bar{x}$ chart with three-sigma limits, where the process mean shifts by $k\sigma$. (Adapted from *Modern Methods for Quality Control and Improvement*, by H. M. Wadsworth, K. S. Stephens, and A. B. Godfrey, 2nd edition, John Wiley & Sons, 2002.)

6.3 Control Charts for $\bar{x}$ and s

Although $\bar{x}$ and R charts are widely used, it is occasionally desirable to estimate the process standard deviation directly instead of indirectly through the use of the range R. This leads to control charts for $\bar{x}$ and s, where s is the sample standard deviation.[1] Generally, $\bar{x}$ and s charts are preferable to their more familiar counterparts, $\bar{x}$ and R charts, when either

1. The sample size n is moderately large—say, $n > 10$ or 12. (Recall that the range method for estimating σ loses statistical efficiency for moderate to large samples.)
2. The sample size n is variable.

In this section, we illustrate the construction and operation of $\bar{x}$ and s control charts. We also show how to deal with variable sample size and discuss an alternative to the s chart.

6.3.1 Construction and Operation of $\bar{x}$ and s Charts

Setting up and operating control charts for $\bar{x}$ and s requires about the same sequence of steps as those for $\bar{x}$ and R charts, except that for each sample we must calculate the sample average $\bar{x}$ and the sample standard deviation s. Table 6.3 presents the inside diameter measurements of forged automobile engine piston rings. Each sample or subgroup consists of five piston rings. We have calculated the sample average and sample standard deviation for each of the 25 samples. We will use these data to illustrate the construction and operation of $\bar{x}$ and s charts.

If σ^2 is the unknown variance of a probability distribution, then an unbiased estimator of σ^2 is the sample variance

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

[1]Some authors refer to the s chart as the σ chart.

■ TABLE 6.3
Inside Diameter Measurements (mm) for Automobile Engine Piston Rings

Sample Number	Observations					$\bar{x}_i$	s_i
1	74.030	74.002	74.019	73.992	74.008	74.010	0.0148
2	73.995	73.992	74.001	74.011	74.004	74.001	0.0075
3	73.988	74.024	74.021	74.005	74.002	74.008	0.0147
4	74.002	73.996	73.993	74.015	74.009	74.003	0.0091
5	73.992	74.007	74.015	73.989	74.014	74.003	0.0122
6	74.009	73.994	73.997	73.985	73.993	73.996	0.0087
7	73.995	74.006	73.994	74.000	74.005	74.000	0.0055
8	73.985	74.003	73.993	74.015	73.988	73.997	0.0123
9	74.008	73.995	74.009	74.005	74.004	74.004	0.0055
10	73.998	74.000	73.990	74.007	73.995	73.998	0.0063
11	73.994	73.998	73.994	73.995	73.990	73.994	0.0029
12	74.004	74.000	74.007	74.000	73.996	74.001	0.0042
13	73.983	74.002	73.998	73.997	74.012	73.998	0.0105
14	74.006	73.967	73.994	74.000	73.984	73.990	0.0153
15	74.012	74.014	73.998	73.999	74.007	74.006	0.0073
16	74.000	73.984	74.005	73.998	73.996	73.997	0.0078
17	73.994	74.012	73.986	74.005	74.007	74.001	0.0106
18	74.006	74.010	74.018	74.003	74.000	74.007	0.0070
19	73.984	74.002	74.003	74.005	73.997	73.998	0.0085
20	74.000	74.010	74.013	74.020	74.003	74.009	0.0080
21	73.982	74.001	74.015	74.005	73.996	74.000	0.0122
22	74.004	73.999	73.990	74.006	74.009	74.002	0.0074
23	74.010	73.989	73.990	74.009	74.014	74.002	0.0119
24	74.015	74.008	73.993	74.000	74.010	74.005	0.0087
25	73.982	73.984	73.995	74.017	74.013	73.998	0.0162
						$\Sigma =$ 1850.028	0.2351
						$\bar{\bar{x}} =$ 74.001	$\bar{s} = 0.0094$

However, the sample standard deviation s is *not* an unbiased estimator of σ. In Chapter 4 (Section 4.2) we observed that if the underlying distribution is normal, then s actually estimates $c_4\sigma$, where c_4 is a constant that depends on the sample size n. Furthermore, the standard deviation of s is $\sigma\sqrt{1 - c_4^2}$. This information can be used to establish control charts on $\bar{x}$ and s.

Consider the case where a standard value is given for σ. Since $E(s) = c_4\sigma$, the center line for the chart is $c_4\sigma$. The three-sigma control limits for s are then

$$\text{UCL} = c_4\sigma + 3\sigma\sqrt{1 - c_4^2}$$

$$\text{LCL} = c_4\sigma - 3\sigma\sqrt{1 - c_4^2}$$

It is customary to define the two constants

$$B_5 = c_4 - 3\sqrt{1 - c_4^2} \quad \text{and} \quad B_6 = c_4 + 3\sqrt{1 - c_4^2} \tag{6.24}$$

Consequently, the parameters of the s chart with a standard value for σ given become

$$\begin{aligned} \text{UCL} &= B_6\sigma \\ \text{Center line} &= c_4\sigma \\ \text{LCL} &= B_5\sigma \end{aligned} \tag{6.25}$$

Values of B_5 and B_6 are tabulated for various sample sizes in Appendix Table VI. The parameters of the corresponding $\bar{x}$ chart are given in equation (6.15), Section 6.2.3.

If no standard is given for σ, then it must be estimated by analyzing past data. Suppose that m preliminary samples are available, each of size n, and let s_i be the standard deviation of the ith sample. The average of the m standard deviations is

$$\bar{s} = \frac{1}{m}\sum_{i=1}^{m} s_i$$

The statistic $\bar{s}/c_4$ is an unbiased estimator of σ. Therefore, the parameters of the s chart would be

$$\begin{aligned} \text{UCL} &= \bar{s} + 3\frac{\bar{s}}{c_4}\sqrt{1-c_4^2} \\ \text{Center line} &= \bar{s} \\ \text{LCL} &= \bar{s} - 3\frac{\bar{s}}{c_4}\sqrt{1-c_4^2} \end{aligned}$$

We usually define the constants

$$B_3 = 1 - \frac{3}{c_4}\sqrt{1-c_4^2} \quad \text{and} \quad B_4 = 1 + \frac{3}{c_4}\sqrt{1-c_4^2} \tag{6.26}$$

Consequently, we may write the parameters of the s chart as

$$\begin{aligned} \text{UCL} &= B_4\bar{s} \\ \text{Center line} &= \bar{s} \\ \text{LCL} &= B_3\bar{s} \end{aligned} \tag{6.27}$$

Note that $B_4 = B_6/c_4$ and $B_3 = B_5/c_4$.

When $\bar{s}/c_4$ is used to estimate σ, we may define the control limits on the corresponding $\bar{x}$ chart as

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + \frac{3\bar{s}}{c_4\sqrt{n}} \\ \text{Center line} &= \bar{\bar{x}} \\ \text{LCL} &= \bar{\bar{x}} - \frac{3\bar{s}}{c_4\sqrt{n}} \end{aligned}$$

Let the constant $A_3 = 3/(c_4\sqrt{n})$. Then the $\bar{x}$ chart parameters become

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + A_3\bar{s} \\ \text{Center line} &= \bar{\bar{x}} \\ \text{LCL} &= \bar{\bar{x}} - A_3\bar{s} \end{aligned} \tag{6.28}$$

The constants B_3, B_4, and A_3 for construction of $\bar{x}$ and s charts from past data are listed in Appendix Table VI for various sample sizes.

Note that we have assumed that the sample standard deviation is defined as

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}} \tag{6.29}$$

Some authors define s with n in the denominator of equation (6.29) instead of $n - 1$. When this is the case, the definitions of the constants c_4, B_3, B_4, and A_3 are altered. The corresponding constants based on the use of n in calculating s are called c_2, B_1, B_2, and A_1, respectively. See Bowker and Lieberman (1972) for their definitions.

Traditionally, quality engineers have preferred the R chart to the s chart because of the simplicity of calculating R from each sample. The availability of hand-held calculators with automatic calculation of s and computers at workstations to implement control charts on site have eliminated any computational difficulty.

EXAMPLE 6.3 $\bar{x}$ and s Charts for the Piston Ring Data

Construct and interpret $\bar{x}$ and s charts using the piston ring inside diameter measurements in Table 6.3.

SOLUTION

The grand average and the average standard deviation are

$$\bar{\bar{x}} = \frac{1}{25}\sum_{i=1}^{25}\bar{x}_i = \frac{1}{25}(1850.028) = 74.001$$

and

$$\bar{s} = \frac{1}{25}\sum_{i=1}^{25}s_i = \frac{1}{25}(0.2351) = 0.0094$$

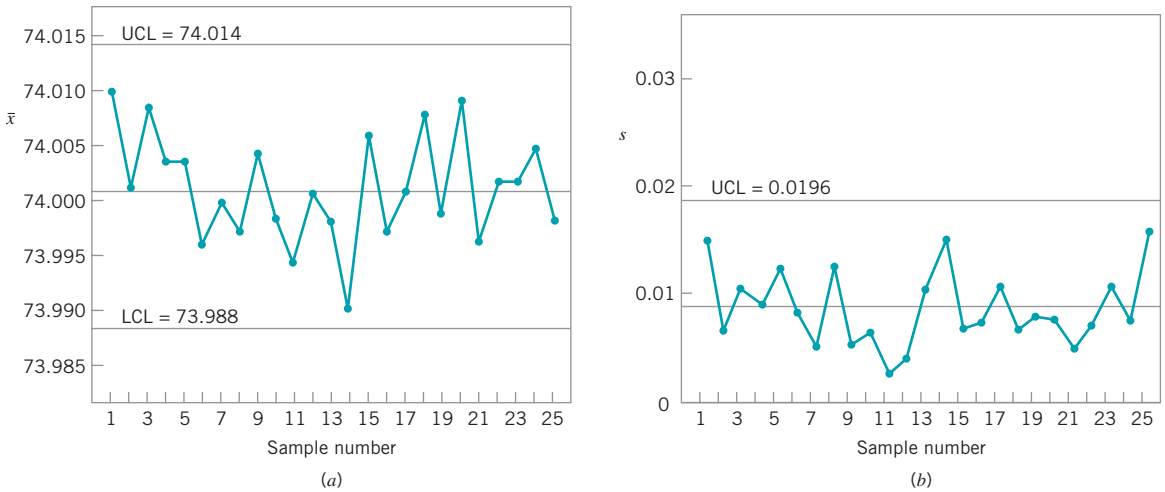
respectively. Consequently, the parameters for the $\bar{x}$ chart are

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + A_3\bar{s} = 74.001 + (1.427)(0.0094) = 74.014 \\ \text{CL} &= \bar{\bar{x}} = 74.001 \\ \text{LCL} &= \bar{\bar{x}} - A_3\bar{s} = 74.001 - (1.427)(0.0094) = 73.988 \end{aligned}$$

and for the s chart

$$\begin{aligned} \text{UCL} &= B_4\bar{s} = (2.089)(0.0094) = 0.0196 \\ \text{CL} &= \bar{s} = 0.0094 \\ \text{LCL} &= B_3\bar{s} = (0)(0.0094) = 0 \end{aligned}$$

The control charts are shown in Fig. 6.17. There is no indication that the process is out of control, so those limits could be adopted for phase II monitoring of the process.



■ **FIGURE 6.17** The $\bar{x}$ and s control charts for Example 6.3. (*a*) The $\bar{x}$ chart with control limits based on $\bar{s}$. (*b*) The s control chart.

Estimation of σ. We can estimate the process standard deviation using the fact that s/c_4 is an unbiased estimate of σ. Therefore, since $c_4 = 0.9400$ for samples of size 5, our estimate of the process standard deviation is

$$\hat{\sigma} = \frac{\bar{s}}{c_4} = \frac{0.0094}{0.9400} = 0.01$$

6.3.2 The $\bar{x}$ and s Control Charts with Variable Sample Size

The $\bar{x}$ and s control charts are relatively easy to apply in cases where the sample sizes are variable. In this case, we should use a weighted average approach in calculating $\bar{\bar{x}}$ and $\bar{s}$. If n_i is the number of observations in the ith sample, then use

$$\bar{\bar{x}} = \frac{\sum_{i=1}^{m} n_i \bar{x}_i}{\sum_{i=1}^{m} n_i} \tag{6.30}$$

and

$$\bar{s} = \left[\frac{\sum_{i=1}^{m} (n_i - 1) s_i^2}{\sum_{i=1}^{m} n_i - m} \right]^{1/2} \tag{6.31}$$

as the center lines on the $\bar{x}$ and s control charts, respectively. The control limits would be calculated from equations (6.27) and (6.28), respectively, but the constants A_3, B_3, and B_4 will depend on the sample size used in each individual subgroup.

EXAMPLE 6.4 $\bar{x}$ and s Chart for the Piston Rings, Variable Sample Size

Consider the data in Table 6.4, which is a modification of the piston-ring data used in Example 6.3. Note that the sample sizes vary from $n = 3$ to $n = 5$. Use the procedure described on page 255 to set up the $\bar{x}$ and s control charts.

SOLUTION

The weighted grand mean and weighted average standard deviation are computed from equations (6.30) and (6.31) as follows:

$$\bar{\bar{x}} = \frac{\sum_{i=1}^{25} n_i \bar{x}_i}{\sum_{i=1}^{25} n_i} = \frac{5(74.010) + 3(73.996) + \cdots + 5(73.998)}{5 + 3 + \cdots + 5}$$

$$= \frac{8362.075}{113} = 74.001$$

■ TABLE 6.4
Inside Diameter Measurements (mm) on Automobile Engine Piston Rings

Sample Number	Observations					$\bar{x}_i$	s_i
1	74.030	74.002	74.019	73.992	74.008	74.010	0.0148
2	73.995	73.992	74.001			73.996	0.0046
3	73.988	74.024	74.021	74.005	74.002	74.008	0.0147
4	74.002	73.996	73.993	74.015	74.009	74.003	0.0091
5	73.992	74.007	74.015	73.989	74.014	74.003	0.0122
6	74.009	73.994	73.997	73.985		73.996	0.0099
7	73.995	74.006	73.994	74.000		73.999	0.0055
8	73.985	74.003	73.993	74.015	73.988	73.997	0.0123
9	74.008	73.995	74.009	74.005		74.004	0.0064
10	73.998	74.000	73.990	74.007	73.995	73.998	0.0063
11	73.994	73.998	73.994	73.995	73.990	73.994	0.0029
12	74.004	74.000	74.007	74.000	73.996	74.001	0.0042
13	73.983	74.002	73.998			73.994	0.0100
14	74.006	73.967	73.994	74.000	73.984	73.990	0.0153
15	74.012	74.014	73.998			74.008	0.0087
16	74.000	73.984	74.005	73.998	73.996	73.997	0.0078
17	73.994	74.012	73.986	74.005		73.999	0.0115
18	74.006	74.010	74.018	74.003	74.000	74.007	0.0070
19	73.984	74.002	74.003	74.005	73.997	73.998	0.0085
20	74.000	74.010	74.013			74.008	0.0068
21	73.982	74.001	74.015	74.005	73.996	74.000	0.0122
22	74.004	73.999	73.990	74.006	74.009	74.002	0.0074
23	74.010	73.989	73.990	74.009	74.014	74.002	0.0119
24	74.015	74.008	73.993	74.000	74.010	74.005	0.0087
25	73.982	73.984	73.995	74.017	74.013	73.998	0.0162

and

$$\bar{s} = \left[\frac{\sum_{i=1}^{25}(n_i - 1)s_i^2}{\sum_{i=1}^{25} n_i - 25}\right]^{1/2} = \left[\frac{4(0.0148)^2 + 2(0.0046)^2 + \cdots + 4(0.0162)^2}{5 + 3 + \cdots + 5 - 25}\right]^{1/2}$$
$$= \left[\frac{0.009324}{88}\right]^{1/2} = 0.0103$$

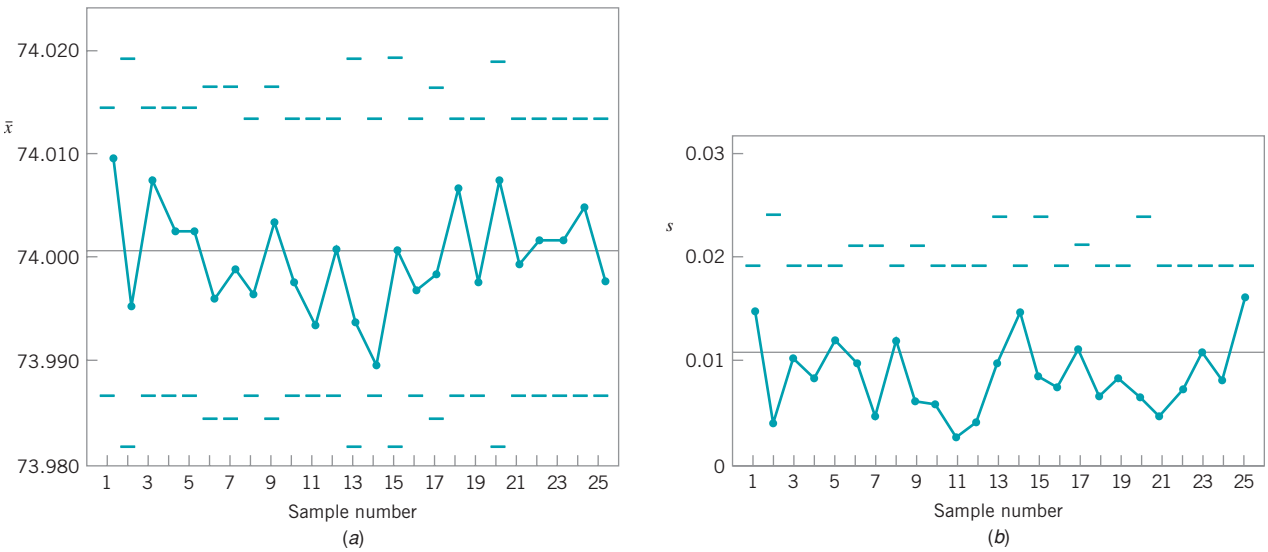
Therefore, the center line of the $\bar{x}$ chart is $\bar{\bar{x}} = 74.001$, and the center line of the s chart is $\bar{s} = 0.0103$. The control limits may now be easily calculated. To illustrate, consider the first sample. The limits for the $\bar{x}$ chart are

$$\text{UCL} = 74.001 \quad 1.427 \quad 0.0103 \quad = 74.016$$
$$\text{CL} = 74.001$$
$$\text{LCL} = 74.001 - \ 1.427 \quad 0.0103 \quad = 73.986$$

The control limits for the s chart are

$$\text{UCL} = \ 2.089 \quad 0.0103 \quad = 0.022$$
$$\text{CL} = 0.0103$$
$$\text{LCL} = 0 \ 0.0103 \quad = 0$$

Note that we have used the values of A_3, B_3, and B_4 for $n_1 = 5$. The limits for the second sample would use the values of these constants for $n_2 = 3$. The control limit calculations for all 25 samples are summarized in Table 6.5. The control charts are plotted in Fig. 6.18.



■ **FIGURE 6.18** The (*a*) $\bar{x}$ and (*b*) s control charts for piston-ring data with variable sample size, Example 6.4.

(*continued*)

■ TABLE 6.5
Computation of Control Limits for $\bar{x}$ and s Charts with Variable Sample Size

					$\bar{x}$ Chart				s Chart	
Sample	n	$\bar{x}$	s	A_3	LCL	UCL	B_3	B_4	LCL	UCL
1	5	74.010	0.0148	1.427	73.986	74.016	0	2.089	0	0.022
2	3	73.996	0.0046	1.954	73.981	74.021	0	2.568	0	0.026
3	5	74.008	0.0147	1.427	73.986	74.016	0	2.089	0	0.022
4	5	74.003	0.0091	1.427	73.986	74.016	0	2.089	0	0.022
5	5	74.003	0.0122	1.427	73.986	74.016	0	2.089	0	0.022
6	4	73.996	0.0099	1.628	73.984	74.018	0	2.266	0	0.023
7	4	73.999	0.0055	1.628	73.984	74.018	0	2.266	0	0.023
8	5	73.997	0.0123	1.427	73.986	74.016	0	2.089	0	0.022
9	4	74.004	0.0064	1.628	73.984	74.018	0	2.266	0	0.023
10	5	73.998	0.0063	1.427	73.986	74.016	0	2.089	0	0.022
11	5	73.994	0.0029	1.427	73.986	74.016	0	2.089	0	0.022
12	5	74.001	0.0042	1.427	73.986	74.016	0	2.089	0	0.022
13	3	73.994	0.0100	1.954	73.981	74.021	0	2.568	0	0.026
14	5	73.990	0.0153	1.427	73.986	74.016	0	2.089	0	0.022
15	3	74.008	0.0087	1.954	73.981	74.021	0	2.568	0	0.026
16	5	73.997	0.0078	1.427	73.986	74.016	0	2.089	0	0.022
17	4	73.999	0.0115	1.628	73.984	74.018	0	2.226	0	0.023
18	5	74.007	0.0070	1.427	73.986	74.016	0	2.089	0	0.022
19	5	73.998	0.0085	1.427	73.986	74.016	0	2.089	0	0.022
20	3	74.008	0.0068	1.954	73.981	74.021	0	2.568	0	0.026
21	5	74.000	0.0122	1.427	73.986	74.016	0	2.089	0	0.022
22	5	74.002	0.0074	1.427	73.986	74.016	0	2.089	0	0.022
23	5	74.002	0.0119	1.427	73.986	74.016	0	2.089	0	0.022
24	5	74.005	0.0087	1.427	73.986	74.016	0	2.089	0	0.022
25	5	73.998	0.0162	1.427	73.986	74.016	0	2.089	0	0.022

An alternative to using variable-width control limits on the $\bar{x}$ and s control charts is to base the control limit calculations on an **average sample size $\bar{n}$.** If the n_i are not very different, this approach may be satisfactory in some situations; it is particularly helpful if the charts are to be used in a presentation to management. Since the average sample size n_i may not be an integer, a useful alternative is to base these approximate control limits on a modal (most common) sample size.

Estimation of σ. We may estimate the process standard deviation, σ, from the individual sample values s_i. First, average all the values of s_i for which $n_i = 5$ (the most frequently occurring value of n_i). This gives

$$\bar{s} = \frac{0.1715}{17} = 0.0101$$

The estimate of the process σ is then

$$\hat{\sigma} = \frac{\bar{s}}{c_4} = \frac{0.0101}{0.9400} = 0.01$$

where the value of c_4 used is for samples of size $n = 5$.

6.3.3 The s^2 Control Chart

Most quality engineers use either the R chart or the s chart to monitor process variability, with s preferable to R for moderate to large sample sizes. Some practitioners recommend a control chart based directly on the sample variance s^2, the ***s*² control chart.** The parameters for the s^2 control chart are

$$\begin{aligned} \text{UCL} &= \frac{\bar{s}^2}{n-1}\chi^2_{\alpha/2,n-1} \\ \text{Center line} &= \bar{s}^2 \\ \text{LCL} &= \frac{\bar{s}^2}{n-1}\chi^2_{1-(\alpha/2),n-1} \end{aligned} \tag{6.32}$$

where $\chi^2_{\alpha/2,}$ and $\chi^2_{(\alpha/2),n-1}$ denote the upper and lower $\alpha/2$ percentage points of the chi-square distribution with $n - 1$ degrees of freedom, and $\bar{s}^2$ is an average sample variance obtained from the analysis of preliminary data. A standard value σ^2 could be used in equation (6.32) instead of $\bar{s}^2$ if one were available. Note that this control chart is defined with probability limits.

6.4 The Shewhart Control Chart for Individual Measurements

There are many situations in which the sample size used for process monitoring is $n = 1$; that is, the sample consists of an individual unit. Some examples of these situations are as follows:

1. Automated inspection and measurement technology is used, and every unit manufactured is analyzed so there is no basis for rational subgrouping.
2. Data comes available relatively slowly, and it is inconvenient to allow sample sizes of $n > 1$ to accumulate before analysis. The long interval between observations will cause problems with rational subgrouping. This occurs frequently in both manufacturing and nonmanufacturing situations.
3. Repeat measurements on the process differ only because of laboratory or analysis error, as in many chemical processes.
4. Multiple measurements are taken on the same unit of product, such as measuring oxide thickness at several different locations on a wafer in semiconductor manufacturing.
5. In process plants, such as papermaking, measurements on some parameter such as coating thickness *across* the roll will differ very little and produce a standard deviation that is much too small if the objective is to control coating thickness *along* the roll.

In such situations, the **control chart for individual units** is useful. (The cumulative sum and exponentially weighted moving-average control charts discussed in Chapter 9 will be a better alternative in phase II or when the magnitude of the shift in process mean that is of interest is small.) In many applications of the *individuals control chart*, we use the *moving range* two successive observations as the basis of estimating the process variability. The moving range is defined as

$$MR_i = |x_i - x_{i-1}|$$

It is also possible to establish a **moving range control chart.** The procedure is illustrated in the following example.

EXAMPLE 6.5 Loan Processing Costs

The mortgage loan processing unit of a bank monitors the costs of processing loan applications. The quantity tracked is the average weekly processing costs, obtained by dividing total weekly costs by the number of loans processed during the week. The processing costs for the most recent 20 weeks are shown in Table 6.6. Set up individual and moving range control charts for these data.

SOLUTION

To set up the control chart for individual observations, note that the sample average cost of the 20 observations is $\bar{x} = 300.5$ and that the average of the moving ranges of two observations is $\overline{MR} = 7.79$. To set up the moving range chart, we use $D_3 = 0$ and $D_4 = 3.267$ for $n = 2$. Therefore, the moving range chart has center line $\overline{MR} = 7.79$, LCL = 0, and UCL $= D_4\overline{MR} = (3.267)7.79 = 25.45$. The control chart (from Minitab) is shown in Fig. 6.19*b*. Notice that no points are out of control.

For the control chart for individual measurements, the parameters are

$$\begin{aligned} \text{UCL} &= \bar{x} + 3\frac{\overline{MR}}{d_2} \\ \text{Center line} &= \bar{x} \\ \text{LCL} &= \bar{x} - 3\frac{\overline{MR}}{d_2} \end{aligned} \qquad (6.33)$$

■ TABLE 6.6
Costs of Processing Mortgage Loan Applications

Weeks	Cost x	Moving Range MR
1	310	
2	288	22
3	297	9
4	298	1
5	307	9
6	303	4
7	294	9
8	297	3
9	308	11
10	306	2
11	294	12
12	299	5
13	297	2
14	299	2
15	314	15
16	295	19
17	293	2
18	306	13
19	301	5
20	304	3
	$\bar{x} = 300.5$	$\overline{MR} = 7.79$

If a moving range of $n = 2$ observations is used, then $d_2 = 1.128$. For the data in Table 6.6, we have

$$\text{UCL} = \bar{x} + 3\frac{\overline{\text{MR}}}{d_2} = 300.5 + 3\frac{7.79}{1.128} = 321.22$$

$$\text{Center line} = \bar{x} = 34.088$$

$$\text{LCL} = \bar{x} - 3\frac{\overline{\text{MR}}}{d_2} = 300.5 - 3\frac{7.79}{1.128} = 279.78$$

The control chart for individual cost values is shown in Fig. 6.19*a*. There are no out of control observations on the individuals control chart.

The interpretation of the individuals control chart is very similar to the interpretation of the ordinary $\bar{x}$ control chart. A shift in the process mean will result in a single point or a series of points that plot outside the control limits on the control chart for individuals. Sometimes a point will plot outside the control limits on both the individuals chart and the moving range chart. This will often occur because a large value of x will also lead to a large value of the moving range for that sample. This is very typical behavior for the individuals and moving range control charts. It is most likely an indication that the mean is out of control and not an indication that both the mean and the variance of the process are out of control.

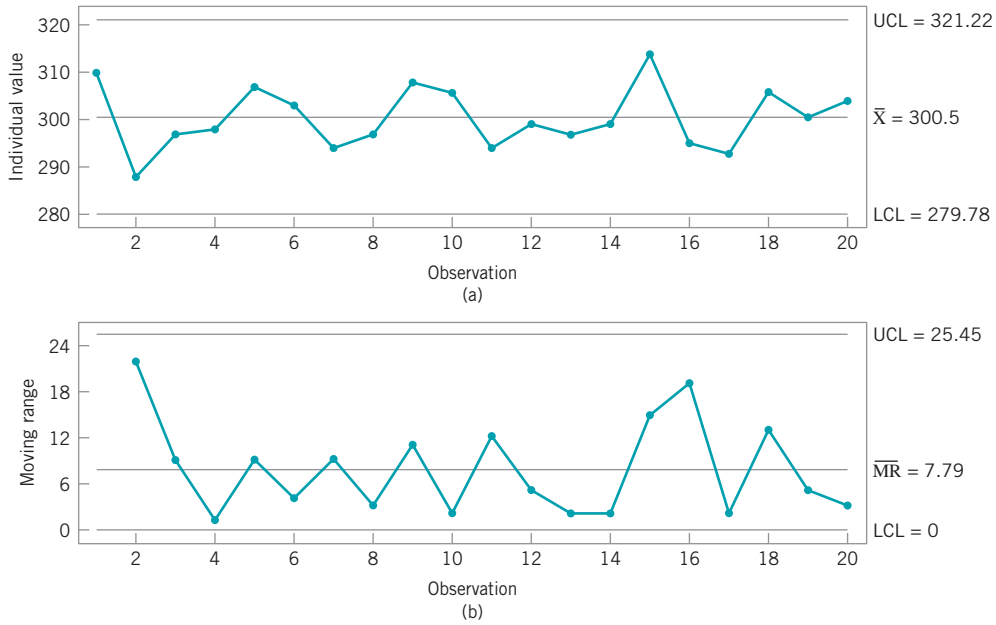


FIGURE 6.19 Control charts for (*a*) individual observations on cost and for (*b*) the moving range.

Phase II Operation and Interpretation of the Charts. Table 6.7 contains data on mortgage application processing costs for weeks 21–40. These data are plotted in Fig. 6.20 on the continuation of the control chart for individuals and the moving range control chart developed in Example 6.5. As this figure makes clear, an upward shift in cost has occurred around week 39, since there is an obvious "shift in process level" pattern on the chart for individuals followed by another out-of-control signal at week 40. Note that the moving range chart also reacts to this level shift with a single large spike at week 39. This spike on the moving range chart is sometimes helpful in identifying exactly where a process shift in the mean has occurred. Clearly one should look for possible assignable causes around weeks 39. Possible causes could include an unusual number of applications requiring additional manual underwriting work, or possibly new underwriters working in the process, or possibly temporary underwriters replacing regular employees taking vacations.

Some care should be exercised in interpreting patterns on the moving range chart. The moving ranges are correlated, and this correlation may often induce a pattern of runs or cycles



■ **FIGURE 6.20** Continuation of the control chart for individuals and the moving range using the additional data in Table 6.7.

■ **TABLE 6.7**
Costs of Processing Mortgage Loan Applications, Weeks 21–40

Week	Cost x	Week	Cost x
21	305	31	310
22	282	32	292
23	305	33	305
24	296	34	299
25	314	35	304
26	295	36	310
27	287	37	304
28	301	38	305
29	298	39	333
30	311	40	328

on the chart. Such a pattern is evident on the moving range chart in Fig. 6.21. The individual measurements on the x chart are assumed to be uncorrelated, however, and any apparent pattern on this chart should be carefully investigated.

Some authorities have recommended not constructing and plotting the moving range chart. They point out (correctly, we add) that the moving range chart cannot really provide useful information about a shift in process variability. For example, see Rigdon, Cruthis, and Champ (1994). In effect, shifts in the process mean also show up on the moving range chart. Our feeling is that as long as the analyst is careful in interpretation and relies primarily on the individuals chart, little trouble will ensue from plotting both charts.

Average Run Lengths. Crowder (1987b) has studied the average run length of the combined control chart for individuals and moving range chart. He produces ARLs for various settings of the control limits and shifts in the process mean and standard deviation. In general, his work shows that the ARL_0 of the combined procedure will generally be much less than the ARL_0 of a standard Shewhart control chart when the process is in control (recall that ARL_0 for a

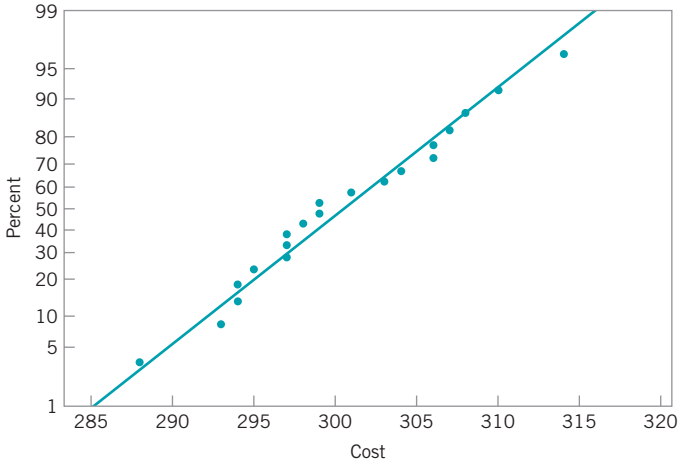


FIGURE 6.21
Normal probability plot of the mortgage application processing cost data from Table 6.6, Example 6.5.

Shewhart chart is 370 samples), if we use the conventional three-sigma limits on the charts. In general, results closer to the Shewhart in-control ARL are obtained if we use three-sigma limits on the chart for individuals and compute the upper control limit on the moving range chart from

$$\text{UCL} = D\overline{\text{MR}}$$

where the constant D should be chosen such that $4 \le D \le 5$.

One can get a very good idea about the ability of the individuals control chart to detect process shifts by looking at the OC curves in Fig. 6.13 or the ARL curves in Fig. 6.15. For an individuals control chart with three-sigma limits, we can compute the following:

Size of Shift	β	ARL_1
1σ	0.9772	43.96
2σ	0.8413	6.30
3σ	0.5000	2.00

Note that the ability of the individuals control chart to detect small shifts is very poor. For instance, consider a continuous chemical process in which samples are taken every hour. If a shift in the process mean of about one standard deviation occurs, the information above tells us that it will take about 44 samples, on the average, to detect the shift. This is nearly two full days of continuous production in the out-of-control state, a situation that has potentially devastating economic consequences. This limits the usefulness of the individuals control chart in phase II process monitoring.

Some individuals have suggested that control limits narrower than three-sigma be used on the chart for individuals to enhance its ability to detect small process shifts. This is a dangerous suggestion, as narrower limits will dramatically reduce the value of ARL_0 and increase the occurrence of false alarms to the point where the charts are ignored and hence become useless. If we are interested in detecting small shifts in phase II, then the correct approach is to use either the cumulative sum control chart or the exponentially weighted moving average control chart (see Chapter 9).

Normality. Our discussion in this section has made an assumption that the observations follow a normal distribution. Borror, Montgomery, and Runger (1999) have studied the phase II performance of the Shewhart control chart for individuals when the process data are not normal. They investigated various gamma distributions to represent skewed process data and t distributions to represent symmetric normal-like data. They found that the in-control ARL is dramatically

affected by non-normal data. For example, if the individuals chart has three-sigma limits so that for normal data $ARL_0 = 370$, the actual ARL_0 for gamma-distributed data is between 45 and 97, depending on the shape of the gamma distribution (more highly skewed distributions yield poorer performance). For the t distribution, the ARL_0 values range from 76 to 283 as the degrees of freedom increase from 4 to 50 (that is, as the t becomes more like the normal distribution).

In the face of these results, we conclude that if the process shows evidence of even moderate departure from normality, the control limits given here may be entirely inappropriate for phase II process monitoring. One approach to dealing with the problem of non-normality would be to determine the control limits for the individuals control chart based on the percentiles of the correct underlying distribution. These percentiles could be obtained from a histogram if a large sample (at least 100 but preferably 200 observations) were available, or from a probability distribution fit to the data. See Willemain and Runger (1996) for details on designing control charts from empirical reference distributions. Another approach would be to transform the original variable to a new variable that is approximately normally distributed, and then apply control charts to the new variable. Borror, Montgomery, and Runger (1999) show how a properly designed EWMA control chart is very insensitive to the normality assumption. This approach will be discussed in Chapter 9.

It is important to check the normality assumption when using the control chart for individuals. A simple way to do this is with the normal probability plot. Figure 6.21 is the normal probability plot for the mortgage application processing cost data in Table 6.6. There is no obvious problem with the normality assumption in these data. However, remember that the normal probability plot is but a **crude check** of the normality assumption and the individuals control chart is very sensitive to non-normality. Furthermore, mean shifts could show up as a problem with normality on the normal probability plot. Process stability is needed to properly interpret the plot. We suggest that the Shewhart individuals chart be used with extreme caution, particularly in phase II process monitoring.

EXAMPLE 6.6 The Use of Transformations

Table 6.8 presents consecutive measurements on the resistivity of 25 silicon wafers after an epitaxial layer is deposited in a single-wafer deposition process. Construct an individuals control chart for this process.

TABLE 6.8
Resistivity Data for Example 6.6

Sample, i	Resistivity (x_i)	ln (x_i)	MR	Sample, i	Resistivity (x_i)	ln (x_i)	MR
1	216	5.37528		14	242	5.48894	0.23791
2	290	5.66988	0.29460	15	168	5.12396	0.36498
3	236	5.46383	0.20605	16	360	5.88610	0.76214
4	228	5.42935	0.03448	17	226	5.42053	0.46557
5	244	5.49717	0.06782	18	253	5.53339	0.11286
6	210	5.34711	0.15006	19	380	5.94017	0.40678
7	139	4.93447	0.41264	20	131	4.87520	1.06497
8	310	5.73657	0.80210	21	173	5.15329	0.27809
9	240	5.48064	0.25593	22	224	5.41165	0.25836
10	211	5.35186	0.12878	23	195	5.27300	0.13865
11	175	5.16479	0.18707	24	199	5.29330	0.02030
12	447	6.10256	0.93777	25	226	5.42053	0.12723
13	307	5.72685	0.37571				
						$\overline{\ln(x_i)} = 5.44402$	$\overline{MR} = 0.33712$

SOLUTION

A normal probability plot of the resistivity measurements is shown in Fig. 6.22. This plot was constructed by Minitab, which fits the line to the points by least squares (not the best method). It is clear from inspection of the normal probability plot that the normality assumption for resistivity is at best questionable, so it would be dangerous to apply an individuals control chart to the original process data.

Figure 6.22 indicates that the distribution of resistivity has a long tail to the right, and consequently we would expect the log transformation (or a similar transformation) to produce a distribution that is closer to normal. The natural log of resistivity is shown in column three of Table 6.8, and the normal probability plot of the natural log of resistivity is shown in Fig. 6.23. Clearly the log transformation has resulted in a new variable that is more nearly approximated by a normal distribution than were the original resistivity measurements.

The last column of Table 6.8 shows the moving ranges of the natural log of resistivity. Figure 6.24 presents the individuals and moving range control charts for the natural log of resistivity. Note that there is no indication of an out-of-control process.

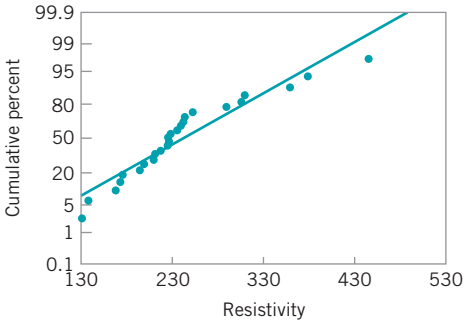


FIGURE 6.22 Normal probability plot of resistivity.

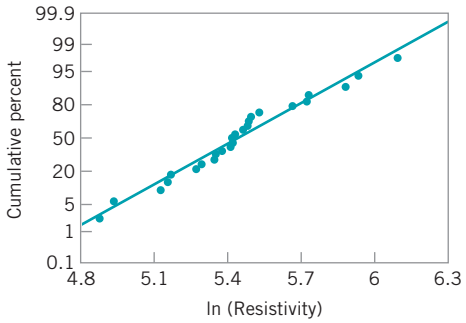


FIGURE 6.23 Normal probability plot of ln (resistivity).

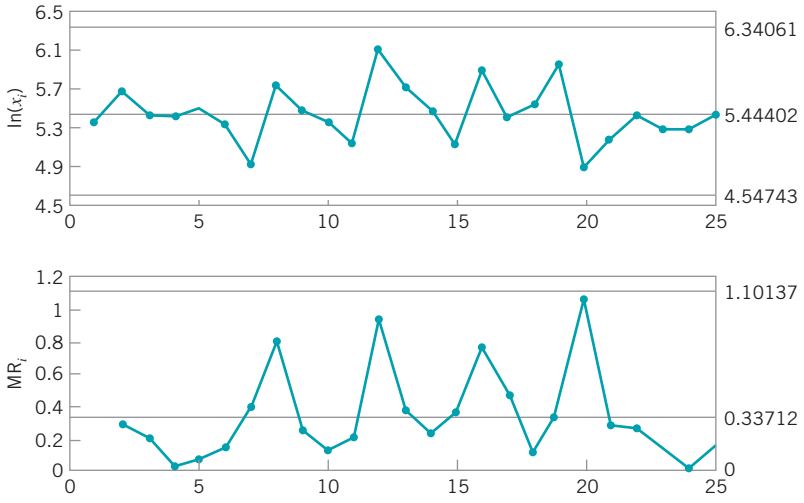


FIGURE 6.24 Individuals and moving range control charts on ln (resistivity), Example 6.6

More about Estimating σ. Very often in practice we use moving ranges in estimating σ for the individuals control chart. Recall that moving ranges are defined as $MR_i = |(x_i - x_{i-1})|$, $i = 2, 3 \ldots, m$. More properly, this statistic should be called a moving range of **span two** since the number of observations used to calculate the range in the moving window could be increased. The most common estimator is the one we used in Example 6.5, based on the **average moving range** $\overline{MR} = \sum_{i=2}^{m} MR_i/(m-1)$ and can be written as

$$\hat{\sigma}_1 = 0.8865\overline{MR}$$

where the constant 0.8865 is the reciprocal of d_2 for samples of size 2. For in-control processes, Cryer and Ryan (1990), among others, have pointed out that a more efficient estimator is one based on the sample standard deviation

$$\hat{\sigma}_2 = \frac{s}{c_4}$$

Both of these estimators are unbiased, assuming that no assignable causes are present in the sequence of m individual observations.

If assignable causes are present, then both $\hat{\sigma}_1$ and $\hat{\sigma}_2$ result in biased estimates of the process standard deviation. To illustrate, suppose that in the sequence of individual observations

$$x_1, x_2, \ldots, x_t, x_{t+1}, \ldots, x_m$$

the process is in control with mean μ_0 and standard deviation σ for the first t observations, but between x_t and x_{t+1} an assignable cause occurs that results in a sustained shift in the process mean to a new level $\mu = \mu_0 + \delta\sigma$ and the mean remains at this new level for the remaining sample observations $x_t + 1, \ldots, x_m$. Under these conditions, Woodall and Montgomery (2000–01) show that

$$E\left(s^2\right) = \sigma^2 + \frac{t(m-1)}{m(m-1)}(\delta\sigma)^2$$

In fact, this result holds for any case in which the mean of t of the observations is μ_0 and the mean of the remaining observations is $\mu_0 + \delta\sigma$, since the order of the observations is not relevant in computing s^2. Therefore, s^2 is biased upwards, and consequently $\hat{\sigma}_2 = S/c_4$ will tend to overestimate σ. Note that the extent of the bias in $\hat{\sigma}_2$ depends on the magnitude of the shift in the mean ($\delta\sigma$), the time period following which the shift occurs (t), and the number of available observations (m). Now the moving range is only impacted by the shift in the mean during one period ($t + 1$), so the bias in $\hat{\sigma}_1$ depends only on the shift magnitude and m. If $1 < t < m - 1$, the bias in $\hat{\sigma}_1$ will always be smaller than the bias in $\hat{\sigma}_2$. Cruthis and Rigdon (1992–93) show how the ratio

$$F^* = \left(\frac{\hat{\sigma}_1}{\hat{\sigma}_2}\right)^2$$

can be used to determine whether the process was in control when both estimates were calculated. They use simulation to obtain the approximate 90th, 95th, 99th, and 99.9th percentiles of the distribution of F^* for sample sizes $m = 10(5)100$, assuming that the process is in control. Since this is an empirical reference distribution, observed values of F^* that exceed one of these percentiles is an indication that the process was not in control over the time period during which the m observations were collected.

One way to reduce or possibly eliminate the bias in estimating σ when a sustained shift in the mean is present is to base the estimator on the **median** of the moving ranges of span two, as suggested by Clifford (1959) and Bryce, Gaudard, and Joiner (1997–98). This estimator is

$$\hat{\sigma}_3 = 1.047\overline{\text{MR}}$$

where $\overline{\text{MR}}$ is the median of the span-two moving ranges, and 1.047 is the reciprocal of the control chart constant d_4 for subgroups of size two defined such that $E(\tilde{R}) = d_4\sigma$ and $\tilde{R}$ is the median range. A table of d_4 values is in Wadsworth, Stephens, and Godfrey (2002). Essentially, only one of the span-two moving ranges should be affected by the sustained shift, and this single large moving range will have little impact on the value of the median moving range, certainly much less impact than it will have on the average moving range. Constructing an individuals control chart using the median moving range to estimate σ is an option in Minitab.

Now suppose that the assignable cause affects a single observation rather than causing a sustained shift in the mean. If there is a single observation that has mean $\mu_0 + \delta\sigma$, then

$$E\left(s^2\right) = \sigma^2 + \frac{1}{m}(\delta\sigma)^2$$

and this observation will affect two of the span-two moving ranges. If there are two adjacent such observations, then

$$E\left(s^2\right) = \sigma^2 + \frac{2(m-2)}{m(m-1)}(\delta\sigma)^2$$

and two of the span-two moving ranges will be affected by the out-of-control observations. Thus, when the assignable cause affects one or only a few adjacent observations, we expect the bias in s^2 to be smaller than when a sustained shift occurs. However, if an assignable cause producing a sustained shift in the mean occurs either very early in the sequence of observations or very late, it will produce much the same effect as an assignable cause affecting only one or a few adjacent points.

Some authors have suggested basing the estimate of σ on moving ranges of span greater than two, and some computer programs for control charts offer this as an option. It is easy to show that this will always lead to potentially increased bias in the estimate of σ when assignable causes are present. Note that if one uses a span-three moving range and there is a single observation whose mean is affected by the assignable cause, then this single observation will affect up to three of the moving ranges. Thus, a span-three moving range will result in more bias in the estimate of σ than will the moving range of span two. Furthermore, two span-three moving ranges will be affected by a sustained shift. In general, if one uses a span-w moving range and there is a single observation whose mean is affected by the assignable cause, up to w of these moving ranges will be impacted by this observation. Furthermore, if there is a sustained shift in the mean, up to $w - 1$ of the moving ranges will be affected by the shift in the mean. Consequently, increasing the span of a moving average beyond two results in increasing the bias in the estimate of σ if assignable causes that either produce sustained shifts in the process mean or that affect the mean of a single observation (or a few adjacent ones) are present. In fact, Wetherill and Brown (1991) advise plotting the estimate of σ versus the span of the moving average used to obtain the estimate. A sharply rising curve indicates the presence of assignable causes. For more discussion of using ranges to estimate process variability, see Woodall and Montgomery (2000–01).

■ TABLE 6.9
Formulas for Control Charts, Standards Given

Chart	Center Line	Control Limits
$\bar{x}$ (μ and σ given)	μ	$\mu \pm A\sigma$
R (σ given)	$d_2\sigma$	UCL = $D_2\sigma$, LCL = $D_1\sigma$
s (σ given)	$c_4\sigma$	UCL = $B_6\sigma$, LCL = $B_5\sigma$

■ TABLE 6.10
Formulas for Control Charts, Control Limits Based on Past Data (No Standards Given)

Chart	Center Line	Control Limits
$\bar{x}$ (using R)	$\bar{\bar{x}}$	$\bar{\bar{x}} \pm A_2\bar{R}$
$\bar{x}$ (using s)	$\bar{\bar{x}}$	$\bar{\bar{x}} \pm A_3 S$
R	$\bar{R}$	UCL = $D_4\bar{R}$, LCL = $D_3\bar{R}$
s	$\bar{s}$	UCL = $B_4\bar{s}$, LCL = $B_3\bar{s}$

6.5 Summary of Procedures for $\bar{x}$, R, And s Charts

It is convenient to summarize in one place the various computational formulas for the major types of variables control charts discussed in this chapter. Table 6.9 summarizes the formulas for $\bar{x}$, R, and s charts when standard values for μ and σ are given. Table 6.10 provides the corresponding summary when no standard values are given and trial control limits must be established from analysis of past data. The constants given for the s chart assume that $n - 1$ is used in the denominator of s. All constants are tabulated for various sample sizes in Appendix Table VI.

6.6 Applications of Variables Control Charts

There are many interesting applications of variables control charts. In this section, a few of them will be described to give additional insights into how the control chart works, as well as ideas for further applications.

EXAMPLE 6.7 Using Control Charts to Improve Suppliers' Processes

A large aerospace manufacturer purchased an aircraft component from two suppliers. These components frequently exhibited excessive variability on a key dimension that made it impossible to assemble them into the final product. This problem always resulted in expensive rework costs and occasionally caused delays in finishing the assembly of an airplane.

The materials receiving group performed 100% inspection of these parts in an effort to improve the situation. They maintained $\bar{x}$ and R charts on the dimension of interest for both suppliers. They found that the fraction of nonconforming units was about the same for both suppliers, but for very different reasons. Supplier A could produce parts with mean dimension equal to the required value, but the process was out of statistical control. Supplier B could maintain good statistical control and, in general, produced a part that exhibited considerably less variability than parts from supplier A, but its process was centered so far off the nominal required dimension that many parts were out of specification.

This situation convinced the procurement organization to work with both suppliers, persuading supplier A to implement SPC and to begin working at continuous improvement, and assisting supplier B to find out why his process was consistently centered incorrectly. Supplier B's problem was ultimately tracked to some incorrect code in an NC (numerical-controlled) machine, and the use of SPC at supplier A resulted in considerable reduction in variability over a six-month period. As a result of these actions, the problem with these parts was essentially eliminated.

EXAMPLE 6.8 Using SPC to Purchase a Machine Tool

An article in *Manufacturing Engineering* ("Picking a Marvel at Deere," January 1989, pp. 74–77) describes how the John Deere Company uses SPC methods to help choose production equipment. When a machine tool is purchased, it must go through the company capability demonstration prior to shipment to demonstrate that the tool has the ability to meet or exceed the established performance criteria. The procedure was applied to a programmable controlled bandsaw. The bandsaw supplier cut 45 pieces that were analyzed using $\bar{x}$ and R charts to demonstrate statistical control and to provide the basis for process capability analysis. The saw proved capable, and the supplier learned many useful things about the performance of his equipment. Control and capability tests such as this one are a basic part of the equipment selection and acquisition process in many companies.

EXAMPLE 6.9 SPC Implementation in a Short-Run Job-Shop

One of the more interesting aspects of SPC is the successful implementation of control charts in a job-shop manufacturing environment. Most job-shops are characterized by short production runs, and many of these shops produce parts on production runs of fewer than 50 units. This situation can make the routine use of control charts appear to be somewhat of a challenge, as not enough units are produced in any one batch to establish the control limits.

This problem can usually be easily solved. Since statistical process-control methods are most frequently applied to a characteristic of a product, we can extend SPC to the job-shop environment by focusing on the **process characteristic** in each unit of product. To illustrate, consider a drilling operation in a job-shop. The operator drills holes of various sizes in each part passing through the machine center. Some parts require one hole, and others several holes of different sizes. It is almost impossible to construct an $\bar{x}$ and R chart on hole diameter, since each part is potentially different. The correct approach is to focus on the characteristic of interest in the *process.* In this case, the manufacturer is interested in drilling holes that have the correct diameter, and therefore wants to reduce the variability in hole diameter as much as possible. This may be accomplished by control charting the *deviation* of the actual hole diameter from the nominal diameter. Depending on the process production rate and the mix of parts produced, either a control chart for individuals with a moving range control chart or a conventional $\bar{x}$ and R chart can be used. In these applications, it is usually important to mark the start of each lot or to batch carefully on the control chart, so that if changing the size, position, or number of holes drilled on each part affects the process the resulting pattern on the control charts will be easy to interpret.

EXAMPLE 6.10 Use of $\bar{x}$ and R Charts in Transactional and Service Businesses

Variables control charts have found frequent application in both manufacturing and nonmanufacturing settings. A fairly widespread but erroneous notion about these charts is that they do not apply to the nonmanufacturing environment because the "product is different." Actually, if we can make measurements on the product that are reflective of quality, function, or performance, then the *nature* of the product has no bearing on the general applicability of control charts. There are, however, two commonly encountered differences between manufacturing and transactional/service business situations: (1) In the nonmanufacturing environment, specification limits rarely apply to the product, so the notion of process capability is often undefined; and (2) more imagination may be required to select the proper variable or variables for measurement.

One application of $\bar{x}$ and R control charts in a transactional business environment involved the efforts of a finance group to reduce the time required to process its accounts payable. The division of the company in which the problem occurred had recently experienced a considerable increase in business volume, and along with this expansion came a gradual lengthening of the time the finance department needed to process check requests. As a result, many suppliers were being paid beyond the normal 30-day period, and the company was failing to capture the discounts available from its suppliers for prompt payment. The quality-improvement team assigned to this project used the flow time through the finance department as the variable for control chart analysis. Five completed check requests were selected each day, and the average and range of flow time were plotted on $\bar{x}$ and R charts. Although management and operating personnel had addressed this problem before, the use of $\bar{x}$ and R charts was responsible for substantial improvements. Within nine months, the finance department had reduced the percentage of invoices paid late from over 90% to under 3%, resulting in an annual savings of several hundred thousand dollars in realized discounts to the company.

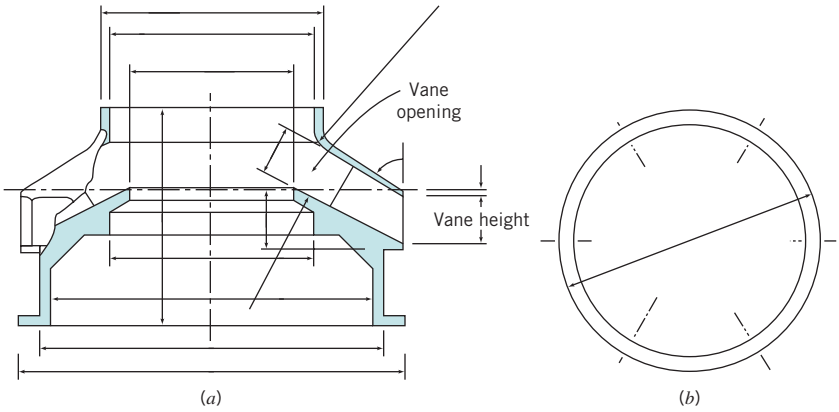
EXAMPLE 6.11 The Need for Care in Selecting Rational Subgroups

Figure 6.25*a* shows a casting used in a gas turbine jet aircraft engine. This part is typical of those produced by both casting and machining processes for use in gas turbine engines and auxiliary power units in the aerospace industry—cylindrical parts created by rotating the cross section around a central axis. The vane height on this part is a critical quality characteristic.

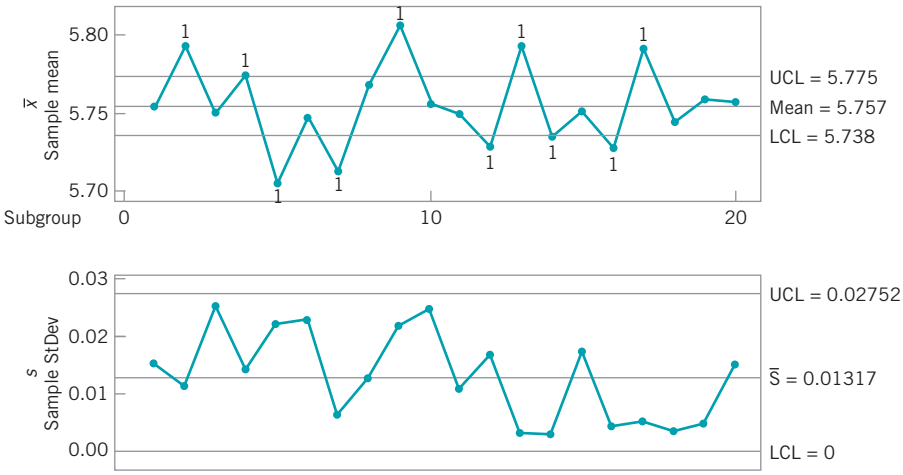
Data on vane heights are collected by randomly selecting five vanes on each casting produced. Initially, the company constructed $\bar{x}$ and s control charts on these data to control and improve the process. This usually produced many out-of-control points on the $\bar{x}$ chart, with an occasional out-of-control point on the s chart. Figure 6.26 shows typical $\bar{x}$ and s charts for 20 castings. A more careful analysis of the control-charting procedure revealed that the chief problem was the use of the five measurements on a single part as a rational subgroup, and that the out-of-control conditions on the $\bar{x}$ chart did not provide a valid basis for corrective action.

Remember that the control chart for $\bar{x}$ deals with the issue of whether or not the between-sample variability is consistent with the within-sample variability. In this case it is not: The vanes on a single casting are formed together in a common wax mold assembly. It is likely that the vane heights on a specific casting will be very similar, and it is reasonable to believe that there will be more variation in average vane height between the castings.

This situation was handled by using the s chart in the ordinary way to measure variation in vane height. However, as this



■ FIGURE 6.25 An aerospace casting.



■ FIGURE 6.26 Typical $\bar{x}$ and s control charts (from Minitab) for the vane heights of the castings in Fig. 6.26.

standard deviation is clearly too small to provide a valid basis for control of $\bar{x}$, the quality engineer at the company decided to treat the average vane height on each casting as an *individual measurement* and to control average vane height by using a control chart for individuals with a moving range chart. This solution worked extremely well in practice, and the group of three control charts provided an excellent basis for process improvement.

Figure 6.27 shows this set of three control charts as generated by Minitab. The Minitab package generates these charts automatically, referring to them as "between/within" control charts. Note that the individuals chart exhibits control, whereas the $\bar{x}$ chart in Fig. 6.26 did not. Essentially, the moving range of the average vane heights provides a much more reasonable estimate of the variability in height **between parts.** The s chart can be thought of as a measure of the variability in vane height on a single casting. We want this variability to be as small as possible, so that all vanes on the same part will be nearly identical. The paper by Woodall and Thomas (1995) is a good reference on this general subject.

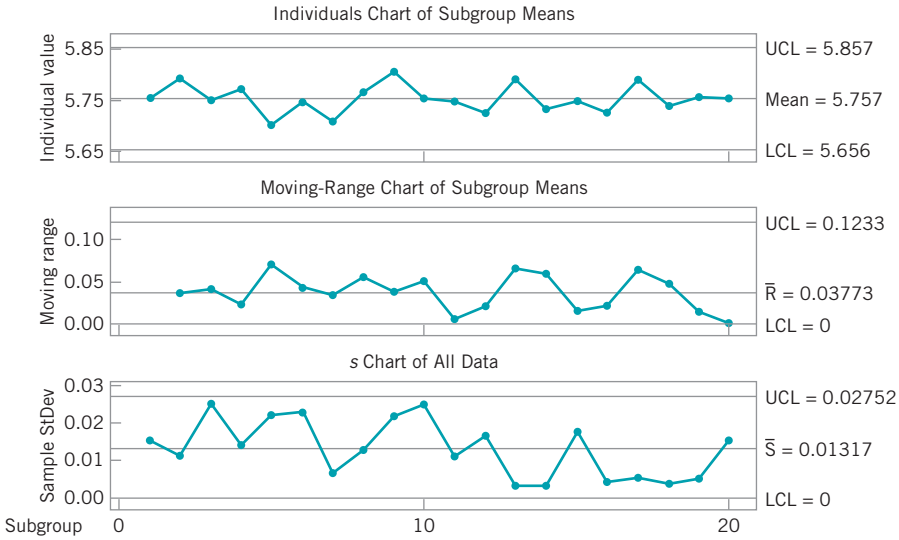


FIGURE 6.27 Individuals, moving-range, and s control charts for the vane heights of the castings in Fig. 6.25.

Situations such as the one described in Example 6.11 occur frequently in the application of control charts. For example, there are many similar problems in the semiconductor industry. In such cases, it is important to study carefully the behavior of the variables being measured and to have a clear understanding of the purpose of the control charts. For instance, if the variation in vane height on a specific casting were completely unrelated, using the average height as an individual measurement could be very inappropriate. It would be necessary to (1) use a control chart on each individual vane included in the sample, (2) investigate the use of a control chart technique for multistream processes, or (3) use some multivariate control chart technique. Some of these possibilities are discussed in the chapters in Part IV of the text.

Important Terms and Concepts

Average run length
Control chart for individuals units
Control limits
Interpretation of control charts
Moving-range control chart
Natural tolerance limits of a process
Normality and control charts
Operating-characteristic (OC) curve for the $\bar{x}$ control chart
Patterns on control charts
Phase I control chart usage

Phase II control chart usage
Probability limits for control charts
Process capability
Process capability ratio (PCR) C_p
R control chart
Rational subgroups
s control chart
s^2 control chart
Shewhart control charts
Specification limits
Three-sigma control limits
Tier chart or tolerance diagram
Trial control limits
Variable sample size on control charts
Variables control charts
$\bar{x}$ control chart

Exercises



The Student Resource Manual presents comprehensive annotated solutions to the odd-numbered exercises included in the Answers to Selected Exercises section in the back of this book.

6.1. The data shown in Table 6E.1 are $\bar{x}$ and R values for 24 samples of size $n = 5$ taken from a process producing bearings. The measurements are made on the inside diameter of the bearing, with only the last three decimals recorded (i.e., 34.5 should be 0.50345).

(a) Set up $\bar{x}$ and R charts on this process. Does the process seem to be in statistical control? If necessary, revise the trial control limits.

(b) If specifications on this diameter are 0.5030 ± 0.0010, find the percentage of nonconforming bearings produced by this process. Assume that diameter is normally distributed.

6.2. A high-voltage power supply should have a nominal output voltage of 350 V. A sample of four units is selected each day and tested for process-control purposes. The data shown in Table 6E.2 give the difference between the observed reading on each unit and the nominal voltage times ten; that is,

x_i = (observed voltage on unit i - 350)10

(a) Set up $\bar{x}$ and R charts on this process. Is the process in statistical control?

(b) If specifications are at 350 V ± 5 V, what can you say about process capability?

(c) Is there evidence to support the claim that voltage is normally distributed?

6.3. The data shown in Table 6E.3 are the deviations from nominal diameter for holes drilled in a carbon-fiber composite material used in aerospace manufacturing.

■ **TABLE 6E.1**
Bearing Diameter Data

Sample Number	$\bar{x}$	R	Sample Number	$\bar{x}$	R
1	34.5	3	13	35.4	8
2	34.2	4	14	34.0	6
3	31.6	4	15	37.1	5
4	31.5	4	16	34.9	7
5	35.0	5	17	33.5	4
6	34.1	6	18	31.7	3
7	32.6	4	19	34.0	8
8	33.8	3	20	35.1	4
9	34.8	7	21	33.7	2
10	33.6	8	22	32.8	1
11	31.9	3	23	33.5	3
12	38.6	9	24	34.2	2

■ **TABLE 6E.2**
Voltage Data for Exercise 6.2.

Sample Number	x_1	x_2	x_3	x_4
1	6	9	10	15
2	10	4	6	11
3	7	8	10	5
4	8	9	6	13
5	9	10	7	13
6	12	11	10	10
7	16	10	8	9
8	7	5	10	4
9	9	7	8	12
10	15	16	10	13
11	8	12	14	16
12	6	13	9	11
13	16	9	13	15
14	7	13	10	12
15	11	7	10	16
16	15	10	11	14
17	9	8	12	10
18	15	7	10	11
19	8	6	9	12
20	13	14	11	15

■ TABLE 6E.3
Hole Diameter Data for Exercise 6.3

Sample Number	x_1	x_2	x_3	x_4	x_5
1	−30	+50	−20	+10	+30
2	0	+50	−60	−20	+30
3	−50	+10	+20	+30	+20
4	−10	−10	+30	−20	+50
5	+20	−40	+50	+20	+10
6	0	0	+40	−40	+20
7	0	0	+20	−20	−10
8	+70	−30	+30	−10	0
9	0	0	+20	−20	+10
10	+10	+20	+30	+10	+50
11	+40	0	+20	0	+20
12	+30	+20	+30	+10	+40
13	+30	−30	0	+10	+10
14	+30	−10	+50	−10	−30
15	+10	−10	+50	+40	0
16	0	0	+30	−10	0
17	+20	+20	+30	+30	−20
18	+10	−20	+50	+30	+10
19	+50	−10	+40	+20	0
20	+50	0	0	+30	+10

The values reported are deviations from nominal in ten-thousandths of an inch.

(a) Set up $\bar{x}$ and R charts on the process. Is the process in statistical control?

(b) Estimate the process standard deviation using the range method.

(c) If specifications are at nominal ±100, what can you say about the capability of this process? Calculate the PCR C_p.

6.4. The thickness of a printed circuit board is an important quality parameter. Data on board thickness (in inches) are given in Table 6E.4 for 25 samples of three boards each.

(a) Set up $\bar{x}$ and R control charts. Is the process in statistical control?

(b) Estimate the process standard deviation.

(c) What are the limits that you would expect to contain nearly all the process measurements?

(d) If the specifications are at 0.0630 in. ± 0.0015 in., what is the value of the PCR C_p?

6.5. The fill volume of soft-drink beverage bottles is an important quality characteristic. The volume is measured (approximately) by placing a gauge over the crown and comparing the height of the liquid in the neck of the bottle against a coded scale. On this scale, a reading of zero corresponds to the correct fill height. Fifteen samples of size $n = 10$ have been analyzed, and the fill heights are shown in Table 6E.5.

(a) Set up $\bar{x}$ and s control charts on this process. Does the process exhibit statistical control? If necessary, construct revised control limits.

(b) Set up an R chart, and compare with the s chart in part (a).

(c) Set up an s^2 chart and compare with the s chart in part (a).

6.6. The net weight (in oz) of a dry bleach product is to be monitored by $\bar{x}$ and R control charts using a sample size of $n = 5$. Data for 20 preliminary samples are shown in Table 6E.6.

(a) Set up $\bar{x}$ and R control charts using these data. Does the process exhibit statistical control?

(b) Estimate the process mean and standard deviation.

(c) Does fill weight seem to follow a normal distribution?

■ TABLE 6E.4
Printed Circuit Board Thickness for Exercise 6.4

Sample Number	x_1	x_2	x_3
1	0.0629	0.0636	0.0640
2	0.0630	0.0631	0.0622
3	0.0628	0.0631	0.0633
4	0.0634	0.0630	0.0631
5	0.0619	0.0628	0.0630
6	0.0613	0.0629	0.0634
7	0.0630	0.0639	0.0625
8	0.0628	0.0627	0.0622
9	0.0623	0.0626	0.0633
10	0.0631	0.0631	0.0633
11	0.0635	0.0630	0.0638
12	0.0623	0.0630	0.0630
13	0.0635	0.0631	0.0630
14	0.0645	0.0640	0.0631
15	0.0619	0.0644	0.0632
16	0.0631	0.0627	0.0630
17	0.0616	0.0623	0.0631
18	0.0630	0.0630	0.0626
19	0.0636	0.0631	0.0629
20	0.0640	0.0635	0.0629
21	0.0628	0.0625	0.0616
22	0.0615	0.0625	0.0619
23	0.0630	0.0632	0.0630
24	0.0635	0.0629	0.0635
25	0.0623	0.0629	0.0630

■ TABLE 6E.5
Fill Height Data for Exercise 6.5

Sample Number	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
1	2.5	0.5	2.0	−1.0	1.0	−1.0	0.5	1.5	0.5	−1.5
2	0.0	0.0	0.5	1.0	1.5	1.0	−1.0	1.0	1.5	−1.0
3	1.5	1.0	1.0	−1.0	0.0	−1.5	−1.0	−1.0	1.0	−1.0
4	0.0	0.5	−2.0	0.0	−1.0	1.5	−1.5	0.0	−2.0	−1.5
5	0.0	0.0	0.0	−0.5	0.5	1.0	−0.5	−0.5	0.0	0.0
6	1.0	−0.5	0.0	0.0	0.0	0.5	−1.0	1.0	−2.0	1.0
7	1.0	−1.0	−1.0	−1.0	0.0	1.5	0.0	1.0	0.0	0.0
8	0.0	−1.5	−0.5	1.5	0.0	0.0	0.0	−1.0	0.5	−0.5
9	−2.0	−1.5	1.5	1.5	0.0	0.0	0.5	1.0	0.0	1.0
10	−0.5	3.5	0.0	−1.0	−1.5	−1.5	−1.0	−1.0	1.0	0.5
11	0.0	1.5	0.0	0.0	2.0	−1.5	0.5	−0.5	2.0	−1.0
12	0.0	−2.0	−0.5	0.0	−0.5	2.0	1.5	0.0	0.5	−1.0
13	−1.0	−0.5	−0.5	−1.0	0.0	0.5	0.5	−1.5	−1.0	−1.0
14	0.5	1.0	−1.0	−0.5	−2.0	−1.0	−1.5	0.0	1.5	1.5
15	1.0	0.0	1.5	1.5	1.0	−1.0	0.0	1.0	−2.0	−1.5

■ TABLE 6E.6
Data for Exercise 6.6.

Sample Number	x_1	x_2	x_3	x_4	x_5
1	15.8	16.3	16.2	16.1	16.6
2	16.3	15.9	15.9	16.2	16.4
3	16.1	16.2	16.5	16.4	16.3
4	16.3	16.2	15.9	16.4	16.2
5	16.1	16.1	16.4	16.5	16.0
6	16.1	15.8	16.7	16.6	16.4
7	16.1	16.3	16.5	16.1	16.5
8	16.2	16.1	16.2	16.1	16.3
9	16.3	16.2	16.4	16.3	16.5
10	16.6	16.3	16.4	16.1	16.5
11	16.2	16.4	15.9	16.3	16.4
12	15.9	16.6	16.7	16.2	16.5
13	16.4	16.1	16.6	16.4	16.1
14	16.5	16.3	16.2	16.3	16.4
15	16.4	16.1	16.3	16.2	16.2
16	16.0	16.2	16.3	16.3	16.2
17	16.4	16.2	16.4	16.3	16.2
18	16.0	16.2	16.4	16.5	16.1
19	16.4	16.0	16.3	16.4	16.4
20	16.4	16.4	16.5	16.0	15.8

(d) If the specifications are at 16.2 ± 0.5, what conclusions would you draw about process capability?
(e) What fraction of containers produced by this process is likely to be below the lower specification limit of 15.7 oz?

6.7. Rework Exercise 6.2 using the s chart.

6.8. Rework Exercise 6.3 using the s chart.

6.9. Consider the piston ring data shown in Table 6.3. Assume that the specifications on this component are $74.000 + 0.05$ mm.
(a) Set up $\bar{x}$ and R control charts on this process. Is the process in statistical control?
(b) Note that the control limits on the $\bar{x}$ chart in part (a) are identical to the control limits on the $\bar{x}$ chart in Example 6.3, where the limits were based on s. Will this always happen?
(c) Estimate process capability for the piston-ring process. Estimate the percentage of piston rings produced that will be outside of the specification.

6.10. Table 6E.7 shows 15 additional samples for the piston ring process (Table 6.3), taken after the initial control charts were established. Plot these data on the $\bar{x}$ and R chart developed in Exercise 6.9. Is the process in control?

6.11. Control charts on $\bar{x}$ and s are to be maintained on the torque readings of a bearing used in a wingflap actuator assembly. Samples of size $n = 10$ are to be used, and we know from past experience that when the process is in control, bearing torque has a normal

■ TABLE 6E.7
Piston Ring Diameter Data for Exercise 6.10

Sample Number, i	Observations					$\bar{x}_i$	R_i
26	74.012	74.015	74.030	73.986	74.000	74.009	0.044
27	73.995	74.010	73.990	74.015	74.001	74.002	0.025
28	73.987	73.999	73.985	74.000	73.990	73.992	0.015
29	74.008	74.010	74.003	73.991	74.006	74.004	0.019
30	74.003	74.000	74.001	73.986	73.997	73.997	0.017
31	73.994	74.003	74.015	74.020	74.004	74.007	0.026
32	74.008	74.002	74.018	73.995	74.005	74.006	0.023
33	74.001	74.004	73.990	73.996	73.998	73.998	0.014
34	74.015	74.000	74.016	74.025	74.000	74.011	0.025
35	74.030	74.005	74.000	74.016	74.012	74.013	0.030
36	74.001	73.990	73.995	74.010	74.024	74.004	0.034
37	74.015	74.020	74.024	74.005	74.019	74.017	0.019
38	74.035	74.010	74.012	74.015	74.026	74.020	0.025
39	74.017	74.013	74.036	74.025	74.026	74.023	0.023
40	74.010	74.005	74.029	74.000	74.020	74.013	0.029

distribution with mean $\mu = 80$ inch-pounds and standard deviation $\sigma = 10$ inch-pounds. Find the center line and control limits for these control charts.

6.12. Samples of $n = 6$ items each are taken from a process at regular intervals. A quality characteristic is measured, and $\bar{x}$ and R values are calculated for each sample. After 50 samples, we have

$$\sum_{i=1}^{50} \bar{x}_i = 2000 \quad \text{and} \quad \sum_{i=1}^{50} R_i = 200$$

Assume that the quality characteristic is normally distributed.

(a) Compute control limits for the $\bar{x}$ and R control charts.

(b) All points on both control charts fall between the control limits computed in part (a). What are the natural tolerance limits of the process?

(c) If the specification limits are 41 ± 5.0, what are your conclusions regarding the ability of the process to produce items within these specifications?

(d) Assuming that if an item exceeds the upper specification limit it can be reworked, and if it is below the lower specification limit it must be scrapped, what percent scrap and rework is the process producing?

(e) Make suggestions as to how the process performance could be improved.

6.13. Samples of $n = 4$ items are taken from a process at regular intervals. A normally distributed quality characteristic is measured and $\bar{x}$ and s values are calculated for each sample. After 50 subgroups have been analyzed, we have

$$\sum_{i=1}^{50} \bar{x}_i = 1000 \quad \text{and} \quad \sum_{i=1}^{50} s_i = 72$$

(a) Compute the control limit for the $\bar{x}$ and s control charts.

(b) Assume that all points on both charts plot within the control limits. What are the natural tolerance limits of the process?

(c) If the specification limits are 19 ± 4.0, what are your conclusions regarding the ability of the process to produce items conforming to specifications?

(d) Assuming that if an item exceeds the upper specification limit it can be reworked, and if it is below the lower specification limit it must be scrapped, what percent scrap and rework is the process now producing?

(e) If the process were centered at $\mu = 19.0$, what would be the effect on percent scrap and rework?

TABLE 6E.8
Data for Exercise 6.14

Sample Number	x_1	x_2	x_3	x_4	x_5	$\bar{x}$	R
1	138.1	110.8	138.7	137.4	125.4	130.1	27.9
2	149.3	142.1	105.0	134.0	92.3	124.5	57.0
3	115.9	135.6	124.2	155.0	117.4	129.6	39.1
4	118.5	116.5	130.2	122.6	100.2	117.6	30.0
5	108.2	123.8	117.1	142.4	150.9	128.5	42.7
6	102.8	112.0	135.0	135.0	145.8	126.1	43.0
7	120.4	84.3	112.8	118.5	119.3	111.0	36.1
8	132.7	151.1	124.0	123.9	105.1	127.4	46.0
9	136.4	126.2	154.7	127.1	173.2	143.5	46.9
10	135.0	115.4	149.1	138.3	130.4	133.6	33.7
11	139.6	127.9	151.1	143.7	110.5	134.6	40.6
12	125.3	160.2	130.4	152.4	165.1	146.7	39.8
13	145.7	101.8	149.5	113.3	151.8	132.4	50.0
14	138.6	139.0	131.9	140.2	141.1	138.1	9.2
15	110.1	114.6	165.1	113.8	139.6	128.7	54.8
16	145.2	101.0	154.6	120.2	117.3	127.6	53.3
17	125.9	135.3	121.5	147.9	105.0	127.1	42.9
18	129.7	97.3	130.5	109.0	150.5	123.4	53.2
19	123.4	150.0	161.6	148.4	154.2	147.5	38.3
20	144.8	138.3	119.6	151.8	142.7	139.4	32.2

6.14. Table 6E.8 presents 20 subgroups of five measurements on the critical dimension of a part produced by a machining process.

(a) Set up $\bar{x}$ and R control charts on this process. Verify that the process is in statistical control.

(b) Following the establishment of control charts in part (a) above, 10 new samples in Table 6E.9

TABLE 6E.9
Additional Data for Exercise 6.14, part (b)

Sample Number	x_1	x_2	x_3	x_4	x_5	$\bar{x}$	R
1	131.0	184.8	182.2	143.3	212.8	170.8	81.8
2	181.3	193.2	180.7	169.1	174.3	179.7	24.0
3	154.8	170.2	168.4	202.7	174.4	174.1	48.0
4	157.5	154.2	169.1	142.2	161.9	157.0	26.9
5	216.3	174.3	166.2	155.5	184.3	179.3	60.8
6	186.9	180.2	149.2	175.2	185.0	175.3	37.8
7	167.8	143.9	157.5	171.8	194.9	167.2	51.0
8	178.2	186.7	142.4	159.4	167.6	166.9	44.2
9	162.6	143.6	132.8	168.9	177.2	157.0	44.5
10	172.1	191.7	203.4	150.4	196.3	182.8	53.0

■ TABLE 6E.10
New Data for Exercise 6.14, part (c)

Sample Number	x_1	x_2	x_3	x_4	x_5	$\bar{x}$	R
1	131.5	143.1	118.5	103.2	121.6	123.6	39.8
2	111.0	127.3	110.4	91.0	143.9	116.7	52.8
3	129.8	98.3	134.0	105.1	133.1	120.1	35.7
4	145.2	132.8	106.1	131.0	99.2	122.8	46.0
5	114.6	111.0	108.8	177.5	121.6	126.7	68.7
6	125.2	86.4	64.4	137.1	117.5	106.1	72.6
7	145.9	109.5	84.9	129.8	110.6	116.1	61.0
8	123.6	114.0	135.4	83.2	107.6	112.8	52.2
9	85.8	156.3	119.7	96.2	153.0	122.2	70.6
10	107.4	148.7	127.4	125.0	127.5	127.2	41.3

were collected. Plot the $\bar{x}$ and R values on the control chart you established in part (a) and draw conclusions.

(c) Suppose that the assignable cause responsible for the action signals generated in part (b) has been identified and adjustments made to the process to correct its performance. Plot the $\bar{x}$ and R values from the new subgroups shown in Table 6E.10 which were taken following the adjustment, against the control chart limits established in part (a). What are your conclusions?

6.15. Parts manufactured by an injection molding process are subjected to a compressive strength test. Twenty samples of five parts each are collected, and the compressive strengths (in psi) are shown in Table 6E.11.

(a) Establish $\bar{x}$ and R control charts for compressive strength using these data. Is the process in statistical control?

(b) After establishing the control charts in part (a), 15 new subgroups were collected and the compressive strengths are shown in Table 6E.12. Plot the $\bar{x}$ and R values against the control units from part (a) and draw conclusions.

6.16. Reconsider the data presented in Exercise 6.15.

(a) Rework both parts (a) and (b) of Exercise 6.15 using the $\bar{x}$ and s charts.

(b) Does the s chart detect the shift in process variability more quickly than the R chart did originally in part (b) of Exercise 6.15?

6.17. Consider the $\bar{x}$ and R charts you established in Exercise 6.1 using $n = 5$.

(a) Suppose that you wished to continue charting this quality characteristic using $\bar{x}$ and R charts based on a sample size of $n = 3$. What limits would be used on the $\bar{x}$ and R charts?

■ TABLE 6E.11
Strength Data for Exercise 6.15

Sample Number	x_1	x_2	x_3	x_4	x_5	$\bar{x}$	R
1	83.0	81.2	78.7	75.7	77.0	79.1	7.3
2	88.6	78.3	78.8	71.0	84.2	80.2	17.6
3	85.7	75.8	84.3	75.2	81.0	80.4	10.4
4	80.8	74.4	82.5	74.1	75.7	77.5	8.4
5	83.4	78.4	82.6	78.2	78.9	80.3	5.2
6	75.3	79.9	87.3	89.7	81.8	82.8	14.5
7	74.5	78.0	80.8	73.4	79.7	77.3	7.4
8	79.2	84.4	81.5	86.0	74.5	81.1	11.4
9	80.5	86.2	76.2	64.1	80.2	81.4	9.9
10	75.7	75.2	71.1	82.1	74.3	75.7	10.9
11	80.0	81.5	78.4	73.8	78.1	78.4	7.7
12	80.6	81.8	79.3	73.8	81.7	79.4	8.0
13	82.7	81.3	79.1	82.0	79.5	80.9	3.6
14	79.2	74.9	78.6	77.7	75.3	77.1	4.3
15	85.5	82.1	82.8	73.4	71.7	79.1	13.8
16	78.8	79.6	80.2	79.1	80.8	79.7	2.0
17	82.1	78.2	75.5	78.2	82.1	79.2	6.6
18	84.5	76.9	83.5	81.2	79.2	81.1	7.6
19	79.0	77.8	81.2	84.4	81.6	80.8	6.6
20	84.5	73.1	78.6	78.7	80.6	79.1	11.4

■ TABLE 6E.12
New Data for Exercise 6.15, part (b)

Sample Number	x_1	x_2	x_3	x_4	x_5	$\bar{x}$	R
1	68.9	81.5	78.2	80.8	81.5	78.2	12.6
2	69.8	68.6	80.4	84.3	83.9	77.4	15.7
3	78.5	85.2	78.4	80.3	81.7	80.8	6.8
4	76.9	86.1	86.9	94.4	83.9	85.6	17.5
5	93.6	81.6	87.8	79.6	71.0	82.7	22.5
6	65.5	86.8	72.4	82.6	71.4	75.9	21.3
7	78.1	65.7	83.7	93.7	93.4	82.9	27.9
8	74.9	72.6	81.6	87.2	72.7	77.8	14.6
9	78.1	77.1	67.0	75.7	76.8	74.9	11.0
10	78.7	85.4	77.7	90.7	76.7	81.9	14.0
11	85.0	60.2	68.5	71.1	82.4	73.4	24.9
12	86.4	79.2	79.8	86.0	75.4	81.3	10.9
13	78.5	99.0	78.3	71.4	81.8	81.7	27.6
14	68.8	62.0	82.0	77.5	76.1	73.3	19.9
15	83.0	83.7	73.1	82.2	95.3	83.5	22.2

(b) What would be the impact of the decision you made in part (a) on the ability of the $\bar{x}$ chart to detect a 2σ shift in the mean?

(c) Suppose you wished to continue charting this quality characteristic using $\bar{x}$ and R charts based on a sample size of $n = 8$. What limits would be used on the $\bar{x}$ and R charts?

(d) What is the impact of using $n = 8$ on the ability of the $\bar{x}$ chart to detect a 2σ shift in the mean?

6.18. Consider the $\bar{x}$ and R chart that you established in Exercise 6.9 for the piston ring process. Suppose that you want to continue control charting piston ring diameter using $n = 3$. What limits would be used on the $\bar{x}$ and R chart?

6.19. Control charts for $\bar{x}$ and R are maintained for an important quality characteristic. The sample size is $n = 7$; $\bar{x}$ and R are computed for each sample. After 35 samples, we have found that

$$\sum_{i=1}^{35} \bar{x}_i = 7805 \quad \text{and} \quad \sum_{i=1}^{35} R_i = 1200$$

(a) Set up $\bar{x}$ and R charts using these data.

(b) Assuming that both charts exhibit control, estimate the process mean and standard deviation.

(c) If the quality characteristic is normally distributed and if the specifications are 220 ± 35, can the process meet the specifications? Estimate the fraction nonconforming.

(d) Assuming the variance to remain constant, state where the process mean should be located to minimize the fraction nonconforming. What would be the value of the fraction nonconforming under these conditions?

6.20. Samples of size $n = 5$ are taken from a manufacturing process every hour. A quality characteristic is measured, and $\bar{x}$ and R are computed for each sample. After 25 samples have been analyzed, we have

$$\sum_{i=1}^{25} \bar{x}_i = 662.50 \quad \text{and} \quad \sum_{i=1}^{25} R_i = 9.00$$

The quality characteristic is normally distributed.

(a) Find the control limits for the $\bar{x}$ and R charts.

(b) Assume that both charts exhibit control. If the specifications are 26.40 ± 0.50, estimate the fraction nonconforming.

(c) If the mean of the process were 26.40, what fraction nonconforming would result?

6.21. Samples of size $n = 5$ are collected from a process every half hour. After 50 samples have been collected, we calculate $\bar{\bar{x}} = 20.0$ and $\bar{s} = 1.5$. Assume that both charts exhibit control and that the quality characteristic is normally distributed.

(a) Estimate the process standard deviation.

(b) Find the control limits on the $\bar{x}$ and s charts.

(c) If the process mean shifts to 22, what is the probability of concluding that the process is still in control?

6.22. An $\bar{x}$ chart is used to control the mean of a normally distributed quality characteristic. It is known that $\sigma = 6.0$ and $n = 4$. The center line = 200, UCL = 209, and LCL = 191. If the process mean shifts to 188, find the probability that this shift is detected on the first subsequent sample.

6.23. A critical dimension of a machined part has specifications 100 ± 10. Control chart analysis indicates that the process is in control with $\bar{\bar{x}} = 104$ and $\bar{R} = 9.30$. The control charts use samples of size $n = 5$. If we assume that the characteristic is normally distributed, can the mean be located (by adjusting the tool position) so that all output meets specifications? What is the present capability of the process?

6.24. A process is to be monitored with standard values $\mu = 10$ and $\sigma = 2.5$. The sample size is $n = 2$.

(a) Find the center line and control limits for the $\bar{x}$ chart.

(b) Find the center line and control limits for the R chart.

(c) Find the center line and control limits for the s chart.

6.25. Samples of $n = 5$ units are taken from a process every hour. The $\bar{x}$ and R values for a particular quality characteristic are determined. After 25 samples have been collected, we calculate $\bar{\bar{x}} = 20$ and $\bar{R} = 4.56$.

(a) What are the three-sigma control limits for $\bar{x}$ and R?

(b) Both charts exhibit control. Estimate the process standard deviation.

(c) Assume that the process output is normally distributed. If the specifications are 19 ± 5, what are your conclusions regarding the process capability?

(d) If the process mean shifts to 24, what is the probability of not detecting this shift on the first subsequent sample?

6.26. A TiW layer is deposited on a substrate using a sputtering tool. Table 6E.13 contains layer thickness measurements (in Angstroms) on 20 subgroups of four substrates.

(a) Set up $\bar{x}$ and R control charts on this process. Is the process in control? Revise the control limits as necessary.

(b) Estimate the mean and standard deviation of the process.

(c) Is the layer thickness normally distributed?

(d) If the specifications are at 450 ± 30, estimate the process capability.

■ TABLE 6E.13
Layer Thickness Data for Exercise 6.26

Subgroup	x_1	x_2	x_3	x_4
1	459	449	435	450
2	443	440	442	442
3	457	444	449	444
4	469	463	453	438
5	443	457	445	454
6	444	456	456	457
7	445	449	450	445
8	446	455	449	452
9	444	452	457	440
10	432	463	463	443
11	445	452	453	438
12	456	457	436	457
13	459	445	441	447
14	441	465	438	450
15	460	453	457	438
16	453	444	451	435
17	451	460	450	457
18	422	431	437	429
19	444	446	448	467
20	450	450	454	454

6.27. **Continuation of Exercise 6.26.** Table 6E.14 contains 10 new subgroups of thickness data. Plot this data on the control charts constructed in Exercise 6.26 (a). Is the process in statistical control?

6.28. **Continuation of Exercise 6.26.** Suppose that following the construction of the $\bar{x}$ and R control charts in Exercise 6.26, the process engineers decided to change the subgroup size to $n = 2$. Table 6E.15 contains 10 new subgroups of thickness data. Plot this data on the control charts from Exercise 6.26 (a) based on the new subgroup size. Is the process in statistical control?

■ TABLE 6E.14
Additional Thickness Data for Exercise 6.27.

Subgroup	x_1	x_2	x_3	x_4
21	454	449	443	461
22	449	441	444	455
23	442	442	442	450
24	443	452	438	430
25	446	459	457	457
26	454	448	445	462
27	458	449	453	438
28	450	449	445	451
29	443	440	443	451
30	457	450	452	437

■ TABLE 6E.15
Additional Thickness Data for Exercise 6.28.

Subgroup	x_1	x_2
21	454	449
22	449	441
23	442	442
24	443	452
25	446	459
26	454	448
27	458	449
28	450	449
29	443	440
30	457	450

6.29. Rework Exercises 6.26 and 6.27 using $\bar{x}$ and s control charts.

6.30. Control charts for $\bar{x}$ and R are to be established to control the tensile strength of a metal part. Assume that tensile strength is normally distributed. Thirty samples of size $n = 6$ parts are collected over a period of time with the following results:

$$\sum_{i=1}^{30} \bar{x}_i = 6000 \quad \text{and} \quad \sum_{i=1}^{30} R_i = 150$$

(a) Calculate control limits for $\bar{x}$ and R.
(b) Both charts exhibit control. The specifications on tensile strength are 200 ± 5. What are your conclusions regarding process capability?
(c) For the above $\bar{x}$ chart, find the β-risk when the true process mean is 199.

6.31. An $\bar{x}$ chart has a center line of 100, uses three-sigma control limits, and is based on a sample size of four. The process standard deviation is known to be six. If the process mean shifts from 100 to 92, what is the probability of detecting this shift on the first sample following the shift?

6.32. The data in Table 6E.16 were collected from a process manufacturing power supplies. The variable of interest is output voltage, and $n = 5$.
(a) Compute center lines and control limits suitable for controlling future production.
(b) Assume that the quality characteristic is normally distributed. Estimate the process standard deviation.
(c) What are the apparent three-sigma natural tolerance limits of the process?

TABLE 6E.16
Voltage Data for Exercise 6.32.

Sample Number	$\bar{x}$	R	Sample Number	$\bar{x}$	R
1	103	4	11	105	4
2	102	5	12	103	2
3	104	2	13	102	3
4	105	11	14	105	4
5	104	4	15	104	5
6	106	3	16	105	3
7	102	7	17	106	5
8	105	2	18	102	2
9	106	4	19	105	4
10	104	3	20	103	2

(d) What would be your estimate of the process fraction nonconforming if the specifications on the characteristic were 103 ± 4?
(e) What approaches to reducing the fraction nonconforming can you suggest?

6.33. Control charts on $\bar{x}$ and R for samples of size $n = 5$ are to be maintained on the tensile strength in pounds of a yarn. To start the charts, 30 samples were selected, and the mean and range of each computed. This yields

$$\sum_{i=1}^{30} \bar{x}_i = 607.8 \quad \text{and} \quad \sum_{i=1}^{30} R_i = 144$$

(a) Compute the center line and control limits for the $\bar{x}$ and R control charts.
(b) Suppose both charts exhibit control. There is a single lower specification limit of 16 lb. If strength is normally distributed, what fraction of yarn would fail to meet specifications?

6.34. Specifications on a cigar lighter detent are 0.3220 and 0.3200 in. Samples of size 5 are taken every 45 min with the results shown in Table 6E.17 (measured as deviations from 0.3210 in 0.0001 in.).
(a) Set up an R chart and examine the process for statistical control.
(b) What parameters would you recommend for an R chart for on-line control?
(c) Estimate the standard deviation of the process.
(d) What is the process capability?

6.35. **Continuation of Exercise 6.34.** Reconsider the data from Exercise 6.34 and establish $\bar{x}$ and R charts with appropriate trial control limits. Revise these trial limits as necessary to produce a set of control charts for monitoring future production. Suppose that the new data in Table 6E.18 are observed.

TABLE 6E.17
Data for Exercise 6.34.

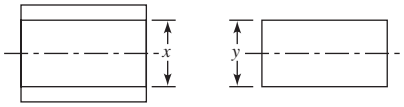
Sample Number	x_1	x_2	x_3	x_4	x_5
1	1	9	6	9	6
2	9	4	3	0	3
3	0	9	0	3	2
4	1	1	0	2	1
5	−3	0	−1	0	−4
6	−7	2	0	0	2
7	−3	−1	−1	0	−2
8	0	−2	−3	−3	−2
9	2	0	−1	−3	−1
10	0	2	−1	−1	2
11	−3	−2	−1	−1	2
12	−16	2	0	−4	−1
13	−6	−3	0	0	−8
14	−3	−5	5	0	5
15	−1	−1	−1	−2	−1

TABLE 6E.18
New Data for Exercise 6.35

Sample Number	x_1	x_2	x_3	x_4	x_5
16	2	10	9	6	5
17	1	9	5	9	4
18	0	9	8	2	5
19	−3	0	5	1	4
20	2	10	9	3	1
21	−5	4	0	6	−1
22	0	2	−5	4	6
23	10	0	3	1	5
24	−1	2	5	6	−3
25	0	−1	2	5	−2

(a) Plot these new observations on the control chart. What conclusions can you draw about process stability?
(b) Use all 25 observations to revise the control limits for the $\bar{x}$ and R charts. What conclusions can you draw now about the process?

6.36. Two parts are assembled as shown in Figure 6.28. Assume that the dimensions x and y are normally distributed with means μ_x and μ_y and standard deviations σ_x and σ_y, respectively. The parts are produced on different machines and are assembled at random. Control charts are maintained on each dimension for



■ FIGURE 6.28 Parts for Exercise 6.36.

the range of each sample ($n = 5$). Both range charts are in control.

(a) Given that for 20 samples on the range chart controlling x and 10 samples on the range chart controlling y, we have

$$\sum_{i=1}^{20} R_{x_i} = 18.608 \quad \text{and} \quad \sum_{i=1}^{10} R_{y_i} = 6.978$$

Estimate σ_x and σ_y.

(b) If it is desired that the probability of a smaller clearance (i.e., $x - y$) than 0.09 should be 0.006, what distance between the average dimensions (i.e., $\mu_x - \mu_y$) should be specified?

6.37. Control charts for $\bar{x}$ and R are maintained on the tensile strength of a metal fastener. After 30 samples of size $n = 6$ are analyzed, we find that

$$\sum_{i=1}^{30} \bar{x}_i = 12{,}870 \quad \text{and} \quad \sum_{i=1}^{30} R_i = 1350$$

(a) Compute control limits on the R chart.

(b) Assuming that the R chart exhibits control, estimate the parameters μ and σ.

(c) If the process output is normally distributed, and if the specifications are 440 ± 40, can the process meet the specifications? Estimate the fraction nonconforming.

(d) If the variance remains constant, where should the mean be located to minimize the fraction nonconforming?

6.38. Control charts for $\bar{x}$ and s are maintained on a quality characteristic. The sample size is $n = 4$. After 30 samples, we obtain

$$\sum_{i=1}^{30} \bar{x}_i = 12{,}870 \quad \text{and} \quad \sum_{i=1}^{30} s_i = 410$$

(a) Find the three-sigma limits for the s chart.

(b) Assuming that both charts exhibit control, estimate the parameters μ and σ.

6.39. An $\bar{x}$ chart on a normally distributed quality characteristic is to be established with the standard values $\mu = 100$, $\sigma = 8$, and $n = 4$. Find the following:

(a) The two-sigma control limits.

(b) The 0.005 probability limits.

6.40. An $\bar{x}$ chart with three-sigma limits has parameters as follows:

$$\text{UCL} = 104$$
$$\text{Center line} = 100$$
$$\text{LCL} = 96$$
$$n = 5$$

Suppose the process quality characteristic being controlled is normally distributed with a true mean of 98 and a standard deviation of 8. What is the probability that the control chart would exhibit lack of control by at least the third point plotted?

6.41. Consider the $\bar{x}$ chart defined in Exercise 6.40. Find the ARL_1 for the chart.

6.42. Control charts for $\bar{x}$ and s with $n = 4$ are maintained on a quality characteristic. The parameters of these charts are as follows:

$\bar{x}$ Chart	s Chart
UCL = 201.88	UCL = 2.266
Center line = 200.00	Center line = 1.000
LCL = 198.12	LCL = 0

Both charts exhibit control. Specifications on the quality characteristic are 197.50 and 202.50. What can be said about the ability of the process to produce product that conforms to specifications?

6.43. Statistical monitoring of a quality characteristic uses both an $\bar{x}$ and an s chart. The charts are to be based on the standard values $\mu = 200$ and $\sigma = 10$, with $n = 4$.

(a) Find three-sigma control limits for the s chart.

(b) Find a center line and control limits for the $\bar{x}$ chart such that the probability of a type I error is 0.05.

6.44. Specifications on a normally distributed dimension are 600 ± 20. $\bar{x}$ and R charts are maintained on this dimension and have been in control over a long period of time. The parameters of these control charts are as follows ($n = 9$).

$\bar{x}$ Chart	s Chart
UCL = 616	UCL = 32.36
Center line = 610	Center line = 17.82
LCL = 604	LCL = 3.28

(a) What are your conclusions regarding the capability of the process to produce items within specifications?

(b) Construct an OC curve for the $\bar{x}$ chart assuming that σ is constant.

6.45. Thirty samples each of size 7 have been collected to establish control over a process. The following data were collected:

$$\sum_{i=1}^{30} \bar{x}_i = 2700 \quad \text{and} \quad \sum_{i=1}^{30} R_i = 120$$

(a) Calculate trial control limits for the two charts.
(b) On the assumption that the R chart is in control, estimate the process standard deviation.
(c) Suppose an s chart were desired. What would be the appropriate control limits and center line?

6.46. An $\bar{x}$ chart is to be established based on the standard values $\mu = 600$ and $\sigma = 12$, with $n = 9$. The control limits are to be based on an α-risk of 0.01. What are the appropriate control limits?

6.47. $\bar{x}$ and R charts with $n = 4$ are used to monitor a normally distributed quality characteristic. The control chart parameters are

$\bar{x}$ Chart	R Chart
UCL = 815	UCL = 46.98
Center line = 800	Center line = 20.59
LCL = 785	LCL = 0

Both charts exhibit control. What is the probability that a shift in the process mean to 790 will be detected on the first sample following the shift?

6.48. Consider the $\bar{x}$ chart in Exercise 6.47. Find the average run length for the chart.

6.49. Control charts for $\bar{x}$ and R are in use with the following parameters:

$\bar{x}$ Chart	R Chart
UCL = 363.0	UCL = 16.18
Center line = 360.0	Center line = 8.91
LCL = 357.0	LCL = 1.64

The sample size is $n = 9$. Both charts exhibit control. The quality characteristic is normally distributed.
(a) What is the α-risk associated with the $\bar{x}$ chart?
(b) Specifications on this quality characteristic are 358 ± 6. What are your conclusions regarding the ability of the process to produce items within specifications?
(c) Suppose the mean shifts to 357. What is the probability that the shift will not be detected on the first sample following the shift?
(d) What would be the appropriate control limits for the $\bar{x}$ chart if the type I error probability were to be 0.01?

6.50. A normally distributed quality characteristic is monitored through use of an $\bar{x}$ and an R chart. These charts have the following parameters ($n = 4$):

$\bar{x}$ Chart	R Chart
UCL = 626.0	UCL = 18.795
Center line = 620.0	Center line = 8.236
LCL = 614.0	LCL = 0

Both charts exhibit control.
(a) What is the estimated standard deviation of the process?
(b) Suppose an s chart were to be substituted for the R chart. What would be the appropriate parameters of the s chart?
(c) If specifications on the product were 610 ± 15, what would be your estimate of the process fraction nonconforming?
(d) What could be done to reduce this fraction nonconforming?
(e) What is the probability of detecting a shift in the process mean to 610 on the first sample following the shift (σ remains constant)?
(f) What is the probability of detecting the shift in part (e) by at least the third sample after the shift occurs?

6.51. Control charts for $\bar{x}$ and s have been maintained on a process and have exhibited statistical control. The sample size is $n = 6$. The control chart parameters are as follows:

$\bar{x}$ Chart	s Chart
UCL = 708.20	UCL = 3.420
Center line = 706.00	Center line = 1.738
LCL = 703.80	LCL = 0.052

(a) Estimate the mean and standard deviation of the process.
(b) Estimate the natural tolerance limits for the process.
(c) Assume that the process output is well modeled by a normal distribution. If specifications are 703 and 709, estimate the fraction nonconforming.
(d) Suppose the process mean shifts to 702.00 while the standard deviation remains constant. What is the probability of an out-of-control signal occurring on the first sample following the shift?

(e) For the shift in part (d), what is the probability of detecting the shift by at least the third subsequent sample?

6.52. The following $\bar{x}$ and s charts based on $n = 4$ have shown statistical control:

$\bar{x}$ Chart	s Chart
UCL = 710	UCL = 18.08
Center line = 700	Center line = 7.979
LCL = 690	LCL = 0

(a) Estimate the process parameters μ and σ.
(b) If the specifications are at 705 ± 15, and the process output is normally distributed, estimate the fraction nonconforming.
(c) For the $\bar{x}$ chart, find the probability of a type I error, assuming σ is constant.
(d) Suppose the process mean shifts to 693 and the standard deviation simultaneously shifts to 12. Find the probability of detecting this shift on the $\bar{x}$ chart on the first subsequent sample.
(e) For the shift of part (d), find the average run length.

6.53. One-pound coffee cans are filled by a machine, sealed, and then weighed automatically. After adjusting for the weight of the can, any package that weighs less than 16 oz is cut out of the conveyor. The weights of 25 successive cans are shown in Table 6E.19. Set up a moving range control chart and a control chart for individuals. Estimate the mean and standard deviation of the amount of coffee packed in each can. Is it reasonable to assume that can weight is normally distributed? If the process remains in control at this level, what percentage of cans will be underfilled?

■ TABLE 6E.19
Can Weight Data for Exercise 6.53

Can Number	Weight	Can Number	Weight
1	16.11	14	16.12
2	16.08	15	16.10
3	16.12	16	16.08
4	16.10	17	16.13
5	16.10	18	16.15
6	16.11	19	16.12
7	16.12	20	16.10
8	16.09	21	16.08
9	16.12	22	16.07
10	16.10	23	16.11
11	16.09	24	16.13
12	16.07	25	16.10
13	16.13		

■ TABLE 6E.20
Hardness Data for Exercise 6.54

Heat	Hardness (coded)	Heat	Hardness (coded)
1	52	9	58
2	51	10	51
3	54	11	54
4	55	12	59
5	50	13	53
6	52	14	54
7	50	15	55
8	51		

6.54. Fifteen successive heats of a steel alloy are tested for hardness. The resulting data are shown in Table 6E.20. Set up a control chart for the moving range and a control chart for individual hardness measurements. Is it reasonable to assume that hardness is normally distributed?

6.55. The viscosity of a polymer is measured hourly. Measurements for the last twenty hours are shown in Table 6E.21.
(a) Does viscosity follow a normal distribution?
(b) Set up a control chart on viscosity and a moving range chart. Does the process exhibit statistical control?
(c) Estimate the process mean and standard deviation.

6.56. **Continuation of Exercise 6.55.** The next five measurements on viscosity are 3163, 3199, 3054, 3147, and 3156. Do these measurements indicate that the process is in statistical control?

■ TABLE 6E.21
Viscosity Data for Exercise 6.55

Test	Viscosity	Test	Viscosity
1	2838	11	3174
2	2785	12	3102
3	3058	13	2762
4	3064	14	2975
5	2996	15	2719
6	2882	16	2861
7	2878	17	2797
8	2920	18	3078
9	3050	19	2964
10	2870	20	2805

■ TABLE 6E.22
Data for Exercise 6.57

Wafer	Oxide Thickness	Wafer	Oxide Thickness
1	45.4	16	58.4
2	48.6	17	51.0
3	49.5	18	41.2
4	44.0	19	47.1
5	50.9	20	45.7
6	55.2	21	60.6
7	45.5	22	51.0
8	52.8	23	53.0
9	45.3	24	56.0
10	46.3	25	47.2
11	53.9	26	48.0
12	49.8	27	55.9
13	46.9	28	50.0
14	49.8	29	47.9
15	45.1	30	53.4

6.57. (a) Thirty observations on the oxide thickness of individual silicon wafers are shown in Table 6E.22. Use these data to set up a control chart on oxide thickness and a moving range chart. Does the process exhibit statistical control? Does oxide thickness follow a normal distribution?

(b) Following the establishment of the control charts in part (a), ten new wafers were observed. The oxide thickness measurements are as follows:

Wafer	Oxide Thickness	Wafer	Oxide Thickness
1	54.3	6	51.5
2	57.5	7	58.4
3	64.8	8	67.5
4	62.1	9	61.1
5	59.6	10	63.3

Plot these observations against the control limits determined in part (a). Is the process in control?

(c) Suppose the assignable cause responsible for the out-of-control signal in part (b) is discovered and removed from the process. Twenty additional wafers are subsequently sampled. Plot the oxide thickness against the part (a) control limits. What conclusions can you draw? The new data are shown in Table 6E.23.

6.58. Thirty observations on concentration (in g/l) of the active ingredient in a liquid cleaner produced in a continuous chemical process are shown in Table 6E.24.

■ TABLE 6E.23
Additional Data for Exercise 6.57, part (c)

Wafer	Oxide Thickness	Wafer	Oxide Thickness
1	43.4	11	50.0
2	46.7	12	61.2
3	44.8	13	46.9
4	51.3	14	44.9
5	49.2	15	46.2
6	46.5	16	53.3
7	48.4	17	44.1
8	50.1	18	47.4
9	53.7	19	51.3
10	45.6	20	42.5

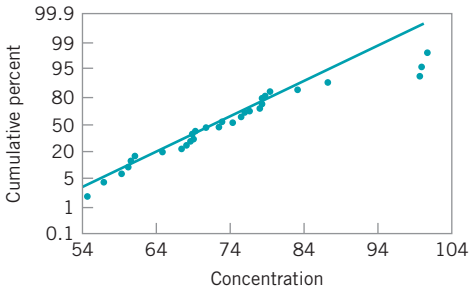
(a) A normal probability plot of the concentration data is shown in Figure 6.29. The straight line was fit by eye to pass approximately through the twentieth and eightieth percentiles. Does the normality assumption seem reasonable here?

(b) Set up an individuals and moving range control chart for the concentration data. Interpret the charts.

(c) Construct a normal probability plot for the natural log of concentration. Is the transformed variable normally distributed?

■ TABLE 6E.24
Data for Exercise 6E.58

Observation	Concentration	Observation	Concentration
1	60.4	16	99.9
2	69.5	17	59.3
3	78.4	18	60.0
4	72.8	19	74.7
5	78.2	20	75.8
6	78.7	21	76.6
7	56.9	22	68.4
8	78.4	23	83.1
9	79.6	24	61.1
10	100.8	25	54.9
11	99.6	26	69.1
12	64.9	27	67.5
13	75.5	28	69.2
14	70.4	29	87.2
15	68.1	30	73.0



■ FIGURE 6.29 Normal Probability Plot of the Concentration Data for Exercise 6.58

(d) Repeat part (b), using the natural log of concentration as the charted variable. Comment on any differences in the charts you note in comparison to those constructed in part (b).

6.59. In 1879, A. A. Michelson measured the velocity of light in air using a modification of a method proposed by the French physicist Foucauld. Twenty of these measurements are in Table 6E.25 (the value reported is in kilometers per second and has 299,000 subtracted from it). Use these data to set up an individuals and a moving range control chart. Is there some evidence that the measurements of the velocity of light are normally distributed? Do the measurements exhibit statistical control? Revise the control limits if necessary.

6.60. **Continuation of Exercise 6.59.** Michelson actually made 100 measurements on the velocity of light in five trials of 20 observations each. The second set of 20 measurements is shown in Table 6E.26.

(a) Plot these new measurements on the control charts constructed in Exercise 6.59. Are these new measurements in statistical control? Give a practical interpretation of the control charts.

■ TABLE 6E.25
Velocity of Light Data for Exercise 6E.59

Measurement	Velocity	Measurement	Velocity
1	850	11	850
2	1000	12	810
3	740	13	950
4	980	14	1000
5	900	15	980
6	930	16	1000
7	1070	17	980
8	650	18	960
9	930	19	880
10	760	20	960

■ TABLE 6E.26
Additional Velocity of Light Data for Exercise 6.60.

Measurement	Velocity	Measurement	Velocity
21	960	31	800
22	830	32	830
23	940	33	850
24	790	34	800
25	960	35	880
26	810	36	790
27	940	37	900
28	880	38	760
29	880	39	840
30	880	40	800

(b) Is there evidence that the variability in the measurements has decreased between trial 1 and trial 2?

6.61. The uniformity of a silicon wafer following an etching process is determined by measuring the layer thickness at several locations and expressing uniformity as the range of the thicknesses. Table 6E.27 presents uniformity determinations for 30 consecutive wafers processed through the etching tool.

(a) Is there evidence that uniformity is normally distributed? If not, find a suitable transformation for the data.

(b) Construct a control chart for individuals and a moving range control chart for uniformity for the etching process. Is the process in statistical control?

■ TABLE 6E.27
Uniformity Data for Exercise 6.61

Wafer	Uniformity	Wafer	Uniformity
1	11	16	15
2	16	17	16
3	22	18	12
4	14	19	11
5	34	20	18
6	22	21	14
7	13	22	13
8	11	23	18
9	6	24	12
10	11	25	13
11	11	26	12
12	23	27	15
13	14	28	21
14	12	29	21
15	7	30	14

■ TABLE 6E.28
Purity Data for Exercise 6.62.

Batch	Purity	Batch	Purity
1	0.81	11	0.81
2	0.82	12	0.83
3	0.81	13	0.81
4	0.82	14	0.82
5	0.82	15	0.81
6	0.83	16	0.85
7	0.81	17	0.83
8	0.80	18	0.87
9	0.81	19	0.86
10	0.82	20	0.84

6.62. The purity of a chemical product is measured on each batch. Purity determinations for 20 successive batches are shown in Table 6E.28.
(a) Is purity normally distributed?
(b) Is the process in statistical control?
(c) Estimate the process mean and standard deviation.

6.63. Reconsider the situation in Exercise 6.53. Construct an individuals control chart using the median of the span-two moving ranges to estimate variability. Compare this control chart to the one constructed in Exercise 6.53 and discuss.

6.64. Reconsider the hardness measurements in Exercise 6.54. Construct an individuals control chart using the median of the span-two moving ranges to estimate variability. Compare this control chart to the one constructed in Exercise 6.54 and discuss.

6.65. Reconsider the polymer viscosity data in Exercise 6.55. Use the median of the span-two moving ranges to estimate σ and set up the individuals control chart. Compare this chart to the one originally constructed using the average moving range method to estimate σ.

6.66. **Continuation of Exercise 6.57.** Use all 60 observations on oxide thickness.
(a) Set up an individuals control chart with σ estimated by the average moving range method.
(b) Set up an individuals control chart with σ estimated by the median moving range method.
(c) Compare and discuss the two control charts.

6.67. Consider the individuals measurement data shown in Table 6E.29.
(a) Estimate σ using the average of the moving ranges of span two.
(b) Estimate σ using s/c_4.
(c) Estimate σ using the median of the span-two moving ranges.

■ TABLE 6E.29
Data for Exercise 6.67

Observation	x	Observation	x
1	10.07	14	9.58
2	10.47	15	8.80
3	9.45	16	12.94
4	9.44	17	10.78
5	8.99	18	11.26
6	7.74	19	9.48
7	10.63	20	11.28
8	9.78	21	12.54
9	9.37	22	11.48
10	9.95	23	13.26
11	12.04	24	11.10
12	10.93	25	10.82
13	11.54		

(d) Estimate σ using the average of the moving ranges of span 3, 4, . . . , 20.
(e) Discuss the results you have obtained.

6.68. The vane heights for 20 of the castings from Fig. 6.25 are shown in Table 6E.30. Construct the "between/within" control charts for these process

■ TABLE 6E.30
Vane Heights for Exercise 6.68.

Casting	Vane 1	Vane 2	Vane 3	Vane 4	Vane 5
1	5.77799	5.74907	5.76672	5.74836	5.74122
2	5.79090	5.78043	5.79163	5.79393	5.81158
3	5.77314	5.71216	5.74810	5.77292	5.75591
4	5.77030	5.75903	5.77157	5.79687	5.78063
5	5.72047	5.68587	5.73302	5.70472	5.68116
6	5.77265	5.76426	5.74373	5.71338	5.74765
7	5.70581	5.70835	5.71866	5.71252	5.72089
8	5.76466	5.78766	5.76115	5.77523	5.75590
9	5.79397	5.83308	5.77902	5.81122	5.82335
10	5.78671	5.76411	5.75941	5.75619	5.71787
11	5.75352	5.74144	5.74109	5.76817	5.75019
12	5.72787	5.70716	5.75349	5.72389	5.73488
13	5.79707	5.79231	5.79022	5.79694	5.79805
14	5.73765	5.73615	5.73249	5.74006	5.73265
15	5.72477	5.76565	5.76963	5.74993	5.75196
16	5.73199	5.72926	5.72963	5.72259	5.73513
17	5.79166	5.79516	5.79903	5.78548	5.79826
18	5.74973	5.74863	5.73994	5.74405	5.74682
19	5.76449	5.75632	5.76197	5.76684	5.75474
20	5.75168	5.75579	5.73979	5.77963	5.76933

data using a range chart to monitor the within-castings vane height. Compare these to the control charts shown in Fig. 6.27.

6.69. The diameter of the casting in Fig. 6.25 is also an important quality characteristic. A coordinate measuring machine is used to measure the diameter of each casting at five different locations. Data for 20 castings are shown in the Table 6E.31.

(a) Set up $\bar{x}$ and R charts for this process, assuming the measurements on each casting form a rational subgroup.

(b) Discuss the charts you have constructed in part (a).

(c) Construct "between/within" charts for this process.

(d) Do you believe that the charts in part (c) are more informative than those in part (a)? Discuss why.

(e) Provide a practical interpretation of the "within" chart.

6.70. In the semiconductor industry, the production of microcircuits involves many steps. The wafer fabrication process typically builds these microcircuits on silicon wafers, and there are many microcircuits per wafer. Each production lot consists of between 16 and 48 wafers. Some processing steps treat each wafer separately, so that the batch size for that step is one wafer. It is usually necessary to estimate several components of variation: within-wafer, between-wafer, between-lot, and the total variation.

(a) Suppose that one wafer is randomly selected from each lot and that a single measurement on a critical dimension of interest is taken. Which components of variation could be estimated with these data? What type of control charts would you recommend?

(b) Suppose that each wafer is tested at five fixed locations (say, the center and four points at the circumference). The average and range of these within-wafer measurements are $\bar{x}_{ww}$ and R_{ww}, respectively. What components of variability are estimated using control charts based on these data?

(c) Suppose that one measurement point on each wafer is selected and that this measurement is recorded for five consecutive wafers. The average and range of these between-wafer measurements are $\bar{x}_{BW}$ and R_{BW}, respectively. What components of variability are estimated using control charts based on these data? Would it be necessary to run separate $\bar{x}$ and R charts for all five locations on the wafer?

(d) Consider the question in part (c). How would your answer change if the test sites on each wafer were randomly selected and varied from wafer to wafer?

(e) What type of control charts and rational subgroup scheme would you recommend to control the batch-to-batch variability?

6.71. Consider the situation described in Exercise 6.70. A critical dimension (measured in μm) is of interest to the process engineer. Suppose that five fixed positions are used on each wafer (position 1 is the center) and that two consecutive wafers are selected from each batch. The data that result from several batches are shown in Table 6E.32.

(a) What can you say about overall process capability?

(b) Can you construct control charts that allow within-wafer variability to be evaluated?

(c) What control charts would you establish to evaluate variability between wafers? Set up these charts and use them to draw conclusions about the process.

(d) What control charts would you use to evaluate lot-to-lot variability? Set up these charts and use them to draw conclusions about lot-to-lot variability.

■ **TABLE 6E.31**
Diameter Data for Exercise 6.69.

	Diameter				
Casting	**1**	**2**	**3**	**4**	**5**
1	11.7629	11.7403	11.7511	11.7474	11.7374
2	11.8122	11.7506	11.7787	11.7736	11.8412
3	11.7742	11.7114	11.7530	11.7532	11.7773
4	11.7833	11.7311	11.7777	11.8108	11.7804
5	11.7134	11.6870	11.7305	11.7419	11.6642
6	11.7925	11.7611	11.7588	11.7012	11.7611
7	11.6916	11.7205	11.6958	11.7440	11.7062
8	11.7109	11.7832	11.7496	11.7496	11.7318
9	11.7984	11.8887	11.7729	11.8485	11.8416
10	11.7914	11.7613	11.7356	11.7628	11.7070
11	11.7260	11.7329	11.7424	11.7645	11.7571
12	11.7202	11.7537	11.7328	11.7582	11.7265
13	11.8356	11.7971	11.8023	11.7802	11.7903
14	11.7069	11.7112	11.7492	11.7329	11.7289
15	11.7116	11.7978	11.7982	11.7429	11.7154
16	11.7165	11.7284	11.7571	11.7597	11.7317
17	11.8022	11.8127	11.7864	11.7917	11.8167
18	11.7775	11.7372	11.7241	11.7773	11.7543
19	11.7753	11.7870	11.7574	11.7620	11.7673
20	11.7572	11.7626	11.7523	11.7395	11.7884

■ TABLE 6E.32
Data for Exercise 6.71

Lot Number	Wafer Number	Position 1	2	3	4	5	Lot Number	Wafer Number	Position 1	2	3	4	5
1	1	2.15	2.13	2.08	2.12	2.10	11	1	2.15	2.13	2.14	2.09	2.08
	2	2.13	2.10	2.04	2.08	2.05		2	2.11	2.13	2.10	2.14	2.10
2	1	2.02	2.01	2.06	2.05	2.08	12	1	2.03	2.06	2.05	2.01	2.00
	2	2.03	2.09	2.07	2.06	2.04		2	2.04	2.08	2.03	2.10	2.07
3	1	2.13	2.12	2.10	2.11	2.08	13	1	2.05	2.03	2.05	2.09	2.08
	2	2.03	2.08	2.03	2.09	2.07		2	2.08	2.01	2.03	2.04	2.10
4	1	2.04	2.01	2.10	2.11	2.09	14	1	2.08	2.04	2.05	2.01	2.08
	2	2.07	2.14	2.12	2.08	2.09		2	2.09	2.11	2.06	2.04	2.05
5	1	2.16	2.17	2.13	2.18	2.10	15	1	2.14	2.13	2.10	2.10	2.08
	2	2.17	2.13	2.10	2.09	2.13		2	2.13	2.10	2.09	2.13	2.15
6	1	2.04	2.06	1.97	2.10	2.08	16	1	2.06	2.08	2.05	2.03	2.09
	2	2.03	2.10	2.05	2.07	2.04		2	2.03	2.01	1.99	2.06	2.05
7	1	2.04	2.02	2.01	2.00	2.05	17	1	2.05	2.03	2.08	2.01	2.04
	2	2.06	2.04	2.03	2.08	2.10		2	2.06	2.05	2.03	2.05	2.00
8	1	2.13	2.10	2.10	2.15	2.13	18	1	2.03	2.08	2.04	2.00	2.03
	2	2.10	2.09	2.13	2.14	2.11		2	2.04	2.03	2.05	2.01	2.04
9	1	1.95	2.03	2.08	2.07	2.08	19	1	2.16	2.13	2.10	2.13	2.12
	2	2.01	2.03	2.06	2.05	2.04		2	2.13	2.15	2.18	2.19	2.13
10	1	2.04	2.08	2.09	2.10	2.01	20	1	2.06	2.03	2.04	2.09	2.10
	2	2.06	2.04	2.07	2.04	2.01		2	2.01	1.98	2.05	2.08	2.06

7 Control Charts for Attributes

Chapter Outline

7.1 INTRODUCTION
7.2 THE CONTROL CHART FOR FRACTION NONCONFORMING
- 7.2.1 Development and Operation of the Control Chart
- 7.2.2 Variable Sample Size
- 7.2.3 Applications in Transactional and Service Businesses
- 7.2.4 The Operating-Characteristic Function and Average Run Length Calculations

7.3 CONTROL CHARTS FOR NONCONFORMITIES (DEFECTS)
- 7.3.1 Procedures with Constant Sample Size
- 7.3.2 Procedures with Variable Sample Size
- 7.3.3 Demerit Systems
- 7.3.4 The Operating-Characteristic Function
- 7.3.5 Dealing with Low Defect Levels
- 7.3.6 Nonmanufacturing Applications

7.4 CHOICE BETWEEN ATTRIBUTES AND VARIABLES CONTROL CHARTS
7.5 GUIDELINES FOR IMPLEMENTING CONTROL CHARTS

Supplemental Material for Chapter 7

- S7.1 Probability Limits on Control Charts

The supplemental material is on the textbook Website www.wiley.com/college/montgomery.

Chapter Overview and Learning Objectives

Many quality characteristics cannot be conveniently represented numerically. In such cases, we usually classify each item inspected as either conforming or nonconforming to the specifications on that quality characteristic. The terminology **defective** or **nondefective** is often used to identify these two classifications of product. More recently, the terminology **conforming** and **nonconforming** has become popular. Quality characteristics of this type are called **attributes.** Some examples of quality characteristics that are attributes are the proportion of warped automobile engine connecting rods in a day's production, the number of nonfunctional semiconductor chips on a wafer, the number of errors or mistakes made in completing a loan application, and the number of medical errors made in a hospital.

This chapter presents three widely used attributes control charts. The first of these relates to the fraction of nonconforming or defective product produced by a manufacturing process, and is called the **control chart for fraction nonconforming,** or p chart. In some situations it is more convenient to deal with the number of **defects** or **nonconformities** observed

rather than the fraction nonconforming. The second type of control chart that we study, called the **control chart for nonconformities,** or the *c* chart, is designed to deal with this case. Finally, we present a **control chart for nonconformities per unit,** or the *u* chart, which is useful in situations where the average number of nonconformities per unit is a more convenient basis for process control. An excellent supplement to this chapter is the paper by Woodall (1997) that summarizes over 250 papers on attributes control charts and provides a comprehensive bibliography. The chapter concludes with some guidelines for implementing control charts.

After careful study of this chapter you should be able to do the following:

1. Understand the statistical basis of attributes control charts
2. Know how to design attributes control charts
3. Know how to set up and use the *p* chart for fraction nonconforming
4. Know how to set up and use the *np* control chart for the number of nonconforming items
5. Know how to set up and use the *c* control chart for defects
6. Know how to set up and use the *u* control chart for defects per unit
7. Use attributes control charts with variable sample size
8. Understand the advantages and disadvantages of attributes versus variables control charts
9. Understand the rational subgroup concept for attributes control charts
10. Determine the average run length for attributes control charts

7.1 Introduction

In Chapter 6, we introduced control charts for quality characteristics that are expressed as variables. Although these control charts enjoy widespread application they are not universally applicable because not all quality characteristics can be expressed with variables data. For example, consider a glass container for a liquid product. Suppose we examine a container and classify it into one of the two categories called *conforming* or *nonconforming,* depending on whether the container meets the requirements on one or more quality characteristics. This is an example of **attributes data,** and a control chart for the fraction of nonconforming containers could be established (we show how to do this in Section 7.2). Alternatively, in some processes we may examine a unit of product and count defects or nonconformities on the unit. These types of data are widely encountered in the semiconductor industry, for example. In Section 7.3 we show how to establish control charts for counts, or for the average number of counts per unit.

Attributes charts are generally not as informative as variables charts because there is typically more information in a numerical measurement than in merely classifying a unit as conforming or nonconforming. However, attribute charts do have important applications. They are particularly useful in service industries and in nonmanufacturing quality-improvement efforts because so many of the quality characteristics found in these environments are not easily measured on a numerical scale.

7.2 The Control Chart for Fraction Nonconforming

The **fraction nonconforming** is defined as the ratio of the number of nonconforming items in a population to the total number of items in that population. The items may have *several* quality characteristics that are examined simultaneously by the inspector. If the item does not conform to standard on one or more of these characteristics, it is classified as nonconforming.

We usually express the fraction nonconforming as a decimal, although occasionally the percent nonconforming (which is simply 100% times the fraction nonconforming) is used. When demonstrating or displaying the control chart to production personnel or presenting results to management, the percent nonconforming is often used, as it has more intuitive appeal. Although it is customary to work with fraction nonconforming, we could also analyze the fraction conforming just as easily, resulting in a control chart on **process yield.** For example, many manufacturing organizations operate a yield-management system at each stage of their manufacturing process, with the first-pass yield tracked on a control chart.

The statistical principles underlying the control chart for fraction nonconforming are based on the binomial distribution. Suppose the production process is operating in a stable manner, such that the probability that any unit will not conform to specifications is p, and that successive units produced are independent. Then each unit produced is a realization of a Bernoulli random variable with parameter p. If a random sample of n units of product is selected, and if D is the number of units of product that are nonconforming, then D has a binomial distribution with parameters n and p; that is,

$$P\{D = x\} = \binom{n}{x} p^x (1-p)^{n-x} \qquad x = 0,\ 1,\ \ldots,\ n \tag{7.1}$$

From Section 3.2.2 we know that the mean and variance of the random variable D are np and $np(1-p)$, respectively.

The **sample fraction nonconforming** is defined as the ratio of the number of nonconforming units in the sample D to the sample size n; that is,

$$\hat{p} = \frac{D}{n} \tag{7.2}$$

As noted in Section 3.2.2, the distribution of the random variable $\hat{p}$ can be obtained from the binomial. Furthermore, the mean and variance of $\hat{p}$ are

$$\mu_{\hat{p}} = p \tag{7.3}$$

and

$$\sigma^2_{\hat{p}} = \frac{p(1-p)}{n} \tag{7.4}$$

respectively. We will now see how this theory can be applied to the development of a control chart for fraction nonconforming. Because the chart monitors the process fraction nonconforming p, it is also called the p chart.

7.2.1 Development and Operation of the Control Chart

In Chapter 5, we discussed the general statistical principles on which the Shewhart control chart is based. If w is a statistic that measures a quality characteristic, and if the mean of w is μ_w and the variance of w is σ^2_w, then the general model for the Shewhart control chart is as follows:

$$\begin{aligned} \text{UCL} &= \mu_w + L\sigma_w \\ \text{Center line} &= \mu_w \\ \text{LCL} &= \mu_w - L\sigma_w \end{aligned} \tag{7.5}$$

where L is the distance of the control limits from the center line, in multiples of the standard deviation of w. It is customary to choose $L = 3$.

Suppose that the true fraction nonconforming p in the production process is known or is a specified **standard value.** Then from equation (7.5), the center line and control limits of the fraction nonconforming control chart would be as follows:

Fraction Nonconforming Control Chart: Standard Given

$$\begin{aligned} \text{UCL} &= p + 3\sqrt{\frac{p(1-p)}{n}} \\ \text{Center line} &= p \\ \text{LCL} &= p - 3\sqrt{\frac{p(1-p)}{n}} \end{aligned} \tag{7.6}$$

Depending on the values of p and n, sometimes the lower control limit LCL < 0. In these cases, we customarily set LCL = 0 and assume that the control chart only has an upper control limit. The actual operation of this chart would consist of taking subsequent samples of n units, computing the sample fraction nonconforming $\hat{p}$, and plotting the statistic $\hat{p}$ on the chart. As long as $\hat{p}$ remains within the control limits and the sequence of plotted points does not exhibit any systematic nonrandom pattern, we can conclude that the process is in control at the level p. If a point plots outside of the control limits, or if a nonrandom pattern in the plotted points is observed, we can conclude that the process fraction nonconforming has most likely shifted to a new level and the process is out of control.

When the process fraction nonconforming p is not known, then it must be estimated from observed data. The usual procedure is to select m preliminary samples, each of size n. As a general rule, m should be at least 20 or 25. Then if there are D_i nonconforming units in sample i, we compute the fraction nonconforming in the ith sample as

$$\hat{p}_i = \frac{D_i}{n} \qquad i = 1, 2, \ldots, m$$

and the average of these individual sample fractions nonconforming is

$$\bar{p} = \frac{\sum_{i=1}^{m} D_i}{mn} = \frac{\sum_{i=1}^{m} \hat{p}_i}{m} \tag{7.7}$$

The statistic $\bar{p}$ estimates the unknown fraction nonconforming p. The center line and control limits of the control chart for fraction nonconforming are computed as follows:

Fraction Nonconforming Control Chart: No Standard Given

$$\begin{aligned} \text{UCL} &= \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \\ \text{Center line} &= \bar{p} \\ \text{LCL} &= \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \end{aligned} \tag{7.8}$$

This control chart is also often called the *p-chart*.

The control limits defined in equation (7.8) should be regarded as **trial control limits.** The sample values of $\hat{p}_i$ from the preliminary subgroups should be plotted against the trial limits to test whether the process was in control when the preliminary data were collected. This is the usual phase I aspect of control chart usage. Any points that exceed the trial control limits should be investigated. If assignable causes for these points are discovered, they should be discarded and new trial control limits determined. Refer to the discussion of trial control limits for the $\bar{x}$ and R charts in Chapter 6.

If the control chart is based on a known or standard value for the fraction nonconforming p, then the calculation of trial control limits is generally unnecessary. However, one should be cautious when working with a standard value for p. Since in practice the true value of p would rarely be known with certainty, we would usually be given a standard value of p that represents a desired or **target value** for the process fraction nonconforming. If this is the case, and future samples indicate an out-of-control condition, we must determine whether the process is out of control at the *target* p but in control at some *other* value of p. For example, suppose we specify a target value of $p = 0.01$, but the process is really in control at a larger value of fraction nonconforming—say, $p = 0.05$. Using the control chart based on $p = 0.01$, we see that many of the points will plot above the upper control limit, indicating an out-of-control condition. However, the process is really out of control only with respect to the target $p = 0.01$. Sometimes it may be possible to "improve" the level of quality by using target values, or to bring a process into control at a particular level of quality performance. In processes where the fraction nonconforming can be controlled by relatively simple process adjustments, target values of p may be useful.

EXAMPLE 7.1 Construction and Operation of a Fraction Nonconforming Control Chart

Frozen orange juice concentrate is packed in 6-oz cardboard cans. These cans are formed on a machine by spinning them from cardboard stock and attaching a metal bottom panel. By inspection of a can, we may determine whether, when filled, it could possibly leak either on the side seam or around the bottom joint. Such a nonconforming can has an improper seal on either the side seam or the bottom panel. Set up a control chart to improve the fraction of nonconforming cans produced by this machine.

SOLUTION

To establish the control chart, 30 samples of $n = 50$ cans each were selected at half-hour intervals over a three-shift period in which the machine was in continuous operation. The data are shown in Table 7.1.

We construct a phase I control chart using this preliminary data to determine if the process was in control when these data were collected. Since the 30 samples contain $\sum_{i=1}^{30} D_i = 347$ nonconforming cans, we find from equation (7.7),

$$\bar{p} = \frac{\sum_{i=1}^{m} D_i}{mn} = \frac{347}{(30)(50)} = 0.2313$$

Using $\bar{p}$ as an estimate of the true process fraction nonconforming, we can now calculate the upper and lower control limits as

$$\bar{p} \pm 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2313 \pm 3\sqrt{\frac{0.2313(0.7687)}{50}}$$
$$= 0.2313 \pm 3(0.0596)$$
$$= 0.2313 \pm 0.1789$$

Therefore,

$$\text{UCL} = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2313 + 0.1789 = 0.4102$$

and

$$\text{LCL} = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2313 - 0.1789 = 0.0524$$

TABLE 7.1

Data for Trial Control Limits, Example 7.1, Sample Size $n = 50$

Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$	Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$
1	12	0.24	17	10	0.20
2	15	0.30	18	5	0.10
3	8	0.16	19	13	0.26
4	10	0.20	20	11	0.22
5	4	0.08	21	20	0.40
6	7	0.14	22	18	0.36
7	16	0.32	23	24	0.48
8	9	0.18	24	15	0.30
9	14	0.28	25	9	0.18
10	10	0.20	26	12	0.24
11	5	0.10	27	7	0.14
12	6	0.12	28	13	0.26
13	17	0.34	29	9	0.18
14	12	0.24	30	6	0.12
15	22	0.44		347	$\bar{p} = 0.2313$
16	8	0.16			

The control chart with center line at $\bar{p} = 0.2313$ and the above upper and lower control limits is shown in Fig. 7.1. The sample fraction nonconforming from each preliminary sample is plotted on this chart. We note that two points, those from samples 15 and 23, plot above the upper control limit, so the process is not in control. These points must be investigated to see whether an assignable cause can be determined.

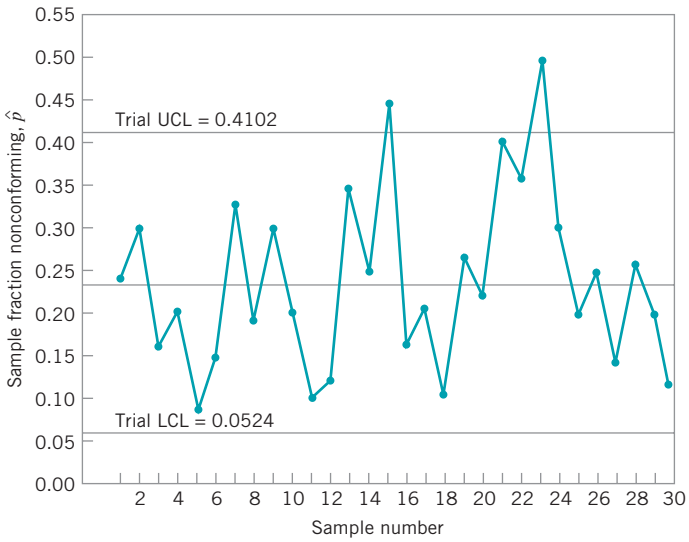


FIGURE 7.1 Initial phase I fraction nonconforming control chart for the data in Table 7.1.

(*continued*)

Analysis of the data from sample 15 indicates that a new batch of cardboard stock was put into production during that half-hour period. The introduction of new batches of raw material sometimes causes irregular production performance, and it is reasonable to believe that this has occurred here. Furthermore, during the half-hour period in which sample 23 was obtained, a relatively inexperienced operator had been temporarily assigned to the machine, and this could account for the high fraction nonconforming obtained from that sample. Consequently, samples 15 and 23 are eliminated, and the new center line and revised control limits are calculated as

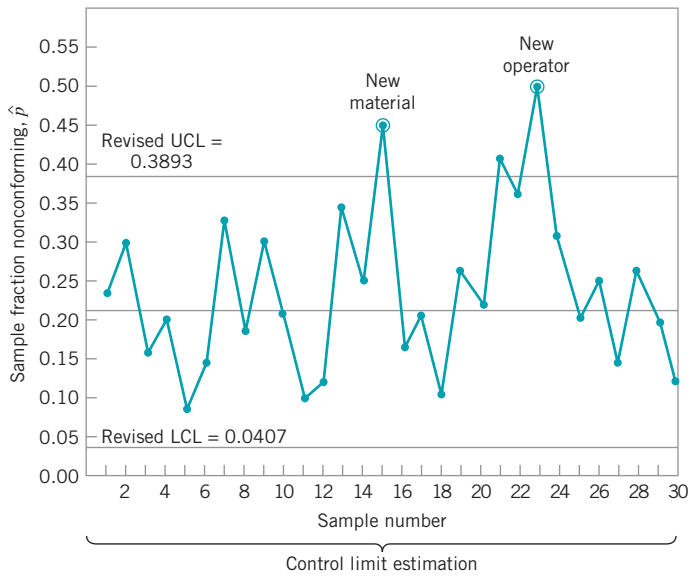
$$\bar{p} = \frac{301}{(28)(50)} = 0.2150$$

$$\text{UCL} = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2150 + 3\sqrt{\frac{0.2150(0.7850)}{50}} = 0.3893$$

$$\text{LCL} = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2150 - 3\sqrt{\frac{0.2150(0.7850)}{50}} = 0.0407$$

The revised center line and control limits are shown on the control chart in Fig. 7.2. Note that we have not dropped samples 15 and 23 from the chart, but they have been excluded from the control limit calculations, and we have noted this directly on the control chart. This annotation of the control chart to indicate unusual points, process adjustments, or the type of investigation made at a particular point in time forms a useful record for future process analysis and should become a standard practice in control chart usage.

Note also that the fraction nonconforming from sample 21 now exceeds the upper control limit. However, analysis of the data does not produce any reasonable or logical assignable cause for this, and we decide to retain the point. Therefore, we conclude that the new control limits in Fig. 7.2 can be used for future samples. Thus, we have concluded the control limit estimation phase (phase I) of control chart usage.



■ FIGURE 7.2 Revised control limits for the data in Table 7.1.

Sometimes examination of control chart data reveals information that affects other points that are not necessarily outside the control limits. For example, if we had found that the temporary operator working when sample 23 was obtained was actually working during the entire two-hour period in which samples 21–24 were obtained, then we should discard all four samples, even if only sample 21 exceeded the control limits, on the grounds that this inexperienced operator probably had some adverse influence on the fraction nonconforming during the entire period.

Before we conclude that the process is in control at this level, we could examine the remaining 28 samples for runs and other nonrandom patterns. The largest run is one of length 5 above the center line, and there are no obvious patterns present in the data. There is no strong evidence of anything other than a random pattern of variation about the center line.

We conclude that the process is in control at the level $p = 0.2150$ and that the revised control limits should be adopted for monitoring current production. However, we note that although the process is in control, the fraction nonconforming is much too high. That is, the process is operating in a stable manner, and no unusual **operator-controllable** problems are present. It is unlikely that the process quality can be improved by action at the workforce level. The nonconforming cans produced are **management controllable** because an intervention by management in the process will be required to improve performance. Plant management agrees with this observation and directs that, in addition to implementing the control chart program, the engineering staff should analyze the process in an effort to improve the process yield. This study indicates that several adjustments can be made on the machine that should improve its performance.

During the next three shifts following the machine adjustments and the introduction of the control chart, an additional 24 samples of $n = 50$ observations each are collected. These data are shown in Table 7.2, and the sample fractions nonconforming are plotted on the control chart in Fig. 7.3.

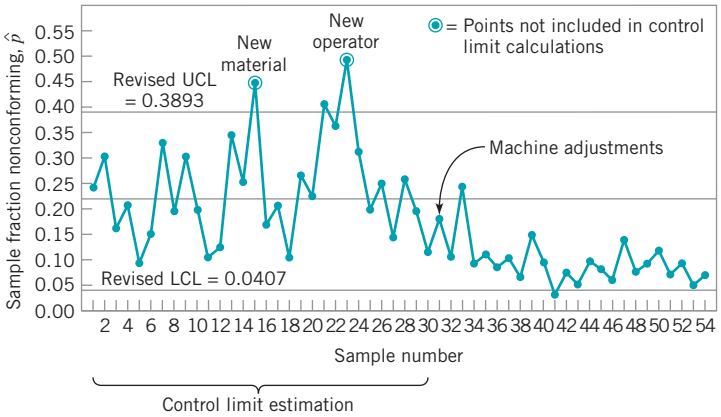
From an examination of Fig. 7.3, our immediate impression is that the process is now operating at a new quality level that is substantially better than the center line level of $\bar{p} = 0.2150$. One point, that from sample 41, is below the lower control limit. No assignable cause for this out-of-control signal can be determined. The only logical reasons for this ostensible change in process performance are the machine adjustments made by the engineering staff and, possibly, the operators themselves. It is not unusual to find that process performance improves following the introduction of formal statistical process-control procedures, often because the operators are more aware of process quality and because the control chart provides a continuing visual display of process performance.

We may formally test the hypothesis that the process fraction nonconforming in this current three-shift period differs

TABLE 7.2
Orange Juice Concentrate Can Data in Samples of Size $n = 50$

Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$	Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$
31	9	0.18	44	6	0.12
32	6	0.12	45	5	0.10
33	12	0.24	46	4	0.08
34	5	0.10	47	8	0.16
35	6	0.12	48	5	0.10
36	4	0.08	49	6	0.12
37	6	0.12	50	7	0.14
38	3	0.06	51	5	0.10
39	7	0.14	52	6	0.12
40	6	0.12	53	3	0.06
41	2	0.04	54	5	0.10
42	4	0.08		133	$\bar{p} = 0.1108$
43	3	0.06			

(*continued*)



■ **FIGURE 7.3** Continuation of the fraction nonconforming control chart, Example 7.1.

from the process fraction nonconforming in the preliminary data, using the procedure given in Section 4.3.4. The hypotheses are

$$H_0: \quad p_1 = p_2$$
$$H_1: \quad p_1 > p_2$$

where p_1 is the process fraction nonconforming from the preliminary data and p_2 is the process fraction nonconforming in the current period. We may estimate p_1 by $\hat{p}_1 = \bar{p} = 0.2150$, and p_2 by

$$\hat{p}_2 = \frac{\sum_{i=31}^{54} D_i}{(50)(24)} = \frac{133}{1200} = 0.1108$$

The (approximate) test statistic for the above hypothesis is, from equation (4.63),

$$Z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where

$$\hat{p} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}$$

In our example, we have

$$\hat{p} = \frac{(1400)(0.2150) + (1200)(0.1108)}{1400 + 1200} = 0.1669$$

and

$$Z_0 = \frac{0.2150 - 0.1108}{\sqrt{(0.1669)(0.8331)\left(\frac{1}{1400} + \frac{1}{1200}\right)}} = 7.10$$

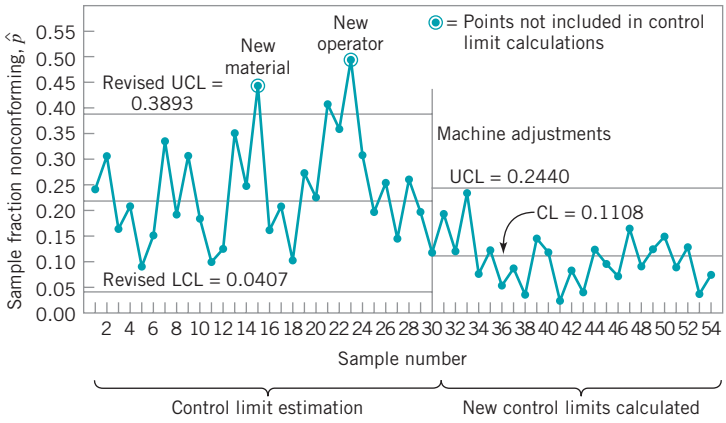
Comparing this to the upper 0.05 point of the standard normal distribution, we find that $Z_0 = 7.10 > Z_{0.05} = 1.645$. Consequently, we reject H_0 and conclude that there has been a significant decrease in the process fallout.

Based on the apparently successful process adjustments, it seems logical to revise the control limits again, using only the most recent samples (numbers 31–54). This results in the new control chart parameters:

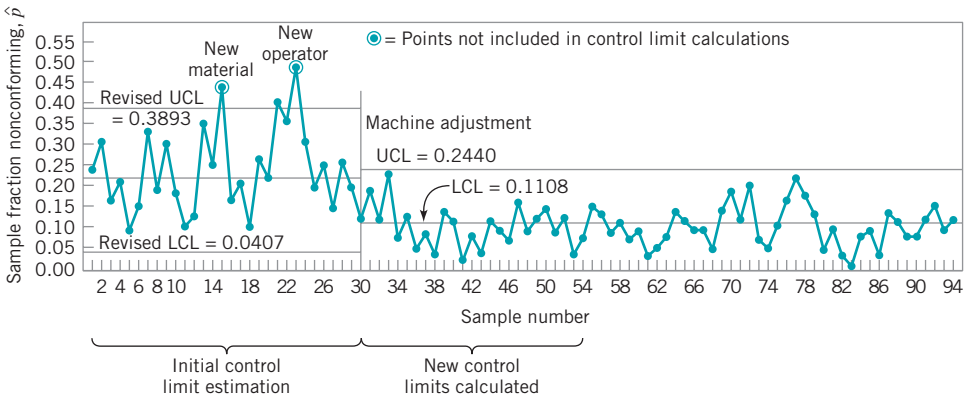
$$\text{Center line} = \bar{p} = 0.1108$$
$$\text{UCL} = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.1108 + 3\sqrt{\frac{(0.1108)(0.8892)}{50}} = 0.2440$$
$$\text{LCL} = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.1108 - 3\sqrt{\frac{(0.1108)(0.8892)}{50}} = -0.0224 = 0$$

Figure 7.4 shows the control chart with these new parameters. Note that since the calculated lower control limit is less than zero, we have set LCL = 0. Therefore, the new control chart will have only an upper control limit. From inspection of Fig. 7.4, we see that all the points would fall inside the revised upper control limit; therefore, we conclude that the process is in control at this new level.

The continued operation of this control chart for the next five shifts is shown in Fig. 7.5. Data for the process during this period are shown in Table 7.3. The control chart does not indicate lack of control. Despite the improvement in yield following the engineering changes in the process and the introduction of the control chart, the process fallout of $\bar{p} = 0.1108$ is still too high. Further analysis and action will be required to improve the yield. These management interventions may be further adjustments to the machine. **Statistically designed experiments** (see Part IV) are an appropriate way to determine which machine adjustments are critical to further process improvement, and the appropriate magnitude and direction of these adjustments. The control chart should be continued during the period in which the adjustments are made. By marking the time scale of the control chart *when* a process change is made, the control chart becomes a **logbook** in which the timing of process interventions and their subsequent effect on process performance are easily seen. This logbook aspect of control chart usage is extremely important.



■ **FIGURE 7.4** New control limits on the fraction nonconforming control chart, Example 7.1.



■ **FIGURE 7.5** Completed fraction nonconforming control chart, Example 7.1.

(*continued*)

■ TABLE 7.3
New Data for the Fraction Nonconforming Control Chart in Fig. 7.5, $n = 50$

Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$	Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$
55	8	0.16	75	5	0.10
56	7	0.14	76	8	0.16
57	5	0.10	77	11	0.22
58	6	0.12	78	9	0.18
59	4	0.08	79	7	0.14
60	5	0.10	80	3	0.06
61	2	0.04	81	5	0.10
62	3	0.06	82	2	0.04
63	4	0.08	83	1	0.02
64	7	0.14	84	4	0.08
65	6	0.12	85	5	0.10
66	5	0.10	86	3	0.06
67	5	0.10	87	7	0.14
68	3	0.06	88	6	0.12
69	7	0.14	89	4	0.08
70	9	0.18	90	4	0.08
71	6	0.12	91	6	0.12
72	10	0.20	92	8	0.16
73	4	0.08	93	5	0.10
74	3	0.06	94	6	0.12

Design of the Fraction Nonconforming Control Chart. The fraction nonconforming control chart has three parameters that must be specified: the sample size, the frequency of sampling, and the width of the control limits. Ideally, we should have some general guidelines for selecting those parameters.

It is relatively common to base a control chart for fraction nonconforming on 100% inspection of *all* process output over some convenient period of time, such as a shift or a day. In this case, both sample size and sampling frequency are interrelated. We would generally select a sampling frequency appropriate for the production rate, and this fixes the sample size. Rational subgrouping may also play a role in determining the sampling frequency. For example, if there are three shifts, and we suspect that shifts differ in their general quality level, then we should use the output of each shift as a subgroup rather than pooling the output of all three shifts together to obtain a daily fraction defective.

If we are to select a sample of process output, then we must choose the sample size n. Various rules have been suggested for the choice of n. If p is very small, we should choose n sufficiently large so that we have a high probability of finding at least one nonconforming unit in the sample. Otherwise, we might find that the control limits are such that the presence of only one nonconforming unit in the sample would indicate an out-of-control condition. For example, if $p = 0.01$ and $n = 8$, we find that the upper control limit is

$$\text{UCL} = p + 3\sqrt{\frac{p(1-p)}{n}} = 0.01 + 3\sqrt{\frac{(0.01)(0.99)}{8}} = 0.1155$$

If there is one nonconforming unit in the sample, then $\hat{p} = \frac{1}{8} = 0.1250$, and we can conclude that the process is out of control. Since for any $p > 0$ there is a positive probability

of producing *some* defectives, it is unreasonable in many cases to conclude that the process is out of control on observing a single nonconforming item.

To avoid this pitfall, we can choose the sample size n so that the probability of finding at least one nonconforming unit per sample is at least γ. For example, suppose that $p = 0.01$, and we want the probability of at least one nonconforming unit in the sample to be at least 0.95. If D denotes the number of nonconforming units, then we want to find n such that $P\{D \geq 1\} \geq 0.95$. Using the Poisson approximation to the binomial, we find from the cumulative Poisson table that $\lambda = np$ must exceed 3.00. Consequently, since $p = 0.01$, this implies that the sample size should be 300.

Duncan (1986) has suggested that the sample size should be large enough that we have approximately a 50% chance of detecting a process shift of some specified amount. For example, suppose that $p = 0.01$, and we want the probability of detecting a shift to $p = 0.05$ to be 0.50. Assuming that the normal approximation to the binomial applies, we should choose n so that the upper control limit exactly coincides with the fraction nonconforming in the out-of-control state.[1] If δ is the magnitude of the process shift, then n must satisfy

$$\delta = L\sqrt{\frac{p(1-p)}{n}} \tag{7.9}$$

Therefore,

$$n = \left(\frac{L}{\delta}\right)^2 p(1-p) \tag{7.10}$$

In our example, $p = 0.01$, $\delta = 0.05 - 0.01 = 0.04$, and if three-sigma limits are used, then from equation (7.10),

$$n = \left(\frac{3}{0.04}\right)^2 (0.01)(0.99) = 56$$

If the in-control value of the fraction nonconforming is small, another useful criterion is to choose n large enough so that the control chart will have a positive lower control limit. This ensures that we will have a mechanism to force us to investigate one or more samples that contain an unusually small number of nonconforming items. Since we wish to have

$$\text{LCL} = p - L\sqrt{\frac{p(1-p)}{n}} > 0 \tag{7.11}$$

this implies that

$$n > \frac{(1-p)}{p} L^2 \tag{7.12}$$

For example, if $p = 0.05$ and three-sigma limits are used, the sample size must be

$$n > \frac{0.95}{0.05}(3)^2 = 171$$

Thus, if $n \geq 172$ units, the control chart will have a positive lower control limit.

[1] If $\hat{p}$ is approximately normal, then the probability that $\hat{p}$ exceeds the UCL is 0.50 if the UCL equals the out-of-control fraction nonconforming p, due to the symmetry of the normal distribution. See Section 3.4.3 for a discussion of the normal approximation to the binomial.

Three-sigma control limits are usually employed on the control chart for fraction nonconforming on the grounds that they have worked well in practice. As discussed in Section 5.3.2, narrower control limits would make the control chart more sensitive to small shifts in p but at the expense of more frequent "false alarms." Occasionally, we have seen narrower limits used in an effort to *force* improvement in process quality. Care must be exercised in this, however, as too many false alarms will destroy the operating personnel's confidence in the control chart program.

We should note that the fraction nonconforming control chart is not a universal model for *all* data on fraction nonconforming. It is based on the binomial probability model; that is, the probability of occurrence of a nonconforming unit is *constant,* and successive units of production are *independent.* In processes where nonconforming units are clustered together, or where the probability of a unit being nonconforming depends on whether or not previous units were nonconforming, the fraction nonconforming control chart is often of little use. In such cases, it is necessary to develop a control chart based on the correct probability model.

Interpretation of Points on the Control Chart for Fraction Nonconforming. Example 7.1 illustrated how points that plot beyond the control limits are treated, both in establishing the control chart and during its routine operation. Care must be exercised in interpreting points that plot *below* the lower control limit. These points often do not represent a real improvement in process quality. Frequently, they are caused by errors in the inspection process resulting from inadequately trained or inexperienced inspectors or from improperly calibrated test and inspection equipment. We have also seen cases in which inspectors deliberately passed nonconforming units or reported fictitious data. The analyst must keep these warnings in mind when looking for assignable causes if points plot below the lower control limits. Not all downward shifts in p are attributable to improved quality.

The* np *Control Chart. It is also possible to base a control chart on the number nonconforming rather than the fraction nonconforming. This is often called an **number nonconforming (*np*) control chart.** The parameters of this chart are as follows.

The *np* Control Chart

$$\begin{aligned} \text{UCL} &= np + 3\sqrt{np(1-p)} \\ \text{Center line} &= np \\ \text{LCL} &= np - 3\sqrt{np(1-p)} \end{aligned} \tag{7.13}$$

If a standard value for p is unavailable, then $\bar{p}$ can be used to estimate p. Many nonstatistically trained personnel find the *np* chart easier to interpret than the usual fraction nonconforming control chart.

EXAMPLE 7.2 An *np* Control Chart

Set up an *np* control chart for the orange juice concentrate can process in Example 7.1.

Using the data in Table 7.1, we found that

$$\bar{p} = 0.2313 \qquad n = 50$$

SOLUTION

Therefore, the parameters of the *np* control chart would be

$$\begin{aligned}
\text{UCL} &= n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})} \\
&= 50(0.2313) + 3\sqrt{(50)(0.2313)(0.7687)} \\
&= 20.510 \\
\text{Center line} &= n\bar{p} = 50(0.2313) = 11.565 \\
\text{LCL} &= n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})} \\
&= 50(0.2313) - 3\sqrt{(50)(0.2313)(0.7687)} \\
&= 2.620
\end{aligned}$$

Now in practice, the number of nonconforming units in each sample is plotted on the *np* control chart, and the number of nonconforming units is an integer. Thus, if 20 units are nonconforming the process is in control, but if 21 occur the process is out of control. Similarly, there are three nonconforming units in the sample and the process is in control, but two nonconforming units would imply an out-of-control process. Some practitioners prefer to use integer values for control limits on the *np* chart instead of their decimal fraction counterparts. In this example we could choose 2 and 21 as the LCL and UCL, respectively, and the process would be considered out of control if a sample value of *np* plotted at or beyond the control limits.

7.2.2 Variable Sample Size

In some applications of the control chart for fraction nonconforming, the sample is a 100% inspection of process output over some period of time. Since different numbers of units could be produced in each period, the control chart would then have a variable sample size. There are three approaches to constructing and operating a control chart with a variable sample size.

Variable-Width Control Limits. The first and perhaps the most simple approach is to determine control limits for each individual sample that are based on the specific sample size. That is, if the *i*th sample is of size n_i, then the upper and lower control limits are $\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n_i}$. Note that the width of the control limits is inversely proportional to the square root of the sample size.

To illustrate this approach, consider the data in Table 7.4. These data came from the purchasing group of a large aerospace company. This group issues purchase orders to the company's suppliers. The sample sizes in Table 7.4 are the total number of purchase orders issued each week. Obviously, this is not constant. A nonconforming unit is a purchase order with an error. Among the most common errors are specifying incorrect part numbers, wrong delivery dates, and wrong supplier information. Any of these mistakes can result in a purchase order change, which takes time and resources and may result in delayed delivery of material.

For the 25 samples, we calculate

$$\bar{p} = \frac{\sum_{i=1}^{25} D_i}{\sum_{i=1}^{25} n_i} = \frac{234}{2450} = 0.096$$

Consequently, the center line is at 0.096, and the control limits are

$$\text{UCL} = \bar{p} + 3\hat{\sigma}_{\hat{p}} = 0.096 + 3\sqrt{\frac{(0.096)(0.904)}{n_i}}$$

and

$$\text{LCL} = \bar{p} - 3\hat{\sigma}_{\hat{p}} = 0.096 - 3\sqrt{\frac{(0.096)(0.904)}{n_i}}$$

■ TABLE 7.4
Purchase Order Data for a Control Chart for Fraction Nonconforming with Variable Sample Size

Sample Number, i	Sample Size, n_i	Number of Nonconforming Units, D_i	Sample Fraction Nonconforming, $\hat{p}_i = D_i/n_i$	Standard Deviation $\hat{\sigma}_{\hat{p}} = \sqrt{\frac{(0.096)(0.904)}{n_i}}$	Control Limits LCL	Control Limits UCL
1	100	12	0.120	0.029	0.009	0.183
2	80	8	0.100	0.033	0	0.195
3	80	6	0.075	0.033	0	0.195
4	100	9	0.090	0.029	0.009	0.183
5	110	10	0.091	0.028	0.012	0.180
6	110	12	0.109	0.028	0.012	0.180
7	100	11	0.110	0.029	0.009	0.183
8	100	16	0.160	0.029	0.009	0.183
9	90	10	0.110	0.031	0.003	0.189
10	90	6	0.067	0.031	0.003	0.189
11	110	20	0.182	0.028	0.012	0.180
12	120	15	0.125	0.027	0.015	0.177
13	120	9	0.075	0.027	0.015	0.177
14	120	8	0.067	0.027	0.015	0.177
15	110	6	0.055	0.028	0.012	0.180
16	80	8	0.100	0.033	0	0.195
17	80	10	0.125	0.033	0	0.195
18	80	7	0.088	0.033	0	0.195
19	90	5	0.056	0.031	0.003	0.189
20	100	8	0.080	0.029	0.009	0.183
21	100	5	0.050	0.029	0.009	0.183
22	100	8	0.080	0.029	0.009	0.183
23	100	10	0.100	0.029	0.009	0.183
24	90	6	0.067	0.031	0.003	0.189
25	90	9	0.100	0.031	0.003	0.189
	2450	234	2.383			

where $\hat{\sigma}_{\hat{p}}$ is the estimate of the standard deviation of the sample fraction nonconforming $\hat{p}$. The calculations to determine the control limits are displayed in the last three columns of Table 7.4. The manually constructed control chart is plotted in Fig. 7.6.

Many popular quality control computer programs will handle the variable sample size case. Figure 7.7 presents the computer-generated control chart corresponding to Fig. 7.6. This control chart was obtained using Minitab.

Control Limits Based on an Average Sample Size. The second approach is to base the control chart on an ***average*** **sample size,** resulting in an **approximate set of control limits.** This assumes that future sample sizes will not differ greatly from those previously observed. If this approach is used, the control limits will be *constant,* and the resulting control chart will not look as formidable to operating personnel as the control chart with variable limits. However, if there is an unusually large variation in the size of a particular sample or if a point plots near the approximate control limits, then the *exact* control limits for that point

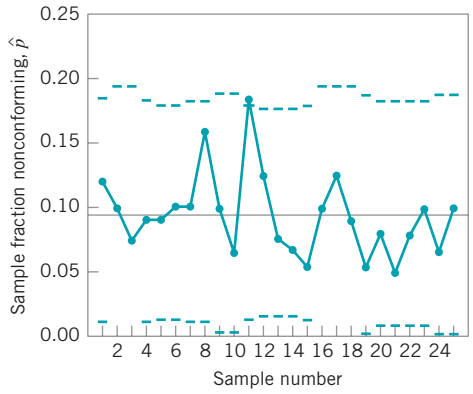


FIGURE 7.6 Control chart for fraction nonconforming with variable sample size.

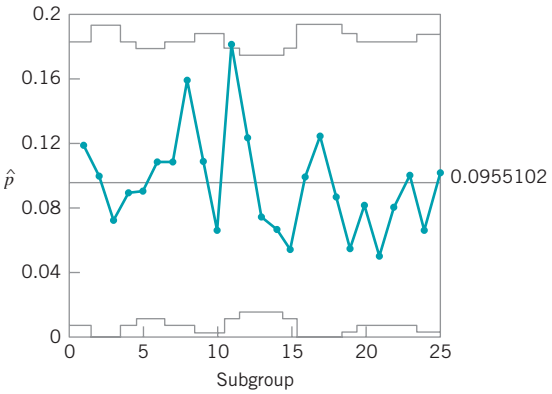


FIGURE 7.7 Control chart for fraction nonconforming with variable sample size using Minitab.

should be determined and the point examined relative to that value. For the purchase order data in Table 7.4, we find that the average sample size is

$$\bar{n} = \frac{\sum_{i=1}^{25} n_i}{25} = \frac{2450}{25} = 98$$

Therefore, the approximate control limits are

$$\text{UCL} = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} = 0.096 + 3\sqrt{\frac{(0.096)(0.904)}{98}} = 0.185$$

and

$$\text{LCL} = \bar{p} - \sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} = 0.096 - 3\sqrt{\frac{(0.096)(0.904)}{98}} = 0.007$$

The resulting control chart is shown in Fig. 7.8. Note that $\hat{p}$ for sample 11 plots close to the approximate upper control limit, yet it appears to be in control. However, when compared to

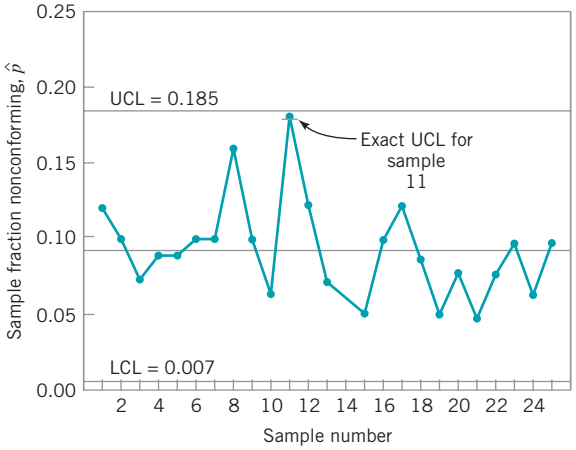


FIGURE 7.8 Control chart for fraction nonconforming based on average sample size.

its exact upper control limit (0.180, from Table 7.4), the point indicates an out-of-control condition. Similarly, points that are outside the approximate control limits may be inside their exact control limits. In general, care must be taken in the interpretation of points near the approximate control limits.

We must also be careful in analyzing runs or other apparently abnormal patterns on control charts with variable sample sizes. The problem is that a change in the sample fraction nonconforming $\hat{p}$ must be interpreted relative to the sample size. For example, suppose that $p = 0.20$ and that two successive sample fractions nonconforming are $\hat{p}_t = 0.28$ and $\hat{p}_{i+1} = 0.24$. The first observation seems to indicate poorer quality than the second, since $\hat{p}_i > \hat{p}_{i+1}$. However, suppose that the sample sizes are $n_i = 50$ and $n_{i+1} = 250$. In *standard deviation units,* the first point is 1.41 units above average and the second point is 1.58 units above average. That is, the second point actually represents a *greater* deviation from the standard of $p = 0.20$ than does the first, even though the second point is the smaller of the two. Clearly, looking for runs or other nonrandom patterns is virtually meaningless here.

The Standardized Control Chart. The third approach to dealing with variable sample size is to use a **standardized control chart,** where the points are plotted in standard deviation units. Such a control chart has the center line at zero, and upper and lower control limits of +3 and −3, respectively. The variable plotted on the chart is

$$Z_i = \frac{\hat{p}_i - p}{\sqrt{\dfrac{p(1-p)}{n_i}}} \tag{7.14}$$

where p (or $\bar{p}$ if no standard is given) is the process fraction nonconforming in the in-control state. The standardized control chart for the purchase order data in Table 7.4 is shown in Fig. 7.9. The calculations associated with this control chart are shown in Table 7.5. Tests for runs and pattern-recognition methods could safely be applied to this chart, because the relative changes from one point to another are all expressed in terms of the same units of measurement.

The standardized control chart is no more difficult to construct or maintain than either of the other two procedures discussed in this section. In fact, many quality control software packages either automatically execute this as a standard feature or can be programmed to plot a standardized control chart. For example, the version of Fig. 7.9 shown in Fig. 7.10 was created using Minitab. Conceptually, however, it may be more difficult for operating personnel to understand and interpret, because reference to the actual process fraction defective has been "lost." However, if there is large variation in sample size, then runs and pattern-recognition methods can only be safely applied to the standardized control chart. In such a case, it might be advisable to maintain a control chart with individual control limits for each sample (as in Fig. 7.6) for the operating personnel, while simultaneously maintaining a standardized control chart for engineering use.

The standardized control chart is also recommended when the length of the production run is short, as in many job-shop settings. Control charts for short production runs are discussed in Chapter 9.

7.2.3 Applications in Transactional and Service Businesses

The control chart for fraction nonconforming is widely used in transactional and service industry applications of statistical process control. In the nonmanufacturing environment,

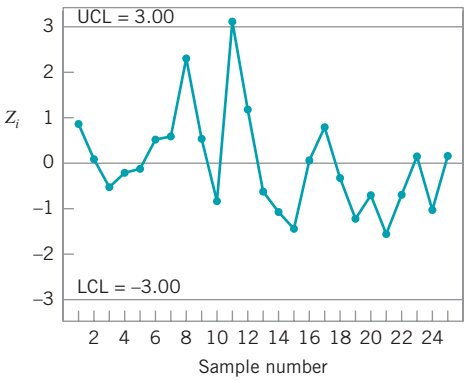


FIGURE 7.9 Standardized control chart for fraction nonconforming.

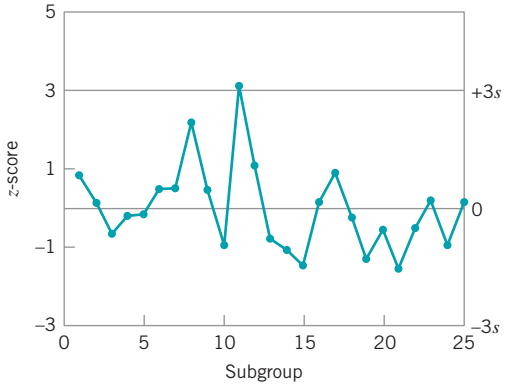


FIGURE 7.10 Standardized control chart from Minitab for fraction nonconforming, Table 7.4.

TABLE 7.5

Calculations for the Standardized Control Chart in Fig. 7.9, $\hat{p} = 0.096$

Sample Number, i	Sample Size, n_i	Number of Nonconforming Units, D_i	Sample Fraction Nonconforming, $\hat{p}_i = D_i/n_i$	Standard Deviation $\hat{\sigma}_p = \sqrt{\frac{(0.096)(0.904)}{n_i}}$	$z_i = \frac{\hat{p}_i - \bar{p}}{\sqrt{\frac{(0.096)(0.904)}{n_i}}}$
1	100	12	0.120	0.029	0.83
2	80	8	0.100	0.033	0.12
3	80	6	0.075	0.033	–0.64
4	100	9	0.090	0.029	–0.21
5	110	10	0.091	0.028	–0.18
6	110	12	0.109	0.028	0.46
7	100	11	0.110	0.029	0.48
8	100	16	0.160	0.029	2.21
9	90	10	0.110	0.031	0.45
10	90	6	0.067	0.031	–0.94
11	110	20	0.182	0.028	3.07
12	120	15	0.125	0.027	1.07
13	120	9	0.075	0.027	–0.78
14	120	8	0.067	0.027	–1.07
15	110	6	0.055	0.028	–1.46
16	80	8	0.100	0.033	0.12
17	80	10	0.125	0.033	0.88
18	80	7	0.088	0.033	–0.24
19	90	5	0.056	0.031	–1.29
20	100	8	0.080	0.029	–0.55
21	100	5	0.050	0.029	–1.59
22	100	8	0.080	0.029	–0.55
23	100	10	0.100	0.029	0.14
24	90	6	0.067	0.031	–0.94
25	90	9	0.100	0.031	0.13

many quality characteristics can be observed on a conforming or nonconforming basis. Examples would include the number of employee paychecks that are in error or distributed late during a pay period, the number of check requests that are not paid within the standard accounting cycle, and the number of deliveries made by a supplier that are not on time.

Many nonmanufacturing applications of the fraction nonconforming control chart will involve the variable sample size case. For example, the total number of check requests during an accounting cycle is most likely not constant, and since information about the timeliness of processing for all check requests is generally available, we would calculate $\hat{p}$ as the ratio of all late checks to the total number of checks processed during the period.

As an illustration consider the purchase order data in Table 7.4. The sample sizes in Table 7.4 are the actual number of purchase orders issued each week. It would be very unusual for this to be exactly the same from week to week. Consequently, a fraction nonconforming control chart with variable sample size was an ideal approach for this situation. The use of this control chart was a key initial step in identifying many of the root causes of the errors on purchase orders and in developing the corrective actions necessary to improve the process.

7.2.4 The Operating-Characteristic Function and Average Run Length Calculations

The operating-characteristic (or OC) function of the fraction nonconforming control chart is a graphical display of the probability of incorrectly accepting the hypothesis of statistical control (i.e., a type II or β-error) against the process fraction nonconforming. The OC curve provides a measure of the **sensitivity** of the control chart—that is, its ability to detect a shift in the process fraction nonconforming from the nominal value $\overline{p}$ to some other value p. The probability of type II error for the fraction nonconforming control chart may be computed from

$$\begin{aligned}\beta &= P\{\hat{p} < \text{UCL}|p\} - P\{\hat{p} \le \text{LCL}|p\} \\ &= P\{D < n\text{UCL}|p\} - P\{D \le n\text{LCL}|p\}\end{aligned} \tag{7.15}$$

Since D is a binomial random variable with parameters n and p, the β-error defined in equation (7.15) can be obtained from the cumulative binomial distribution. Note that when the LCL is negative, the second term on the right-hand side of equation (7.15) should be dropped.

Table 7.6 illustrates the calculations required to generate the OC curve for a control chart for fraction nonconforming with parameters $n = 50$, LCL = 0.0303, and UCL = 0.3697. Using these parameters, equation (7.15) becomes

$$\begin{aligned}\beta &= P\{D < (50)(0.3697)|p\} - P\{D \le (50)(0.0303)|p\} \\ &= P\{D < 18.49|p\} - P\{D \le 1.52|p\}\end{aligned}$$

However, since D must be an integer, we find that

$$\beta = P\{D \le 18|p\} - P\{D \le 1|p\}$$

The OC curve is plotted in Fig. 7.11.

TABLE 7.6

Calculations[a] for Constructing the OC Curve for a Control Chart for Fraction Nonconforming with $n = 50$, LCL = 0.0303, and UCL = 0.3697

p	$P\{D \le 18 \mid p\}$	$P\{D \le 1 \mid p\}$	$\beta = P\{D \le 18 \mid p\} - P\{D \le 1 \mid p\}$
0.01	1.0000	0.9106	0.0894
0.03	1.0000	0.5553	0.4447
0.05	1.0000	0.2794	0.7206
0.10	1.0000	0.0338	0.9662
0.15	0.9999	0.0029	0.9970
0.20	0.9975	0.0002	0.9973
0.25	0.9713	0.0000	0.9713
0.30	0.8594	0.0000	0.8594
0.35	0.6216	0.0000	0.6216
0.40	0.3356	0.0000	0.3356
0.45	0.1273	0.0000	0.1273
0.50	0.0325	0.0000	0.0325
0.55	0.0053	0.0000	0.0053

[a]The probabilities in this table were found by evaluating the cumulative binomial distribution. For small p ($p < 0.1$, say) the Poisson approximation could be used, and for larger values of p the normal approximation could be used.

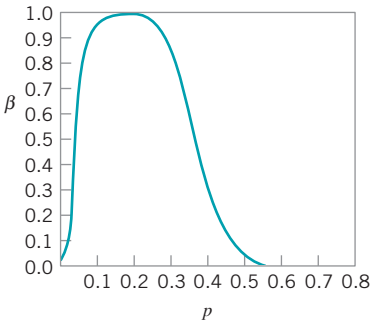


FIGURE 7.11 Operating-characteristic curve for the fraction nonconforming control chart with $\bar{p} = 0.20$, LCL = 0.0303, and UCL = 0.3697.

We may also calculate average run lengths (ARLs) for the fraction nonconforming control chart. Recall from Chapter 5 that for uncorrelated process data the ARL for any Shewhart control chart can be written as

$$\text{ARL} = \frac{1}{P(\text{sample point plots out of control})}$$

Thus, if the process is in control, ARL_0 is

$$\text{ARL}_0 = \frac{1}{\alpha}$$

and if it is out of control, then

$$\text{ARL}_1 = \frac{1}{1-\beta}$$

These probabilities (α, β) can be calculated directly from the binomial distribution or read from an OC curve.

To illustrate, consider the control chart for fraction nonconforming used in the OC curve calculations in Table 7.6. This chart has parameters $n = 50$, UCL = 0.3697, LCL = 0.0303, and the center line is $\bar{p} = 0.20$. From Table 7.6 (or the OC curve in Fig. 7.11) we find that if the process is in control with $p = \bar{p}$, the probability of a point plotting in control is 0.9973. Thus, in this case $\alpha = 1 - \beta = 0.0027$, and the value of ARL_0 is

$$\text{ARL}_0 = \frac{1}{\alpha} = \frac{1}{0.0027} \simeq 370$$

Therefore, if the process is really in control, we will experience a false out-of-control signal about every 370 samples. (This will be approximately true, in general, for any Shewhart control chart with three-sigma limits.) This in-control ARL_0 is generally considered to be satisfactorily large. Now suppose that the process shifts out of control to $p = 0.3$. Table 7.6 indicates that if $p = 0.3$, then $\beta = 0.8594$. Therefore, the value of ARL_1 is

$$\text{ARL}_1 = \frac{1}{1-\beta} = \frac{1}{1-0.8594} \simeq 7$$

and it will take about seven samples, on the average, to detect this shift with a point outside of the control limits. If this is unsatisfactory, then action must be taken to reduce the out-of-control ARL_1. Increasing the sample size would result in a smaller value of β and a shorter out-of-control ARL_1. Another approach would be to reduce the interval *between* samples. That is, if we are currently sampling every hour, it will take about seven hours, on the average, to detect the shift. If we take the sample every half hour, it will require only three and a half hours, on the average, to detect the shift. Another approach is to use a control chart that is more responsive to small shifts, such as the cumulative sum charts in Chapter 9.

7.3 Control Charts for Nonconformities (Defects)

A nonconforming item is a unit of product that does not satisfy one or more of the specifications for that product. Each specific point at which a specification is not satisfied results in a **defect** or **nonconformity.** Consequently, a nonconforming item will contain at least one nonconformity. However, depending on their nature and severity, it is quite possible for a unit to contain several nonconformities and *not* be classified as nonconforming. As an example, suppose we are manufacturing personal computers. Each unit could have one or more very minor flaws in the cabinet finish, and since these flaws do not seriously affect the unit's functional operation, it could be classified as conforming. However, if there are too many of these flaws, the personal computer should be classified as nonconforming, since the flaws would be very noticeable to the customer and might affect the sale of the unit. There are many practical situations in which we prefer to work directly with the number of defects or nonconformities rather than the fraction nonconforming. These include the number of defective welds in 100 m of oil pipeline, the number of broken rivets in an aircraft wing, the number of functional defects in an electronic logic device, the number of errors on a document, and so forth.

It is possible to develop control charts for either the **total number of nonconformities** in a unit or the **average number of nonconformities per unit.** These control charts usually assume that the occurrence of nonconformities in samples of constant size is well modeled by the Poisson distribution. Essentially, this requires that the number of opportunities or potential locations for nonconformities be infinitely large and that the probability of occurrence of a nonconformity at any location be small and constant. Furthermore, the **inspection unit** must be the same for each sample. That is, each inspection unit must always represent an identical **area of opportunity** for the occurrence of nonconformities. In addition, we can count nonconformities of several different types on one unit, as long as the above conditions are satisfied for each class of nonconformity.

In most practical situations, these conditions will not be satisfied exactly. The number of opportunities for the occurrence of nonconformities may be finite, or the probability of occurrence of nonconformities may not be constant. As long as these departures from the assumptions are not severe, the Poisson model will usually work reasonably well. There are cases, however, in which the Poisson model is completely inappropriate. These situations are discussed in more detail at the end of Section 7.3.1.

7.3.1 Procedures with Constant Sample Size

Consider the occurrence of nonconformities in an inspection unit of product. In most cases, the inspection unit will be a single unit of product, although this is not necessarily always so. The inspection unit is simply an entity for which it is convenient to keep records. It could be a group of 5 units of product, 10 units of product, and so on. Suppose that defects or nonconformities occur in this inspection unit according to the Poisson distribution; that is,

$$p(x) = \frac{e^{-c}c^{x}}{x!} \qquad x = 0,\ 1,\ 2,\ \ldots$$

where x is the number of nonconformities and $c > 0$ is the parameter of the Poisson distribution. From Section 3.2.3 we recall that both the mean and variance of the Poisson distribution are the parameter c. Therefore, a **control chart for nonconformities, or *c* chart** with three-sigma limits would be defined as follows,[2]

Control Chart for Nonconformities: Standard Given

$$\begin{aligned} \text{UCL} &= c + 3\sqrt{c} \\ \text{Center line} &= c \\ \text{LCL} &= c - 3\sqrt{c} \end{aligned} \tag{7.16}$$

assuming that a standard value for c is available. Should these calculations yield a negative value for the LCL, set LCL = 0.

If no standard is given, then c may be estimated as the observed average number of nonconformities in a preliminary sample of inspection units—say, $\bar{c}$. In this case, the control chart has parameters defined as follows.

[2]The α-risk for three-sigma limits is not equally allocated above the UCL and below the LCL, because the Poisson distribution is asymmetric. Some authors recommend the use of probability limits for this chart, particularly when c is small.

Control Chart for Nonconformities: No Standard Given

$$\begin{aligned} \text{UCL} &= \bar{c} + 3\sqrt{\bar{c}} \\ \text{Center line} &= \bar{c} \\ \text{LCL} &= \bar{c} - 3\sqrt{\bar{c}} \end{aligned} \tag{7.17}$$

When no standard is given, the control limits in equation (7.17) should be regarded as *trial* control limits, and the preliminary samples examined for lack of control in the usual phase I analysis. The control chart for nonconformities is also sometimes called the *c* chart.

EXAMPLE 7.3 Nonconformities in Printed Circuit Boards

Table 7.7 presents the number of nonconformities observed in 26 successive samples of 100 printed circuit boards. Note that, for reasons of convenience, the inspection unit is defined as 100 boards. Set up a *c* chart for these data.

■ TABLE 7.7
Data on the Number of Nonconformities in Samples of 100 Printed Circuit Boards

Sample Number	Number of Nonconformities	Sample Number	Number of Nonconformities
1	21	14	19
2	24	15	10
3	16	16	17
4	12	17	13
5	15	18	22
6	5	19	18
7	28	20	39
8	20	21	30
9	31	22	24
10	25	23	16
11	20	24	19
12	24	25	17
13	16	26	15

SOLUTION

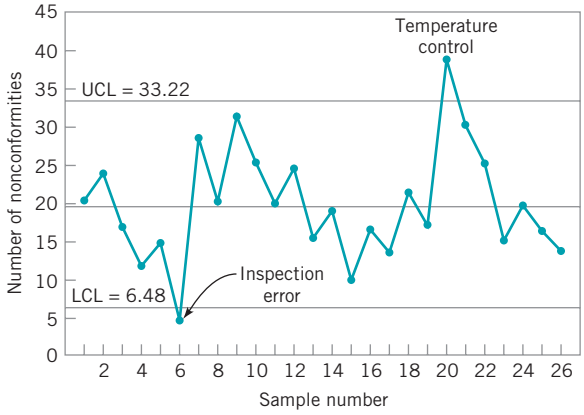
Since the 26 samples contain 516 total nonconformities, we estimate *c* by

$$\bar{c} = \frac{516}{26} = 19.85$$

Therefore, the trial control limits are given by

$$\begin{aligned} \text{UCL} &= \bar{c} + 3\sqrt{\bar{c}} = 19.85 + 3\sqrt{19.85} = 33.22 \\ \text{Center line} &= \bar{c} = 19.85 \\ \text{LCL} &= \bar{c} - 3\sqrt{\bar{c}} = 19.85 - 3\sqrt{19.85} = 6.48 \end{aligned}$$

The control chart is shown in Fig. 7.12. The number of observed nonconformities from the preliminary samples is plotted on this chart. Two points plot outside the control limits, samples 6 and 20. Investigation of sample 6 revealed that a new inspector had examined the boards in this sample and that he did not recognize several of the types of nonconformities that could have been present. Furthermore, the unusually large number of nonconformities in sample 20 resulted from a temperature control problem in the wave soldering machine, which was subsequently repaired. Therefore, it seems reasonable to exclude these two



■ **FIGURE 7.12** Control chart for nonconformities for Example 7.3.

samples and revise the trial control limits. The estimate of c is now computed as

$$\bar{c} = \frac{472}{24} = 19.67$$

and the revised control limits are

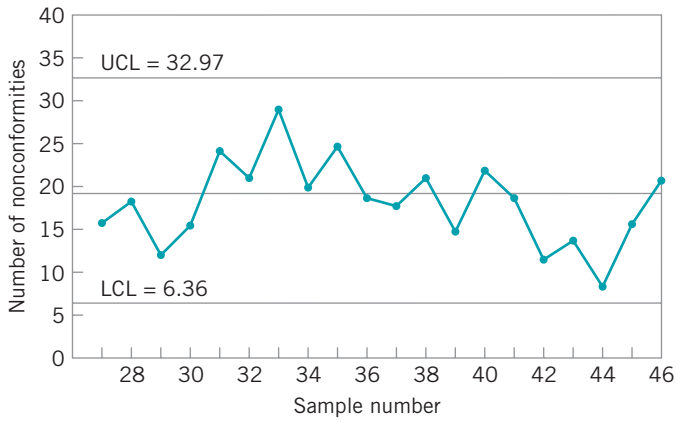
$$\text{UCL} = \bar{c} + 3\sqrt{\bar{c}} = 19.67 + 3\sqrt{19.67} = 32.97$$
$$\text{Center line} = \bar{c} = 19.67$$
$$\text{LCL} = \bar{c} - 3\sqrt{\bar{c}} = 19.67 - 3\sqrt{19.67} = 6.36$$

These become the standard values against which production in the next period can be compared.

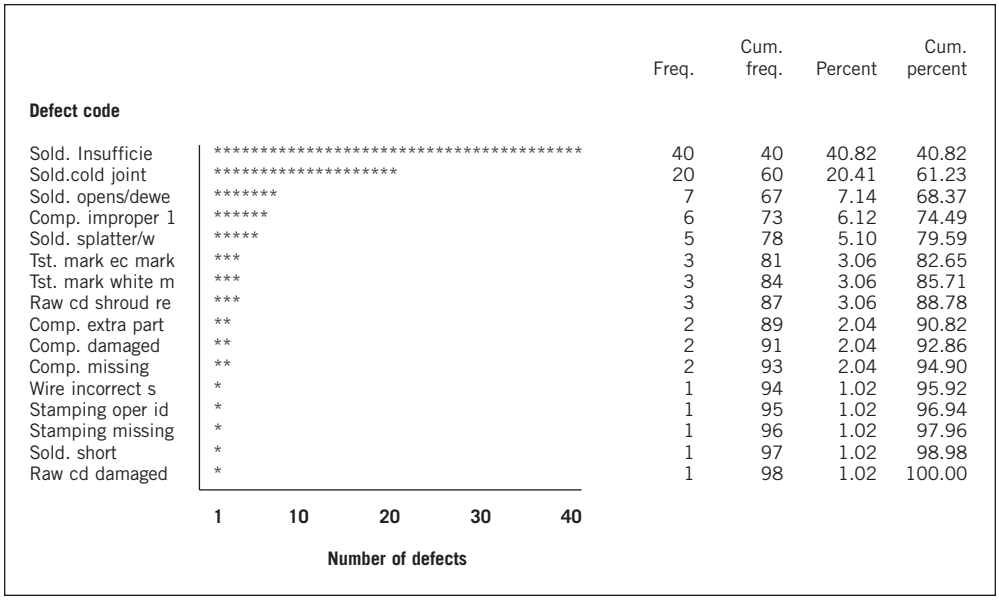
Twenty new samples, each consisting of one inspection unit (i.e., 100 boards), are subsequently collected. The number of nonconformities in each sample is noted and recorded in Table 7.8. These points are plotted on the control chart in Fig. 7.13. No lack of control is indicated; however, the number of nonconformities per board is still unacceptably high. Further action is necessary to improve the process.

■ **TABLE 7.8**
Additional Data for the Control Chart for Nonconformities, Example 7.3

Sample Number	Number of Nonconformities	Sample Number	Number of Nonconformities
27	16	37	18
28	18	38	21
29	12	39	16
30	15	40	22
31	24	41	19
32	21	42	12
33	28	43	14
34	20	44	9
35	25	45	16
36	19	46	21



■ **FIGURE 7.13** Continuation of the control chart for nonconformities. Example 7.3.



Defect code	Freq.	Cum. freq.	Percent	Cum. percent
Sold. Insufficie	40	40	40.82	40.82
Sold.cold joint	20	60	20.41	61.23
Sold. opens/dewe	7	67	7.14	68.37
Comp. improper l	6	73	6.12	74.49
Sold. splatter/w	5	78	5.10	79.59
Tst. mark ec mark	3	81	3.06	82.65
Tst. mark white m	3	84	3.06	85.71
Raw cd shroud re	3	87	3.06	88.78
Comp. extra part	2	89	2.04	90.82
Comp. damaged	2	91	2.04	92.86
Comp. missing	2	93	2.04	94.90
Wire incorrect s	1	94	1.02	95.92
Stamping oper id	1	95	1.02	96.94
Stamping missing	1	96	1.02	97.96
Sold. short	1	97	1.02	98.98
Raw cd damaged	1	98	1.02	100.00

■ **FIGURE 7.14** Pareto analysis of nonconformities for the printed circuit board process.

Further Analysis of Nonconformities. Defect or nonconformity data are always more informative than fraction nonconforming, because there will usually be several different *types* of nonconformities. By analyzing the nonconformities by type, we can often gain considerable insight into their cause. This can be of considerable assistance in developing the **out-of-control-action plans** (OCAPs) that must accompany control charts.

For example, in the printed circuit board process, there are sixteen different types of defects. Defect data for 500 boards are plotted on a Pareto chart in Fig. 7.14. Note that over 60% of the total number of defects is due to *two defect types:* solder insufficiency and solder cold joints. This points to further problems with the wave soldering process. If these problems can be isolated and eliminated, there will be a dramatic increase in process yield. Notice that the nonconformities follow the Pareto distribution; that is, most of the defects are attributable to a few (in this case, two) defect types.

This process manufactures several different types of printed circuit boards. Therefore, it may be helpful to examine the occurrence of defect type by type of printed circuit board (part number). Table 7.9 presents this information. Note that all 40 solder insufficiencies and all 20 solder cold joints occurred on the same part number, 0001285. This implies that this particular type of board is very susceptible to problems in wave soldering, and special attention must be directed toward improving this step of the process for this part number.

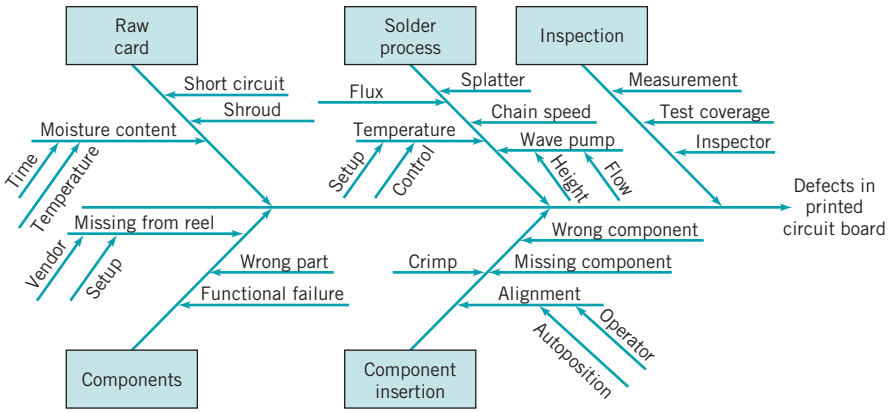
Another useful technique for further analysis of nonconformities is the **cause-and-effect diagram** discussed in Chapter 5. The cause-and-effect diagram is used to illustrate the various sources of nonconformities in products and their interrelationships. It is useful in focusing the attention of operators, manufacturing engineers, and managers on quality problems. Developing a good cause-and-effect diagram usually advances the level of technological understanding of the process.

A cause-and-effect diagram for the printed circuit board assembly process is shown in Fig. 7.15. Since most of the defects in this example were solder-related, the cause-and-effect diagram could help choose the variables for a designed experiment to optimize the wave

■ TABLE 7.9
Table of Defects Classified by Part Number and Defect Code

Part Number Frequency Percent Row Percent Column Percent	Defect								
	Component Component Missing	Damaged (NO)	Code Component Extra Part	Component Improper I	Raw Raw Card Shroud RE	Card Damaged	Solder Short	Solder Solder Opens/DEWE	Cold Joint
0001285	1	0	0	0	0	1	0	5	20
	1.02	0.00	0.00	0.00	0.00	1.02	0.00	5.10	20.41
	1.41	0.00	0.00	0.00	0.00	1.41	0.00	7.04	28.17
	50.00	0.00	0.00	0.00	0.00	100.00	0.00	71.43	100.00
0001481	1	2	2	6	3	0	1	2	0
	1.02	2.04	6.12	3.06	0.00	0.00	1.02	2.04	0.00
	3.70	7.41	22.22	11.11	0.00	0.00	3.70	7.41	0.00
	50.00	100.00	100.00	100.00	100.00	0.00	100.00	28.57	0.00
0006429	0	0	0	0	0	0	0	0	0
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total	2	2	2	6	3	1	1	7	20
	2.04	2.04	2.04	6.12	3.06	1.02	1.02	7.14	20.41

Part Number Frequency Percent Row Percent Column Percent	Defect								
	Solder Insufficiencies	Solder Splatter	Code Stamping Missing	Test Stamping Operator ID	Test Mark White *M*	Wire Mark EC Mark	Incorrect 5	Good Unit(s)	Total
0001285	40	0	0	0	2	1	1	0	71
	40.82	0.00	0.00	0.00	2.04	1.02	1.02	0.00	72.45
	56.32	0.00	0.00	0.00	2.82	1.41	1.41	0.00	
	100.00	0.00	0.00	0.00	66.67	33.33	100.00	0.00	
0001481	0	5	1	1	1	2	0	0	27
	0.00	5.10	1.02	1.02	1.02	2.04	0.00	0.00	27.55
	0.00	18.52	3.70	3.70	3.70	7.41	0.00	0.00	
	0.00	100.00	100.00	100.00	33.33	66.67	0.00	0.00	
0006429	0	0	0	0	0	0	0	0	0
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
Total	40	5	1	1	3	3	1	0	98
	40.82	5.10	1.02	1.02	3.06	3.06	1.02	0.00	100.00



■ **FIGURE 7.15** Cause-and-effect diagram.

soldering process. There are several ways to draw the diagram. This one focuses on the three main generic sources of nonconformities: materials, operators, and equipment. Another useful approach is to organize the diagram according to the flow of material through the process.

Choice of Sample Size: The* u *Chart. Example 7.3 illustrates a control chart for nonconformities with the sample size exactly equal to one inspection unit. The inspection unit is chosen for operational or data-collection simplicity. However, there is no reason why the sample size must be restricted to one inspection unit. In fact, we would often prefer to use *several* inspection units in the sample, thereby increasing the area of opportunity for the occurrence of nonconformities. The sample size should be chosen according to statistical considerations, such as specifying a sample size large enough to ensure a positive lower control limit or to obtain a particular probability of detecting a process shift. Alternatively, economic factors could enter into sample-size determination.

Suppose we decide to base the control chart on a sample size of n inspection units. Note that n does not have to be an integer. To illustrate this, suppose that in Example 7.3 we were to specify a subgroup size of $n = 2.5$ inspection units. Then the sample size becomes (2.5)(100) = 250 boards. There are two general approaches to constructing the revised chart once a new sample size has been selected. One approach is simply to redefine a new inspection unit that is equal to n times the old inspection unit. In this case, the center line on the new control chart is $n\bar{c}$ and the control limits are located at $n\bar{c} \pm 3\sqrt{n\bar{c}}$, where $\bar{c}$ is the observed mean number of nonconformities in the *original* inspection unit. Suppose that in Example 7.3, after revising the trial control limits, we decided to use a sample size of $n = 2.5$ inspection units. Then the center line would have been located at $n\bar{c}$ = (2.5)(19.67) = 49.18 and the control limits would have been $49.18 \pm 3\sqrt{49.18}$ or LCL = 28.14 and UCL = 70.22.

The second approach involves setting up a control chart based on the average number of nonconformities per inspection unit. If we find x *total* nonconformities in a sample of n inspection units, then the *average* number of nonconformities per inspection unit is

$$u = \frac{x}{n} \tag{7.18}$$

Note that x is a Poisson random variable; consequently, the parameters of the control chart for the average number of nonconformities per unit are as follows,

Control Chart for Average Number of Nonconformities per Unit

$$\text{UCL} = \bar{u} + 3\sqrt{\frac{\bar{u}}{n}}$$
$$\text{Center line} = \bar{u} \tag{7.19}$$
$$\text{LCL} = \bar{u} - 3\sqrt{\frac{\bar{u}}{n}}$$

where $\bar{u}$ represents the observed average number of nonconformities per unit in a preliminary set of data. Control limits found from equation (7.19) would be regarded as trial control limits. This per-unit chart often is called the **control chart for nonconformities, or *u* chart.**

EXAMPLE 7.4 Control Charts in Supply Chain Operations

A supply chain engineering group monitors shipments of materials through the company distribution network. Errors on either the delivered material or the accompanying documentation are tracked on a weekly basis. Fifty randomly selected shipments are examined and the errors recorded. Data for twenty weeks are shown in Table 7.10. Set up a u control chart to monitor this process.

SOLUTION

From the data in Table 7.10, we estimate the number of errors (nonconformities) per unit (shipment) to be:

$$\bar{u} = \frac{\sum_{i=1}^{20} u_i}{20} = \frac{1.48}{20} = 0.0740$$

Therefore, the parameters of the control chart are

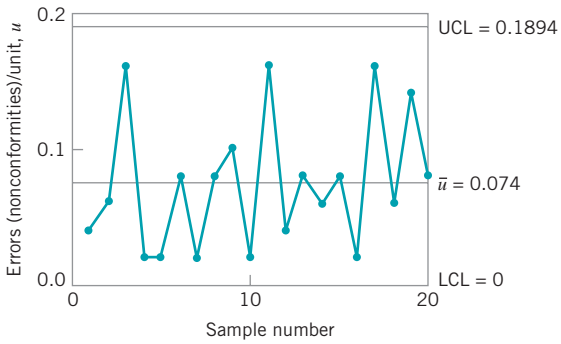
$$\text{UCL} = \bar{u} + 3\sqrt{\frac{\bar{u}}{n}} = 0.0740 + 3\sqrt{\frac{0.0740}{50}} = 0.1894$$
$$\text{Center line} = \bar{u} = 1.93$$
$$\text{LCL} = \bar{u} - 3\sqrt{\frac{\bar{u}}{n}} = 0.0740 - 3\sqrt{\frac{0.0740}{50}} = -0.0414$$

Since the LCL < 0, we would set LCL = 0 for the u chart. The control chart is shown in Fig. 7.16. The preliminary data do not exhibit lack of statistical control; therefore, the trial control limits given here would be adopted for phase II monitoring of future operations. Once again, note that, although the process is in control, the average number of errors per shipment is high. Action should be taken to improve the supply chain system.

(*continued*)

■ TABLE 7.10
Data on Number of Shipping Errors in a Supply Chain Network

Sample Number (week), i	Sample Size, n	Total Number of Errors (Nonconformities), x_i	Average Number of Errors (Nonconformities) per Unit, $u_i = x_i/n$
1	50	2	0.04
2	50	3	0.06
3	50	8	0.16
4	50	1	0.02
5	50	1	0.02
6	50	4	0.08
7	50	1	0.02
8	50	4	0.08
9	50	5	0.10
10	50	1	0.02
11	50	8	0.16
12	50	2	0.04
13	50	4	0.08
14	50	3	0.06
15	50	4	0.08
16	50	1	0.02
17	50	8	0.16
18	50	3	0.06
19	50	7	0.14
20	50	4	0.08
		74	1.48



■ FIGURE 7.16 The control chart for nonconformities per unit from Minitab for Example 7.4.

Alternative Probability Models for Count Data. Most applications of the c chart assume that the Poisson distribution is the correct probability model underlying the process. However, it is not the only distribution that could be utilized as a model of "count" or nonconformities per unit-type data. Various types of phenomena can produce distributions of defects that are not well modeled by the Poisson distribution. In the Poisson

distribution, the mean and the variance are equal. When the sample data indicate that the sample variance is substantially different from the mean, the Poisson assumption is likely to be inappropriate.

The situation where the Poisson assumption is likely to be inappropriate is when nonconformities tend to occur in **clusters;** that is, if there is one nonconformity in some part of a product, then it is likely that there will be others. Note that there are at least two random processes at work here: one generating the number and location of clusters, and the second generating the number of nonconformities within each cluster. If the number of clusters has a Poisson distribution and the number of nonconformities within each cluster has a common distribution (say, f), then the total number of nonconformities has a **compound Poisson distribution.** Many types of compound or generalized distributions could be used as a model for count-type data. As an illustration, if the number of clusters has a Poisson distribution and the number of nonconformities within each cluster is also Poisson, then Neyman's type-A distribution models the total number of nonconformities. Alternatively, if the cluster distribution is gamma and the number of nonconformities within each cluster is Poisson, the negative binomial distribution results. Johnson and Kotz (1969) give a good summary of these and also discuss other discrete distributions that could be useful in modeling count-type data.

Mixtures of various types of nonconformities can lead to situations in which the total number of nonconformities is not adequately modeled by the Poisson distribution. Similar situations occur when the count data have either too many or too few zeros. A good discussion of this general problem is the paper by Jackson (1972). The use of the negative binomial distribution to model count data in inspection units of varying size has been studied by Sheaffer and Leavenworth (1976). The dissertation by Gardiner (1987) describes the use of various discrete distributions to model the occurrence of defects in integrated circuits.

As we noted in Section 3.2.4, the geometric distribution can also be useful as a model for count or "event" data. Kaminski et al. (1992) have proposed control charts for counts based on the geometric distribution. The probability model that they use for the geometric distribution is

$$p(x) = p(1-p)^{x-a} \text{ for } x = a, a+1, a+2, \ldots$$

where a is the known minimum possible number of events. Suppose that the data from the process is available as a subgroup of size n, say $x_1, x_2, \ldots . x_n$. These observations are independently and identically distributed observations from a geometric distribution when the process is stable (in control). The two statistics that can be used to form a control chart are the ***total*** number of events

$$T = x_1 + x_2 + \ldots + x_n$$

and the ***average*** number of events

$$\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n}$$

It turns out that the sum of independently and identically distributed geometric random variables is a negative binomial random variable. This would be useful information in constructing OC curves or calculating ARLs for the control charts for T or $\bar{x}$.

The mean and variance of the total number of events T are

$$\mu_T = n\left(\frac{1-p}{p} + a\right)$$

and

$$\sigma_T^2 = \frac{n(1-p)}{p^2}$$

and the mean and variance of the average number of events are

$$\mu_{\bar{x}} = \frac{1-p}{p} + a$$

and

$$\sigma^2_{\bar{x}} = \frac{1-p}{np^2}$$

Consequently, the control charts can be constructed in the usual manner for Shewhart charts. Kaminski et al. (1992) refer to the control chart for the total number of events as a "*g* chart" and the control chart for the average number of events as an "*h* chart." The center lines and control limits for each chart are shown in the following display.

g and h Control Charts, Standards Given

	Total number of events chart, *g* chart	Average number of events chart, *h* chart
Upper control limit (UCL)	$n\left(\frac{1-p}{p} + a\right) + L\sqrt{\frac{n(1-p)}{p^2}}$	$\frac{1-p}{p} + a + L\sqrt{\frac{1-p}{np^2}}$
Center line (CL)	$n\left(\frac{1-p}{p} + a\right)$	$\frac{1-p}{p} + a$
Lower control limit (LCL)	$n\left(\frac{1-p}{p} + a\right) - L\sqrt{\frac{n(1-p)}{p^2}}$	$\frac{1-p}{p} + a - L\sqrt{\frac{1-p}{np^2}}$

While we have assumed that a is known, in most situations the parameter p will likely be unknown. The estimator for p is

$$\hat{p} = \frac{1}{\bar{\bar{x}} - a + 1}$$

where $\bar{\bar{x}}$ is the average of all of the count data. Suppose that there are m subgroups available, each of size n, and let the total number of events in each subgroup be $t_1, t_2, \ldots, t_m$. The average number of events per subgroup is

$$\bar{t} = \frac{t_1 + t_2 + \cdots + t_m}{m}$$

Therefore,

$$\bar{\bar{x}} = \frac{\bar{t}}{n} = \frac{1-\hat{p}}{\hat{p}} + a$$

and

$$\frac{1-\hat{p}}{\hat{p}^2} = \left(\frac{\bar{t}}{n} - a\right)\left(\frac{\bar{t}}{n} - a + 1\right)$$

The center line and control limits for the g chart and the h chart based on an estimate of p are shown below.

g and h Control Charts, No Standards Given

	Total number of events chart, g chart	Average number of events chart, h chart
Upper control limit (UCL)	$\bar{t} + L\sqrt{n\left(\frac{\bar{t}}{n} - a\right)\left(\frac{\bar{t}}{n} - a + 1\right)}$	$\frac{\bar{t}}{n} + \frac{L}{\sqrt{n}}\sqrt{\left(\frac{\bar{t}}{n} - a\right)\left(\frac{\bar{t}}{n} - a + 1\right)}$
Center line (CL)	$\bar{t}$	$\frac{\bar{t}}{n}$
Lower control limit (LCL)	$\bar{t} - L\sqrt{n\left(\frac{\bar{t}}{n} - a\right)\left(\frac{\bar{t}}{n} - a + 1\right)}$	$\frac{\bar{t}}{n} - \frac{L}{\sqrt{n}}\sqrt{\left(\frac{\bar{t}}{n} - a\right)\left(\frac{\bar{t}}{n} - a + 1\right)}$

7.3.2 Procedures with Variable Sample Size

Control charts for nonconformities are occasionally formed using 100% inspection of the product. When this method of sampling is used, the number of inspection units in a sample will usually not be constant. For example, the inspection of rolls of cloth or paper often leads to a situation in which the size of the sample varies, because not all rolls are exactly the same length or width. If a control chart for nonconformities (c chart) is used in this situation, both the center line and the control limits will vary with the sample size. Such a control chart would be very difficult to interpret. The correct procedure is to use a control chart for nonconformities per unit (u chart). This chart will have a constant center line; however, the control limits will vary inversely with the square root of the sample size n.

EXAMPLE 7.5 Constructing a u chart

In a textile finishing plant, dyed cloth is inspected for the occurrence of defects per 50 square meters. The data on ten rolls of cloth are shown in Table 7.11. Use these data to set up a control chart for nonconformities per unit.

TABLE 7.11
Occurrence of Nonconformities in Dyed Cloth

Roll Number	Number of Square Meters	Total Number of Nonconformities	Number of Inspection Units in Roll, n	Number of Nonconformities per Inspection Unit
1	500	14	10.0	1.40
2	400	12	8.0	1.50
3	650	20	13.0	1.54
4	500	11	10.0	1.10
5	475	7	9.5	0.74
6	500	10	10.0	1.00
7	600	21	12.0	1.75
8	525	16	10.5	1.52
9	600	19	12.0	1.58
10	625	23	12.5	1.84
		153	107.50	

(*continued*)

SOLUTION

The center line of the chart should be the average number of nonconformities per inspection unit—that is, the average number of nonconformities per 50 square meters, computed as

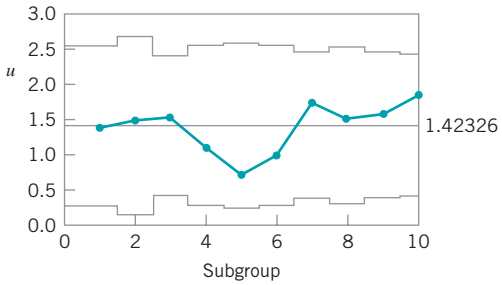
$$\bar{u} = \frac{153}{107.5} = 1.42$$

Note that $\bar{u}$ is the ratio of the total number of observed nonconformities to the total number of inspection units.

The control limits on this chart are computed from equation (7.19) with n replaced by n_i. The width of the control limits will vary inversely with n_i, the number of inspection units in the roll. The calculations for the control limits are displayed in Table 7.12. Figure 7.17 plots the control chart constructed by Minitab.

■ **TABLE 7.12**
Calculation of Control Limits, Example 7.5

Roll Number, i	n_i	UCL $= \bar{u} + 3\sqrt{\bar{u}/n_i}$	LCL $= \bar{u} - 3\sqrt{\bar{u}/n_i}$
1	10.0	2.55	0.29
2	8.0	2.68	0.16
3	13.0	2.41	0.43
4	10.0	2.55	0.29
5	9.5	2.58	0.26
6	10.0	2.55	0.29
7	12.0	2.45	0.39
8	10.5	2.52	0.32
9	12.0	2.45	0.39
10	12.5	2.43	0.41



■ **FIGURE 7.17** Computer-generated (Minitab) control chart for Example 7.5.

As noted previously, the u chart should always be used when the sample size is variable. The most common implementation involves variable control limits, as illustrated in Example 7.5. There are, however, two other possible approaches:

1. Use control limits based on an average sample size

$$\bar{n} = \sum_{i=1}^{m} n_i / m$$

2. Use a standardized control chart (this is the preferred option). This second alternative would involve plotting a standardized statistic

$$Z_i = \frac{u_i - \bar{u}}{\sqrt{\frac{\bar{u}}{n_i}}} \tag{7.20}$$

on a control chart with LCL = –3 and UCL = +3 and the center line at zero. This chart is appropriate if tests for runs and other pattern-recognition methods are to be used in conjunction with the chart. Figure 7.18 shows the standardized version of the control chart in Example 7.5. This standardized control chart could also be useful in the short production run situation (see Chapter 10, Section 10.1).

7.3.3 Demerit Systems

With complex products such as automobiles, computers, or major appliances, we usually find that many different types of nonconformities or defects can occur. Not all of these types of defects are equally important. A unit of product having one very serious defect would probably be classified as nonconforming to requirements, but a unit having several minor defects might not necessarily be nonconforming. In such situations, we need a method to classify nonconformities or defects according to severity and to weight the various types of defects in a reasonable manner. **Demerit systems for attribute data** can be of value in these situations.

One possible demerit scheme is defined as follows.

Class A Defects—Very Serious. The unit is either completely unfit for service, or will fail in service in such a manner that cannot be easily corrected in the field, or will cause personal injury or property damage.

Class B Defects—Serious. The unit will possibly suffer a Class A operating failure, or will certainly cause somewhat less serious operating problems, or will certainly have reduced life or increased maintenance cost.

Class C Defects—Moderately Serious. The unit will possibly fail in service, or cause trouble that is less serious than operating failure, or possibly have reduced life or increased maintenance costs, or have a major defect in finish, appearance, or quality of work.

Class D Defects—Minor. The unit will not fail in service but has minor defects in finish, appearance, or quality of work.

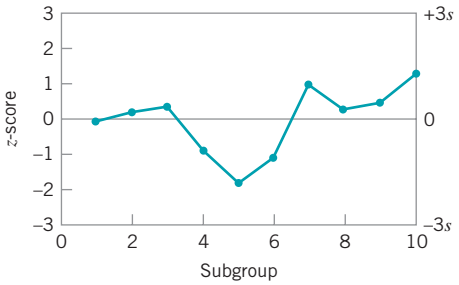


FIGURE 7.18 Standardized control chart for nonconformities per unit, Example 7.5.

Let c_{iA}, c_{iB}, c_{iC}, and c_{iD} represent the number of Class A, Class B, Class C, and Class D defects, respectively, in the ith inspection unit. We assume that each class of defect is independent, and the occurrence of defects in each class is well modeled by a Poisson distribution. Then we define the number of **demerits** in the inspection unit as

$$d_i = 100c_{iA} + 50c_{iB} + 10c_{iC} + c_{iD} \tag{7.21}$$

The demerit weights of Class A—100, Class B—50, Class C—10, and Class D—1 are used fairly widely in practice. However, any reasonable set of weights appropriate for a specific problem may also be used.

Suppose that a sample of n inspection units is used. Then the number of demerits per unit is

$$u_i = \frac{D}{n} \tag{7.22}$$

where $D = \sum_{i=1}^{n} d_i$ is the total number of demerits in all n inspection units. Since u_i is a linear combination of independent Poisson random variables, the statistics u_i could be plotted on a control chart with the following parameters:

$$\begin{aligned} \text{UCL} &= \bar{u} + 3\hat{\sigma}_u \\ \text{Center line} &= \bar{u} \\ \text{LCL} &= \bar{u} - 3\hat{\sigma}_u \end{aligned} \tag{7.23}$$

where

$$\bar{u} = 100\bar{u}_A + 50\bar{u}_B + 10\bar{u}_C + \bar{u}_D \tag{7.24}$$

and

$$\hat{\sigma}_u = \left[\frac{(100)^2 \bar{u}_A + (50)^2 \bar{u}_B + (10)^2 \bar{u}_C + \bar{u}_D}{n} \right]^{1/2} \tag{7.25}$$

In the preceding equations, $\bar{u}_A$, $\bar{u}_B$, $\bar{u}_C$, and $\bar{u}_D$ represent the average number of Class A, Class B, Class C, and Class D defects per unit. The values of $\bar{u}_A$, $\bar{u}_B$, $\bar{u}_C$, and $\bar{u}_D$ are obtained from the analysis of preliminary data, taken when the process is supposedly operating in control. Standard values for u_A, u_B, u_C, and u_D may also be used, if they are available.

Jones, Woodall, and Conerly (1999) provide a very thorough discussion of demerit-based control charts. They show how probability-based limits can be computed as alternatives to the traditional three-sigma limits used above. They also show that, in general, the probability limits give superior performance. They are, however, more complicated to compute.

Many variations of this idea are possible. For example, we can classify nonconformities as either **functional defects** or **appearance defects** if a two-class system is preferred. It is also fairly common practice to maintain separate control charts on each defect class rather than combining them into one chart.

7.3.4 The Operating-Characteristic Function

The operating-characteristic (OC) curves for both the c chart and the u chart can be obtained from the Poisson distribution. For the c chart, the OC curve plots the probability of type II error β against the true mean number of defects c. The expression for β is

$$\beta = P\{x < \text{UCL}|c\} - P\{x \le \text{LCL}|c\} \tag{7.26}$$

TABLE 7.13
Calculation of the OC Curve for a c Chart with UCL = 33.22 and LCL = 6.48

c	$P\{x \le 33 \mid c\}$	$P\{x \le 6 \mid c\}$	$\beta = P\{x \le 33 \mid c\} - P\{x \le 6 \mid c\}$
1	1.000	0.999	0.001
3	1.000	0.966	0.034
5	1.000	0.762	0.238
7	1.000	0.450	0.550
10	1.000	0.130	0.870
15	0.999	0.008	0.991
20	0.997	0.000	0.997
25	0.950	0.000	0.950
30	0.744	0.000	0.744
33	0.546	0.000	0.546
35	0.410	0.000	0.410
40	0.151	0.000	0.151
45	0.038	0.000	0.038

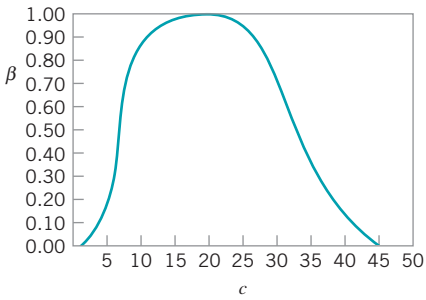


FIGURE 7.19 OC curve of a c chart with LCL = 6.48 and UCL = 33.22.

where x is a Poisson random variable with parameter c. Note that if the LCL < 0 the second term on the right-hand side of equation (7.26) should be dropped.

We will generate the OC curve for the c chart in Example 7.3. For this example, since the LCL = 6.48 and the UCL = 33.22, equation (7.26) becomes

$$\beta = P\{x < 33.22|c\} - P\{x \le 6.48|c\}$$

Since the number of nonconformities must be integer, this is equivalent to

$$\beta = P\{x \le 33|c\} - P\{x \le 6|c\}$$

These probabilities are evaluated in Table 7.13. The OC curve is shown in Fig. 7.19.

For the u chart, we may generate the OC curve from

$$\begin{aligned}\beta &= P\{x < \text{UCL}|u\} - P\{x \le \text{LCL}|u\} \\ &= P\{c < n\text{UCL}|u\} - P\{c \le n\text{LCL}|u\} \\ &= P\{n\text{LCL} < x \le n\text{UCL}|u\} \\ &= \sum_{x=\langle n\text{LCL}\rangle}^{[n\text{UCL}]} \frac{e^{-nu}(nu)^x}{x!}\end{aligned} \tag{7.27}$$

where $\langle n\text{LCL}\rangle$ denotes the smallest integer greater than or equal to nLCL and $[n\text{UCL}]$ denotes the largest integer less than or equal to nUCL. The limits on the summation in equation (7.26) follow from the fact that the total number of nonconformities observed in a sample of n inspection units must be an integer. Note that n need not be an integer.

7.3.5 Dealing with Low Defect Levels

When defect levels or in general, ***count rates***, in a process become very low—say, under 1000 occurrences per million—there will be very long periods of time between the occurrence of a nonconforming unit. In these situations, many samples will have zero defects, and a control chart with the statistic consistently plotting at zero will be relatively uninformative. Thus,

conventional c and u charts become ineffective as count rates are driven into the low parts per million (ppm) range.

One way to deal with this problem is adopt a **time between occurrence control chart,** which charts a new variable: the time between the successive occurrences of the count. The time-between-events control chart has been very effective as a process-control procedure for processes with low defect levels.

Suppose that defects or counts or "events" of interest occur according to a Poisson distribution. Then the probability distribution of the time between events is the exponential distribution. Therefore, constructing a time-between-events control chart is essentially equivalent to control charting an exponentially distributed variable. However, the exponential distribution is highly skewed, and as a result, the corresponding control chart would be very asymmetric. Such a control chart would certainly look unusual, and might present some difficulties in interpretation for operating personnel.

Nelson (1994) has suggested solving this problem by transforming the exponential random variable to a Weibull random variable such that the resulting Weibull distribution is well approximated by the normal distribution. If y represents the original exponential random variable, the appropriate transformation is

$$x = y^{1/3.6} = y^{0.2777} \tag{7.28}$$

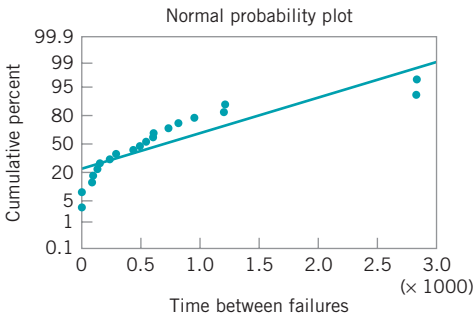
One would now construct a control chart on x, assuming that x follows a normal distribution.

EXAMPLE 7.6

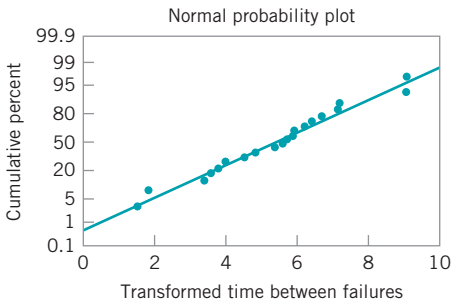
A chemical engineer wants to set up a control chart for monitoring the occurrence of failures of an important valve. She has decided to use the number of hours between failures as the variable to monitor. Table 7.14 shows the number of hours between failures for the last twenty failures of this valve. Figure 7.20 is a normal probability plot of the time between failures.

SOLUTION

Set up a time-between-events control chart for this process. Clearly, time between failures is not normally distributed. Table 7.14 also shows the values of the transformed time between events, computed from equation (7.27). Figure 7.21 is a normal probability plot of the transformed time between failures. Note that the plot indicates that the distribution of this transformed variable is well approximated by the normal.



■ FIGURE 7.20 Normal probability plot of time between failures, Example 7.6.



■ FIGURE 7.21 Normal probability plot for the transformed failure data.

TABLE 7.14
Time between Failure Data, Example 7.6

Failure	Time between Failures, y (hr)	Transformed Value of Time between Failures, $x = y^{0.2777}$
1	286	4.80986
2	948	6.70903
3	536	5.72650
4	124	3.81367
5	816	6.43541
6	729	6.23705
7	4	1.46958
8	143	3.96768
9	431	5.39007
10	8	1.78151
11	2837	9.09619
12	596	5.89774
13	81	3.38833
14	227	4.51095
15	603	5.91690
16	492	5.59189
17	1199	7.16124
18	1214	7.18601
19	2831	9.09083
20	96	3.55203

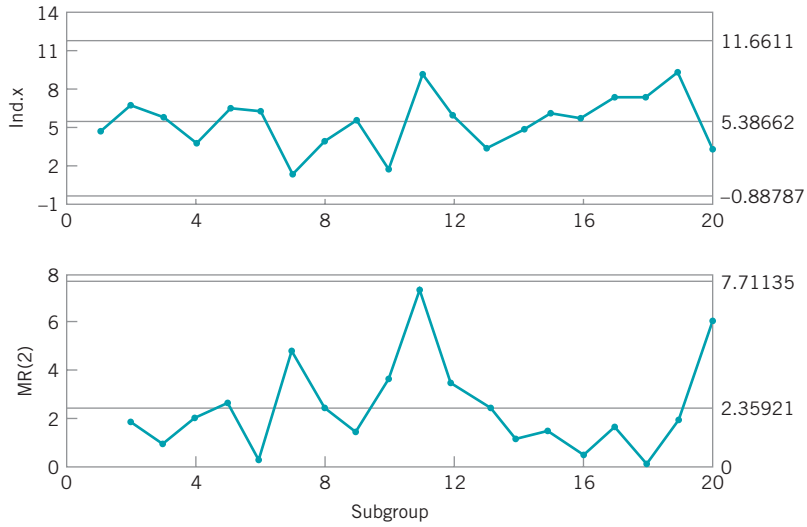


FIGURE 7.22 Control charts for individuals and moving-range control chart for the transformed time between failures, Example 7.6.

Figure 7.22 is a control chart for individuals and a moving range control chart for the transformed time between failures. Note that the control charts indicate a state of control, implying that the failure mechanism for this valve is constant. If a process change is made that improves the failure rate (such as a different type of maintenance action), then we would expect to see the mean time between failures get longer. This would result in points plotting above the upper control limit on the individuals control chart in Fig. 7.22.

The previous example illustrated the use of the individuals control chart with time-between-events data. In many cases, the cusum and EWMA control charts in Chapter 4 would be better alternatives, because they are more effective in detecting small shifts in the mean.

Kittlitz (1999) has also investigated transforming the exponential distribution for control charting purposes. He notes that a log transformation will stabilize the variance of the exponential distribution, but produces a rather negatively skewed distribution. Kittlitz suggests using the transformation $x = y^{0.25}$, noting that it is very similar to Nelson's recommendation and it is also very easy to compute (simply push the square root key on the calculator twice!).

7.3.6 Nonmanufacturing Applications

The *c* chart and *u* chart are widely used in transactional and service business applications of statistical process control. In effect, we can treat **errors** in those environments the same as we treat defects or nonconformities in the manufacturing world. To give just a few examples, we can plot errors on engineering drawings, errors on plans and documents, and errors in computer software as *c* or *u* charts. An example using *u* charts to monitor errors in computer software during product development is given in Gardiner and Montgomery (1987).

7.4 Choice Between Attributes and Variables Control Charts

In many applications, the analyst will have to choose between using a variables control chart, such as the $\bar{x}$ and *R* charts, and an attributes control chart, such as the *p* chart. In some cases, the choice will be clear-cut. For example, if the quality characteristic is the color of the item, such as might be the case in carpet or cloth production, then attributes inspection would often be preferred over an attempt to quantify the quality characteristic "color." In other cases, the choice will not be obvious, and the analyst must take several factors into account in choosing between attributes and variables control charts.

Attributes control charts have the advantage that several quality characteristics can be considered jointly and the unit classified as nonconforming if it fails to meet the specification on any one characteristic. On the other hand, if the several quality characteristics are treated as variables, then each one must be measured, and either a separate $\bar{x}$ and *R* chart must be maintained on each or some multivariate control technique that considers all the characteristics must simultaneously be employed. There is an obvious simplicity associated with the attributes chart in this case. Furthermore, expensive and time-consuming measurements may sometimes be avoided by attributes inspection.

Variables control charts, in contrast, provide much more useful information about process performance than does an attributes control chart. Specific information about the process mean and variability is obtained directly. In addition, when points plot out of control on variables control charts, usually much more information is provided relative to the potential *cause* of that out-of-control signal. For a process capability study, variables control charts are almost always preferable to attributes control charts. The exceptions to this are studies relative to nonconformities produced by machines or operators in which there are a very limited number of sources of nonconformities, or studies directly concerned with process yields and fallouts.

Perhaps the most important advantage of the $\bar{x}$ and *R* control charts is that they often provide an indication of impending trouble and allow operating personnel to take corrective action *before* any defectives are actually produced. Thus, $\bar{x}$ and *R* charts are *leading indicators* of trouble, whereas *p* charts (or *c* and *u* charts) will not react unless the process has already changed

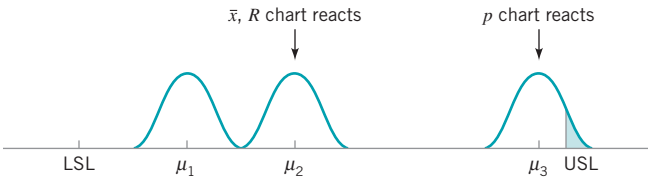


FIGURE 7.23 Why the $\bar{x}$ and R charts can warn of impending trouble.

so that *more* nonconforming units are being produced. This increased efficiency of the $\bar{x}$ and R charts is much more pronounced when p is small, but less so when p is close to 0.5.

To illustrate, consider the production process depicted in Fig. 7.23. When the process mean is at μ_1, few nonconforming units are produced. Suppose the process mean begins to shift upward. By the time it has reached μ_2, the $\bar{x}$ and R charts will have reacted to the change in the mean by generating a strong nonrandom pattern and possibly several out-of-control points. However, a p chart would not react until the mean had shifted all the way to μ_3, or until the actual number of nonconforming units produced had increased. Thus, the $\bar{x}$ and R charts are more powerful control tools than the p chart.

For a specified level of protection against process shifts, variables control charts usually require a much smaller sample size than does the corresponding attributes control chart. Thus, although variables-type inspection is usually more expensive and time-consuming on a per unit basis than attributes inspection, many fewer units must be examined. This is an important consideration, particularly in cases where inspection is destructive (such as opening a can to measure the volume of product within or to test chemical properties of the product). The following example demonstrates the economic advantage of variables control charts.

EXAMPLE 7.7 The Advantage of Variables Control Chart

The nominal value of the mean of a quality characteristic is 50, and the standard deviation is 2. The process is controlled by an $\bar{x}$ chart. Specification limits on the process are established at ± three-sigma, such that the lower specification limit is 44 and the upper specification limit is 56. When the process is in control at the nominal level of 50, the fraction of nonconforming product produced, assuming that the quality characteristic is normally distributed, is 0.0027. Suppose that the process mean were to shift to 52. The fraction of nonconforming product produced following the shift is approximately 0.0228. Suppose that we want the probability of detecting this shift on the first subsequent sample to be 0.50. Find the appropriate sample size for the $\bar{x}$ chart and compare it to the sample size for a p chart that has the same probability of detecting the shift.

SOLUTION

The sample size on the $\bar{x}$ chart must be large enough for the upper three-sigma control limit to be 52. This implies that

$$50 + \frac{3(2)}{\sqrt{n}} = 52$$

or $n = 9$. If a p chart is used, then we may find the required sample size to give the same probability of detecting the shift from equation (7.10), that is,

$$n = \left(\frac{L}{\delta}\right)^2 p(1-p)$$

where $L = 3$ is the width of the control limits, $p = 0.0027$ is the in-control fraction nonconforming, and $\delta = 0.0228 - 0.0027 = 0.0201$ is the magnitude of the shift. Consequently, we find

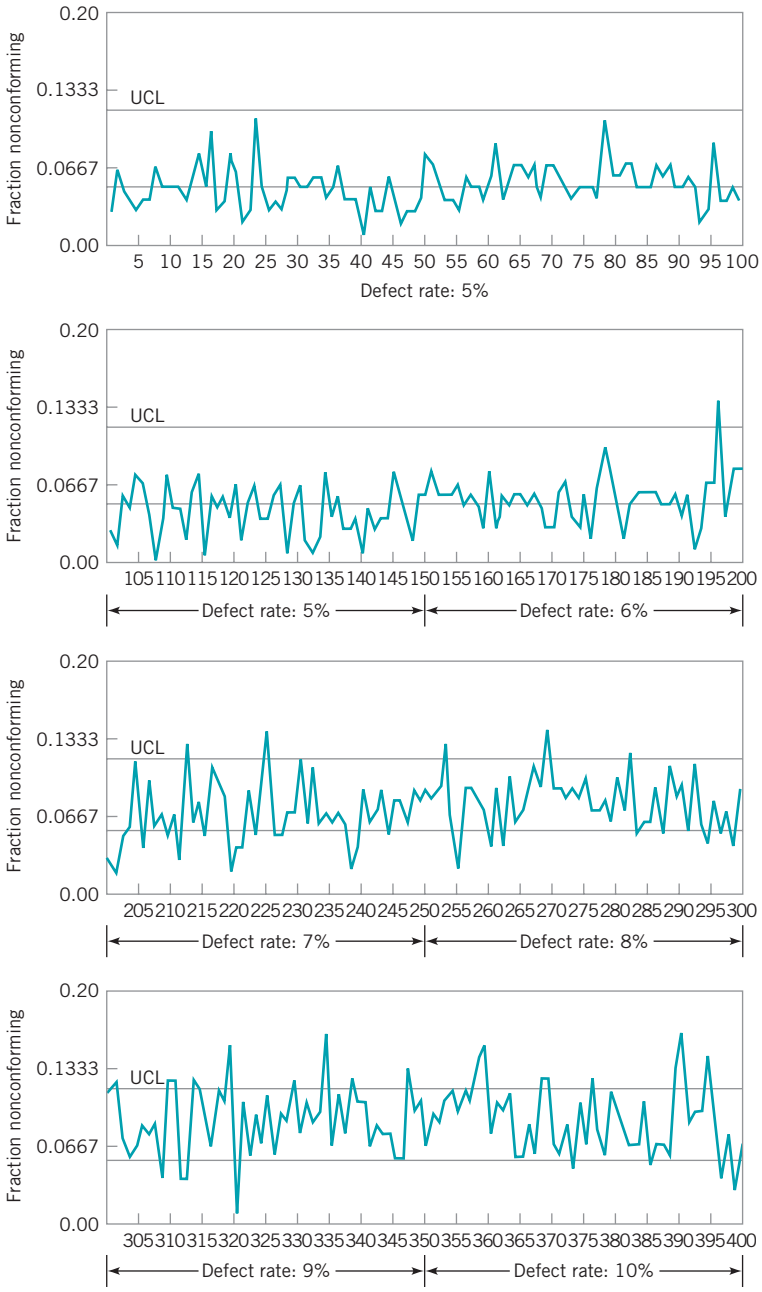
$$n = (3/0.0201)^2(0.0027)(0.9973) = 59.98$$

or $n \simeq 60$ would be required for the p chart. Unless the cost of measurements inspection is more than seven times as costly as attributes inspection, the $\bar{x}$ chart is less expensive to operate.

Generally speaking, variables control charts are preferable to attributes. However, this logic can be carried to an illogical extreme, as shown in Example 7.8.

EXAMPLE 7.8 A Misapplication of $\bar{x}$ and R Charts

This example illustrates a misapplication of $\bar{x}$ and R charts that the author encountered in the electronics industry. A company manufacturing a box-level product inspected a sample of the production units several times each shift using attributes inspection. The output of each sample inspection was an estimate of the process fraction nonconforming $\hat{p}_i$. The company personnel were well aware that attributes data did not contain as much information about the process as variables data, and



■ FIGURE 7.24 Fraction nonconforming control chart for Example 7.8.

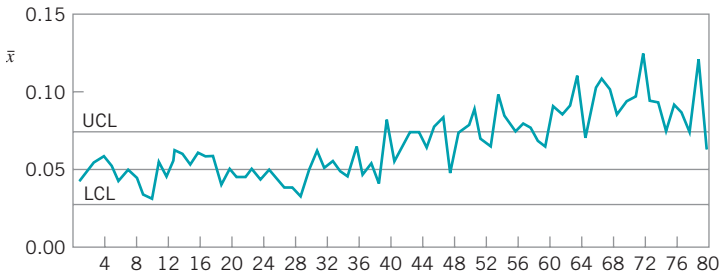


FIGURE 7-25 $\bar{x}$ chart for Example 7.8.

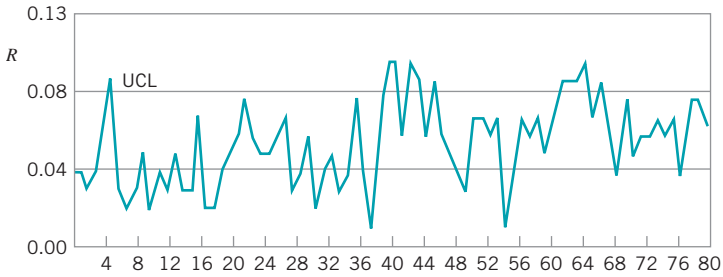


FIGURE 7.26 *R* chart for Example 7.8.

were exploring ways to get more useful information about their process. A consultant to the company (*not* the author) had suggested that they could achieve this objective by converting their fraction nonconforming data into $\bar{x}$ and R charts. To do so, each group of five successive values of $\hat{p}_i$ was treated as if it were a sample of five *variables* measurements; then the average and range were computed as

$$\bar{x} = \frac{1}{5}\sum_{i=1}^{5} \hat{p}_i$$

and

$$R = \max(\hat{p}_i) - \min(\hat{p}_i)$$

and these values were plotted on $\bar{x}$ and R charts. The consultant claimed that this procedure would provide more information than the fraction nonconforming control chart.

This suggestion was incorrect. If the inspection process actually produces attributes data governed by the binomial distribution with fixed n, then the sample fraction nonconforming contains *all* the information in the sample (this is an application of the concept of minimal sufficient statistics) and forming two new functions of $\hat{p}_i$ will not provide any additional information.

To illustrate this idea, consider the control chart for fraction nonconforming in Fig. 7.24. This chart was produced by drawing 100 samples (each of size 200) from a process for which $p = 0.05$ and by using these data to compute the control limits. Then the sample draws were continued until sample 150, where the population fraction nonconforming was increased to $p = 0.06$. At each subsequent 50-sample interval, the value of p was increased by 0.01. Note that the control chart reacts to the shift in p at sample number 196. Figures 7.25 and 7.26 present the $\bar{x}$ and R charts obtained by subgrouping the sample values of $\hat{p}_i$ as suggested above. The first twenty of those subgroups were used to compute the center line and control limits on the $\bar{x}$ and R charts. Note that the $\bar{x}$ chart reacts to the shift in $\hat{p}$ at about subgroup number 40. (This would correspond to *original* samples 196–200.) This result is to be expected, as the $\bar{x}$ chart is really monitoring the fraction nonconforming p. The R chart in Fig. 7.26 is misleading, however. One subgroup within the original set used to construct the control limits is out of control. (This is a false alarm, since $p = 0.05$ for all 100 original samples.) Furthermore, the out-of-control points beginning at about subgroup 40 do not contribute any additional useful information about the process because when $\hat{p}$ shifts from 0.05 to 0.06 (say), the standard deviation of p will *automatically* increase. Therefore, in this case there is no added benefit to the user from $\bar{x}$ and R charts.

This is not to say that the conventional fraction nonconforming control chart based on the binomial probability distribution is the right control chart for all fraction nonconforming data, just as the c chart (based on the Poisson distribution) is not always the right control chart for defect data. If the variability in $\hat{p}_i$ from sample to sample is greater than that which could plausibly be explained by the binomial model, then the analyst should determine the *correct* underlying probability model and base the control chart on that distribution.

7.5 Guidelines for Implementing Control Charts

Almost any process will benefit from SPC, including the use of control charts. In this section, we present some general guidelines helpful in implementing control charts. Specifically, we deal with the following:

1. Determining **which** process characteristics to control
2. Determining **where** the charts should be implemented in the process
3. Choosing the proper **type** of control charts
4. Taking **actions** to improve processes as the result of SPC/control chart analysis
5. Selecting data-collection systems and computer software

The guidelines are applicable to both variables and attributes control charts. Remember, control charts are not only for process surveillance; they should be used as an active, on-line method for reduction of process variability.

Determining Which Characteristics to Control and Where to Put the Control Charts. At the start of a control chart program, it is usually difficult to determine which product or process characteristics should be controlled and at which points in the process to apply control charts. Some useful guidelines follow.

1. At the beginning, control charts should be applied to any product characteristics or manufacturing operations believed to be important. The charts will provide immediate feedback as to whether they are actually needed.
2. The control charts found to be unnecessary should be removed, and others that engineering and operator judgment indicates may be required should be added. More control charts will usually be employed until the process has stabilized.
3. Information on the number and types of control charts on the process should be kept current. It is best to keep separate records on the variables and attributes charts. In general, after the control charts are first installed, the number of control charts tends to increase rather steadily. After that, it will usually decrease. When the process stabilizes, we typically find that it has the same number of charts from one year to the next. However, they are not necessarily the same charts.
4. If control charts are being used effectively and if new knowledge is being gained about the key process variables, we should find that the number of $\bar{x}$ and R charts increases and the number of attributes control charts decreases.
5. At the beginning of a control chart program there will usually be more attributes control charts, applied to semifinished or finished units near the *end* of the manufacturing process. As we learn more about the process, these charts will be replaced with $\bar{x}$ and R charts applied *earlier* in the process to the critical parameters and operations that result in nonconformities in the finished product. Generally, **the earlier that process control can be established, the better.** In a complex assembly process, this may imply that process controls need to be implemented at the vendor or supplier level.
6. Control charts are an on-line, process-monitoring procedure. They should be implemented and maintained as close to the work center as possible, so that feedback will be rapid. Furthermore, the process operators and process engineering should have direct responsibility for collecting the process data, maintaining the charts, and interpreting the results. The operators and engineers have the detailed knowledge of the process required to correct process upsets and use the control chart to improve process performance.

Microcomputers can speed up the feedback and should be an integral part of any modern, on-line, process-control procedure.

7. The out-of-control-action plan (OCAP) is a vital part of the control chart. Operating and engineering personnel should strive to keep OCAPs up-to-date and valid.

Choosing the Proper Type of Control Chart.

A. $\bar{x}$ and R (or $\bar{x}$ and s) charts. Consider using variables control charts in these situations:

1. A new process is coming on stream, or a new product is being manufactured by an existing process.
2. The process has been in operation for some time, but it is chronically in trouble or unable to hold the specified tolerances.
3. The process is in trouble, and the control chart can be useful for diagnostic purposes (troubleshooting).
4. Destructive testing (or other expensive testing procedures) is required.
5. It is desirable to reduce acceptance-sampling or other downstream testing to a minimum when the process can be operated in control.
6. Attributes control charts have been used, but the process is either out of control or in control but the yield is unacceptable.
7. There are very tight specifications, overlapping assembly tolerances, or other difficult manufacturing problems.
8. The operator must decide whether or not to adjust the process, or when a setup must be evaluated.
9. A change in product specifications is desired.
10. Process stability and capability must be continually demonstrated, such as in regulated industries.

B. Attributes Charts (p charts, c charts, and u charts). Consider using attributes control charts in these situations:

1. Operators control the assignable causes, and it is necessary to reduce process fallout.
2. The process is a complex assembly operation and product quality is measured in terms of the occurrence of nonconformities, successful or unsuccessful product function, and so forth. (Examples include computers, office automation equipment, automobiles, and the major subsystems of these products.)
3. Process control is necessary, but measurement data cannot be obtained.
4. A historical summary of process performance is necessary. Attributes control charts, such as p charts, c charts, and u charts, are very effective for summarizing information about the process for management review.
5. Remember that attributes charts are generally inferior to charts for variables. Always use $\bar{x}$ and R or $\bar{x}$ and s charts whenever possible.

C. Control Charts for Individuals. Consider using the control chart for individuals in conjunction with a moving-range chart in these situations:

1. It is inconvenient or impossible to obtain more than one measurement per sample, or repeat measurements will only differ by laboratory or analysis error. Examples often occur in chemical processes.

2. Automated testing and inspection technology allow measurement of every unit produced. In these cases, also consider the cumulative sum control chart and the exponentially weighted moving average control chart discussed in Chapter 9.
3. The data become available very slowly, and waiting for a larger sample will be impractical or make the control procedure too slow to react to problems. This often happens in nonproduct situations; for example, accounting data may become available only monthly.
4. Generally, once we are in phase II, individuals charts have poor performance in shift detection and can be very sensitive to departures from normality. Always use the EWMA and CUSUM charts of Chapter 9 in phase II instead of individuals charts whenever possible.

Actions Taken to Improve the Process. Process improvement is the primary objective of statistical process control. The application of control charts will give information on two key aspects of the process: statistical control and capability. Figure 7.27 shows the possible states in which the process may exist with respect to these two issues. Technically speaking, the capability of a process cannot be adequately assessed until statistical control has been established, but we will use a less precise definition of capability that is just a qualitative assessment of whether or not the level of nonconforming units produced is low enough to warrant no immediate *additional* effort to further improve the process.

Figure 7.27 gives the answers to two questions: "Is the process in control?" and "Is the process capable?" (in the sense of the previous paragraph). Each of the four cells in the figure contains some recommended courses of action that depend on the answers to these two questions. The box in the northwest corner is the ideal state: the process is in statistical control and exhibits adequate capability for present business objectives. In this case, SPC methods are valuable for process monitoring and for warning against the occurrence of any new assignable causes that could cause slippage in performance. The northeast corner implies that the process exhibits statistical control but has poor capability. Perhaps the PCR is lower than the value required by the customer, or there is sufficient variability remaining to result in excessive scrap or rework. In this case, SPC methods may be useful for process diagnosis and improvement, primarily through the recognition of *patterns* on the control chart, but the control charts will not produce very many out-of-control signals. It will usually be necessary to intervene actively in the process to improve it. Experimental design methods are helpful in this regard [see Montgomery (2005)]. Usually, it is also helpful to reconsider the

		IS THE PROCESS CAPABLE?	
		Yes	No
IS THE PROCESS IN CONTROL?	Yes	SPC	SPC Experimental design Investigate specifications Change process
	No	SPC	SPC Experimental design Investigate specifications Change process

■ FIGURE 7.27 Actions taken to improve a process.

specifications: They may have been set at levels tighter than necessary to achieve function or performance from the part. As a last resort, we may have to consider changing the process; that is, investigating or developing new technology that has less variability with respect to this quality characteristic than the existing process.

The lower two boxes in Fig. 7.27 deal with the case of an out-of-control process. The southeast corner presents the case of a process that is out of control and not capable. (Remember our nontechnical use of the term *capability.*) The actions recommended here are identical to those for the box in the northeast corner, except that SPC would be expected to yield fairly rapid results now, because the control charts should be identifying the presence of assignable causes. The other methods of attack will warrant consideration and use in many cases, however. Finally, the southwest corner treats the case of a process that exhibits lack of statistical control but does not produce a meaningful number of defectives because the specifications are very wide. SPC methods should still be used to establish control and reduce variability in this case, for the following reasons:

1. Specifications can change without notice.
2. The customer may require both **control** and **capability.**
3. The fact that the process experiences assignable causes implies that unknown forces are at work; these unknown forces could result in poor capability in the near future.

Selection of Data-Collection Systems and Computer Software. The past few years have produced an explosion of quality control software and electronic data-collection devices. Some SPC consultants recommended against using the computer, noting that it is unnecessary, since most applications of SPC in Japan emphasized the *manual* use of control charts. If the Japanese were successful in the 1960s and 1970s using manual control charting methods, then does the computer truly have a useful role in SPC?

The answer to this question is yes, for several reasons:

1. Although it can be helpful to begin with manual methods of control charting at the start of an SPC implementation, it is necessary to move successful applications to the computer very soon. The computer is a great productivity improvement device. We don't drive cars with the passenger safety systems of the 1960s and we don't fly airplanes with 1960s avionics technology. We shouldn't use 1960s technology with control charts either.
2. The computer will enable the SPC data to become part of the companywide manufacturing database, and in that form, the data will be useful (and hence more likely to be used) to everyone—management, engineering, marketing, and so on, *not* only manufacturing and quality.
3. A computer-based SPC system can provide more information than any manual system. It permits the user to monitor many quality characteristics and to provide automatic signaling of assignable causes.

What type of software should be used? That is a difficult question to answer, because all applications have unique requirements and the capability of the software is constantly changing. However, several features are necessary for successful results:

1. The software should be capable of stand-alone operation on a personal computer, on a multiterminal local area network, or on a multiterminal minicomputer-host system. SPC packages that are exclusively tied to a large mainframe system are frequently not very useful because they usually cannot produce control charts and other routine reports in a timely manner.

2. The system must be user-friendly. If operating personnel are to use the system, it must have limited options, be easy to use, provide adequate error correction opportunities, and contain many on-line help features. It should ideally be possible to tailor or customize the system for each application, although this installation activity may have to be carried out by engineering/technical personnel.
3. The system should provide display of control charts for at least the last 25 samples. Ideally, the length of record displayed should be controlled by the user. Printed output should be immediately available on either a line printer or a plotter.
4. File storage should be sufficient to accommodate a reasonable amount of process history. Editing and updating files should be straightforward. Provisions to transfer data to other storage media or to transfer the data to a master manufacturing database is critical.
5. The system should be able to handle multiple files simultaneously. Only rarely does a process have only one quality characteristic that needs to be examined.
6. The user should be able to calculate control limits from any subset of the data on the file. The user should have the capability to input center lines and control limits directly.
7. The system should be able to accept a variety of inputs, including manual data entry, input from an electronic instrument, or input from another computer or instrument controller. It is important to have the capability for real-time process monitoring, or to be able to transfer data from a real-time data acquisition system.
8. The system should support other statistical applications, including as a minimum histograms and computation of process capability indices.
9. Service and support from the software supplier after purchase is always an important factor in deciding which software package to use.

The purchase price of commercially available software varies widely. Obviously, the total cost of software is very different from the purchase price. In many cases, a $500 SPC package is really a $10,000 package when we take into account the total costs of making the package work correctly in the intended application. It is also relatively easy to establish control charts with most of the popular spreadsheet software packages. However, it may be difficult to integrate those spreadsheet control charts into the overall manufacturing database or other business systems.

Important Terms and Concepts

Attribute data
Average run length for attribute control charts
Cause-and-effect diagram
Choice between attributes and variables data
Control chart for defects or nonconformities per unit or u chart
Control chart for fraction nonconforming or p chart
Control chart for nonconformities or c chart
Control chart for number nonconforming or np chart
Defect
Defective
Demerit systems for attribute data
Design of attributes control charts
Fraction defective
Fraction nonconforming
Nonconformity
Operating characteristic curve for the c and u charts
Operating characteristic curve for the p chart
Pareto chart
Standardized control charts
Time between occurrence control charts
Variable sample size for attributes control chart

Exercises



The Student Resource Manual presents comprehensive annotated solutions to the odd-numbered exercises included in the Answers to Selected Exercises section in the back of this book.

7.1. The data in Table 7E.1 give the number of nonconforming bearing and seal assemblies in samples of size 100. Construct a fraction nonconforming control chart for these data. If any points plot out of control, assume that assignable causes can be found and determine the revised control limits.

7.2. The number of nonconforming switches in samples of size 150 are shown in Table 7E.2. Construct a fraction nonconforming control chart for these data. Does the process appear to be in control? If not, assume that assignable causes can be found for all points outside the control limits and calculate the revised control limits.

7.3. The data in Table 7E.3 represent the results of inspecting all units of a personal computer produced for the past ten days. Does the process appear to be in control?

7.4. A process that produces titanium forgings for automobile turbocharger wheels is to be controlled through use of a fraction nonconforming chart. Initially, one sample of size 150 is taken each day for twenty days, and the results shown in Table 7E.4 are observed.

(a) Establish a control chart to monitor future production.

(b) What is the smallest sample size that could be used for this process and still give a positive lower control limit on the chart?

TABLE 7E.1
Data for Exercise 7.1

Sample Number	Number of Nonconforming Assemblies	Sample Number	Number of Nonconforming Assemblies
1	7	11	6
2	4	12	15
3	1	13	0
4	3	14	9
5	6	15	5
6	8	16	1
7	10	17	4
8	5	18	5
9	2	19	7
10	7	20	12

TABLE 7E.2
Number of Nonconforming Switches

Sample Number	Number of Nonconforming Switches	Sample Number	Number of Nonconforming Switches
1	8	11	6
2	1	12	0
3	3	13	4
4	0	14	0
5	2	15	3
6	4	16	1
7	0	17	15
8	1	18	2
9	10	19	3
10	6	20	0

TABLE 7E.3
Personal Computer Inspecting Results

Day	Units Inspected	Nonconforming Units	Fraction Nonconforming
1	80	4	0.050
2	110	7	0.064
3	90	5	0.056
4	75	8	0.107
5	130	6	0.046
6	120	6	0.050
7	70	4	0.057
8	125	5	0.040
9	105	8	0.076
10	95	7	0.074

TABLE 7E.4
Nonconforming Unit Data for Exercise 7.4

Day	Nonconforming Units	Day	Nonconforming Units
1	3	11	2
2	2	12	4
3	4	13	1
4	2	14	3
5	5	15	6
6	2	16	0
7	1	17	1
8	2	18	2
9	0	19	3
10	5	20	2

■ TABLE 7E.5
Inspection Data for Exercise 7.5

Lot Number	Number of Nonconforming Belts	Lot Number	Number of Nonconforming Belts
1	230	11	456
2	435	12	394
3	221	13	285
4	346	14	331
5	230	15	198
6	327	16	414
7	285	17	131
8	311	18	269
9	342	19	221
10	308	20	407

7.5. A process produces rubber belts in lots of size 2500. Inspection records on the last 20 lots reveal the data in Table 7E.5.
(a) Compute trial control limits for a fraction nonconforming control chart.
(b) If you wanted to set up a control chart for controlling future production, how would you use these data to obtain the center line and control limits for the chart?

7.6. Based on the data in Table 7E.6 if an *np* chart is to be established, what would you recommend as the center line and control limits? Assume that $n = 500$.

7.7. A control chart indicates that the current process fraction nonconforming is 0.02. If fifty items are inspected each day, what is the probability of detecting a shift in the fraction nonconforming to 0.04 on the first day after the shift? By the end of the third day following the shift?

■ TABLE 7E.6
Data for Exercise 7.6

Day	Number of Nonconforming Units
1	3
2	4
3	3
4	2
5	6
6	12
7	5
8	1
9	2
10	2

7.8. A company purchases a small metal bracket in containers of 5000 each. Ten containers have arrived at the unloading facility, and 250 brackets are selected at random from each container. The fraction nonconforming in each sample are 0, 0, 0, 0.004, 0.008, 0.020, 0.004, 0, 0, and 0.008. Do the data from this shipment indicate statistical control?

7.9. Diodes used on printed circuit boards are produced in lots of size 1000. We wish to control the process producing these diodes by taking samples of size 64 from each lot. If the nominal value of the fraction nonconforming is $p = 0.10$, determine the parameters of the appropriate control chart. To what level must the fraction nonconforming increase to make the β-risk equal to 0.50? What is the minimum sample size that would give a positive lower control limit for this chart?

7.10. A control chart for the number of nonconforming piston rings is maintained on a forging process with $np = 16.0$. A sample of size 100 is taken each day and analyzed.
(a) What is the probability that a shift in the process average to $np = 20.0$ will be detected on the first day following the shift? What is the probability that the shift will be detected by at least the end of the third day?
(b) Find the smallest sample size that will give a positive lower control limit.

7.11. A control chart for the fraction nonconforming is to be established using a center line of $p = 0.10$. What sample size is required if we wish to detect a shift in the process fraction nonconforming to 0.20 with probability 0.50?

7.12. A process is controlled with a fraction nonconforming control chart with three-sigma limits, $n = 100$, UCL = 0.161, center line = 0.080, and LCL = 0.
(a) Find the equivalent control chart for the number nonconforming.
(b) Use the Poisson approximation to the binomial to find the probability of a type I error.
(c) Use the correct approximation to find the probability of a type II error if the process fraction nonconforming shifts to 0.2.
(d) What is the probability of detecting the shift in part (c) by at most the fourth sample after the shift?

7.13. A process is being controlled with a fraction nonconforming control chart. The process average has been shown to be 0.07. Three-sigma control limits are used, and the procedure calls for taking daily samples of 400 items.

■ TABLE 7E.7
Nonconforming Unit Data for Exercise 7.15

Sample Number	Sample Size	Number Nonconforming
1	100	10
2	100	15
3	100	31
4	100	18
5	100	24
6	100	12
7	100	23
8	100	15
9	100	8
10	100	8

(a) Calculate the upper and lower control limits.
(b) If the process average should suddenly shift to 0.10, what is the probability that the shift would be detected on the first subsequent sample?
(c) What is the probability that the shift in part (b) would be detected on the first or second sample taken after the shift?

7.14. In designing a fraction nonconforming chart with center line at $p = 0.20$ and three-sigma control limits, what is the sample size required to yield a positive lower control limit? What is the value of n necessary to give a probability of 0.50 of detecting a shift in the process to 0.26?

7.15. A control chart is used to control the fraction nonconforming for a plastic part manufactured in an injection molding process. Ten subgroups yield the data in Table 7E.7
(a) Set up a control chart for the number nonconforming in samples of $n = 100$.
(b) For the chart established in part (a), what is the probability of detecting a shift in the process fraction nonconforming to 0.30 on the first sample after the shift has occurred?

7.16. A control chart for fraction nonconforming indicates that the current process average is 0.03. The sample size is constant at 200 units.
(a) Find the three-sigma control limits for the control chart.
(b) What is the probability that a shift in the process average to 0.08 will be detected on the first subsequent sample? What is the probability that this shift will be detected at least by the fourth sample following the shift?

7.17. (a) A control chart for the number nonconforming is to be established, based on samples of size 400. To start the control chart, thirty samples were selected and the number nonconforming in each sample determined, yielding $\Sigma_{i=1}^{30} D_i = 1200$. What are the parameters of the *np* chart?
(b) Suppose the process average fraction nonconforming shifted to 0.15. What is the probability that the shift would be detected on the first subsequent sample?

7.18. A fraction nonconforming control chart with center line 0.10, UCL = 0.19, and LCL = 0.01 is used to control a process.
(a) If three-sigma limits are used, find the sample size for the control chart.
(b) Use the Poisson approximation to the binomial to find the probability of type I error.
(c) Use the Poisson approximation to the binomial to find the probability of type II error if the process fraction defective is actually $p = 0.20$.

7.19. Consider the control chart designed in Exercise 7.17. Find the average run length to detect a shift to a fraction nonconforming of 0.15.

7.20. Consider the control chart in Exercise 7.18. Find the average run length if the process fraction nonconforming shifts to 0.20.

7.21. A maintenance group improves the effectiveness of its repair work by monitoring the number of maintenance requests that require a second call to complete the repair. Twenty weeks of data are shown in Table 7E.8.
(a) Find trial control limits for this process.
(b) Design a control chart for controlling future production.

7.22. Analyze the data in Exercise 7.21 using an average sample size.

■ TABLE 7E.8
Data for Exercise 7.21

Week	Total Requests	Second Visit Required	Week	Total Requests	Second Visit Required
1	200	6	11	100	1
2	250	8	12	100	0
3	250	9	13	100	1
4	250	7	14	200	4
5	200	3	15	200	5
6	200	4	16	200	3
7	150	2	17	200	10
8	150	1	18	200	4
9	150	0	19	250	7
10	150	2	20	250	6

7.23. Construct a standardized control chart for the data in Exercise 7.21.

7.24. **Continuation of Exercise 7.21.** Note that in Exercise 7.21 there are only four different sample sizes; $n = 100$, 150, 200, and 250. Prepare a control chart that has a set of limits for each possible sample size and show how it could be used as an alternative to the variable-width control limit method used in Exercise 7.21. How easy would this method be to use in practice?

7.25. A fraction nonconforming control chart has center line 0.01, UCL = 0.0399, LCL = 0, and $n = 100$. If three-sigma limits are used, find the smallest sample size that would yield a positive lower control limit.

7.26. Why is the *np* chart not appropriate with variable sample size?

7.27. A fraction nonconforming control chart with $n = 400$ has the following parameters:

UCL = 0.0809
Center line = 0.0500
LCL = 0.0191

(a) Find the width of the control limits in standard deviation units.
(b) What would be the corresponding parameters for an equivalent control chart based on the number nonconforming?
(c) What is the probability that a shift in the process fraction nonconforming to 0.0300 will be detected on the first sample following the shift?

7.28. A fraction nonconforming control chart with $n = 400$ has the following parameters:

UCL = 0.0962
Center line = 0.0500
LCL = 0.0038

(a) Find the width of the control limits in standard deviation units.
(b) Suppose the process fraction nonconforming shifts to 0.15. What is the probability of detecting the shift on the first subsequent sample?

7.29. A fraction nonconforming control chart is to be established with a center line of 0.01 and two-sigma control limits.
(a) How large should the sample size be if the lower control limit is to be nonzero?
(b) How large should the sample size be if we wish the probability of detecting a shift to 0.04 to be 0.50?

7.30. The following fraction nonconforming control chart with $n = 100$ is used to control a process:

UCL = 0.0750
Center line = 0.0400
LCL = 0.0050

(a) Use the Poisson approximation to the binomial to find the probability of a type I error.
(b) Use the Poisson approximation to the binomial to find the probability of a type II error, if the true process fraction nonconforming is 0.0600.
(c) Draw the OC curve for this control chart.
(d) Find the ARL when the process is in control and the ARL when the process fraction nonconforming is 0.0600.

■ TABLE 7E.9
Data for Exercise 7.31, part (b).

Sample Number	Number Nonconforming	Sample Number	Number Nonconforming
1	5	6	1
2	2	7	2
3	3	8	6
4	8	9	3
5	4	10	4

7.31. A process that produces bearing housings is controlled with a fraction nonconforming control chart, using sample size $n = 100$ and a center line $\bar{p} = 0.02$.
(a) Find the three-sigma limits for this chart.
(b) Analyze the ten new samples ($n = 100$) shown in Table 7E.9 for statistical control. What conclusions can you draw about the process now?

7.32. Consider an *np* chart with *k*-sigma control limits. Derive a general formula for determining the minimum sample size to ensure that the chart has a positive lower control limit.

7.33. Consider the fraction nonconforming control chart in Exercise 7.4. Find the equivalent *np* chart.

7.34. Consider the fraction nonconforming control chart in Exercise 7.5. Find the equivalent *np* chart.

7.35. Construct a standardized control chart for the data in Exercise 7.3.

7.36. Surface defects have been counted on twenty-five rectangular steel plates, and the data are shown in Table 7E.10. Set up a control chart for nonconformities using these data. Does the process producing the plates appear to be in statistical control?

7.37. A paper mill uses a control chart to monitor the imperfection in finished rolls of paper. Production output is inspected for twenty days, and the resulting data are shown here. Use these data to set up a control chart for nonconformities per roll of paper. Does the process appear to be in statistical control? What center line and control limits would you recommend for controlling current production?

7.38. **Continuation of Exercise 7.37.** Consider the papermaking process in Exercise 7.37. Set up a *u* chart

TABLE 7E.10
Data for Exercise 7.36

Plate Number	Number of Nonconformities	Plate Number	Number of Nonconformities
1	1	14	0
2	0	15	2
3	4	16	1
4	3	17	3
5	1	18	5
6	2	19	4
7	5	20	6
8	0	21	3
9	2	22	1
10	1	23	0
11	1	24	2
12	0	25	4
13	8		

TABLE 7E.11
Data on Imperfections in Rolls of Paper

Day	Number of Rolls Produced	Total Number of Imperfections	Day	Number of Rolls Produced	Total Number of Imperfections
1	18	12	11	18	18
2	18	14	12	18	14
3	24	20	13	18	9
4	22	18	14	20	10
5	22	15	15	20	14
6	22	12	16	20	13
7	20	11	17	24	16
8	20	15	18	24	18
9	20	12	19	22	20
10	20	10	20	21	17

based on an average sample size to control this process.

7.39. **Continuation of Exercise 7.37.** Consider the papermaking process in Exercise 7.37. Set up a standardized u chart for this process.

7.40. The number of nonconformities found on final inspection of a tape deck is shown in Table 7E.12. Can you conclude that the process is in statistical control? What center line and control limits would you recommend for controlling future production?

7.41. The data in Table 7E.13 represent the number of nonconformities per 1000 meters in telephone cable. From analysis of these data, would you conclude that the process is in statistical control? What control procedure would you recommend for future production?

TABLE 7E.12
Data on Nonconformities in Tape Decks

Deck Number	Number of Nonconformities	Deck Number	Number of Nonconformities
2412	0	2421	1
2413	1	2422	0
2414	1	2423	3
2415	0	2424	2
2416	2	2425	5
2417	1	2426	1
2418	1	2427	2
2419	3	2428	1
2420	2	2429	1

TABLE 7E.13
Telephone Cable Data for Exercise 7.41

Sample Number	Number of Nonconformities	Sample Number	Number of Nonconformities
1	1	12	6
2	1	13	9
3	3	14	11
4	7	15	15
5	8	16	8
6	10	17	3
7	5	18	6
8	13	19	7
9	0	20	4
10	19	21	9
11	24	22	20

7.42. Consider the data in Exercise 7.40. Suppose we wish to define a new inspection unit of four tape decks.
(a) What are the center line and control limits for a control chart for monitoring future production based on the total number of defects in the new inspection unit?
(b) What are the center line and control limits for a control chart for nonconformities per unit used to monitor future production?

7.43. Consider the data in Exercise 7.41. Suppose a new inspection unit is defined as 2500 m of wire.
(a) What are the center line and control limits for a control chart for monitoring future production based on the total number of nonconformities in the new inspection unit?
(b) What are the center line and control limits for a control chart for average nonconformities per unit used to monitor future production?

■ TABLE 7E.14
Data for Exercise 7.44

Sample Number	Number of Nonconformities	Sample Number	Number of Nonconformities
1	1	9	2
2	3	10	1
3	2	11	0
4	1	12	2
5	0	13	1
6	2	14	1
7	1	15	2
8	5	16	3

7.44. An automobile manufacturer wishes to control the number of nonconformities in a subassembly area producing manual transmissions. The inspection unit is defined as four transmissions, and data from sixteen samples (each of size 4) are shown in Table 7E.14.
(a) Set up a control chart for nonconformities per unit.
(b) Do these data come from a controlled process? If not, assume that assignable causes can be found for all out-of-control points and calculate the revised control chart parameters.
(c) Suppose the inspection unit is redefined as eight transmissions. Design an appropriate control chart for monitoring future production.

7.45. Find the three-sigma control limits for
(a) a c chart with process average equal to four nonconformities.
(b) a u chart with $c = 4$ and $n = 4$.

7.46. Find 0.900 and 0.100 probability limits for a c chart when the process average is equal to sixteen nonconformities.

7.47. Find the three-sigma control limits for
(a) a c chart with process average equal to nine nonconformities.
(b) a u chart with $c = 16$ and $n = 4$.

7.48. Find 0.980 and 0.020 probability limits for a control chart for nonconformities per unit when $u = 6.0$ and $n = 3$.

7.49. Find 0.975 and 0.025 probability limits for a control chart for nonconformities when $c = 7.6$.

7.50. A control chart for nonconformities per unit uses 0.95 and 0.05 probability limits. The center line is at $u = 1.4$. Determine the control limits if the sample size is $n = 10$.

7.51. The number of workmanship nonconformities observed in the final inspection of disk-drive assemblies has been tabulated as shown in Table 7E.15. Does the process appear to be in control?

■ TABLE 7E.15
Data for Exercise 7.51

Day	Number of Assemblies Inspected	Total Number of Imperfections	Day	Number of Assemblies Inspected	Total Number of Imperfections
1	2	10	6	4	24
2	4	30	7	2	15
3	2	18	8	4	26
4	1	10	9	3	21
5	3	20	10	1	8

7.52. A control chart for nonconformities is to be constructed with $c = 2.0$, LCL = 0, and UCL such that the probability of a point plotting outside control limits when $c = 2.0$ is only 0.005.
(a) Find the UCL.
(b) What is the type I error probability if the process is assumed to be out of control only when two consecutive points fall outside the control limits?

7.53. A textile mill wishes to establish a control procedure on flaws in towels it manufactures. Using an inspection unit of 50 units, past inspection data show that 100 previous inspection units had 850 total flaws. What type of control chart is appropriate? Design the control chart such that it has two-sided probability control limits of $\alpha = 0.06$, approximately. Give the center line and control limits.

7.54. The manufacturer wishes to set up a control chart at the final inspection station for a gas water heater. Defects in workmanship and visual quality features are checked in this inspection. For the past 22 working days, 176 water heaters were inspected and a total of 924 nonconformities reported.
(a) What type of control chart would you recommend here and how would you use it?
(b) Using two water heaters as the inspection unit, calculate the center line and control limits that are consistent with the past 22 days of inspection data.
(c) What is the probability of type I error for the control chart in part (b)?

7.55. Assembled portable television sets are subjected to a final inspection for surface defects. A total procedure is established based on the requirement that if the average number of nonconformities per unit is 4.0, the probability of concluding that the process is in control will be 0.99. There is to be no lower control limit. What is the appropriate type of control chart and what is the required upper control limit?

7.56. A control chart is to be established on a process producing refrigerators. The inspection unit is one

refrigerator, and a common chart for nonconformities is to be used. As preliminary data, 16 nonconformities were counted in inspecting 30 refrigerators.

(a) What are the three-sigma control limits?
(b) What is the α-risk for this control chart?
(c) What is the β-risk if the average number of defects is actually two (i.e., if $c = 2.0$)?
(d) Find the average run length if the average number of defects is actually two.

7.57. Consider the situation described in Exercise 7.56.

(a) Find two-sigma control limits and compare these with the control limits found in part (a) of Exercise 7.56.
(b) Find the α-risk for the control chart with two-sigma control limits and compare with the results of part (b) of Exercise 7.56.
(c) Find the β-risk for $c = 2.0$ for the chart with two-sigma control limits and compare with the results of part (c) of Exercise 7.56.
(d) Find the ARL if $c = 2.0$ and compare with the ARL found in part (d) of Exercise 7.56.

7.58. A control chart for nonconformities is to be established in conjunction with final inspection of a radio. The inspection unit is to be a group of ten radios. The average number of nonconformities per radio has, in the past, been 0.5. Find three-sigma control limits for a c chart based on this size inspection unit.

7.59. A control chart for nonconformities is maintained on a process producing desk calculators. The inspection unit is defined as two calculators. The average number of nonconformities per machine when the process is in control is estimated to be two.

(a) Find the appropriate three-sigma control limits for this size inspection unit.
(b) What is the probability of type I error for this control chart?

7.60. A production line assembles electric clocks. The average number of nonconformities per clock is estimated to be 0.75. The quality engineer wishes to establish a c chart for this operation, using an inspection unit of six clocks. Find the three-sigma limits for this chart.

7.61. Suppose that we wish to design a control chart for nonconformities per unit with L-sigma limits. Find the minimum sample size that would result in a positive lower control limit for this chart.

7.62. Kittlitz (1999) presents data on homicides in Waco, Texas, for the years 1980–1989 (data taken from the *Waco Tribune-Herald,* December 29, 1989). There were 29 homicides in 1989. Table 7E.16 gives the dates of the 1989 homicides and the number of days between each homicide.

TABLE 7E.16
Homicide Data from Waco, Texas, for Exercise 7.62

Month	Date	Days Between	Month	Date	Days Between
Jan.	20		July	8	2
Feb.	23	34	July	9	1
Feb.	25	2	July	26	17
March	5	8	Sep.	9	45
March	10	5	Sep.	22	13
April	4	25	Sep.	24	2
May	7	33	Oct.	1	7
May	24	17	Oct.	4	3
May	28	4	Oct.	8	4
June	7	10	Oct.	19	11
June	16*	9.25	Nov.	2	14
June	16*	0.50	Nov.	25	23
June	22*	5.25	Dec.	28	33
June	25	3	Dec.	29	1
July	6	11			

The * refers to the fact that two homicides occurred on June 16 and were determined to have occurred 12 hours apart.

(a) Plot the days-between-homicides data on a normal probability plot. Does the assumption of a normal distribution seem reasonable for these data?
(b) Transform the data using the 0.2777 root of the data. Plot the transformed data on a normal probability plot. Does this plot indicate that the transformation has been successful in making the new data more closely resemble data from a normal distribution?
(c) Transform the data using the fourth root (0.25) of the data. Plot the transformed data on a normal probability plot. Does this plot indicate that the transformation has been successful in making the new data more closely resemble data from a normal distribution? Is the plot very different from the one in part (b)?
(d) Construct an individuals control chart using the transformed data from part (b).
(e) Construct an individuals control chart using the transformed data from part (c). How similar is it to the one you constructed in part (d)?
(f) Is the process stable? Provide a practical interpretation of the control chart.

7.63. Suggest at least two nonmanufacturing scenarios in which attributes control charts could be useful for process monitoring.

7.64. What practical difficulties could be encountered in monitoring time-between-events data?

■ TABLE 7E.17
Data for Exercise 7.65

Month	NSCANB	MMSB	Days	NYRSB
Jan. 94	50	26838	31	2.31105
Feb. 94	44	26903	28	2.09246
March 94	71	26895	31	2.31596
Apr. 94	53	26289	30	2.19075
May 94	53	26149	31	2.25172
Jun. 94	40	26185	30	2.18208
July 94	41	26142	31	2.25112
Aug. 94	57	26092	31	2.24681
Sept. 94	49	25958	30	2.16317
Oct. 94	63	25957	31	2.23519
Nov. 94	64	25920	30	2.16000
Dec. 94	62	25907	31	2.23088
Jan. 95	67	26754	31	2.30382
Feb. 95	58	26696	28	2.07636
March 95	89	26565	31	2.28754

■ TABLE 7E.18
Data for Exercise 7.66

Month	Phase	NVISITE	NYRSE	Days	MMSE
Jan. 94	1	1421	0.66099	31	7676
Feb. 94	1	1303	0.59718	28	7678
Mar. 94	1	1569	0.66219	31	7690
Apr. 94	1	1576	0.64608	30	7753
May 94	1	1567	0.66779	31	7755
Jun. 94	1	1450	0.65575	30	7869
July 94	1	1532	0.68105	31	7909
Aug. 94	1	1694	0.68820	31	7992
Sep. 94	2	1721	0.66717	30	8006
Oct. 94	2	1762	0.69612	31	8084
Nov. 94	2	1853	0.68233	30	8188
Dec. 94	2	1770	0.70809	31	8223
Jan. 95	2	2024	0.78215	31	9083
Feb. 95	2	1975	0.70684	28	9088
Mar. 95	2	2097	0.78947	31	9168

7.65. A paper by R. N. Rodriguez ("Health Care Applications of Statistical Process Control: Examples Using the SAS® System," *SAS Users Group International: Proceedings of the 21st Annual Conference,* 1996) illustrated several informative applications of control charts to the health care environment. One of these showed how a control chart was employed to analyze the rate of CAT scans performed each month at a clinic. The data used in this example are shown in Table 7E.17. NSCANB is the number of CAT scans performed each month and MMSB is the number of members enrolled in the health care plan each month, in units of member months. DAYS is the number of days in each month. The variable NYRSB converts MMSB to units of thousand members per year, and is computed as follows: NYRSB = MMSB(Days/30)/12000. NYRSB represents the "area of opportunity." Construct an appropriate control chart to monitor the rate at which CAT scans are performed at this clinic.

7.66. A paper by R. N. Rodriguez ("Health Care Applications of Statistical Process Control: Examples Using the SAS® System," *SAS Users Group International: Proceedings of the 21st Annual Conference,* 1996) illustrated several informative applications of control charts to the health care environment. One of these showed how a control chart was employed to analyze the number of office visits by health care plan members. The data for clinic E are shown in Table 7E.18. The variable NVISITE is the number of visits to clinic E each month, and MMSE is the number of members enrolled in the health care plan each month, in units of member months. DAYS is the number of days in each month. The variable NYRSE converts MMSE to units of thousand members per year, and is computed as follows: NYRSE = MMSE(Days/30)/12000. NYRSE represents the "area of opportunity." The variable PHASE separates the data into two time periods.

(a) Use the data from P 1 to construct a control chart for monitoring the rate of office visits performed at clinic E. Does this chart exhibit control?

(b) Plot the data from P 2 on the chart constructed in part (a). Is there a difference in the two phases?

(c) Consider only the P 2 data. Do these data exhibit control?

7.67. The data in Table 7E.19 are the number of information errors found in customer records in a marketing company database. Five records were sampled each day.

(a) Set up a c chart for the total number of errors. Is the process in control?

(b) Set up a t chart for the total number of errors, assuming a geometric distribution with $a = 1$. Is the process in control?

(c) Discuss the findings from parts (a) and (b). Is the Poisson distribution a good model for the customer error data? Is there evidence of this in the data?

■ TABLE 7E.19
Customer Error Data for Exercise 7.67

Day	Record 1	Record 2	Record 3	Record 4	Record 5
1	8	7	1	11	17
2	11	1	11	2	9
3	1	1	8	2	5
4	3	2	5	1	4
5	3	2	13	6	5
6	6	3	3	3	1
7	8	8	2	1	5
8	4	10	2	6	4
9	1	6	1	3	2
10	15	1	3	2	8
11	1	7	13	5	1
12	6	7	9	3	1
13	7	6	3	3	1
14	2	9	3	8	7
15	6	14	7	1	8
16	2	9	4	2	1
17	11	1	1	3	2
18	5	5	19	1	3
19	6	15	5	6	6
20	2	7	9	2	8
21	7	5	6	14	10
22	4	3	8	1	2
23	4	1	4	20	5
24	15	2	7	10	17
25	2	15	3	11	2

■ TABLE 7E.20
Number of Orders per Truck for Exercise 7.69

Truck	No. of Orders	Truck	No. of Orders	Truck	No. of Orders	Truck	No. of Orders
1		9		17		25	
2		10		18		26	
3		11		19		27	
4		12		20		28	
5		13		21		29	
6		14		22		30	
7		15		23		31	
8		16		24		32	

7.69. Kaminski et al. (1992) present data on the number of orders per truck at a distribution center. Some of this data is shown in Table 7E.20.

(a) Set up a c chart for the number of orders per truck. Is the process in control?

(b) Set up a t chart for the number of orders per truck, assuming a geometric distribution with $a = 1$. Is the process in control?

(c) Discuss the findings from parts (a) and (b). Is the Poisson distribution a good model for these data? Is there evidence of this in the data?

8 Process and Measurement System Capability Analysis

Chapter Outline

8.1 INTRODUCTION

8.2 PROCESS CAPABILITY ANALYSIS USING A HISTOGRAM OR A PROBABILITY PLOT

- 8.2.1 Using the Histogram
- 8.2.2 Probability Plotting

8.3 PROCESS CAPABILITY RATIOS

- 8.3.1 Use and Interpretation of C_p
- 8.3.2 Process Capability Ratio for an Off-Center Process
- 8.3.3 Normality and the Process Capability Ratio
- 8.3.4 More about Process Centering
- 8.3.5 Confidence Intervals and Tests on Process Capability Ratios

8.4 PROCESS CAPABILITY ANALYSIS USING A CONTROL CHART

8.5 PROCESS CAPABILITY ANALYSIS USING DESIGNED EXPERIMENTS

8.6 PROCESS CAPABILITY ANALYSIS WITH ATTRIBUTE DATA

8.7 GAUGE AND MEASUREMENT SYSTEM CAPABILITY STUDIES

- 8.7.1 Basic Concepts of Gauge Capability
- 8.7.2 The Analysis of Variance Method
- 8.7.3 Confidence Intervals in Gauge R & R Studies
- 8.7.4 False Defectives and Passed Defectives
- 8.7.5 Attribute Gauge Capability

8.8 SETTING SPECIFICATION LIMITS ON DISCRETE COMPONENTS

- 8.8.1 Linear Combinations
- 8.8.2 Nonlinear Combinations

8.9 ESTIMATING THE NATURAL TOLERANCE LIMITS OF A PROCESS

- 8.9.1 Tolerance Limits Based on the Normal Distribution
- 8.9.2 Nonparametric Tolerance Limits

Supplemental Material for Chapter 8

- S8.1 Fixed versus Random Factors in the Analysis of Variance
- S8.2 More about Analysis of Variance Methods for Measurement Systems Capability Studies

The supplemental material is on the textbook Website www.wiley.com/college/montgomery.

Chapter Overview and Learning Objectives

In Chapter 6, we formally introduced the concept of process capability, or how the inherent variability in a process compares with the specifications or requirements for the product. Process capability analysis is an important tool in the DMAIC process, with application in both the analyze and improve steps. This chapter provides more extensive discussion of process capability, including several ways to study or analyze the capability of a process. We believe that the control chart is a simple and effective process capability analysis technique. We also extend the presentation of process capability ratios that we began in Chapter 6, showing how to interpret these ratios and discussing their potential dangers. The chapter also contains information on evaluating measurement system performance, illustrating graphical methods, as well as techniques based on the analysis of variance. Measurement systems analysis is used extensively in DMAIC, principally during the measure step. We also discuss setting specifications on individual discrete parts or components and estimating the natural tolerance limits of a process.

After careful study of this chapter you should be able to do the following:

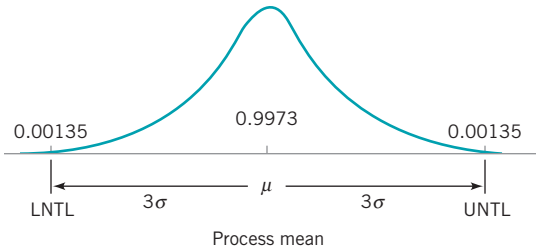
1. Investigate and analyze process capability using control charts, histograms, and probability plots
2. Understand the difference between process capability and process potential
3. Calculate and properly interpret process capability ratios
4. Understand the role of the normal distribution in interpreting most process capability ratios
5. Calculate confidence intervals on process capability ratios
6. Know how to conduct and analyze a measurement systems capability (or gauge R & R) experiment
7. Know how to estimate the components of variability in a measurement system
8. Know how to set specifications on components in a system involving interaction components to ensure that overall system requirements are met
9. Estimate the natural limits of a process from a sample of data from that process

8.1 Introduction

Statistical techniques can be helpful throughout the product cycle, including development activities prior to manufacturing, in quantifying process variability, in analyzing this variability relative to product requirements or specifications, and in assisting development and manufacturing in eliminating or greatly reducing this variability. This general activity is called **process capability analysis.**

Process capability refers to the **uniformity** of the process. Obviously, the variability of critical-to-quality characteristics in the process is a measure of the uniformity of output. There are two ways to think of this variability:

1. The natural or inherent variability in a critical-to-quality characteristic at a specified time; that is, "instantaneous" variability
2. The variability in a critical-to-quality characteristic over time



■ **FIGURE 8.1** Upper and lower natural tolerance limits in the normal distribution.

We present methods for investigating and assessing both aspects of process capability. Determining process capability is an important part of the DMAIC process. It is used primarily in the analyze step, but it also can be useful in other steps, such as improve.

It is customary to take the six-sigma spread in the distribution of the product quality characteristic as a measure of process capability. Figure 8.1 shows a process for which the quality characteristic has a normal distribution with mean μ and standard deviation σ. The upper and lower **natural tolerance limits of the process** fall at $\mu + 3\sigma$ and $\mu - 3\sigma$, respectively; that is,

$$\text{UNTL} = \mu + 3\sigma$$
$$\text{LNTL} = \mu - 3\sigma$$

For a normal distribution, the natural tolerance limits include 99.73% of the variable, or put another way, only 0.27% of the process output will fall outside the natural tolerance limits. Two points should be remembered:

1. 0.27% outside the natural tolerances sounds small, but this corresponds to 2700 nonconforming parts per million.
2. If the distribution of process output is non-normal, then the percentage of output falling outside $\mu \pm 3\sigma$ may differ considerably from 0.27%.

We define **process capability analysis** as a formal study to estimate **process capability.** The estimate of process capability may be in the form of a probability distribution having a specified shape, center (mean), and spread (standard deviation). For example, we may determine that the process output is normally distributed with mean $\mu = 1.0$ cm and standard deviation $\sigma = 0.001$ cm. In this sense, a process capability analysis may be performed **without regard to specifications on the quality characteristic.** Alternatively, we may express process capability as a percentage outside of specifications. However, specifications are not *necessary* to process capability analysis.

A process capability study usually measures functional parameters or critical-to-quality characteristics on the product, not the process itself. When the analyst can directly observe the process and can control or monitor the data-collection activity, the study is a true process capability study, because by controlling the data collection and knowing the time sequence of the data, inferences can be made about the stability of the process over time. However, when we have available only sample units of product, perhaps obtained from the supplier, and there is no direct observation of the process or time history of production, then the study is more properly called **product characterization.** In a product characterization study we can only estimate the distribution of the product quality characteristic or the process yield (fraction conforming to specifications); we can say nothing about the dynamic behavior of the process or its state of statistical control.

Process capability analysis is a vital part of an overall quality-improvement program. Among the major uses of data from a process capability analysis are the following:

1. Predicting how well the process will hold the tolerances
2. Assisting product developers/designers in selecting or modifying a process
3. Assisting in establishing an interval between sampling for process monitoring
4. Specifying performance requirements for new equipment
5. Selecting between competing suppliers and other aspects of supply chain management
6. Planning the sequence of production processes when there is an interactive effect of processes on tolerances
7. Reducing the variability in a process

Thus, process capability analysis is a technique that has application in many segments of the product cycle, including product and process design, supply chain management, production or manufacturing planning, and manufacturing.

Three primary techniques are used in process capability analysis: **histograms** or **probability plots, control charts,** and **designed experiments.** We will discuss and illustrate each of these methods in the next three sections. We will also discuss the process capability ratio (PCR) introduced in Chapter 6 and some useful variations of this ratio.

8.2 Process Capability Analysis Using a Histogram or a Probability Plot

8.2.1 Using the Histogram

The histogram can be helpful in estimating process capability. Alternatively, a stem-and-leaf plot may be substituted for the histogram. At least 100 or more observations should be available for the histogram (or the stem-and-leaf plot) to be moderately stable so that a reasonably reliable estimate of process capability may be obtained. If the quality engineer has access to the process and can control the data-collection effort, the following steps should be followed prior to data collection:

1. Choose the machine or machines to be used. If the results based on one (or a few) machines are to be extended to a larger population of machines, the machine selected should be representative of those in the population. Furthermore, if the machine has multiple workstations or heads, it may be important to collect the data so that head-to-head variability can be isolated. This may imply that designed experiments should be used.
2. Select the process operating conditions. Carefully define conditions, such as cutting speeds, feed rates, and temperatures, for future reference. It may be important to study the effects of varying these factors on process capability.
3. Select a representative operator. In some studies, it may be important to estimate *operator* variability. In these cases, the operators should be selected at random from the population of operators.
4. Carefully monitor the data-collection process, and record the time order in which each unit is produced.

The histogram, along with the sample average $\bar{x}$ and sample standard deviation s, provides information about process capability. You may wish to review the guidelines for constructing histograms in Chapter 3.

EXAMPLE 8.1 Estimating Process Capability with a Histogram

Figure 8.2 presents a histogram of the bursting strength of 100 glass containers. The data are shown in Table 8.1. What is the capability of the process?

SOLUTION

Analysis of the 100 observations gives

$$\bar{x} = 264.06 \qquad s = 32.02$$

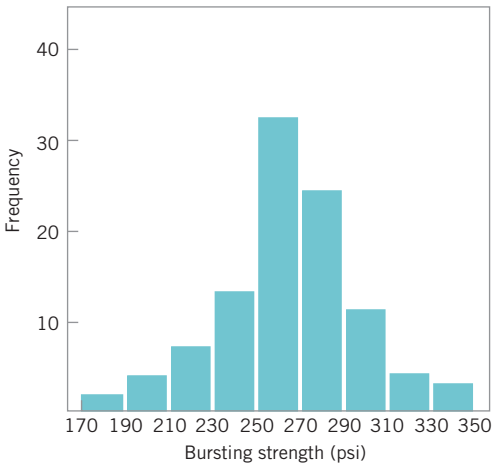
Consequently, the process capability would be estimated as

$$\bar{x} \pm 3s$$

or

$$264.06 \pm 3(32.02) \simeq 264 \pm 96 \text{ psi}$$

Furthermore, the shape of the histogram implies that the distribution of bursting strength is approximately normal. Thus, we can estimate that approximately 99.73% of the bottles manufactured by this process will burst between 168 and 360 psi. Note that we can estimate process capability *independent of the specifications on bursting strength.*

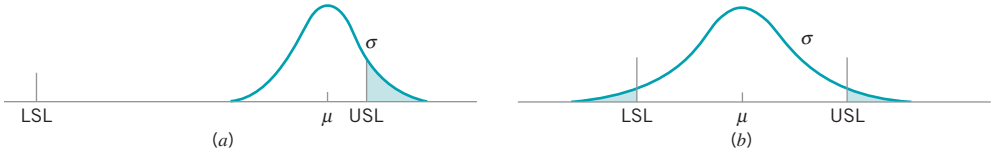


■ FIGURE 8.2 Histogram for the bursting-strength data.

■ TABLE 8.1
Bursting Strengths for 100 Glass Containers

265	197	346	280	265	200	221	265	261	278
205	286	317	242	254	235	176	262	248	250
263	274	242	260	281	246	248	271	260	265
307	243	258	321	294	328	263	245	274	270
220	231	276	228	223	296	231	301	337	298
268	267	300	250	260	276	334	280	250	257
260	281	208	299	308	264	280	274	278	210
234	265	187	258	235	269	265	253	254	280
299	214	264	267	283	235	272	287	274	269
215	318	271	293	277	290	283	258	275	251

An advantage of using the histogram to estimate process capability is that it gives an immediate, visual impression of process performance. It may also immediately show the reason for poor process performance. For example, Fig. 8.3*a* shows a process with adequate potential capability, but the process target is poorly located, whereas Fig. 8.3*b* shows a process with poor capability resulting from excess variability.



■ FIGURE 8.3 Some reasons for poor process capability. (*a*) Poor process centering. (*b*) Excess process variability.

8.2.2 Probability Plotting

Probability plotting is an alternative to the histogram that can be used to determine the shape, center, and spread of the distribution. It has the advantage that it is unnecessary to divide the range of the variable into class intervals, and it often produces reasonable results for moderately small samples (which the histogram will not). Generally, a probability plot is a graph of the ranked data versus the sample cumulative frequency on special paper with a vertical scale chosen so that the cumulative distribution of the assumed type is a straight line. In Chapter 3 we discussed and illustrated **normal probability plots.** These plots are very useful in process capability studies.

To illustrate the use of a normal probability plot in a process capability study, consider the following 20 observations on glass container bursting strength: 197, 200, 215, 221, 231, 242, 245, 258, 265, 265, 271, 275, 277, 278, 280, 283, 290, 301, 318, and 346. Figure 8.4 is the normal probability plot of strength. Note that the data lie nearly along a straight line, implying that the distribution of bursting strength is normal. Recall from Chapter 4 that the mean of the normal distribution is the fiftieth percentile, which we may estimate from Fig. 8.4 as approximately 265 psi, and the standard deviation of the distribution is the *slope* of the straight line. It is convenient to estimate the standard deviation as the difference between the eighty-fourth and the fiftieth percentiles. For the strength data shown above and using Fig. 8.4, we find that

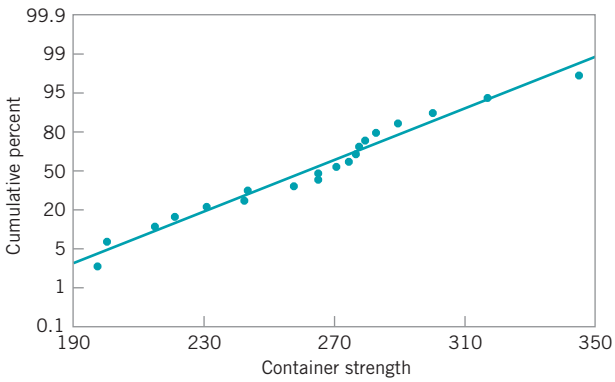
$$\hat{\sigma} = 84\text{th percentile} - 50\text{th percentile} = 298 - 265 \text{ psi} = 33 \text{ psi}$$

Note that $\hat{\mu} = 265$ psi and $\hat{\sigma} = 33$ psi are not far from the sample average $\bar{x} = 264.06$ and standard deviation $s = 32.02$.

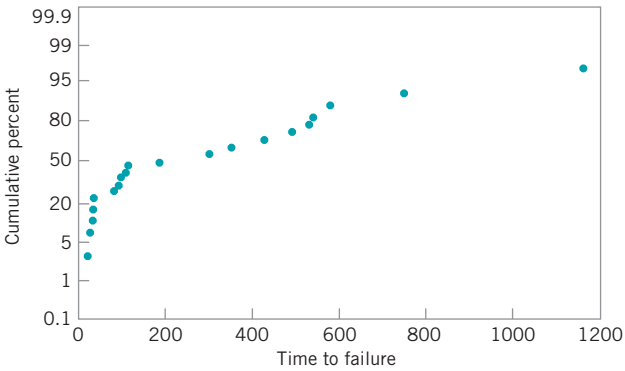
The normal probability plot can also be used to estimate process yields and fallouts. For example, the specification on container strength is LSL = 200 psi. From Fig. 8.4, we would estimate that about 5% of the containers manufactured by this process would burst below this limit.

Care should be exercised in using probability plots. If the data do not come from the assumed distribution, inferences about process capability drawn from the plot may be seriously in error. Figure 8.5 presents a normal probability plot of times to failure (in hours) of a valve in a chemical plant. From examining this plot, we can see that the distribution of failure time is not normal.

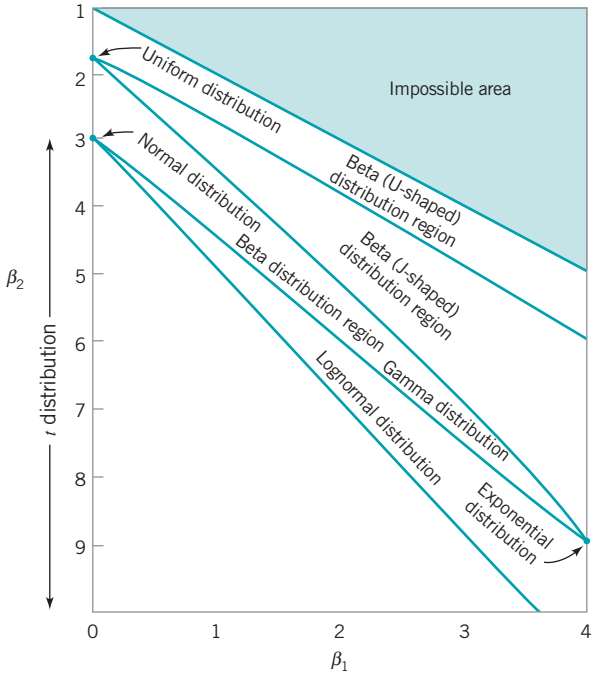
An obvious disadvantage of probability plotting is that it is not an objective procedure. It is possible for two analysts to arrive at different conclusions using the same data. For this reason, it is often desirable to supplement probability plots with more formal statistically



■ **FIGURE 8.4** Normal probability plot of the container-strength data.



■ **FIGURE 8.5** Normal probability plot of time to failure of a valve.



■ **FIGURE 8.6** Regions in the β_1, β_2 plane for various distributions. (From *Statistical Models in Engineering* by G. J. Hahn and S. S. Shapiro, John Wiley: New York, 1967.)

based goodness-of-fit tests. A good introduction to these tests is in Shapiro (1980). Augmenting the interpretation of a normal probability plot with the Shapiro-Wilk test for normality can make the procedure much more powerful and objective.

Choosing the distribution to fit the data is also an important step in probability plotting. Sometimes we can use our knowledge of the physical phenomena or past experience to suggest the choice of distribution. In other situations, the display in Fig. 8.6 may be useful in selecting a distribution that describes the data. This figure shows the regions in the β_1, β_2 plane for several standard probability distributions, where β_1 and β_2 are the measures of *skewness* and *kurtosis,* respectively. To use Fig. 8.6, calculate estimates of skewness and kurtosis from the sample—say,

$$\sqrt{\hat{\beta}_1} = \frac{M_3}{(M_2)^{3/2}} \tag{8.1}$$

and

$$\hat{\beta}_2 = \frac{M_4}{M_2^2} \tag{8.2}$$

where

$$M_j = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^j}{n} \qquad j = 1,\ 2,\ 3,\ 4 \tag{8.3}$$

and plot the point $(\hat{\beta}_1, \hat{\beta}_2)$ on the graph. If the plotted point falls close to a point, line, or region that corresponds to one of the distributions in the figure, then this distribution is a logical choice to use as a model for the data. If the point falls in regions of the β_1, β_2 plane where *none* of the distributions seems appropriate, other, more general probability distributions,

such as the Johnson or Pearson families of distributions, may be required. A note of caution should be sounded here; the skewness and kurtosis statistics are not reliable unless they are computed from very large samples. Procedures similar to that in Fig. 8.6 for fitting these distributions and graphs are in Hahn and Shapiro (1967).

8.3 Process Capability Ratios

8.3.1 Use and Interpretation of C_p

It is frequently convenient to have a simple, quantitative way to express process capability. One way to do so is through the **process capability ratio (PCR)** C_p first introduced in Chapter 6. Recall that

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma} \tag{8.4}$$

where USL and LSL are the upper and lower specification limits, respectively. C_p and other process capability ratios are used extensively in industry. They are also widely *misused*. We will point out some of the more common abuses of process capability ratios. An excellent recent book on process capability ratios that is highly recommended is Kotz and Lovelace (1998). There is also extensive technical literature on process capability analysis and process capability ratios. The review paper by Kotz and Johnson (2002) and the bibliography by Spiring, Leong, Cheng, and Yeung (2003) are excellent sources.

In a practical application, the process standard deviation σ is almost always unknown and must be replaced by an estimate $\hat{\sigma}$. To estimate σ we typically use either the *sample standard deviation s* or $\bar{R}/d_2$ (when variables control charts are used in the capability study). This results in an estimate of C_p—say,

$$\hat{C}_p = \frac{\text{USL} - \text{LSL}}{6\hat{\sigma}} \tag{8.5}$$

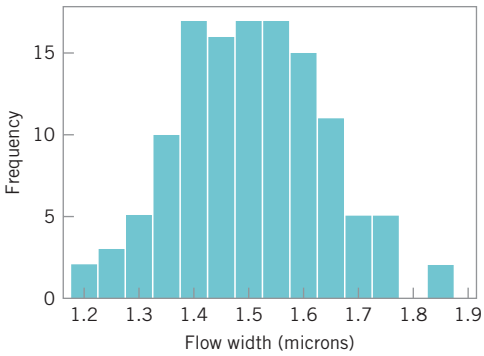
To illustrate the calculation of C_p, recall the semiconductor hard-bake process first analyzed in Example 6.1 using $\bar{x}$ and R charts. The specifications on flow width are USL = 1.00 microns and LSL = 2.00 microns, and from the R chart we estimated $\hat{\sigma} = \bar{R}/d_2 = 0.1398$. Thus, our estimate of the PCR C_p is

$$\hat{C}_p = \frac{\text{USL} - \text{LSL}}{6\hat{\sigma}} = \frac{2.00 - 1.00}{6(0.1398)} = 1.192$$

In Chapter 6, we assumed that flow width is approximately normally distributed (a reasonable assumption, based on the histogram in Fig. 8.7) and the cumulative normal distribution table in the Appendix was used to estimate that the process produces approximately 350 ppm (parts per million) defective.

The PCR C_p in equation (8.4) has a useful practical interpretation—namely,

$$P = \left(\frac{1}{C_p}\right)100 \tag{8.6}$$



■ **FIGURE 8.7** Histogram of flow width from Example 6.1.

is the percentage of the specification band used up by the process. The hard-bake process uses

$$P = \left(\frac{1}{1.192}\right)100 = 83.89$$

percent of the specification band.

Equations (8.4) and (8.5) assume that the process has both upper and lower specification limits. For one-sided specifications, **one-sided process-capability ratios** are used. One-sided PCRs are defined as follows.

$$C_{pu} = \frac{\text{USL} - \mu}{3\sigma} \quad \text{(upper specification only)} \tag{8.7}$$

$$C_{pl} = \frac{\mu - \text{LSL}}{3\sigma} \quad \text{(lower specification only)} \tag{8.8}$$

Estimates $\hat{C}_{pu}$ and $\hat{C}_{pl}$ would be obtained by replacing μ and σ in equations (8.7) and (8.8) by estimates $\hat{\mu}$ and $\hat{\sigma}$, respectively.

EXAMPLE 8.2 One-Sided Process-Capability Ratios

Construct a one-sided process-capability ratio for the container bursting-strength data in Example 8.1. Suppose that the lower specification limit on bursting strength is 200 psi.

SOLUTION

We will use $\bar{x} = 264$ and $s = 32$ as estimates of μ and σ, respectively, and the resulting estimate of the one-sided lower process-capability ratio is

$$\hat{C}_{pl} = \frac{\hat{\mu} - \text{LSL}}{3\hat{\sigma}} = \frac{264 - 200}{3(32)} = 0.67$$

The fraction of defective containers produced by this process is estimated by finding the area to the left of $Z = (\text{LSL} - \mu)/\sigma = (200 - 264)/32 = -2$ under the standard normal distribution. The estimated fallout is about 2.28% defective, or about 22,800 nonconforming containers per million. Note that if the normal distribution were an inappropriate model for strength, then this last calculation would have to be performed using the appropriate probability distribution.

TABLE 8.2

Values of the Process Capability Ratio (C_p) and Associated Process Fallout for a Normally Distributed Process (in Defective ppm) That Is in Statistical Control

	Process Fallout (in defective ppm)	
PCR	**One-Sided Specifications**	**Two-Sided Specifications**
0.25	226,628	453,255
0.50	66,807	133,614
0.60	35,931	71,861
0.70	17,865	35,729
0.80	8,198	16,395
0.90	3,467	6,934
1.00	1,350	2,700
1.10	484	967
1.20	159	318
1.30	48	96
1.40	14	27
1.50	4	7
1.60	1	2
1.70	0.17	0.34
1.80	0.03	0.06
2.00	0.0009	0.0018

The process capability ratio is a measure of the ability of the process to manufacture product that meets the specifications. Table 8.2 presents several values of the PCR C_p along with the associated values of process fallout, expressed in defective parts or nonconforming units of product per million (ppm). To illustrate the use of Table 8.2, notice that a PCR for a normally distributed stable process of $C_p = 1.00$ implies a fallout rate of 2700 ppm for two-sided specifications, whereas a PCR of $C_p = 1.50$ for this process implies a fallout rate of 4 ppm for one-sided specifications.

The ppm quantities in Table 8.2 were calculated using the following **important assumptions:**

1. The quality characteristic has a normal distribution.
2. The process is in statistical control.
3. In the case of two-sided specifications, the process mean is centered between the lower and upper specification limits.

These assumptions are absolutely **critical** to the **accuracy** and **validity** of the reported numbers, and if they are not valid, then the reported quantities may be seriously in error. For example, Somerville and Montgomery (1996) report an extensive investigation of the errors in using the normality assumption to make inferences about the ppm level of a process when in fact the underlying distribution is non-normal. They investigated various non-normal distributions and observed that errors of several orders of magnitude can result in predicting ppm by erroneously making the normality assumption. Even when using a t distribution with as many as 30 degrees of freedom, substantial errors result. Thus even though a t distribution

■ TABLE 8.3
Recommended Minimum Values of the Process Capability Ratio

	Two-Sided Specifications	One-Sided Specifications
Existing processes	1.33	1.25
New processes	1.50	1.45
Safety, strength, or critical parameter, existing process	1.50	1.45
Safety, strength, or critical parameter, new process	1.67	1.60

with 30 degrees of freedom is symmetrical and almost visually indistinguishable from the normal, the longer and heavier tails of the t distribution make a significant difference when estimating the ppm. Consequently, symmetry in the distribution of process output alone is insufficient to ensure that any PCR will provide a reliable prediction of process ppm. We will discuss the non-normality issue in more detail in Section 8.3.3.

Stability or **statistical control** of the process is also **essential** to the correct interpretation of any PCR. Unfortunately, it is fairly common practice to compute a PCR from a sample of historical process data without any consideration of whether or not the process is in statistical control. If the process is not in control, then of course its parameters are unstable, and the value of these parameters in the future is uncertain. Thus the predictive aspects of the PCR regarding process ppm performance are lost.

Finally, remember that what we actually observe in practice is an **estimate** of the PCR. This estimate is subject to error in estimation, since it depends on sample statistics. English and Taylor (1993) report that large errors in estimating PCRs from sample data can occur, so the estimate one actually has at hand may not be very reliable. It is always a good idea to report the estimate of any PCR in terms of a **confidence interval.** We will show how to do this for some of the commonly used PCRs in Section 8.3.5.

Table 8.3 presents some recommended **guidelines for minimum values** of the PCR. The bottle-strength characteristic is a parameter closely related to the safety of the product; bottles with inadequate pressure strength may fail and injure consumers. This implies that the PCR should be at least 1.45. Perhaps one way the PCR could be improved would be by increasing the mean strength of the containers—say, by pouring more glass in the mold.

We point out that the values in Table 8.3 are only **minimums.** In recent years, many companies have adopted criteria for evaluating their processes that include process capability objectives that are more stringent than those of Table 8.3. For example, a six-sigma company would require that when the process mean is in control, it will not be closer than six standard deviations from the nearest specification limit. This, in effect, requires that the minimum acceptable value of the process capability ratio will be at least 2.0.

8.3.2 Process Capability Ratio for an Off-Center Process

The process capability ratio C_p does not take into account *where* the process mean is located relative to the specifications. C_p simply measures the spread of the specifications relative to the six-sigma spread in the process. For example, the top two normal distributions in Fig. 8.8 both have $C_p = 2.0$, but the process in panel (b) of the figure clearly has lower capability than the process in panel (a) because it is not operating at the midpoint of the interval between the specifications.

This situation may be more accurately reflected by defining a new process capability ratio—**PRC C_{pk}**—that takes process centering into account. This quantity is

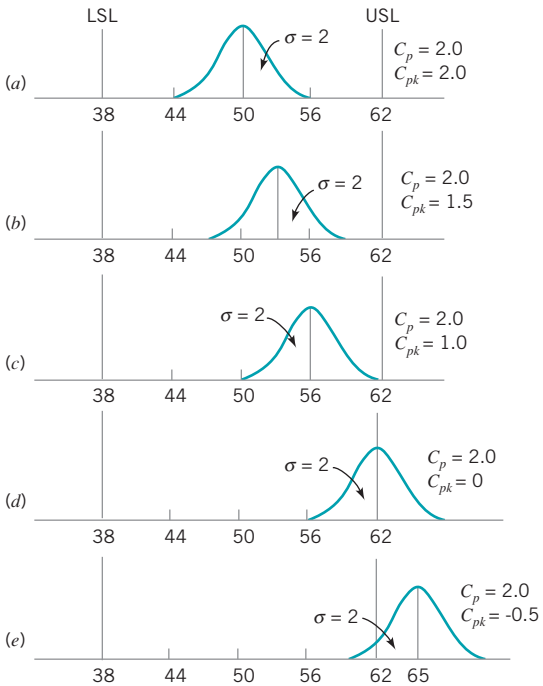
$$C_{pk} = \min\left(C_{pu}, C_{pl}\right) \tag{8.9}$$

Note that C_{pk} is simply the one-sided PCR for the specification limit nearest to the process average. For the process shown in Fig. 8.8*b*, we would have

$$\begin{aligned} C_{pk} &= \min\left(C_{pu}, C_{pl}\right) \\ &= \min\left(C_{pu} = \frac{\text{USL} - \mu}{3\sigma}, C_{pl} = \frac{\mu - \text{LSL}}{3\sigma}\right) \\ &= \min\left(C_{pu} = \frac{62 - 53}{3(2)} = 1.5, C_{pl} = \frac{53 - 38}{3(2)} = 2.5\right) \\ &= 1.5 \end{aligned}$$

Generally, if $C_p = C_{pk}$, the process is centered at the midpoint of the specifications, and when $C_{pk} < C_p$ the process is off-center.

The magnitude of C_{pk} relative to C_p is a direct measure of how off-center the process is operating. Several commonly encountered cases are illustrated in Fig. 8.8. Note in panel (*c*) of Fig. 8.8 that $C_{pk} = 1.0$ while $C_p = 2.0$. One can use Table 8.2 to get a quick estimate of potential improvement that would be possible by centering the process. If we take $C_p = 1.0$ in Table 8.2 and read the fallout from the one-sided specifications column, we can estimate the *actual* fallout as 1350 ppm. However, if we can center the process, then $C_p = 2.0$ can be achieved, and Table 8.2 (using $C_p = 2.0$ and two-sided specifications) suggests that the *potential*



■ **FIGURE 8.8** Relationship of C_p and C_{pk}.

fallout is 0.0018 ppm, an improvement of several orders of magnitude in process performance. Thus, we usually say that C_p measures **potential capability** in the process, whereas C_{pk} measures **actual capability.**

Panel (*d*) of Fig. 8.8 illustrates the case in which the process mean is exactly equal to one of the specification limits, leading to $C_{pk} = 0$. As panel (*e*) illustrates, when $C_{pk} < 0$ the implication is that the process mean lies outside the specifications. Clearly, if $C_{pk} < -1$, the entire process lies outside the specification limits. Some authors define C_{pk} to be nonnegative, so that values less than zero are defined as zero.

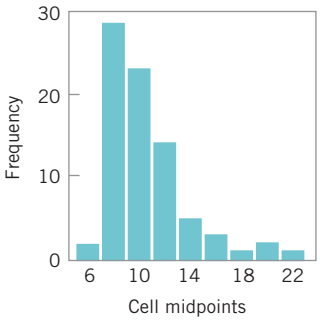
Many quality-engineering authorities have advised against the **routine use** of process capability ratios such as C_p and C_{pk} (or the others discussed later in this section) on the grounds that they are an oversimplification of a complex phenomenon. Certainly, any statistic that combines information about both location (the mean and process centering) and variability and that requires the assumption of normality for its meaningful interpretation is likely to be misused (or abused). Furthermore, as we will see, point estimates of process capability ratios are virtually useless if they are computed from small samples. Clearly, these ratios need to be used and interpreted very carefully.

8.3.3 Normality and the Process Capability Ratio

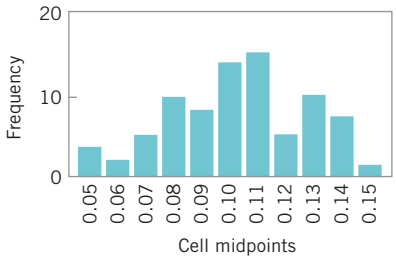
An important assumption underlying our discussion of process capability and the ratios C_p and C_{pk} is that their usual interpretation is based on a **normal distribution** of process output. If the underlying distribution is non-normal, then as we previously cautioned, the statements about expected process fallout attributed to a particular value of C_p or C_{pk} may be in error.

To illustrate this point, consider the data in Fig. 8.9, which is a histogram of 80 measurements of surface roughness on a machined part (measured in microinches). The upper specification limit is at USL = 32 microinches. The sample average and standard deviation are $\bar{x} = 10.44$ and $S = 3.053$, implying that $\hat{C}_{pu} = 2.35$, and Table 8.2 would suggest that the fallout is less than one part per billion. However, since the histogram is highly skewed, we are fairly certain that the distribution is non-normal. Thus, this estimate of capability is unlikely to be correct.

One approach to dealing with this situation is to **transform the data** so that in the new, transformed metric the data have a normal distribution appearance. There are various graphical and analytical approaches to selecting a transformation. In this example, a reciprocal transformation was used. Figure 8.10 presents a histogram of the reciprocal values $x^* = 1/x$. In the transformed scale, $\bar{x}^* = 0.1025$ and $s^* = 0.0244$, and the original upper specification limit becomes $1/32 = 0.03125$. This results in a value of $\hat{C}_{pl} = 0.97$, which implies that about 1350 ppm are outside of specifications. This estimate of process performance is clearly much more realistic than the one resulting from the usual "normal theory" assumption.



■ **FIGURE 8.9** Surface roughness in microinches for a machined part.



■ **FIGURE 8.10** Reciprocals of surface roughness. (Adapted from data in the "Statistics Corner" column in *Quality Progress,* March 1989, with permission of the American Society for Quality Control.)

Other approaches have been considered in dealing with non-normal data. There have been various attempts to extend the definitions of the standard capability indices to the case of non-normal distributions. Luceño (1996) introduced the index C_{pc}, defined as

$$C_{pc} = \frac{\text{USL} - \text{LSL}}{6\sqrt{\frac{\pi}{2}}E|X - T|} \tag{8.10}$$

where the process target value $T = \frac{1}{2}(\text{USL} + \text{LSL})$. Luceño uses the second subscript in C_{pc} to stand for *confidence,* and he stresses that the confidence intervals based on C_{pc} are reliable; of course, this statement should be interpreted cautiously. The author has also used the constant $6\sqrt{\pi/2} = 7.52$ in the denominator, to make it equal to 6σ when the underlying distribution is normal. We will give the confidence interval for C_{pc} in Section 8.3.5.

There have also been attempts to modify the usual capability indices so that they are appropriate for two **general families** of distributions, the Pearson and Johnson families. This would make PCRs broadly applicable for both normal and non-normal distributions. Good discussions of these approaches are in Rodriguez (1992) and Kotz and Lovelace (1998).

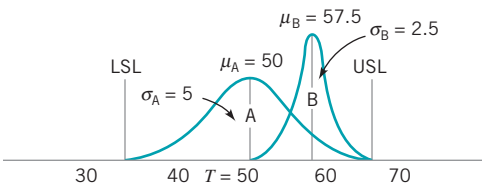
The general idea is to use appropriate **quantiles** of the process distribution—say, $x_{0.00135}$ and $x_{0.99865}$—to define a quantile-based PCR—say,

$$C_p(q) = \frac{\text{USL} - \text{LSL}}{x_{0.99865} - x_{0.00135}} \tag{8.11}$$

Now since in the normal distribution $x_{0.00135} = \mu - 3\sigma$ and $x_{0.99865} = \mu + 3\sigma$, we see that in the case of a normal distribution $C_p(q)$ reduces to C_p. Clements (1989) proposed a method for determining the quantiles based on the Pearson family of distributions. In general, however, we could fit *any* distribution to the process data, determine its quantiles $x_{0.99865}$ and $x_{0.00135}$, and apply equation (8.11). Refer to Kotz and Lovelace (1998) for more information.

8.3.4 More about Process Centering

The process capability ratio C_{pk} was initially developed because C_p does not adequately deal with the case of a process with mean μ that is not centered between the specification limits. However, C_{pk} alone is still an inadequate measure of process centering. For example, consider the two processes shown in Fig. 8.11. Both processes A and B have $C_{pk} = 1.0$, yet their centering is clearly different. To characterize process centering satisfactorily, C_{pk} must be compared to C_p. For process A, $C_{pk} = C_p = 1.0$, implying that the process is centered, whereas for process B, $C_p = 2.0 > C_{pk} = 1.0$, implying that the process is off-center. For any fixed value of μ in the interval from LSL to USL, C_{pk} depends inversely on σ and becomes large as σ



■ **FIGURE 8.11** Two processes with $C_{pk} = 1.0$.

approaches zero. This characteristic can make C_{pk} unsuitable as a measure of centering. That is, a large value of C_{pk} does not really tell us anything about the location of the mean in the interval from LSL to USL.

One way to address this difficulty is to use a process capability ratio that is a better indicator of centering. PCR $\boldsymbol{C_{pm}}$ is one such ratio,

$$C_{pm} = \frac{\text{USL} - \text{LSL}}{6\tau} \tag{8.12}$$

and τ is the square root of expected squared deviation from target $T = \frac{1}{2}(\text{USL} + \text{LSL})$,

$$\begin{aligned}\tau^2 &= E\left[(x-T)^2\right] \\ &= E\left[(x-\mu)^2\right] + (\mu - T)^2 \\ &= \sigma^2 + (\mu - T)^2\end{aligned}$$

Thus, equation (8.12) can be written as

$$C_{pm} = \frac{\text{USL} - \text{LSL}}{6\sqrt{\sigma^2 + (\mu - T)^2}} = \frac{C_p}{\sqrt{1+\xi^2}} \tag{8.13}$$

where

$$\xi = \frac{\mu - T}{\sigma} \tag{8.14}$$

A logical way to estimate C_{pm} is by

$$\hat{C}_{pm} = \frac{\hat{C}_p}{\sqrt{1+V^2}} \tag{8.15}$$

where

$$V = \frac{\bar{x} - T}{s} \tag{8.16}$$

Chan et al. (1988) discussed this ratio, various estimators of C_{pm}, and their sampling properties. Boyles (1991) has provided a definitive analysis of C_{pm} and its usefulness in measuring process centering. He notes that both C_{pk} and C_{pm} coincide with C_p when $\mu = T$ and decrease as μ moves away from T. However, $C_{pk} < 0$ for $\mu >$ USL or $\mu <$ LSL, whereas C_{pm} approaches zero asymptotically as $|(\mu - T)| \to \infty$. Boyles also shows that the C_{pm} of a process with $|(\mu - T)| = \Delta > 0$ is strictly bounded above by the C_p value of a process with $\sigma = \Delta$. That is,

$$C_{pm} < \frac{\text{USL} - \text{LSL}}{6|\mu - T|} \tag{8.17}$$

Thus, a necessary condition for $C_{pm} \geq 1$ is

$$|\mu - T| < \tfrac{1}{6}(\text{USL} - \text{LSL})$$

This statistic says that if the target value T is the midpoint of the specifications, a C_{pm} of one or greater implies that the mean μ lies within the middle third of the specification band. A similar statement can be made for any value of C_{pm}. For instance, $C_{pm} \geq \frac{4}{3}$ implies that $|(\mu - T)| < \frac{1}{8}(\text{USL} - \text{LSL})$. Thus, a given value of C_{pm} places a constraint on the difference between μ and the target value T.

EXAMPLE 8.3 Process Centering

To illustrate the use of C_{pm}, consider the two processes A and B in Fig. 8.11. For process A we find that

$$C_{pm} = \frac{C_p}{\sqrt{1+\xi^2}} = \frac{1.0}{\sqrt{1+0}} = 1.0$$

since process A is centered at the target value $T = 50$. Note that $C_{pm} = C_{pk}$ for process A. Now consider process B:

$$C_{pm} = \frac{C_p}{\sqrt{1+\xi^2}} = \frac{2}{\sqrt{1+(3)^2}} = 0.63$$

If we use equation (8.17), this is equivalent to saying that the process mean lies approximately within the middle half of the specification range. Visual examination of Fig. 8.11 reveals this to be the case.

Pearn et al. (1992) proposed the process capability ratio

$$C_{pkm} = \frac{C_{pk}}{\sqrt{1+\left(\dfrac{\mu - T}{\sigma}\right)^2}} = \frac{C_{pk}}{\sqrt{1+\xi^2}} \tag{8.18}$$

This is sometimes called a "third generation" process capability ratio, since it is constructed from the "second generation" ratios C_{pk} and C_{pm} in the same way that they were generated from the "first generation" ratio C_p. The motivation of this new ratio is increased sensitivity to departures of the process mean μ from the desired target T. For more details, see Kotz and Lovelace (1998).

8.3.5 Confidence Intervals and Tests on Process Capability Ratios

Confidence Intervals on Process Capability Ratios. Much of the industrial use of process capability ratios focuses on computing and interpreting the **point estimate** of the desired quantity. It is easy to forget that $\hat{C}_p$ or $\hat{C}_{pk}$ (for examples) are simply point estimates, and, as such, are subject to statistical fluctuation. An alternative that should become standard practice is to report **confidence intervals for process capability ratios.**

It is easy to find a confidence interval for the "first generation" ratio C_p. If we replace σ by s in the equation for C_p, we produce the usual point estimator $\hat{C}_p$. If the quality characteristic follows a normal distribution, then a $100(1-\alpha)\%$ confidence interval on C_p is obtained from

$$\frac{\text{USL} - \text{LSL}}{6s}\sqrt{\frac{\chi^2_{1-\alpha/2,n-1}}{n-1}} \le C_p \le \frac{\text{USL} - \text{LSL}}{6s}\sqrt{\frac{\chi^2_{\alpha/2,n-1}}{n-1}} \tag{8.19}$$

or

$$\hat{C}_p\sqrt{\frac{\chi^2_{1-\alpha/2,n-1}}{n-1}} \le C_p \le \hat{C}_p\sqrt{\frac{\chi^2_{\alpha/2,n-1}}{n-1}} \tag{8.20}$$

where $\chi^2_{1-\alpha/2,n-1}$ and $\chi^2_{\alpha/2,n-1}$ are the lower $\alpha/2$ and upper $\alpha/2$ percentage points of the chi-square distribution with $n-1$ degrees of freedom. These percentage points are tabulated in Appendix Table III.

EXAMPLE 8.4 A Confidence Interval in C_p

Suppose that a stable process has upper and lower specifications at USL = 62 and LSL = 38. A sample of size $n = 20$ from this process reveals that the process mean is centered approximately at the midpoint of the specification interval and that the sample standard deviation $s = 1.75$. Find a 95% confidence interval on C_p.

SOLUTION

A point estimate of C_p is

$$\hat{C}_p = \frac{\text{USL} - \text{LSL}}{6s} = \frac{62 - 38}{6(1.75)} = 2.29$$

The 95% confidence interval on C_p is found from equation (8.20) as follows:

$$\hat{C}_p\sqrt{\frac{\chi^2_{1-0.025,n-1}}{n-1}} \le C_p \le \hat{C}_p\sqrt{\frac{\chi^2_{0.025,n-1}}{n-1}}$$

$$2.29\sqrt{\frac{8.91}{19}} \le C_p \le 2.29\sqrt{\frac{32.85}{19}}$$

$$1.57 \le C_p \le 3.01$$

where $\chi^2_{0.975,19} = 8.91$ and $\chi^2_{0.025,19} = 32.85$ were taken from Appendix Table III.

The confidence interval on C_p in Example 8.4 is relatively wide because the sample standard deviation s exhibits considerable fluctuation in small to moderately large samples. This means, in effect, that confidence intervals on C_p based on small samples will be wide.

Note also that the confidence interval uses s rather than $\bar{R}/d_2$ to estimate σ. This further emphasizes that the process must be **in statistical control** for PCRs to have any real meaning. If the process is not in control, s and $\bar{R}/d_2$ could be very different, leading to very different values of the PCR.

For more complicated ratios such as C_{pk} and C_{pm}, various authors have developed approximate confidence intervals; for example, see Zhang, Stenbeck, and Wardrop (1990), Bissell (1990), Kushler and Hurley (1992), and Pearn et al. (1992). If the quality characteristic is normally distributed, then an approximate $100(1 - \alpha)\%$ confidence interval on C_{pk} is given as follows.

$$\hat{C}_{pk}\left[1 - Z_{\alpha/2}\sqrt{\frac{1}{9n\hat{C}^2_{pk}} + \frac{1}{2(n-1)}}\right] \le C_{pk} \le \hat{C}_{pk}\left[1 + Z_{\alpha/2}\sqrt{\frac{1}{9n\hat{C}^2_{pk}} + \frac{1}{2(n-1)}}\right] \tag{8.21}$$

Kotz and Lovelace (1998) give an extensive summary of confidence intervals for various PCRs.

EXAMPLE 8.5 A Confidence Interval in C_{pk}

A sample of size $n = 20$ from a stable process is used to estimate C_{pk}, with the result that $\hat{C}_{pk} = 1.33$. Find an approximate 95% confidence interval on C_{pk}.

SOLUTION

Using equation (8.21), an approximate 95% confidence interval on C_{pk} is

$$\hat{C}_{pk}\left[1 - Z_{\alpha/2}\sqrt{\frac{1}{9n\hat{C}_{pk}^2} + \frac{1}{2(n-1)}}\right] \le C_{pk} \le \hat{C}_{pk}\left[1 + Z_{\alpha/2}\sqrt{\frac{1}{9n\hat{C}_{pk}^2} + \frac{1}{2(n-1)}}\right]$$

$$1.33\left[1 - 1.96\sqrt{\frac{1}{9(20)(1.33)^2} + \frac{1}{2(19)}}\right] \le C_{pk} \le 1.33\left[1 + 1.96\sqrt{\frac{1}{9(20)(1.33)^2} + \frac{1}{2(19)}}\right]$$

or

$$0.88 \le C_{pk} \le 1.78$$

This is an extremely wide confidence interval. Based on the sample data, the ratio C_{pk} could be less than 1 (a very bad situation), or it could be as large as 1.78 (a very good situation). Thus, we have learned very little about actual process capability, because C_{pk} is very imprecisely estimated. The reason for this, of course, is that a very small sample ($n = 20$) has been used.

For non-normal data, the PCR C_{pc} developed by Luceño (1996) can be employed. Recall that C_{pc} was defined in equation (8.10). Luceño developed the confidence interval for C_{pc} as follows: First, evaluate $|(x - T)|$, whose expected value is estimated by

$$\bar{c} = \frac{1}{n}\sum_{i=1}^{n}|x_i - T|$$

leading to the estimator

$$\hat{C}_{pc} = \frac{\text{USL} - \text{LSL}}{6\sqrt{\frac{\pi}{2}}c}$$

A $100(1 - \alpha)\%$ confidence interval for $E|(x - T)|$ is given as

$$\bar{c} \pm t_{\alpha/2,n-1}\frac{s_c}{\sqrt{n}}$$

where

$$s_c^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(|x_i - T| - \bar{c}\right)^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n}|x_i - T|^2 - n\bar{c}^2\right)$$

Therefore, a $100(1-\alpha)\%$ confidence interval for C_{pc} is given by

$$\frac{\hat{C}_{pc}}{1+t_{\alpha/2,n-1}\left[s_c/\left(\bar{c}\sqrt{n}\right)\right]} \le C_{pc} \le \frac{\hat{C}_{pc}}{1-t_{\alpha/2,n-1}\left[s_c/\left(\bar{c}\sqrt{n}\right)\right]} \tag{8.22}$$

Testing Hypotheses about PCRs. A practice that is becoming increasingly common in industry is to require a supplier to **demonstrate** process capability as part of the contractual agreement. Thus, it is frequently necessary to demonstrate that the process capability ratio C_p meets or exceeds some particular target value—say, C_{p0}. This problem may be formulated as a hypothesis testing problem

$$H_0: \quad C_p = C_{p0} \text{ (or the process is not capable)}$$
$$H_1: \quad C_p \ge C_{p0} \text{ (or the process is capable)}$$

We would like to reject H_0 (recall that in statistical hypothesis testing rejection of H_0 is always a strong conclusion), thereby demonstrating that the process is capable. We can formulate the statistical test in terms of $\hat{C}_p$, so that we will reject H_0 if $\hat{C}_p$ exceeds a critical value C.

Kane (1986) has investigated this test, and provides a table of sample sizes and critical values for C to assist in testing process capability. We may define C_p(High) as a process capability that we would like to accept with probability $1-\alpha$ and C_p(Low) as a process capability that we would like to reject with probability $1-\beta$. Table 8.4 gives values of C_p(High)/C_p(Low) and C/C_p(Low) for varying sample sizes and $\alpha=\beta=0.05$ or $\alpha=\beta=0.10$. Example 8.6 illustrates the use of this table.

EXAMPLE 8.6 Supplier Qualification

A customer has told his supplier that, in order to qualify for business with his company, the supplier must demonstrate that his process capability exceeds $C_p = 1.33$. Thus, the supplier is interested in establishing a procedure to test the hypotheses

$$H_0: \quad C_p = 1.33$$
$$H_1: \quad C_p > 1.33$$

The supplier wants to be sure that if the process capability is below 1.33 there will be a high probability of detecting this (say, 0.90), whereas if the process capability exceeds 1.66 there will be a high probability of judging the process capable (again, say, 0.90). This would imply that C_p(Low) = 1.33, C_p(High) = 1.66, and $\alpha=\beta=0.10$. To find the sample size and critical value for C from Table 8.4, compute

$$\frac{C_p(\text{High})}{C_p(\text{Low})} = \frac{1.66}{1.33} = 1.25$$

■ **TABLE 8.4**
Sample Size and Critical Value Determination for Testing H_0: $C_p = C_{p0}$

	(a) $\alpha=\beta=0.10$		(b) $\alpha=\beta=0.05$	
Sample Size, n	**C_p(High)/C_p(Low)**	**C/C_p(Low)**	**C_p(High)/C_p(Low)**	**C/C_p(Low)**
10	1.88	1.27	2.26	1.37
20	1.53	1.20	1.73	1.26
30	1.41	1.16	1.55	1.21
40	1.34	1.14	1.46	1.18
50	1.30	1.13	1.40	1.16
60	1.27	1.11	1.36	1.15
70	1.25	1.10	1.33	1.14
80	1.23	1.10	1.30	1.13
90	1.21	1.10	1.28	1.12
100	1.20	1.09	1.26	1.11

Source: Adapted from Kane (1986), with permission of the American Society for Quality Control.

and enter the table value in panel (*a*) where $\alpha = \beta = 0.10$. This yields

$$n = 70$$

and

$$C/C_p(\text{Low}) = 1.10$$

from which we calculate

$$C = C_p(\text{Low})1.10 = 1.33(1.10) = 1.46$$

Thus, to demonstrate capability, the supplier must take a sample of $n = 70$ parts, and the sample process capability ratio $\hat{C}_p$ must exceed $C \le 1.46$.

This example shows that, in order to demonstrate that process capability is at least equal to 1.33, the *observed sample* $\hat{C}_p$ will have to exceed 1.33 by a considerable amount. This illustrates that some common industrial practices may be questionable statistically. For example, it is fairly common practice to accept the process as capable at the level $C_p \ge 1.33$ if the *sample* $\hat{C}_p \ge 1.33$ based on a sample size of $30 \le n \le 50$ parts. Clearly, this procedure does not account for sampling variation in the estimate of σ, and larger values of n and/or higher acceptable values of $\hat{C}_p$ may be necessary in practice.

Process Performance Indices. In 1991, the Automotive Industry Action Group (AIAG) was formed and consists of representatives of the "big three" (Ford, General Motors, and Chrysler) and the American Society for Quality Control (now the American Society for Quality). One of their objectives was to standardize the reporting requirements from suppliers and in general of their industry. The AIAG recommends using the process capability indices C_p and C_{pk} when the process is in control, with the process standard deviation estimated by $\hat{\sigma} = \bar{R}/d_2$. When the process is *not* in control, the AIAG recommends using **process performance indices P_p and P_{pk},** where, for example,

$$\hat{P}_p = \frac{\text{USL} - \text{LSL}}{6s}$$

and s is the usual sample standard deviation $s = \sqrt{\Sigma_{i=1}^{n}(x_i - \bar{x})^2/(n-1)}$. Even the American National Standards Institute in ANSI Standard Z1 on Process Capability Analysis (1996) states that P_p and P_{pk} should be used when the process is not in control.

Now it is clear that when the process is normally distributed and in control, $\hat{P}_p$ is essentially $\hat{C}_p$ and $\hat{P}_{pk}$ is essentially $\hat{C}_{pk}$ because for a stable process the difference between s and $\hat{\sigma} = \bar{R}/d_2$ is minimal. However, please note that if the process is **not** in control, the indices P_p and P_{pk} have no meaningful interpretation relative to process capability, because they cannot **predict** process performance. Furthermore, their statistical properties are not determinable, and so no valid inference can be made regarding their true (or population) values. Also, P_p and P_{pk} provide no motivation or incentive to the companies that use them to bring their processes into control.

Kotz and Lovelace (1998) strongly recommend **against** the use of P_p and P_{pk}, indicating that these indices are actually a step **backward** in quantifying process capability. They refer to the mandated use of P_p and P_{pk} through quality standards or industry guidelines as undiluted "statistical terrorism" (i.e., the use or misuse of statistical methods along with threats and/or intimidation to achieve a business objective).

This author agrees completely with Kotz and Lovelace. The process performance indices P_p and P_{pk} are actually more than a step backward. **They are a waste of engineering and management effort—they tell you nothing.** Unless the process is stable (in control), no index is going to carry useful predictive information about process capability or convey any

information about future performance. Instead of imposing the use of meaningless indices, organizations should devote effort to developing and implementing an effective process characterization, control, and improvement plan. This is a much more reasonable and effective approach to process improvement.

8.4 Process Capability Analysis Using a Control Chart

Histograms, probability plots, and process capability ratios summarize the performance of the process. They do not necessarily display the **potential capability** of the process because they do not address the issue of **statistical control,** or show systematic patterns in process output that, if eliminated, would reduce the variability in the quality characteristic. Control charts are very effective in this regard. The control chart should be regarded as the primary technique of process capability analysis.

Both attributes and variables control charts can be used in process capability analysis. The $\bar{x}$ and R charts should be used whenever possible, because of the greater power and better information they provide relative to attributes charts. However, both p charts and c (or u) charts are useful in analyzing process capability. Techniques for constructing and using these charts are given in Chapters 6 and 7. Remember that to use the p chart there must be specifications on the product characteristics. The $\bar{x}$ and R charts allow us to study processes without regard to specifications.

The $\bar{x}$ and R control charts allow both the instantaneous variability (short-term process capability) and variability across time (long-term process capability) to be analyzed. It is particularly helpful if the data for a process capability study are collected in two to three different time periods (such as different shifts, different days, etc.).

Table 8.5 presents the container bursting-strength data in 20 samples of five observations each. The calculations for the $\bar{x}$ and R charts are summarized here:

$\bar{x}$ Chart

$$\text{Center line} = \bar{\bar{x}} = 264.06$$
$$\text{UCL} = \bar{\bar{x}} + A_2\bar{R} = 264.06 + (0.577)(77.3) = 308.66$$
$$\text{LCL} = \bar{\bar{x}} - A_2\bar{R} = 264.06 - (0.577)(77.3) = 219.46$$

R Chart

$$\text{Center line} = \bar{R} = 77.3$$
$$\text{UCL} = D_4\bar{R} = (2.115)(77.3) = 163.49$$
$$\text{LCL} = D_3\bar{R} = (0)(77.3) = 0$$

Figure 8.12 presents the $\bar{x}$ and R charts for the 20 samples in Table 8.5. Both charts exhibit statistical control. The process parameters may be estimated from the control chart as

$$\hat{\mu} = \bar{\bar{x}} = 264.06$$
$$\hat{\sigma} = \frac{\bar{R}}{d_2} = \frac{77.3}{2.326} = 33.23$$

TABLE 8.5
Glass Container Strength Data (psi)

Sample	Data					$\bar{x}$	R
1	265	205	263	307	220	252.0	102
2	268	260	234	299	215	255.2	84
3	197	286	274	243	231	246.2	89
4	267	281	265	214	318	269.0	104
5	346	317	242	258	276	287.8	104
6	300	208	187	264	271	246.0	113
7	280	242	260	321	228	266.2	93
8	250	299	258	267	293	273.4	49
9	265	254	281	294	223	263.4	71
10	260	308	235	283	277	272.6	73
11	200	235	246	328	296	261.0	128
12	276	264	269	235	290	266.8	55
13	221	176	248	263	231	227.8	87
14	334	280	265	272	283	286.8	69
15	265	262	271	245	301	268.8	56
16	280	274	253	287	258	270.4	34
17	261	248	260	274	337	276.0	89
18	250	278	254	274	275	266.2	28
19	278	250	265	270	298	272.2	48
20	257	210	280	269	251	253.4	70
						$\bar{\bar{x}} = 264.06$	$\bar{R} = 77.3$

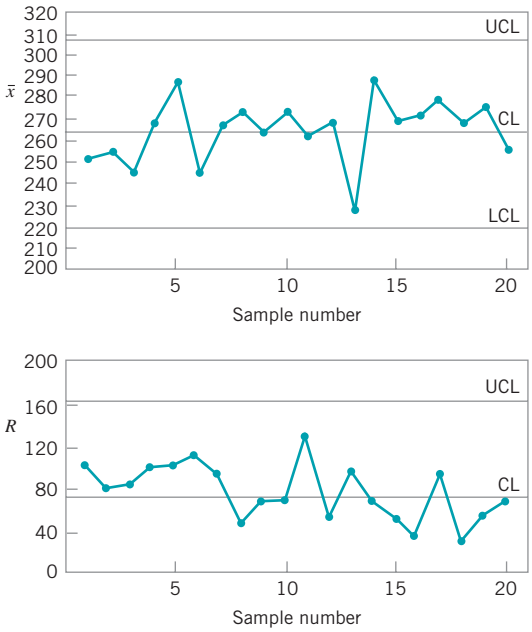


FIGURE 8.12 $\bar{x}$ and R charts for the bottle-strength data.

Thus, the one-sided lower process capability ratio is estimated by

$$\hat{C}_{pl} = \frac{\mu - \text{LSL}}{3\hat{\sigma}} = \frac{264.06 - 200}{3(33.23)} = 0.64$$

Clearly, since strength is a safety-related parameter, the process capability is inadequate.

This example illustrates a process that is in control but operating at an unacceptable level. There is no evidence to indicate that the production of nonconforming units is **operator-controllable.** Engineering and/or management intervention will be required either to improve the process or to change the requirements if the quality problems with the bottles are to be solved. The objective of these interventions is to increase the process capability ratio to at least a minimum acceptable level. The control chart can be used as a monitoring device or logbook to show the effect of changes in the process on process performance.

Sometimes the process capability analysis indicates an out-of-control process. It is **unsafe** to estimate process capability in such cases. The process must be stable in order to produce a reliable estimate of process capability. When the process is out of control in the early stages of process capability analysis, the first objective is finding and eliminating the assignable causes in order to bring the process into an in-control state.

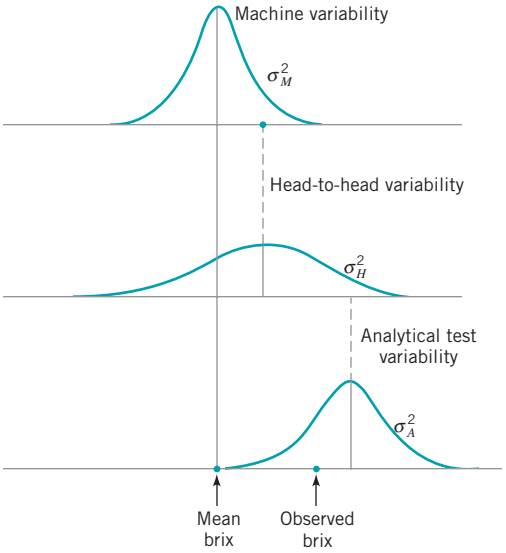
8.5 Process Capability Analysis Using Designed Experiments

A designed experiment is a systematic approach to varying the input **controllable** variables in the process and analyzing the effects of these process variables on the output. Designed experiments are also useful in discovering *which* set of process variables is influential on the output, and at what levels these variables should be held to optimize process performance. Thus, design of experiments is useful in more general problems than merely estimating process capability. For an introduction to design of experiments, see Montgomery (2005). Part V of this textbook provides more information on experimental design methods and on their use in process improvement.

One of the major uses of designed experiments is in isolating and estimating the **sources of variability** in a process. For example, consider a machine that fills bottles with a soft-drink beverage. Each machine has a large number of filling heads that must be independently adjusted. The quality characteristic measured is the syrup content (in degrees brix) of the finished product. There can be variation in the observed brix (σ_B^2) because of machine variability (σ_M^2), head variability (σ_H^2), and analytical test variability (σ_A^2). The variability in the observed brix value is

$$\sigma_B^2 = \sigma_M^2 + \sigma_H^2 + \sigma_A^2$$

An experiment can be designed, involving sampling from several machines and several heads on each machine, and making several analyses on each bottle, which would allow estimation of the variances (σ_M^2), (σ_H^2), and (σ_A^2). Suppose that the results appear as in Fig. 8.13. Since a substantial portion of the total variability in observed brix is due to variability between heads, this indicates that the process can perhaps best be improved by reducing the head-to-head variability. This could be done by more careful setup or by more careful control of the operation of the machine.



■ **FIGURE 8.13** Sources of variability in the bottling line example.

8.6 Process Capability Analysis with Attribute Data

Often process performance is measured in terms of **attribute** data; that is, nonconforming units or defectives, or nonconformities or defects. When a fraction nonconforming is the measure of performance, it is typical to use the parts per million (ppm) defective as a measure of process capability. In some organizations, this ppm defective is converted to an equivalent sigma level. For example, a process producing 2700 ppm defective would be equivalent to a three-sigma process (without the "usual" 1.5 σ shift in the mean that many six-sigma organizations employ in the calculations taken into account).

When dealing with nonconformities or defects, a defects per unit (DPU) statistic is often used as a measure of capability, where

$$\text{DPU} = \frac{\text{Total number of defects}}{\text{Total number of units}}$$

Here the unit is something that is delivered to a customer and can be evaluated or judged as to its suitability. Some examples include:

1. An invoice
2. A shipment
3. A customer order
4. An enquiry or call

The defects or nonconformities are anything that does not meet the customer requirements, such as:

1. An error on an invoice
2. An incorrect or incomplete shipment
3. An incorrect or incomplete customer order
4. A call that is not satisfactorily completed

Obviously, these quantities are estimated from sample data. Large samples need to be used to obtain reliable estimates.

The DPU measure does not directly take the complexity of the unit into account. A widely used way to do this is the defect per million opportunities (DPMO) measure

$$\text{DPMO} = \frac{\text{Total number of defects}}{\text{Number of units} \times \text{Number of opportunities}}$$

Opportunities are the number of potential chances within a unit for a defect to occur. For example, on a purchase order, the number of opportunities would be the number of fields in which information is recorded times two, because each field can either be filled out incorrectly or blank (information is missing). It is important to be consistent about how opportunities are defined, as a process may be artificially improved simply by increasing the number of opportunities over time.

8.7 Gauge and Measurement System Capability Studies

8.7.1 Basic Concepts of Gauge Capability

Determining the capability of the measurement system is an important aspect of many quality and process improvement activities. Generally, in any activity involving measurements, some of the observed variability will be inherent in the units or items that are being measured, and some of the variability will result from the measurement system that is used. The measurement system will consist (minimally) of an instrument or gauge, and it often has other components, such as the operator(s) that uses it and the conditions or different points in time under which the instrument is used. There may also be other factors that impact measurement system performance, such as setup or calibration activities. The purpose of most measurement systems capability studies is to:

1. Determine how much of the total observed variability is due to the gauge or instrument
2. Isolate the components of variability in the measurement system
3. Assess whether the instrument or gauge is capable (that is, is it suitable for the intended application)

Measurements are a significant component of any quality system. Measurement is an integral component of the DMAIC problem-solving process, but it's even more important than that. An ineffective measurement system can dramatically impact business performance because it leads to uninformed (and usually bad) decision making.

In this section we will introduce the two R's of measurement systems capability: **repeatability** (do we get the same observed value if we measure the same unit several times under identical conditions), and **reproducibility** (how much difference in observed values do we experience when units are measured under different conditions, such as different operators, time periods, and so forth).

These quantities answer only indirectly the fundamental question: Is the system able to distinguish between good and bad units? That is, what is the probability that a good unit is judged to be defective and, conversely, that a bad unit is passed along to the customer as good? These misclassification probabilities are fairly easy to calculate from the results of a standard measurement systems capability study, and give reliable, useful, and easy-to-understand information about measurement systems performance.

In addition to repeatability and reproducibility, there are other important aspects of measurement systems capability. The **linearity** of a measurement system reflects the differences in observed accuracy and/or precision experienced over the range of measurements made by the system. A simple linear regression model is often used to describe this feature.

10.2 Modified and Acceptance Control Charts

In most situations in which control charts are used, the focus is on statistical monitoring or control of the process, reduction of variability, and continuous process improvement. When a high level of process capability has been achieved, it is sometimes useful to relax the level of surveillance provided by the standard control chart. One method for doing this with $\bar{x}$ charts uses **modified** (or **reject**) **control limits,** and the second uses the **acceptance control chart.**

10.2.1 Modified Control Limits for the $\bar{x}$ Chart

Modified control limits are generally used in situations where the natural variability or "spread" of the process is considerably smaller than the spread in the specification limits; that is, C_p or C_{pk} is much greater than 1. This situation occurs occasionally in practice. In fact, this should be the natural eventual result of a successful variability reduction effort—reduction of process variability with a corresponding increase in the process capability ratio. The six-sigma approach to variability reduction focuses on improving processes until the minimum value of C_{pk} is 2.0.

Suppose, for example, that the specification limits on the fill volume of a carbonated beverage container are LSL = 10.00 oz and USL = 10.20 oz, but as the result of a program of engineering and operating refinements, the filling machine can operate with a standard deviation of fill volume of approximately $\sigma = 0.01$ oz. Therefore, the distance USL − LSL is approximately twenty-sigma, or much greater than the six-sigma natural tolerance limits on the process, and the process capability ratio is PCR = (USL − LSL)/6σ = 0.20/[6(0.01)] = 3.33. This is clearly a six-sigma process.

In situations where six-sigma is much smaller than the spread in the specifications (USL–LSL), the process mean can sometimes be allowed to vary over an interval without appreciably affecting the overall performance of the process.[1] For example, see Fig. 10.2. When this situation occurs, we can use a **modified control chart** for $\bar{x}$ instead of the usual $\bar{x}$ chart. The modified $\bar{x}$ control chart is concerned only with detecting whether the true process mean μ is located such that the process is producing a fraction nonconforming in excess of some specified value δ. In effect, μ is allowed to vary over an interval—say, $\mu_L \le \mu \le \mu_U$—where μ_L and μ_U are chosen as the smallest and largest permissible values of μ, respectively, consistent with producing a fraction nonconforming of at most δ. We will assume that the process variability σ is in control. Good general discussions of the modified control chart are in Hill (1956) and Duncan (1986). As noted in Duncan (1986), the procedure is sometimes used when a process is subject to tool wear (see Section 10.7.1).

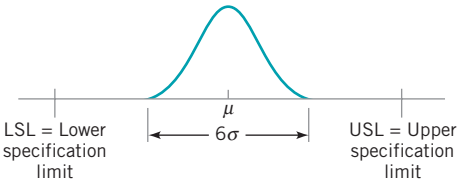
To specify the control limits for a modified $\bar{x}$ chart, we will assume that the process output is normally distributed. For the process fraction nonconforming to be less than δ, we must require that the true process mean is in the interval. Consequently, we see from Fig. 10.3*a* that we must have

$$\mu_L = \text{LSL} + Z_\delta \sigma$$

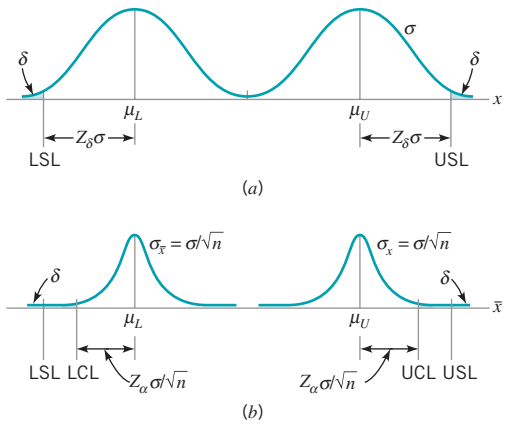
and

$$\mu_U = \text{USL} - Z_\delta \sigma$$

[1]In the original Motorola definition of a six-sigma process, the process mean was assumed to drift about, wandering as far as 1.5σ from the desired target. If this is the actual behavior of the process and this type of behavior is acceptable, then modified control charts are a useful alternative to standard charts. There are also many cases where the mean should not be allowed to vary even if C_p or C_{pk} is large. The conventional control charts should be used in such situations.



■ **FIGURE 10.2** A process with the spread of the natural tolerance limits less than the spread of the specification limits, or $6\sigma < \text{USL} - \text{LSL}$.



■ **FIGURE 10.3** Control limits on a modified control chart. (*a*) Distribution of process output. (*b*) Distribution of the sample mean $\bar{x}$.

where Z_δ is the upper $100(1-\delta)$ percentage point of the standard normal distribution. Now if we specify a type I error of α, the upper and lower control limits are

$$\begin{aligned}\text{UCL} &= \mu_U + \frac{Z_\alpha \sigma}{\sqrt{n}} \\ &= \text{USL} - Z_\delta \sigma + \frac{Z_\alpha \sigma}{\sqrt{n}} \\ &= \text{USL} - \left(Z_\delta - \frac{Z_\alpha}{\sqrt{n}} \right) \sigma \end{aligned} \tag{10.3a}$$

and

$$\begin{aligned}\text{LCL} &= \mu_L - \frac{Z_\alpha \sigma}{\sqrt{n}} \\ &= \text{LSL} + Z_\delta \sigma - \frac{Z_\alpha \sigma}{\sqrt{n}} \\ &= \text{LSL} + \left(Z_\delta - \frac{Z_\alpha}{\sqrt{n}} \right) \sigma \end{aligned} \tag{10.3b}$$

respectively. The control limits are shown on the distribution of $\bar{x}$ in Fig. 10.3*b*. Instead of specifying a type I error, one may use the following:

$$\text{UCL} = \text{USL} - \left(Z_\delta - \frac{3}{\sqrt{n}} \right) \sigma \tag{10.4a}$$

and

$$\text{LCL} = \text{LSL} + \left(Z_\delta - \frac{3}{\sqrt{n}} \right) \sigma \tag{10.4b}$$

Two-sigma limits are sometimes recommended for the modified control chart, based on an argument that the tighter control limits afford better protection (smaller β-risk) against critical shifts in the mean at little loss in the α-risk. A discussion of this subject is in Freund (1957). Note that the modified control chart is equivalent to testing the hypothesis that the process mean lies in the interval $\mu_L \le \mu \le \mu_U$.

To design a modified control chart, we must have a good estimate of σ available. If the process variability shifts, then the modified control limits are not appropriate. Consequently, an R or an s chart should always be used in conjunction with the modified control chart. Furthermore, the initial estimate of σ required to set up the modified control chart would usually be obtained from an R or an s chart.

EXAMPLE 10.1 A Control Chart for a Six-Sigma Process

Consider a normally distributed process with a target value of the mean of 20 and standard deviation $\sigma = 2$. The upper and lower process specifications are at LSL = 8 and USL = 32, so that if the process is centered at the target, $C_p = C_{pk} = 2.0$. This is a process with six-sigma capability. In a six-sigma process it is assumed that the mean may drift as much as 1.5 standard deviations off target without causing serious problems. Set up a control chart for monitoring the mean of this six-sigma process with a sample size of $n = 4$.

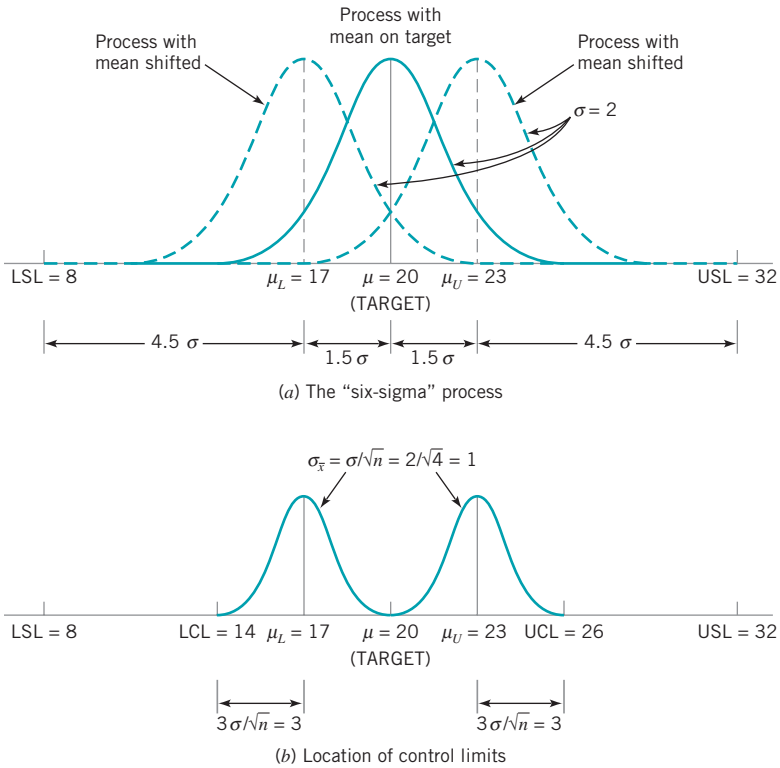
SOLUTION

Figure 10.4*a* shows this six-sigma process and Fig. 10.4*b* illustrates the control limit calculation. Notice that we can use equation (10.4) with Z_δ replaced by 4.5. Therefore, the upper and lower control limits become

$$\begin{aligned}\text{UCL} &= \text{USL} - \left(4.5 - \frac{3}{\sqrt{4}}\right)\sigma \\ &= 32 - (4.5 - 1.5)2 \\ &= 26\end{aligned}$$

and

$$\begin{aligned}\text{LCL} &= \text{LSL} + \left(4.5 - \frac{3}{\sqrt{4}}\right)\sigma \\ &= 8 + (4.5 - 1.5)2 \\ &= 14\end{aligned}$$



■ **FIGURE 10.4** Control limits for a six-sigma process.

10.2.2 Acceptance Control Charts

The second approach to using an $\bar{x}$ chart to monitor the fraction of nonconforming units, or the fraction of units exceeding specifications, is called the acceptance control chart. In the modified control chart design of Section 10.2.1, the chart was based on a specified sample size n, a process fraction nonconforming δ, and type I error probability α. Thus, we interpret δ as a process fraction nonconforming that we will accept with probability $1 - \alpha$. Freund (1957) developed the **acceptance control chart** to take into account both the risk of rejecting a process operating at a satisfactory level (type I error or α-risk) and the risk of accepting a process that is operating at an unsatisfactory level (type II error or β-risk).

There are two ways to design the control chart. In the first approach, we design the control chart based on a specified n and a process fraction nonconforming γ that we would like to reject with probability $1 - \beta$. In this case, the control limits for the chart are

$$\begin{aligned} \text{UCL} &= \mu_U - \frac{Z_\beta \sigma}{\sqrt{n}} \\ &= \text{USL} - Z_\gamma \sigma - \frac{Z_\beta \sigma}{\sqrt{n}} \\ &= \text{USL} - \left(Z_\gamma + \frac{Z_\beta}{\sqrt{n}} \right) \sigma \end{aligned} \tag{10.5a}$$

and

$$\begin{aligned} \text{LCL} &= \mu_L + \frac{Z_\beta \sigma}{\sqrt{n}} \\ &= \text{LSL} + Z_\gamma \sigma + \frac{Z_\beta \sigma}{\sqrt{n}} \\ &= \text{LSL} + \left(Z_\gamma + \frac{Z_\beta}{\sqrt{n}} \right) \sigma \end{aligned} \tag{10.5b}$$

Note that when n, γ, and $1 - \beta$ (or β) are specified, the control limits are inside the μ_L and μ_U values that produce the fraction nonconforming γ. In contrast, when n, δ, and α are specified, the lower control limit falls between μ_L and LSL and the upper control limit falls between μ_U and USL.

It is also possible to choose a sample size for an acceptance control chart so that specified values of δ, α, γ, and β are obtained. By equating the upper control limits (say) for a specified δ and α [equation (10.3a)] and a specified γ and β [equation (10.5a)], we obtain

$$\text{USL} - \left(Z_\delta - \frac{Z_\alpha}{\sqrt{n}} \right) \sigma = \text{USL} - \left(Z_\gamma + \frac{Z_\beta}{\sqrt{n}} \right) \sigma$$

Therefore, a sample size of

$$n = \left(\frac{Z_\alpha + Z_\beta}{Z_\delta - Z_\gamma} \right)^2$$

will yield the required values of δ, α, γ, and β. For example, if $\delta = 0.01$, $\alpha = 0.00135$, $\gamma = 0.05$, and $\beta = 0.20$, we must use a sample of size

$$n = \left(\frac{Z_{0.00135} + Z_{0.20}}{Z_{0.01} - Z_{0.05}} \right)^2 = \left(\frac{3.00 + 0.84}{2.33 - 1.645} \right)^2 = 31.43 \simeq 32$$

on the acceptance control chart. Obviously, to use this approach, n must not be seriously restricted by cost or other factors such as rational subgrouping considerations.

10.3 Control Charts for Multiple-Stream Processes

10.3.1 Multiple-Stream Processes

A **multiple-stream process (MSP)** is a process with data at a point in time consisting of measurements from several individual sources or streams. When the process is in control, the sources or streams are assumed to be identical. Another characteristic of the MSP is that we can monitor and adjust each of the streams individually or in small groups.

The MSP occurs often in practice. For example, a machine may have several heads, with each head producing (we hope) identical units of product. In such situations several possible control procedures may be followed. One possibility is to use separate control charts on each stream. This approach usually results in a prohibitively large number of control charts. If the output streams are highly correlated—say, nearly perfectly correlated—then control charts on only one stream may be adequate. The most common situation is that the streams are only moderately correlated, so monitoring only one of the streams is not appropriate. There are at least two types of situations involving the occurrence of assignable causes in the MSP.

1. The output of one stream (or a few streams) has shifted off target.
2. The output of all streams has shifted off target.

In the first case, we are trying to detect an assignable cause that affects only one stream (or at most a few streams), whereas in the second, we are looking for an assignable cause that impacts *all* streams (such as would result from a change in raw materials).

The standard control chart for the MSP is the group control chart, introduced by Boyd (1950). We will also discuss other approaches to monitoring the MSP. Throughout this section we assume that the process has s streams and that the output quality characteristic from each stream is normally distributed.

10.3.2 Group Control Charts

The **group control chart (GCC)** was introduced by Boyd (1950) and remains the basic procedure for monitoring an MSP. To illustrate the methods of construction and use, suppose that the process has $s = 6$ streams and that each stream has the same target value and inherent variability. Variables measurement is made on the items produced, and the distribution of the measurement is well approximated by the normal. To establish a group control chart, the sampling is performed as if separate control charts were to be set up on each stream. Suppose, for purposes of illustration, that a sample size of $n = 4$ is used. This means that 4 units will be taken from each of the $s = 6$ streams over a short period of time. This will be repeated until about 20 such groups of samples have been taken. At this point we would have $20 \times 6 = 120$ averages of $n = 4$ observations each and 120 corresponding ranges. These averages and ranges would be averaged to produce a grand average $\bar{\bar{x}}$ and an average range $\bar{R}$. The limits on the group control charts would be at

$$\text{UCL} = \bar{\bar{x}} + A_2\bar{R}$$

$$\text{LCL} = \bar{\bar{x}} - A_2\bar{R}$$

for the $\bar{x}$ chart and at

$$\text{UCL} = D_4\bar{R}$$

$$\text{LCL} = D_3\bar{R}$$

for the R chart, with $A_2 = 0.729$, $D_3 = 0$, and $D_4 = 2.282$. Note that the sample size $n = 4$ determines the control chart constants.

When the group control chart is used to control the process, we plot only the largest and smallest of the $s = 6$ means observed at any time period on the $\bar{x}$ chart. If these means are inside the control limits, then all other means will also lie inside the limits. Similarly, only the largest range will be plotted on the range chart. Each plotted point is identified on the chart by the number of the stream that produced it. The process is out of control if a point exceeds a three-sigma limit. Runs tests cannot be applied to these charts, because the conventional runs tests were not developed to test averages or ranges that are the extremes of a group of averages or ranges.

It is useful to examine the stream numbers on the chart. In general, if a stream consistently gives the largest (or smallest) value several times in a row, that may constitute evidence that this stream is different from the others. If the process has s streams and if r is the number of consecutive times that a particular stream is the largest (or smallest) value, then the one-sided in-control average run length for this event is given by Nelson (1986) as

$$\text{ARL}(1)_0 = \frac{s^r - 1}{s - 1} \tag{10.6}$$

if all streams are identical. To illustrate the use of this equation, if $s = 6$ and $r = 4$, then

$$\text{ARL}(1)_0 = \frac{6^4 - 1}{6 - 1} = 259$$

That is, if the process is in control, we will expect to see the same stream producing an extreme value on the group control chart four times in a row only once every 259 samples.

One way to select the value of r to detect the presence of one stream that is different from the others is to use equation (10.6) to find an ARL that is roughly consistent with the ARL of a conventional control chart. The ARL for an in-control process for a single point beyond the upper control limit (say) is 740. Thus, using $r = 4$ for a six-stream process results in an ARL that is too short and that will give too many false alarms. A better choice is $r = 5$, since

$$\text{ARL}(1)_0 = \frac{6^5 - 1}{6 - 1} = 1555$$

Thus, if we have six streams and if the same stream produces an extreme value on the control chart in five consecutive samples, then we have strong evidence that this stream is different from the others.

Using equation (10.6), we can generate some general guidelines for choosing r given the number of streams s. Suitable pairs (s, r) would include (3, 7), (4, 6), (5–6, 5), and (7–10, 4). All of these combinations would give reasonable values of the one-sided ARL when the process is in control.

The two-sided in-control average run length $\text{ARL}(2)_0$ is defined as the expected number of trials until r consecutive largest or r consecutive smallest means come from the same stream while the MSP is in control. Mortell and Runger (1995) and Nelson and Stephenson (1996) used the *Markov chain approach* of Brook and Evans (1972) to compute $\text{ARL}(2)_0$ numerically. A closed-form expression is not possible for $\text{ARL}(2)_0$, but Nelson and Stephenson (1996) give a lower bound on $\text{ARL}(2)_0$ as

$$\text{ARL}(2)_0 = \frac{s^r - 1}{2(r - 1)} \tag{10.7}$$

This approximation agrees closely with the numerical computations from the Markov chain approach.

There are some drawbacks connected with the GCC. These may be summarized as follows:

1. At each point in time where samples are taken, all s streams must be sampled. When s is small, this may be a satisfactory approach. However, some high-speed filling and packaging machines may have $100 \leq s \leq 200$ streams. The GCC is not practical for these situations.
2. There is no information about nonextreme streams at each trial, thus, we cannot utilize past values to form an EWMA or a cusum statistic to improve on the GCC performance.
3. The supplementary run tests fail if more than one stream shifts simultaneously (all up or all down) to the same off-target level, because the shifted streams will most likely alternate having the extreme value, and no single stream shift can be identified.
4. The selection of the run or pattern length r for a specific problem with s streams is a discrete process, and the difference in ARL_0 for $r = r^*$ or $r = r^* + 1$ choice can be substantial, as observed by Mortell and Runger (1995). This problem is evident in the original run rule devised by Nelson (1986), but can be alleviated somewhat by using the modified run rules in Nelson and Stephenson (1996), which are based on an $r - 1$ out of r in a row approach. In fact, it can be easily shown that the difference in the ARL_0 between $r = r^*$ and $r = r^* + 1$ choices equals s^{r^*}, which becomes very large for problems with large numbers of streams.
5. Although using an $r - 1$ out of r in a row run rule allows some flexibility in the choice of the critical value $r = r^*$ and ARL_0, it adds some complexity since it is no longer a simple counting procedure.

10.3.3 Other Approaches

Mortell and Runger (1995) proposed the following model for the MSP to accommodate the practical case of dependent (cross-correlated) streams:

$$x_{tjk} = \mu + A_t + \varepsilon_{tjk} \tag{10.8}$$

where x_{tjk} is the kth measurement on the jth stream at time t, $A_t \sim N(0,\sigma_a^2)$ represents the difference of the mean over all the streams at time t from the MSP target μ, and ε_{tjk} is a $N(0, \sigma^2)$ random variable that represents the difference of the kth measurements on the jth stream from the process mean over all streams at time t. In this representation of MSP data, the total variation is allocated into two sources: σ_a^2 accounting for the variation over time common to all streams, and σ^2 accounting for the variation between the streams at specific time t. From this model, the cross correlation between any pair of streams is given by $\rho = \sigma_a^2/(\sigma_a^2 + \sigma^2)$.

Mortell and Runger proposed monitoring the average at time t of the means across all streams with an individuals control chart to detect an overall assignable cause. Also, they proposed monitoring the range of the stream's means at time t denoted by $R_t = \max(\bar{x}_{tj}) - \min(\bar{x}_{tj})$ or the maximum ***residual*** at time t given by $\max(\bar{x}_{ij}) - \bar{x}_t$, with any individuals control chart to detect a relative assignable cause affecting one or a few streams. These proposed **control charts on residuals** are better than the GCC, especially when the variation in the process means over time is greater than the between-stream variability (i.e., when $\sigma_a > \sigma$). Mortell and Runger compared the performance of these proposed charts with the runs test of the GCC by Nelson (1986) using simulations of various MSPs and concluded that the cusum chart on the range outperforms the Shewhart and EWMA on either the range or the maximum residual. On the other hand, little difference in performance exists between the runs test of Nelson (1986)

and the proposed cusum chart, but it is preferred because of the flexibility in the choice of in-control ARL.

Runger, Alt, and Montgomery (1996a) related control charts for the MSP with correlated data to principal components analysis. (See Chapter 11 for information on principal components.) The first control chart monitors the major principal component variable, which equals the mean of the streams at any time t. The second control chart is based on the Hotelling T^2 statistic (see Chapter 11) of the last $p - 1$ principal components, which is shown to be equivalent to the s^2 chart; thus it is sensitive to relative assignable causes that affect the uniformity across the streams.

Lanning (1998) studied the use of "fractional" and "adaptive" sampling in the multiple-stream process with a large number of identical and independent streams, where simultaneous monitoring of all streams is impractical. In his procedures, only a fraction of the s streams are sampled at each time period, and this fraction is increased when there are indications that an out-of-control condition may be present. (Adaptive sampling schemes are introduced in Section 10.5.) He used the average time to signal (ATS) to compare performance of the competing charts, and focused on the scenarios where assignable causes tend to impact all or most of the process streams together. He concluded that the adaptive fraction approach gives better ATS results than the fixed fraction scheme, and often yields satisfactory results compared to a complete sampling situation. An application of his methodology to a multiple-head filling machine with many filling valves is described in Lanning, Montgomery, and Runger (2002–03).

10.4 SPC with Autocorrelated Process Data

10.4.1 Sources and Effects of Autocorrelation in Process Data

The standard assumptions that are usually cited in justifying the use of control charts are that the data generated by the process when it is in control are normally and independently distributed with mean μ and standard deviation σ. Both μ and σ are considered fixed and unknown. An out-of-control condition is a change or shift in μ or σ (or both) to some different value. Therefore, we could say that when the process is in control the quality characteristic at time t, x_t, is represented by the model

$$x_t = \mu + \varepsilon_t \quad t = 1, 2, \ldots \tag{10.9}$$

where ε_t is normally and independently distributed with mean zero and standard deviation σ. This is often called the **Shewhart process model.**

When these assumptions are satisfied, one may apply conventional control charts and draw conclusions about the state of statistical control of the process. Furthermore, the statistical properties of the control chart, such as the false alarm rate with three-sigma control limits, or the average run length, can be easily determined and used to provide guidance for chart interpretation. Even in situations in which the normality assumption is violated to a slight or moderate degree, these control charts will still work reasonably well.

The most important of the assumptions made concerning control charts is that of independence of the observations, for conventional control charts do not work well if the quality characteristic exhibits even low levels of correlation over time. Specifically, these control charts will give misleading results in the form of too many false alarms if the data are positively correlated. This point has been made by numerous authors, including Berthouex, Hunter, and Pallesen (1978), Alwan and Roberts (1988), Montgomery and Friedman (1989), Alwan (1992), Harris and Ross (1991), Montgomery and Mastrangelo (1991), and Maragah and Woodall (1992).

Unfortunately, the assumption of uncorrelated or independent observations is not even approximately satisfied in some manufacturing processes. Examples include chemical processes

10.10 Control Charts in Health Care Monitoring and Public Health Surveillance

Control charts have many applications in health care and public health monitoring and surveillance. Control charts can be used for monitoring hospital performance with respect to patient infection rates, patient falls or accidents, emergency room waiting times, or surgical outcomes for various types of procedures. There are also applications for monitoring potential outbreaks of infectious diseases, and even bioterrorism events. This is often called *syndromic surveillance,* in which data are obtained from sources such as emergency room records, over-the-counter drug and prescription sales, visits to physicians, and other sources, in an attempt to supplement traditional sentinel surveillance for natural disease outbreaks or bioterrorist attacks. An excellent review paper on control chart applications in public health surveillance and health care monitoring is Woodall (2006). The book by Winkel and Zhang is also recommended. The papers by Fienberg and Shmueli (2005), Buckeridge et al. (2005), Fricker (2007), and Rolka et al. (2007) offer useful insight on syndromic surveillance.

Health Care Applications of Control Charts. There are some important differences between the industrial and business settings where statistical process control and control charts are traditionally applied and the health care and disease-monitoring environments. One of these is data. Attribute data are much more widely encountered in the health care environment than in the industrial and business world. For example, in surveillance for outbreaks of a particular disease, the incidence rate, or number of cases in the population of interest per unit of time, is typically monitored. Furthermore, disease outbreaks are very likely to be transitory, with more gradual increases and decreases as opposed to the distinct shifts usually encountered in industrial applications. In the health care and public health environment, one-sided methods tend to be employed to allow a focus on detecting increases in rates, which are of primary concern. Two-sided methods are more typically used in business and industry. Some of the techniques used in health care monitoring and disease surveillance were developed independently of the industrial statistical process-control field. There have been few comparative studies of the methods unique to health care monitoring and disease surveillance and often the performance measures used are different from those employed in industrial APC. Often the data monitored in these environments is nonstationary, or has some built-in patterns that are part of the process. An example would be influenza outbreaks, which are largely confined to the winter and late spring months. In some public health surveillance applications, the parameter estimates and measures of uncertainty are updated as new observations become available or the baseline performance is adjusted. These are methods to account for the nonstationary nature of the monitored data.

Scan Methods. The public health surveillance community often uses scan statistic methods instead of more conventional control charting methods such as the cumulative sum or EWMA control chart. A scan statistic is a moving window approach that is similar to the moving average control chart, discussed in Chapter 9 (see Section 9.3). A scan method signals an increase in a rate if the count of events in the most recent specified number of time periods is unusually large. For example, a scan method would signal an increased rate at a given time if m or more events of interest have occurred in the most recent n trials. A plot of the scan statistic over time with an upper control limit could be considered as a control charting technique, but this viewpoint is not usually taken in the health care or the disease surveillance filed. In general, there are no specific guidelines on the design of a scan method.

Scan methods were first applied to the surveillance of chronic diseases by the cancer surveillance community and have more recently been adapted to the surveillance of infectious diseases. Scan methods can be applied in either the temporal case, where only the times of incidences are known, or in the spatial-temporal case, where both the times and locations of incidences are known. Most of the work on scan-based methods has been for the phase I situation in which

a set of historical data is analyzed. Comprehensive reviews of many scan-based procedures are in Balakrishnan and Koutras (2002) and Glaz et al. (2001). Also see Kulldorff (1997), (2001), (2003), (2005), Sonesson and Bock (2003), and Naus and Wallenstein (2006).

An important feature of health care monitoring is the human element of the process. Differences between the people in the system are likely to be great, and these differences have great potential impact on the outcomes. For example, when evaluating hospital performance and physician care, we have to take into account that the patients will vary (from patient to patient, from hospital to hospital, from physician to physician) with regard to their general state of health and other demographics and factors. Risk-adjusted control charts can be devised to deal with this situation. Grigg and Farewell (2004) give a review of risk-adjusted monitoring. Grigg and Spiegelhalter (2007) recently proposed a risk-adjusted EWMA method. Risk adjusted control charts are also discussed by Winkel and Zhang (2007).

10.11 Overview of Other Procedures

There are many other process monitoring and control techniques in addition to those presented previously. This section gives a brief overview of some of these methods, along with some basic references. The selection of topics is far from exhaustive but does reflect a collection of ideas that have found practical application.

10.11.1 Tool Wear

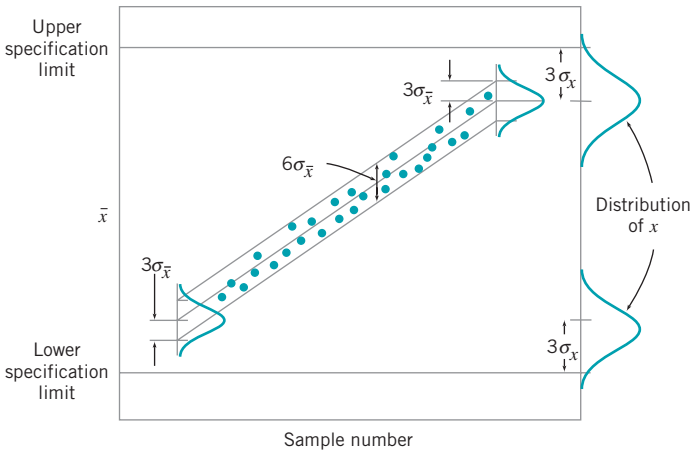
Many production processes are subject to tool wear. When tool wear occurs, we usually find that the process variability at any one point in time is considerably less than the allowable variability over the entire life of the tool. Furthermore, as the tool wears out, there will generally be an upward drift or trend in the mean caused by the worn tool producing larger dimensions. In such cases, the distance between specification limits is generally much greater than, say, 6σ. Consequently, the modified control chart concept can be applied to the tool-wear problem. The procedure is illustrated in Fig. 10.29.

The initial setting for the tool is at some multiple of σ_x above the lower specification limit—say, $3\sigma_x$—and the maximum permissible process average is at the same multiple of σ_x below the upper specification limit. If the rate of wear is known or can be estimated from the data, we can construct a set of slanting control limits about the tool-wear trend line. If the sample values of $\bar{x}$ fall within these limits, the tool wear is in control. When the trend line exceeds the maximum permissible process average, the process should be reset or the tool replaced.

Control charts for tool wear are discussed in more detail by Duncan (1986) and Manuele (1945). The regression control chart [see Mandel (1969)] can also be adapted to the tool-wear problem. Quesenberry (1988) points out that these approaches essentially assume that resetting the process is expensive and that they attempt to minimize the number of adjustments made to keep the parts within specifications rather than reducing overall variability. Quesenberry develops a two-part tool-wear compensator that centers the process periodically and protects against assignable causes, as well as adjusting for the estimated mean tool wear since the previous adjustment.

10.11.2 Control Charts Based on Other Sample Statistics

Some authors have suggested the use of sample statistics other than the average and range (or standard deviation) for construction of control charts. For example, Ferrell (1953) proposed that subgroup midranges and ranges be used, with control limits determined by the median midrange and the median range. The author noted that ease of computation would be



■ **FIGURE 10.29** Control chart for tool wear.

a feature of such control charts and that they would do a better job of detecting "outlier" points than conventional control charts. The median has been used frequently instead of $\bar{x}$ as a center line on charts of individuals when the underlying distribution is skewed. Similarly, medians of R and s have been proposed as the center lines of those charts so that the asymmetrical distribution of these statistics will not influence the number of runs above and below the center line.

The recent interest in robust statistical methods has generated some application of these ideas to control charts. Generally speaking, the presence of assignable causes produces outlier values that stretch or extend the control limits, thereby reducing the sensitivity of the control chart. One approach to this problem has been to develop control charts using statistics that are themselves outlier-resistant. Examples include the median and midrange control charts [see Clifford (1959)] and plotting subgroup boxplots [(see Inglewitz and Hoaglin (1987) and White and Schroeder (1987)]. These procedures are typically not as effective in assignable-cause or outlier detection as are conventional $\bar{x}$ and R (or S) charts.

A better approach is to plot a sample statistic that is sensitive to assignable causes ($\bar{x}$ and R or s), but to base the control limits on some outlier-resistant method. The paper by Ferrell (1953) mentioned above is an example of this approach, as is plotting $\bar{x}$ and R on charts with control limits determined by the trimmed mean of the sample means and the trimmed mean of the ranges, as suggested by Langenberg and Inglewitz (1986).

Rocke (1989) has reported that plotting an outlier-sensitive statistic on a control chart with control limits determined using an outlier-resistant method works well in practice. The suggested procedures in Ferrell (1953), Langenberg and Inglewitz (1986), and his own method are very effective in detecting assignable causes. Interestingly enough, Rocke also notes that the widely used two-stage method of setting control limits, wherein the initial limits are treated as trial control limits, the samples that plot outside these trial limits are then removed, and a final set of limits are then calculated, performs nearly as well as the more complex robust methods. In other words, the use of this two-stage method creates a robust control chart.

In addition to issues of robustness, other authors have suggested control charts for other sample statistics for process-specific reasons. For example, when pairs of measurements are made on each unit or when comparison with a standard unit is desired, one may plot the difference $x_{1j} - x_{2j}$ on a **difference control chart** [see Grubbs (1946) and the supplemental text material]. In some cases, the largest and smallest sample values may be of interest. These charts have been developed by Howell (1949).

10.11.3 Fill Control Problems

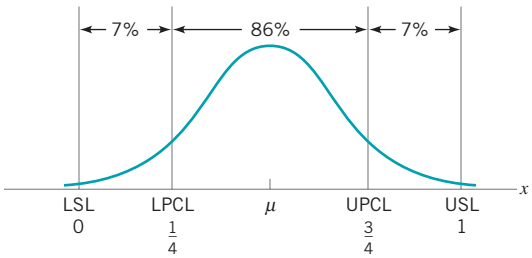
Many products are filled on a high-speed, multiple-head circular filling machine that operates continuously. It is not unusual to find machines in the beverage industry that have from 40 to 72 heads and operate at speeds of from 800 to 1000 bottles per minute. In such cases, it is difficult to sample products from specific heads because there is no automatic method of identifying a filled container with its filling head. Furthermore, in addition to assignable causes that affect all filling heads simultaneously, some assignable causes affect only certain heads. Special sampling and control charting methods are needed for these types of fill control problems. Ott and Snee (1973) present some techniques useful for this problem, particularly for filling machines with a moderate number of heads. Many of the methods described in Section 10.3 for multiple-stream processes are also useful in fill control problems.

10.11.4 Precontrol

Precontrol is a technique that is used to detect shifts or upsets in the process that may result in the production of nonconforming units. The technique differs from statistical process control in that conventional control charts are designed to detect shifts in process parameters that are statistically significant, and precontrol requires no plotting of graphs and no computations. Precontrol uses the normal distribution in determining changes in the process mean or standard deviation that could result in increased production of nonconforming units. Only three units are required to give control information.

To demonstrate the procedure, suppose that the quality characteristic is normally distributed and that the natural tolerance limits ($\mu \pm 3\sigma$) exactly coincide with the specification limits. Furthermore, the process average μ is halfway between the specifications, so that the process is producing 0.27% fallout. Construct two precontrol limits (called upper and lower PC lines), each of which is one-fourth the distance in from the modified limit, as in Fig. 10.30. Since the distribution of the quality characteristic is normal, approximately 86% of the process output will lie inside the PC lines, and approximately 7% will lie in each of the regions between the PC line and specification limit. This means that only about one item in 14 will fall outside a PC line if the process mean and standard deviation are on the target values.

If the probability is one-fourteenth that one unit falls outside a PC line, the probability that two consecutive units fall outside a PC line is $\left(\frac{1}{14}\right)\left(\frac{1}{14}\right) = \frac{1}{196} = \frac{1}{200}$. That is, if the process is operating correctly, the probability of finding two consecutive units outside of a given PC line is only about $\frac{1}{200}$. When two such consecutive units are found, it is likely that the process has shifted to an out-of-control state. Similarly, it is unlikely to find the first unit beyond one PC line and the next beyond the other PC line. In this case, we suspect that process variability has increased.



■ FIGURE 10.30 Location of precontrol lines.

A set of rules follow that describe the operation of precontrol. These rules assume that 1% to 3% nonconforming production is acceptable and that the process capability ratio is at least 1.15.

1. Start the process. If the first item is outside specifications, reset and start again.
2. If an item is inside specifications but outside a PC line, check the next item.
3. If the second item is outside the same PC line, reset the process.
4. If the second item is inside the PC line, continue. The process is reset only when two consecutive items are outside a given PC line.
5. If one item is outside a PC line and the next item is outside the other PC line, the process variability is out of control.
6. When five consecutive units are inside the PC lines, shift to frequency gauging.
7. When frequency gauging, do not adjust process until an item exceeds a PC line. Then examine the next consecutive item, and proceed as in step 4.
8. When the process is reset, five consecutive items must fall inside the PC lines before frequency gauging can be resumed.
9. If the operator samples from the process more than 25 times without having to reset the process, reduce the gauging frequency so that more units are manufactured between samples. If we must reset before 25 samples are taken, increase the gauging frequency. An average of 25 samples to a reset indicates that the sampling frequency is satisfactory.

Precontrol is closely related to a technique called **narrow-limit gauging** (or **compressed-limit gauging**), in which inspection procedures are determined using tightened limits located so as to meet established risks of accepting nonconforming product. Narrow-limit gauging is discussed in more general terms by Ott (1975).

Although precontrol has the advantage of simplicity, it should not be used indiscriminately. The procedure has serious drawbacks. First, because no control chart is usually constructed, all the aspects of pattern recognition associated with the control chart cannot be used. Thus, the diagnostic information about the process contained in the pattern of points on the control chart, along with the logbook aspect of the chart, is lost. Second, the small sample sizes greatly reduce the ability of the procedure to detect even moderate-to-large shifts. Third, precontrol does not provide information that is helpful in bringing the process into control or that would be helpful in reducing variability (which is the goal of statistical process control). Finally, the assumption of an in-control process and adequate process capability is extremely important. Precontrol should only be considered in manufacturing processes where the process capability ratio is much greater than one (perhaps at least two or three), and where a near-zero defects environment has been achieved. Ledolter and Swersey (1997) in a comprehensive analysis of precontrol also observe that its use will likely lead to unnecessary tampering with the process; this can actually increase variability. This author believes that precontrol is a poor substitute for standard control charts and would never recommend it in practice.

10.11.5 Tolerance Interval Control Charts

The usual intent of standard control limits for phase II control charts based on normal distribution is that they contain six standard deviations of the distribution of the statistic that is plotted on the chart. For the normal distribution, this is the familiar quantity $0.9973 = 1 - 2(0.00135)$, where 0.00135 is the probability outside each control limit or the area in the tail of the distribution of the plotted statistic. Standard control limits do not satisfy this because while a normal distribution is assumed, many plotted statistics do not follow the normal distribution. The sample range R and sample standard deviation s are good examples. The

random variables R and s have skewed distributions that lead to zero lower control limits when the subgroup size is six or less for the R chart and five or less for the s chart. Using probability limits would avoid this problem. However, there is another problem. The phase II control limits also depend on the parameters of the distribution that are estimated using the data that was collected and analyzed in phase I.

Hamada (2003) proposes the use of beta-content tolerance intervals as the basis for control limits to more precisely control the probability content of the control limits. He develops these control limits for the $\bar{x}$, R, and s charts, and he provides tables of the constants necessary for their construction and use. His constants control the probability content in each tail at 0.000135, but the formulas provided allow different probability values and potentially different probability contents in each tail. A useful benefit of his approach is that nonzero lower control limits arise naturally for the R and s charts, affording an opportunity to detect downward shifts in process variability more easily.

10.11.6 Monitoring Processes with Censored Data

There are many situations where a process is monitored and the data that is collected is **censored.** For example, Steiner and MacKay (2000) describe manufacturing material used in making automobile interiors. In this process, a vinyl outer layer is bonded to a foam backing; to check the bond strength, a sample of the material is cut and the force required to break the bond is measured. A predetermined maximum force is applied, and most samples do not fail. Therefore, the bond strength measurements are censored data.

When the censoring proportion is not large, say, under 50%, it is fairly standard practice to ignore the censoring. When the censoring proportion is high, say above 95%, it is often possible to use an np control chart to monitor the number of censored observations. However, there are many situations where the proportion of censored observations is between 50% and 95% and these conventional approaches do not apply. Steiner and MacKay (2000) develop conditional expected value (CEV) weight control charts for this problem. To develop the charts, they assume that the measured quantity x is normally distributed with mean and standard deviation μ and σ, respectively. The censoring level is C; that is, the measurement value is not observed exactly if x exceeds C. The probability of censoring is

$$p_c = 1 - \Phi\left(\frac{C-\mu}{\sigma}\right) = Q(C)$$

where Φ is the standard normal cumulative distribution function and $Q(C)$ is called the **survivor function** of the distribution. Their CEV control charts simply replace each censored observation with its conditional expected value. Then the subgroup averages and standard deviations are calculated and plotted in the usual manner. The conditional expected value of a censored observation is

$$w_c = E(x|x > C) = \mu + \sigma\left(\frac{\phi(z_c)}{\Phi(z_c)}\right)$$

where $\phi(z_c)$ is the standard normal density function, and $z_c = (C-\mu)/\sigma$. Therefore, the actual data used to construct the control charts is

$$w = \begin{cases} x, & \text{if } x \le C \\ w_c, & \text{if } x > C \end{cases}$$

The control limits for the CEV control charts depend on the sample size and the censoring proportion p_c. The relationship is complex, and does not lead to a simple rule such as the use

of three-sigma limits. Instead, graphs provided by Steiner and Mackay (2000) that are based on simulation must be used to determine the control limits. Furthermore, in application the mean and standard deviation μ and σ must be estimated. This is done in phase I using preliminary process data. An iterative maximum likelihood procedure is provided by the authors for doing this.

10.11.7 Nonparametric Control Charts

The performance of most control charts depends on the assumption of a particular probability distribution as a model for the process data. The normal distribution is usually assumed, and almost all of the performance analysis reported in the literature assumes that the observations are drawn from a normal distribution. However, much process data is *not* normally distributed and so the robustness of control charts to this assumption has long been an issue in SPC. The robustness of the Shewhart $\bar{x}$ chart to normality has been studied very thoroughly (refer to Section 6.2.5). Researchers have found that some control charts are significantly affected by nonnormality; for example, the Shewhart control chart for individuals.

Because nonnormality can affect control chart performance, some authors have developed nonparametric control charts that do not depend on normality or any other specific distributional assumption. Most nonparametric statistical process-control (NSPC) techniques depend on ranks. Procedures have been developed that are alternatives to many of the standard Shewhart charts, the cusum, and the EWMA. The empirical reference distribution control charts of Willemain and Runger (1996), which are based on order statistics from a large reference sample, are also a form of nonparametric control chart. There has also been some work done on multivariate NSPC. A good review of much of the literature in this area is in Chakraborti, Van Der Laan, and Bakir (2001). There is little evidence that these NSPC techniques have gained any real acceptance in practice. As we have noted previously, a properly designed univariate (and as we will subsequently see, a multivariate) EWMA control chart is very robust to the assumption of normality, and performs quite well for both heavy-tailed symmetric distributions and skewed distributions. Because the univariate EWMA is widely available in standard statistics packages, and the multivariate EWMA discussed in Chapter 11 is also very easy to implement, these charts would seem to be good alternatives to NSPC in many situations.

Important Terms and Concepts

Acceptance control charts
Adaptive (SPC) control charts
Autocorrelation function
Autoregressive integrated moving average (ARIMA) models
Autocorrelated process data
Average run length
Changepoint model for process monitoring
Control charts on residuals
Cuscore statistics and control charts
Deviation from nominal (DNOM) control charts
Economic models of control charts
First-order autoregressive model
First-order integrated moving average model
First-order mixed model
First-order moving average model
Group control charts (GCC)
Health care applications of control charts
Impact of autocorrelation on control charts
Modified control charts
Multiple-stream processes
Positive autocorrelation
Profile monitoring
Sample autocorrelation function
Second-order autoregressive model
Shewhart process model
Standardized $\bar{x}$ and R control charts
Time series model
Unweighted batch means (UBM) control charts